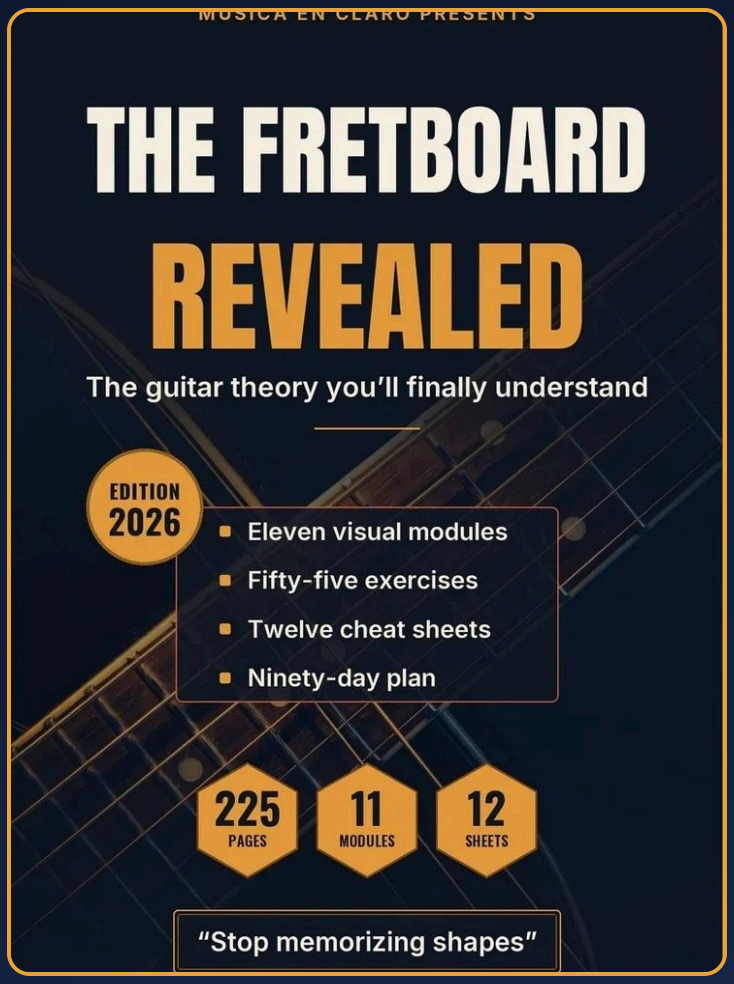
GUITAR THEORY FOR SELF-TAUGHT PLAYERS

The Fretboard Revealed

The guitar theory you will finally understand



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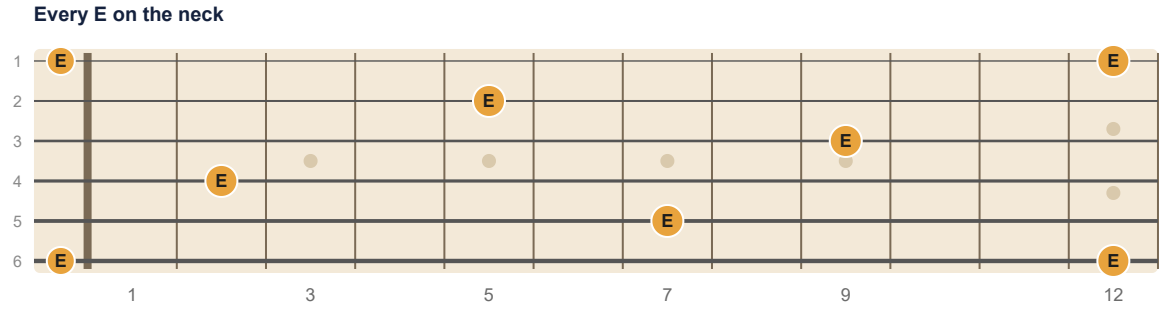
Welcome

> *You already play guitar. You know some chords, you can get through a few songs, and you have probably noticed that something is missing: you play shapes, but you cannot see why they work. This book closes that gap, one diagram at a time.*

Who this book is for

This book is for players who already play. Maybe you learned from videos, from tab sites, from a friend, or from years of playing along with records. You are self-taught, and that is a strength: your ears and your hands are ahead of your knowledge. Theory is the missing map, and with a map you stop memorizing shapes and start understanding the neck.

- You can already play open chords and switch between them without stopping the beat.
- You want to name the notes you are playing, build your own chords and scales, and know where to go next.
- You play acoustic or electric. Everything here works on both.
- You do not read music, and you will not need to.



One note, many homes. The same E lives in six places across the first 12 frets. By Module 1, you will find any note like this in seconds, without memorizing the whole neck.

What this book will not teach

To keep the book focused, we start where most beginner courses stop. This book will not teach you how to hold the guitar, how to hold a pick, or your first chords. If those are still new, spend a few weeks with a beginner course first and come back. Everything else is explained from the ground up, in plain English, with no assumed knowledge of theory.

It also will not ask you to read a single line of standard notation. There are no staffs and no note heads in these pages. Notes are named with letters, scale steps are named with numbers, and everything is drawn as dots on a picture of your fretboard.

Why theory felt hard

Most theory is explained for pianists. On a piano, every note has exactly one key, and the white and black keys draw the pattern of the scale right in front of you. Teachers move to that picture without thinking,

and the guitarist is left translating from a keyboard they do not own.

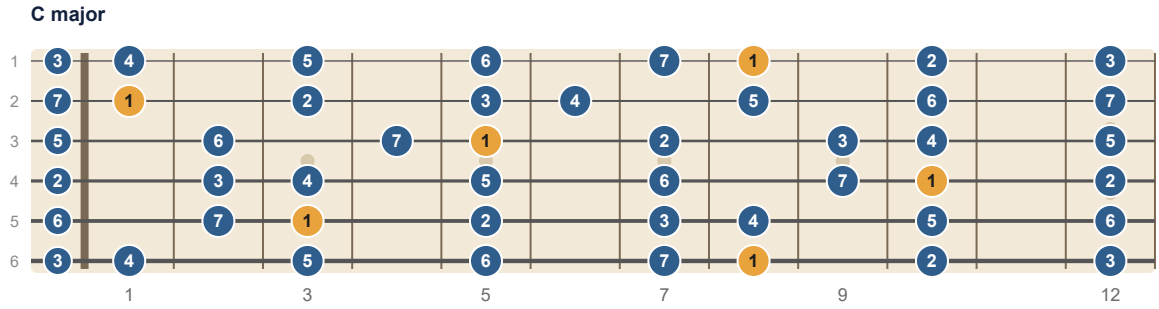
The guitar is built differently. The same note appears in several places, and the strings are tuned mostly in fourths, so the neck looks like a maze. But that design hides a great advantage: patterns on the guitar are shapes, and a shape can slide. Learn a scale or a chord shape once, and you can move it up the neck to change the key. A pianist has to learn every key separately. You will learn a handful of shapes and a few simple rules, and then use them everywhere.

> **The good news**
>
> Most players find theory easier on guitar than they expected once it is drawn on the fretboard, in their own tuning, with their own hands doing the work.

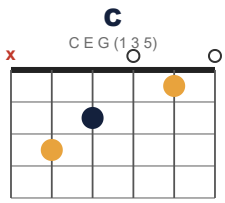
The three principles

Every page follows the same three ideas. They are the opposite of classroom theory.

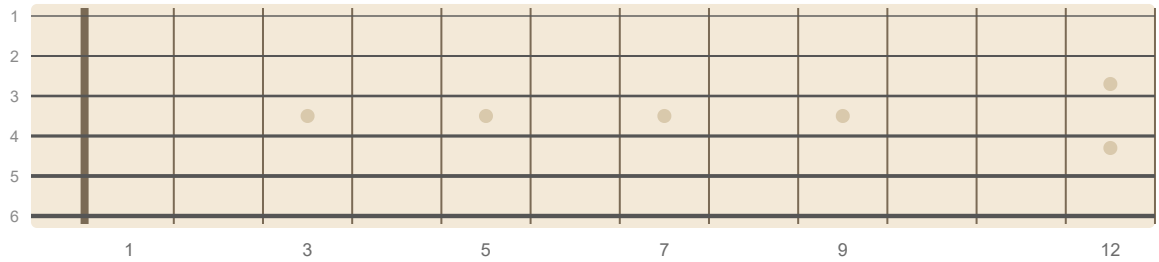
1. **See before you read.** Each idea starts with a picture on the fretboard. Look first, then read the explanation.
2. **Apply on the same page.** Each idea is followed by something to play right away, so the concept lands in your hands while it is fresh.
3. **Check that you have understood.** Every exercise ends with a self-check, a question or a small task that tells you honestly whether you got it, so you never move on guessing.



See before you read. This is the C major scale across the neck, with each note shown as its scale step. Notice the repeating pattern before you know any rule behind it.



Apply on the same page. An open C chord you already know. In Module 5 you will see that its three notes are the 1, 3 and 5 of the C major scale above: C, E and G.



Check that you have understood. Blank necks like this one appear in the exercises. You fill them in, and the answer is on the same page.

How the book is organized

The book has four parts and eleven modules. Each part answers one question, and each module builds on the one before.

PART	QUESTION IT ANSWERS	MODULES
1. SEE	Where is everything on the neck, and how far apart are the notes?	01 The fretboard stops being a maze; 02 Intervals: distance is emotion
2. BUILD	How are scales, modes and chords made from those distances?	03 Scales and keys: the master rule; 04 The modes, without the mysticism; 05 Chords: where they really come from; 06 Sevenths, tensions and color
3. CONNECT	How do the pieces link into whole songs and the whole neck?	07 CAGED: the whole fretboard opens up; 08 Progressions: why songs work; 09 The circle of fifths and modulation
4. CREATE	How do I use all this to play something of my own?	10 Improvising with intention; 11 Working songs out by ear and writing your own

Module 7, CAGED, is the heart of the book. It is the longest module at 22 pages, and it ties everything before it to a single picture of the whole fretboard.

The anatomy of a module

Every module has the same five parts, so you always know where you are.

1. **The cover with a snapshot.** A one-page card that tells you what the module is about and what you will be able to do by the end.
2. **Color fretboard diagrams.** The ideas, drawn on the neck.
3. **A cheat sheet.** One page that holds the module on a single sheet. Print it or keep it on your music stand.
4. **A glossary in plain words.** The new terms of the module, explained without jargon.
5. **Five exercises, On the fretboard.** Each one has steps, a time estimate, and a self-check.

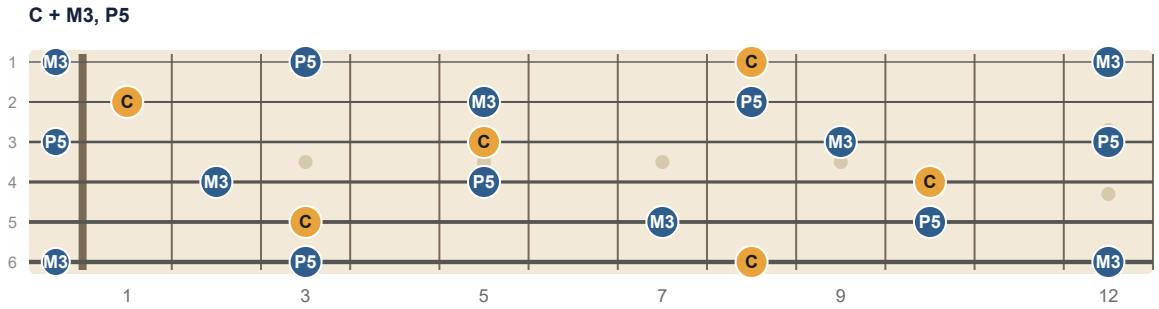
MODULE 3

Scales and keys: the master rule

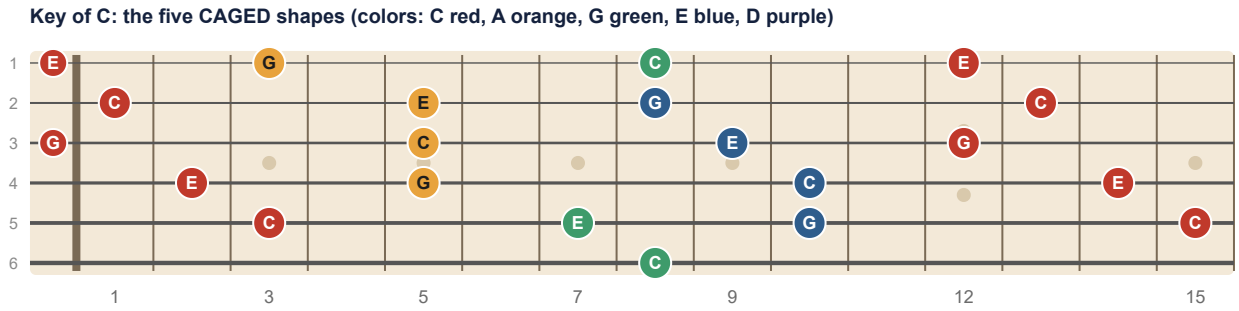
- The major scale is built from a pattern of whole steps and half steps: W W H W W W H.
- The same pattern, started on any note, gives you that note's major key.
- Every major key has a relative minor that shares its notes.

By the end: Build a major scale from any note, name its seven notes, and find the relative minor of any key.

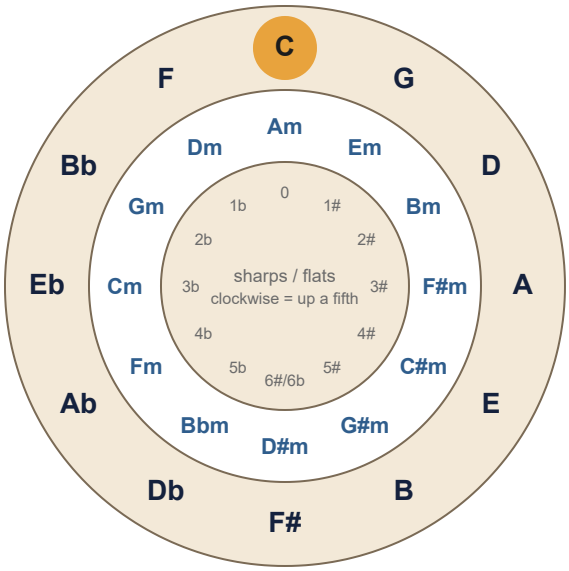
That is what a snapshot looks like. It is the first thing you see in each module, and it is deliberately short.



A taste of Part 1. From the root C, the major third is E and the perfect fifth is G. Those two intervals alone are enough to build the C major chord.



A taste of Part 3. The five CAGED shapes for the key of C, chained across the neck. By the end of Module 7, this picture will make sense at a glance.



A taste of Module 9. The circle of fifths, with C major highlighted. It shows how the twelve keys are related, and which ones are neighbors.

How to use the book in 10 minutes a day

You do not need long practice sessions. Ten focused minutes a day, five days a week, is enough for most players to finish the whole book in about six months, as the plan at the back shows. Short and regular beats long and rare, because theory sticks through repetition.

A 10-minute session

MINUTES	WHAT TO DO
0 to 2	Warm up with the module's cheat sheet. Say the notes or degrees out loud while you play them.
2 to 8	Work one exercise. Follow the steps in order and use the diagram on the page.
8 to 10	Do the self-check. If you can answer it without looking, tick the box. If not, mark it and repeat it tomorrow.

- **Keep the guitar within reach.** The best practice spot is the one you pass every day.
- **Say it as you play it.** Naming notes and degrees aloud links your eyes, ears and hands.
- **Do not skip the self-checks.** They are the point. An exercise you did not check is an exercise you guessed.
- **If you miss a day, just continue.** There is no penalty and no catching up to do.

> **A note on results**
>
> Most players find that after a few weeks of regular practice they start recognizing shapes and patterns on their own. Progress varies from person to person, and with practice the ideas become automatic.

The 12-question diagnostic test

Take this test before you read further. It takes about ten minutes, needs no guitar (although it helps to have one nearby), and tells you where to begin. Write down your answers, then check them against the key. Guessing is fine. The point is to find your honest starting point.

Diagnostic test: where should I start?

1. 1. What is the name of the open 5th string in standard tuning?

A. D

B. A

C. E

2. 2. Which note is on the 3rd fret of the 6th string?

A. F

B. G

C. A

3. 3. Playing the 5th fret of the 6th string gives the same pitch as which other string?

A. The open 5th string

B. The open 4th string

C. The 5th fret of the 5th string

4. 4. What is the interval from C up to G?

A. A perfect fourth

B. A perfect fifth

C. A major sixth

5. 5. What is the interval from C up to Eb?

A. A major third

B. A perfect fourth

C. A minor third

6. 6. Which pattern of whole steps (W) and half steps (H) builds a major scale?

A. W W H W W W H

B. W H W W H W W

C. W W W H W W H

7. 7. What is the 3rd note (scale degree 3) of the D major scale?

A. F

B. F#

C. G

8. 8. Which notes make an A minor chord?

A. A C E

B. A C# E

C. A D E

9. 9. Which note do you add to a C major chord (C E G) to make Cmaj7?

A. Bb

B. D

C. B

10. 10. What is the relative minor of G major?

A. E minor

B. B minor

C. G minor

11. 11. Which five open chord shapes does the CAGED system use?

A. C, A, G, E and D

B. C, A, G, E and B

C. C, A, F, E and D

12. 12. In the key of C major, which chords make the progression I - V - vi - IV?

A. C F Am G

B. C G Am F

C. C G Dm F

ANSWER KEY

1. **1. B** From the thickest string to the thinnest, standard tuning is E A D G B E. String 6 is E, string 5 is A, string 4 is D, string 3 is G, string 2 is B and string 1 is E.
2. **2. B** The open 6th string is E. Counting one half step per fret: fret 1 is F, fret 2 is F#, fret 3 is G.
3. **3. A** E plus five half steps is A (F, F#, G, G#, A), and the open 5th string is A. Most neighboring pairs of strings work this way: the 5th fret of the lower string matches the next open string. The exception is the pair of strings 3 and 2, where it is the 4th fret.
4. **4. B** C D E F G: G is the fifth note counting C as one, and it is 7 half steps above C, which is a perfect fifth.
5. **5. C** Eb is 3 half steps above C. Three half steps is a minor third. Four half steps (C up to E) would be a major third.
6. **6. A** The major scale is W W H W W W H. Starting on C, that gives C D E F G A B with half steps between E and F and between B and C. Option B is the natural minor scale and option C is the Lydian mode.

7. **7. B** D major is D E F# G A B C#. The third degree is F#, a major third above D (4 half steps).
8. **8. A** A minor is the triad 1 b3 5 from A: A, C and E. With C# instead of C, it would be A major.
9. **9. C** A major seventh chord adds the major seventh, 11 half steps above the root. For C that is B. A Bb would give you C7, a dominant seventh chord.
10. **10. A** The relative minor starts on the 6th degree of the major scale. G major is G A B C D E F#, so the 6th degree is E. E minor and G major share exactly the same seven notes.
11. **11. A** CAGED is named after the five open major shapes C, A, G, E and D. Each can be moved up the neck with a barre, and together they cover the whole fretboard for any key.
12. **12. B** The chords of C major are C, Dm, Em, F, G, Am and Bdim. I is C, V is G, vi is Am and IV is F. This is one of the most common progressions in popular songs.

Your score and where to start

Count one point for each correct answer. Then use the table. The ranges are a guide, not a rule: if you feel shaky on a section, start a module earlier.

SCORE	START AT	WHAT IT MEANS
0 to 4	Module 1	The fretboard is still mostly a maze. Start at the very beginning. The first two modules will feel like a quick, relaxed win.
5 to 8	Module 3	You know the neck and basic intervals. Skim Modules 1 and 2 (read the cheat sheets, try one exercise from each), then begin properly at scales and keys.
9 to 12	Module 7	You already know the building blocks. Begin with CAGED, the heart of the book, and go back to earlier modules only when something is unclear.

The express route

If you want the biggest gain in the shortest time, follow the express route: Modules 1, 3, 7 and 10. It takes you from the fretboard, to scales and keys, to the CAGED system, to improvising. Come back to the other modules whenever you want to fill in the details. A week-by-week version is in the practice plan at the back of the book.

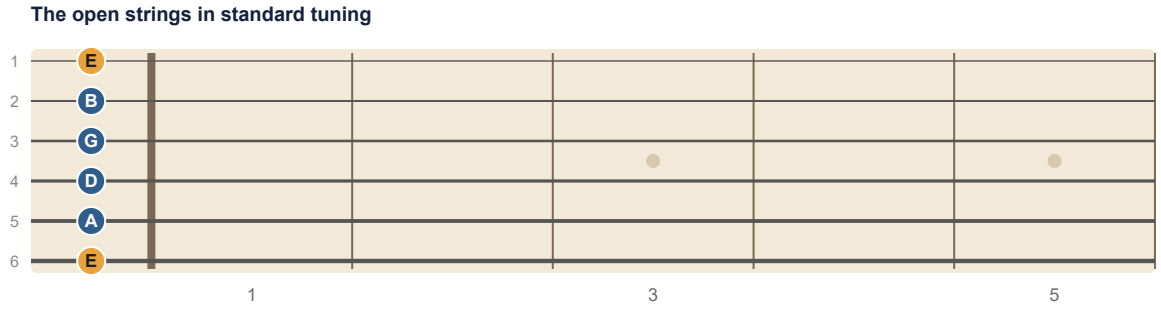
EXPRESS STOP	WHY IT IS ON THE ROUTE
Module 1	Naming any note on the neck.
Module 3	The master rule behind scales and keys.
Module 7	The CAGED system that connects the whole neck.
Module 10	Putting it all to use when you improvise.

Retake the test

Take the same test again after Module 6 and again at the end of the book. Seeing your score rise is the most honest measure of progress.

Standard tuning, a quick reminder

Every diagram in this book assumes standard tuning. From the thickest string to the thinnest, the open strings are E, A, D, G, B, E. The strings are numbered backward from what you might expect: string 6 is the thickest and lowest, and string 1 is the thinnest and highest. A common memory aid is Eddie Ate Dynamite, Good Bye Eddie, but any phrase you like will do.



String 6 (low E) is at the bottom; string 1 (high E) is at the top. The two E strings are two octaves apart: the same note name, very different pitch.

- For most neighboring pairs of strings, the 5th fret of the thicker string gives the pitch of the next open string: A on string 6, D on string 5, G on string 4, and E on string 2 (compared with the open string above it).
- The exception is string 3: its 4th fret, not the 5th, matches the open B string.
- If a chord sounds off, check your tuning first. A tuner takes ten seconds.

Other tunings

If you play in another tuning, the note names in the diagrams will not match your strings. Use standard tuning for this book. The shapes and ideas you learn will carry over later.

How to read the diagrams

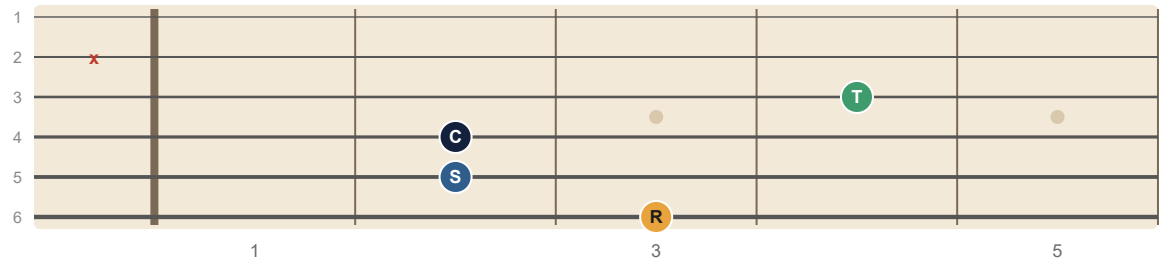
Every diagram in the book is a picture of the neck as if you were looking at your guitar from the front, headstock to the left. Read it like this.

- **The six horizontal lines are the strings.** String 6 (low E) is at the bottom. String 1 (high E) is at the top. This matches the order in a tab.
- **The vertical lines are the frets.** Fret numbers are written along the edge. Fret 0 is the open string (the thick bar on the left is the nut).
- **Each dot is a note.** Its color tells you what role it plays, and its label tells you the name or the scale degree.

- **Degrees count from the root.** The root is 1. A flat sign means one half step lower, so b3 is the minor third.

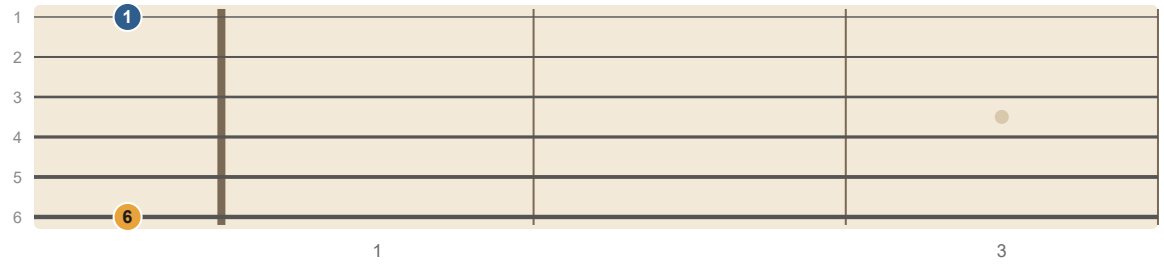
DOT KIND	WHAT IT MEANS
Root	The home note of the scale or chord, in a different color so you always see where you are.
Scale note	Any other note of the scale being shown.
Chord tone	A note that belongs to the chord being shown.
Target	A note to aim for, such as a note to land on when improvising.
Muted	A string you do not play or deaden with your fingers.

The legend, on a real neck



R is a root, S a scale note, C a chord tone, T a target and x a muted string. Letters like these are used only in this legend; in the book the labels will be note names or degrees.

Which string is which



Remember: the bottom line in every diagram is the thickest string, number 6. The top line is the thinnest, number 1.

Left-handed readers

The diagrams show the neck in the usual right-handed view, with the nut on the left. If you play left-handed, mirror each diagram left to right, so that the nut is on the right and the frets count toward the left. String 6 is always the thickest string and string 1 the thinnest.

Next

Take the diagnostic test, find your starting module, and open it. Do not try to learn everything at once.

THE FRETBOARD REVEALED

Part 1 - See

MODULE 1

The fretboard stops being a maze

- Twelve notes make up all the music you will ever play, and they repeat in a fixed loop.
- One fret equals one half step. Two frets equal one whole step. That is the only distance rule you need.
- Three anchors (the 6th string, the 5th string and the octave shapes) let you find any note in a couple of seconds.
- The neck repeats at fret 12, so you only have to learn half of it.

By the end: Name any note at any fret, and find any note anywhere on the neck, without memorizing 150 separate positions.

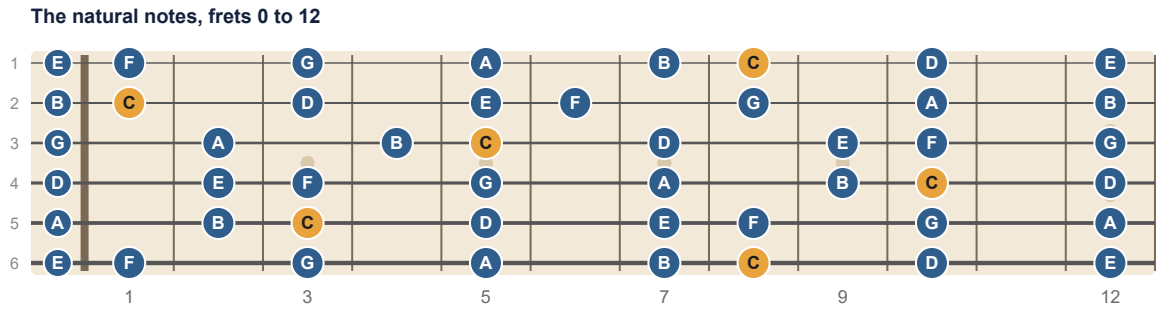
Why the fretboard looks like a maze

> *You can play songs, you know a handful of chords, and yet if someone says 'play an F#' you freeze. That is normal, and it is not a talent problem. It is a map problem.*

A guitar is one of the least obvious instruments ever built. A piano shows you every note in a single straight line: white keys, black keys, left is low, right is high, and each note lives in exactly one place. A guitar does the opposite. The same note lives in several places at once, and the pattern between strings is not the same everywhere. Most self-taught players learn shapes by finger memory (this chord, that riff, this box) and never get shown the map underneath. So the neck stays a maze: lots of paths, no signs.

The good news is that the maze has very few rules, and they are the same rules everywhere on the neck. Twelve notes. One fret equals one half step. The strings are tuned in a fixed pattern. That is nearly all of it. Once you see those rules, the 150 or so positions on the neck collapse into a few repeating ideas.

This module gives you a method instead of a memorization chore. You will not be asked to stare at a chart until it sticks. You will learn **three anchors** and a way of counting from them, and you will test yourself on your own guitar as you go. Here is the map you will be able to see in your head by the end of the module: every natural note (the notes with no sharp or flat) on the first twelve frets. Do not try to memorize it now. Just notice how much of it is a pattern.



Every natural note on the first 12 frets. C is highlighted so you can see how it repeats across the strings.

Look at the highlighted C's. They form a zig-zag that climbs up the neck. Every other note does the same thing, just shifted. That zig-zag is the heart of this module.

> **How to use this module**
>
> Keep your guitar in your hands. Whenever a diagram shows a note, play it and say its name out loud. Saying the name while you hear the pitch wires the map into your memory much faster than reading.

The 12 notes: the musical alphabet

Western music uses only seven letter names: **A B C D E F G**. After G, the alphabet starts over at A, one octave higher. Those seven letters are the natural notes. Between most of them sit five extra notes, named with a sharp (#) or a flat (b). A sharp raises a note by one half step. A flat lowers it by one half step.

Put together, the twelve notes in order are:

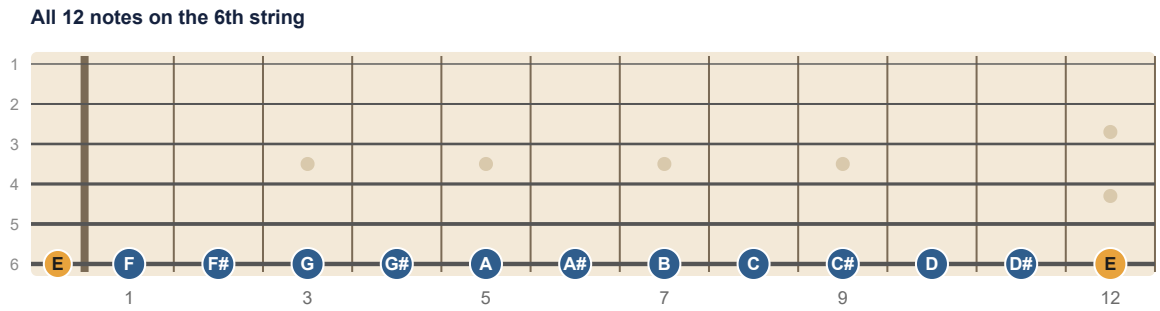
A - A# - B - C - C# - D - D# - E - F - F# - G - G# - (back to A)

Notice the two places where there is **no sharp or flat between two letters**: between **B and C**, and between **E and F**. Every other pair of neighboring letters has a note in the middle. This is the single most important fact about the alphabet. It is why the letter names on a guitar neck are not perfectly regular, and it is the reason most players get confused. Say it aloud once: *no sharp between B and C, no sharp between E and F.*

NOTE	NEXT NOTE UP	DISTANCE
A	A# (or Bb)	half step
A#	B	half step
B	C	half step (no sharp between)
C	C#	half step
C#	D	half step
D	D#	half step

NOTE	NEXT NOTE UP	DISTANCE
D#	E	half step
E	F	half step (no sharp between)
F	F#	half step
F#	G	half step
G	G#	half step
G#	A	half step

The beauty of the guitar is that you do not need a keyboard to see the gaps. Along a single string you walk through all twelve notes one fret at a time, and nothing is skipped. Here are the twelve notes on the 6th string, from the open string up to fret 12:



Open E, F, F#, G, G#, A, A#, B, C, C#, D, D#, then E again at fret 12. One fret = one note, nothing skipped.

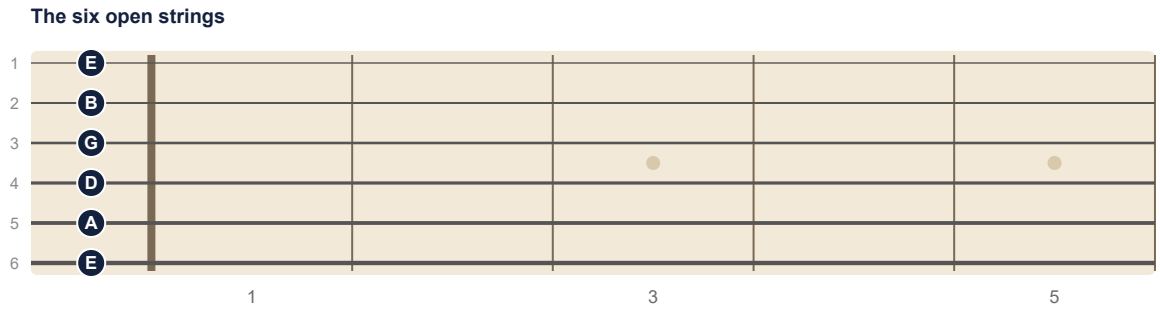
Read that diagram slowly. Between fret 0 (E) and fret 1 (F) there is no sharp: that is the E-F gap. Between fret 7 (B) and fret 8 (C) there is no sharp: that is the B-C gap. At fret 12 you arrive back at E, exactly one octave higher than the open string. The loop is complete. Every string does the same thing starting from its own open note.

The open strings: your first six facts

Everything else on the neck is counted from the open strings, so these are the six facts you need first. In standard tuning, from the thickest string (6th) to the thinnest (1st), the strings are:

6th: E - 5th: A - 4th: D - 3rd: G - 2nd: B - 1st: E

Most players learn this with a silly sentence. Use any one that sticks. A classic is *Eddie Ate Dynamite, Good Bye Eddie* (E A D G B E). Make up your own if that one is too odd; a sentence you invented is easier to remember than one you borrowed.



Open strings: E A D G B E. The 6th and 1st strings are both E, two octaves apart.

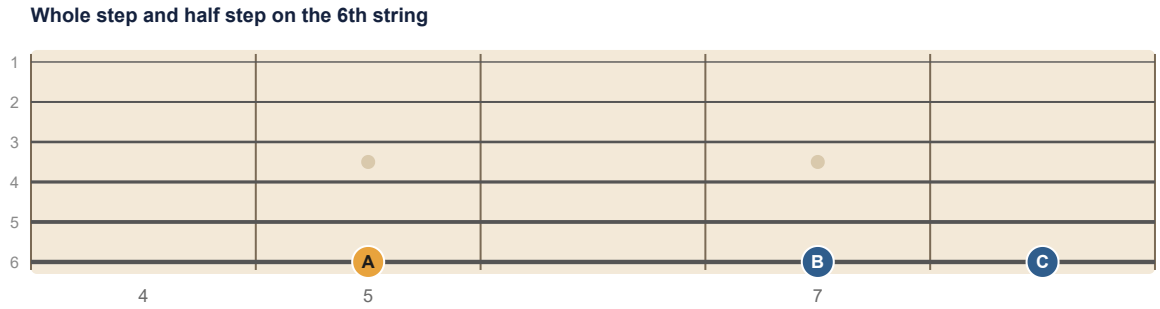
> **Two E strings**
>
> The 6th string and the 1st string are both E, but the 6th is two octaves lower. That means everything you learn on the 6th string works on the 1st string too, at the same fret numbers. You get a free second copy of the map.

Tune up before you go on. Every claim in this module depends on your guitar being in standard tuning. If a string is a half step flat, every note you find will be wrong by a half step, and you will think the method does not work.

Half steps and whole steps on the fretboard

A **half step** is the smallest distance in Western music. On the guitar, it is one fret. A **whole step** is two half steps, so it is two frets. That is the entire distance system. There is nothing else to learn about distance yet.

Because of this, the guitar is wonderfully honest. Move up one fret and the note is a half step higher. Move down one fret and it is a half step lower. This works on every string and at every position, no matter which fingers or which chord shape you are using.



A (fret 5) to B (fret 7) is a whole step: two frets. B (fret 7) to C (fret 8) is a half step: one fret.

- **A to B** is a whole step. There is an A# between them, which you skip over.
- **B to C** is a half step. There is nothing between them.
- **E to F** is a half step. Again, nothing between them.
- **C to D**, **D to E**, **F to G** and **G to A** are all whole steps, with a sharp in the middle.

Here is a way to hear the rule. Play the open 6th string (E), then fret 1 (F), then fret 2 (F#), then fret 3 (G). That is three half steps in a row, each one the smallest step you can play. Now play open E, then fret 2 (F#), then fret 4 (G#). Those are two whole steps. The sound is a bit wider, and that wider feel is what 'whole step' sounds like in your ear.

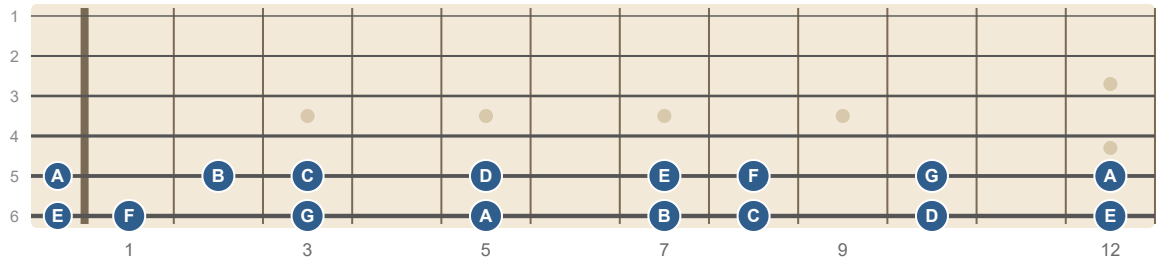
The three anchors method

Counting up from fret 0 every time is slow. If someone asks for an F# and you count 1, 2, 3 from the open string each time, you will get there, but not fast. Instead, keep a few **anchors** in your head, things you know cold, and count a few frets from the nearest one. Here are the three:

1. **Anchor 1: the 6th string.** You know every natural note on it: E0 F1 G3 A5 B7 C8 D10 E12.
2. **Anchor 2: the 5th string.** You know every natural note on it too: A0 B2 C3 D5 E7 F8 G10 A12.
3. **Anchor 3: the octave shapes.** A note and its octave (the same note, higher) always sit in a fixed pattern on the strings.

Anchors 1 and 2 are two single strings. Together they cover the name of every note, because every note exists somewhere on the 6th or 5th string. Anchor 3 is what lets you jump from one string to another without counting.

Anchors 1 and 2: the natural notes on the 6th and 5th strings



6th string: E F G A B C D E at frets 0, 1, 3, 5, 7, 8, 10, 12. 5th string: A B C D E F G A at frets 0, 2, 3, 5, 7, 8, 10, 12.

Look for the patterns. On the 6th string, the naturals sit at 0, 1, 3, 5, 7, 8, 10, 12. On the 5th string they sit at 0, 2, 3, 5, 7, 8, 10, 12. Those two lists are nearly identical: from fret 3 onward the two strings share fret numbers, with G and C at 3, A and D at 5, B and E at 7, C and F at 8, D and G at 10, and E and A at 12. That overlap makes both strings easy to learn together.

How to find any note with an anchor and a count

The recipe is the same for every note, and it takes about two seconds once it is in your fingers:

1. Pick the string you want to use (the 6th or the 5th).
2. Go to the natural note with the same letter. That is your anchor fret.
3. Add one fret for a sharp, or subtract one fret for a flat.

Example: **find F#** on the 6th string. F is at fret 1, and a sharp is one fret higher, so F# is fret 2. Example: **find Bb** on the 5th string. B is at fret 2, and a flat is one fret lower, so Bb is fret 1. Example: **find G#** on the

5th string. G is at fret 10, and the sharp adds one, so G# is fret 11. You can check any of these by playing the fret and comparing it with a note you already know.

> **Never count from zero**
> If you catch yourself counting 1, 2, 3, 4, 5 from the nut, stop. Jump to the closest anchor that you already know and count from there. Counting is only for the last one or two frets.

Octave shapes: how notes repeat across strings

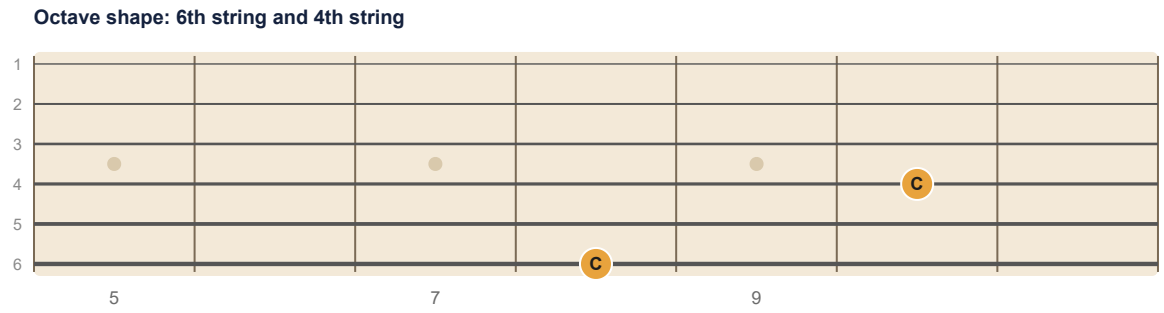
An **octave** is the same note at a higher pitch. Going up an octave means you have passed through all twelve notes and landed back on the same letter. On the guitar, an octave has a very predictable shape across the strings. Learn these four shapes and you can jump between strings without counting.

STARTING STRING	OCTAVE IS ON	FRET SHIFT
6th	4th	two strings up, two frets higher (+2)
5th	3rd	two strings up, two frets higher (+2)
4th	2nd	two strings up, three frets higher (+3)
3rd	1st	two strings up, three frets higher (+3)

The pattern: always two strings up, and either two or three frets higher. The reason is the tuning. The 2nd string (B) is tuned only a major third above the 3rd string (G), instead of the usual perfect fourth, so that one irregular gap changes the count by one fret. You do not need to understand the math to use the shapes. You only need to remember that any shape that lands on the 2nd or 1st string gets +3 instead of +2.

Octave shape 1: 6th string to 4th string

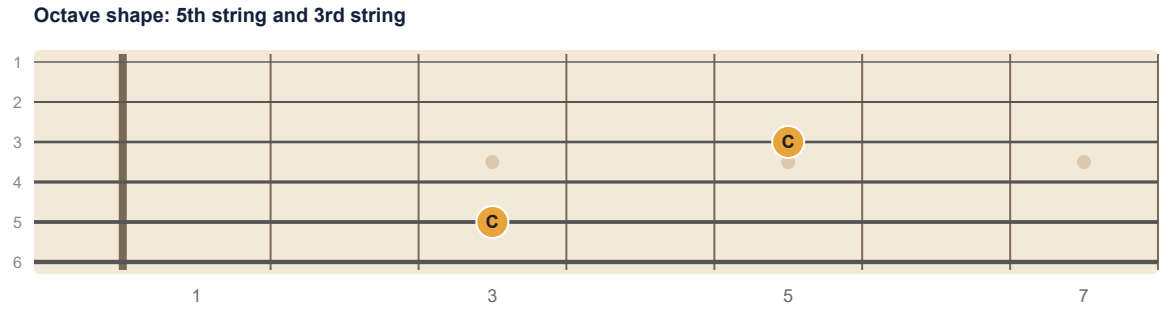
Take any note on the 6th string. Its octave is two strings up (the 4th string) and two frets higher. Take C: it is at fret 8 on the 6th string. Add two strings and two frets and you arrive at fret 10 of the 4th string, and fret 10 of the D string is C.



C on the 6th string, fret 8, and its octave on the 4th string, fret 10. The same shape works for any note on the 6th string.

Octave shape 2: 5th string to 3rd string

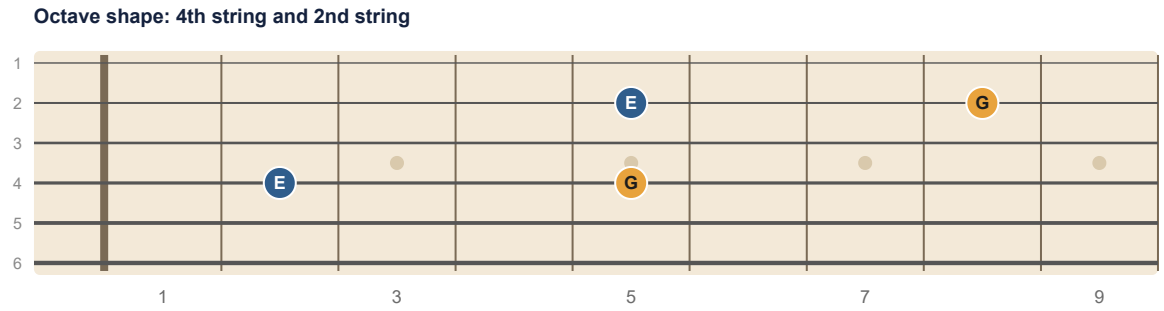
The same shape, shifted over one string. C is at fret 3 on the 5th string. Two strings up and two frets higher gives fret 5 on the 3rd string, which is C.



C on the 5th string, fret 3, and its octave on the 3rd string, fret 5. Exactly the same geometry as the previous shape.

Octave shape 3: 4th string to 2nd string

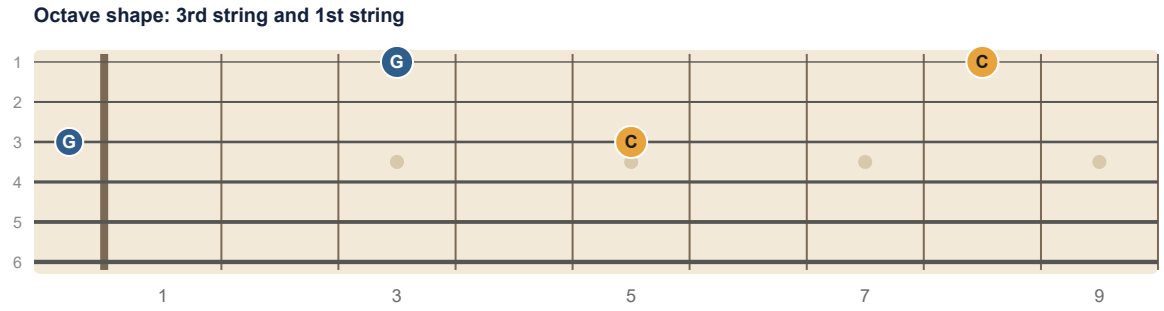
Here is where the extra fret appears. From the 4th string you go two strings up, to the B string, and three frets higher. Take G on the D string at fret 5. Add three frets on the B string and you land at fret 8, which is G, one octave above.



G on the 4th string (fret 5) and G on the 2nd string (fret 8). Also E on the 4th string (fret 2) and E on the 2nd string (fret 5). Always +3.

Octave shape 4: 3rd string to 1st string

The last shape also lands on a string above the irregular gap, so it also uses +3. Open G on the 3rd string and fret 3 on the 1st string are both G. C on the 3rd string at fret 5 and C on the 1st string at fret 8 are the same pair.



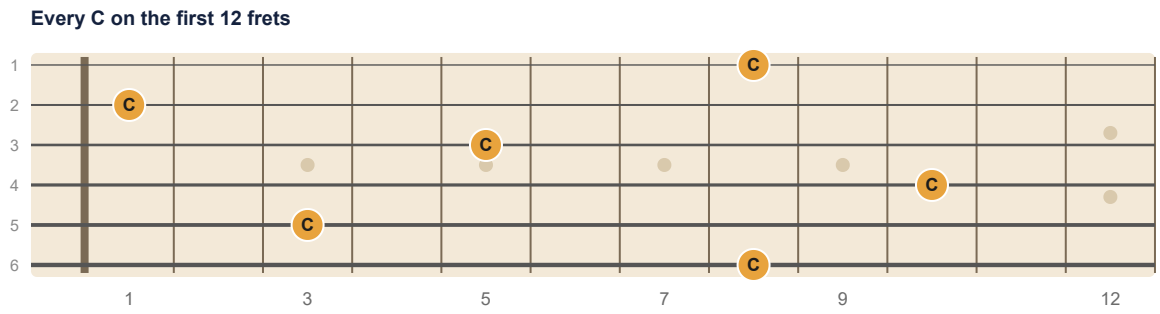
Open G (3rd string) and fret 3 on the 1st string are both G. C at fret 5 (3rd) and fret 8 (1st) are both C. Always +3.

> **Two ideas, not four**
>
> Strings 6-4 and 5-3 use +2 frets. Strings 4-2 and 3-1 use +3 frets. That is only two ideas to remember. If the top note lands on the B or high E string, use +3; otherwise use +2.

Finding every C (and then every note) on the whole neck

Now put the anchors and octave shapes together. Pick one note, C, and find every place it lives within the first twelve frets. Start from the two anchor strings: C is at fret 8 on the 6th string and at fret 3 on the 5th string. Then use the octave shapes and the neighboring open strings to fill in the rest:

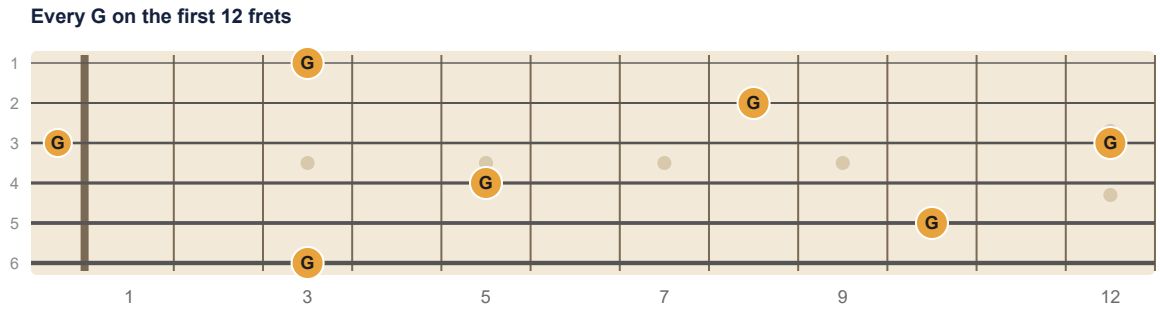
- 6th string, fret 8 (anchor 1)
- 5th string, fret 3 (anchor 2)
- 4th string, fret 10 (octave of the 6th string's C)
- 3rd string, fret 5 (octave of the 5th string's C)
- 2nd string, fret 1 (the open B string is one half step below C)
- 1st string, fret 8 (same fret as the 6th string, two octaves up)



Six C's, one per string. They form a zig-zag that you now know how to build from two anchors.

That diagram is the same C zig-zag you saw on the first map. The difference is that you can now build it yourself in under ten seconds. Do the same for any other note and you have the full map for that note. Try it with G, using the same steps:

- 6th string, fret 3 (anchor 1)
- 5th string, fret 10 (anchor 2)
- 4th string, fret 5 (octave of the 6th string's G: +2 frets)
- 3rd string, fret 0 (the open string is itself a G)
- 2nd string, fret 8 (octave of the 4th string's G: +3 frets)
- 1st string, fret 3 (octave of the open 3rd string: +3 frets)



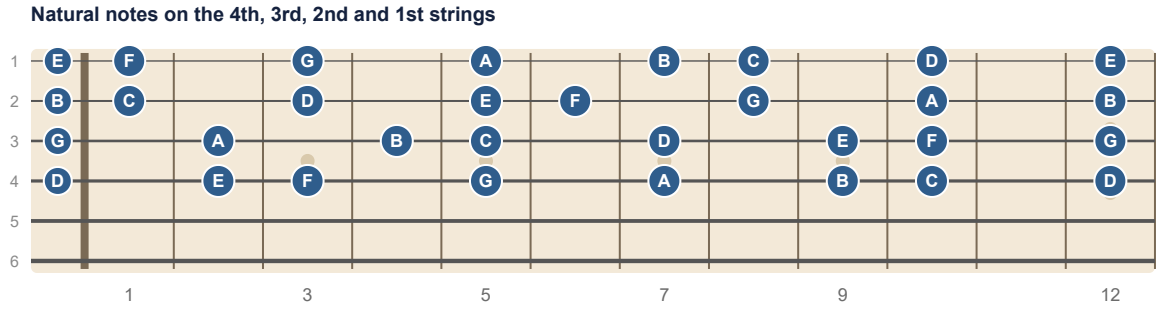
Six G's. The shape of the zig-zag is the same as C, just starting from a different place.

The zig-zag shape is identical for every note. It only slides left or right depending on the note. If you remember one shape, you remember all of them. (The 3rd string also has a G at fret 12, which is the open string again one octave up; see the fret 12 mirror below.)

The natural notes on each string

You already know the 6th and 5th strings. The other four strings are the same alphabet starting from a different open note, so here is the whole neck at a glance:

STRING	OPEN	NATURALS AT FRETS
6th	E	0 E, 1 F, 3 G, 5 A, 7 B, 8 C, 10 D, 12 E
5th	A	0 A, 2 B, 3 C, 5 D, 7 E, 8 F, 10 G, 12 A
4th	D	0 D, 2 E, 3 F, 5 G, 7 A, 9 B, 10 C, 12 D
3rd	G	0 G, 2 A, 4 B, 5 C, 7 D, 9 E, 10 F, 12 G
2nd	B	0 B, 1 C, 3 D, 5 E, 6 F, 8 G, 10 A, 12 B
1st	E	0 E, 1 F, 3 G, 5 A, 7 B, 8 C, 10 D, 12 E



The four thinnest strings, naturals only. The 1st string repeats the 6th string exactly.

Two quick checks to keep you honest. First, the 1st string has the same fret numbers as the 6th string, because both are E. Second, every string is the same alphabet in a different position. Look at where the two half-step gaps (E-F and B-C) fall on each string: they always appear as two neighboring frets with no sharp between them. On the 6th string they are at frets 0-1 and 7-8; on the 5th string at 2-3 and 7-8; on

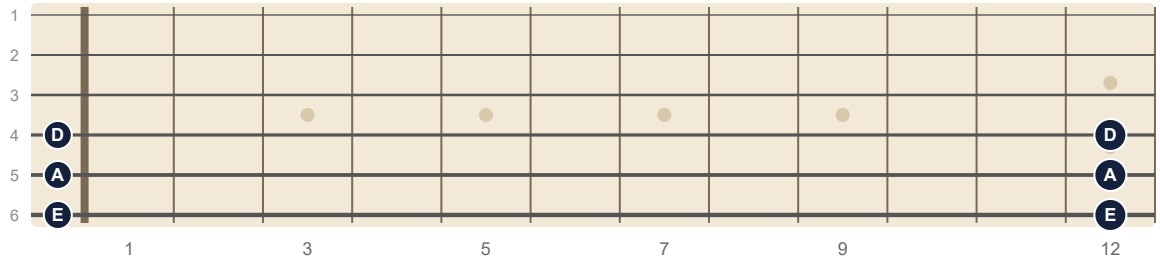
the 4th string at 2-3 and 9-10; on the 3rd string at 4-5 and 9-10; on the 2nd string at 0-1 and 5-6; on the 1st string at 0-1 and 7-8.

> **The fifth-fret shortcut**
> Fret 5 on a string equals the next-thinner open string. 6th string fret 5 is A. 5th string fret 5 is D. 4th string fret 5 is G. 2nd string fret 5 is E. The 3rd string is the exception: its fret 4 is B. This is how guitarists tune by ear, and it is also a quick way to name notes.

The fret 12 mirror

The 12th fret marks the octave of every open string, and the neck begins to repeat. Fret 12 has the same note as fret 0 on the same string, just one octave higher. Fret 13 matches fret 1, fret 14 matches fret 2, and so on. The twelve notes repeat in a loop, and fret 12 is the mirror.

The mirror at fret 12



Open strings and fret 12 on the 6th, 5th and 4th strings: the same note names, one octave apart.

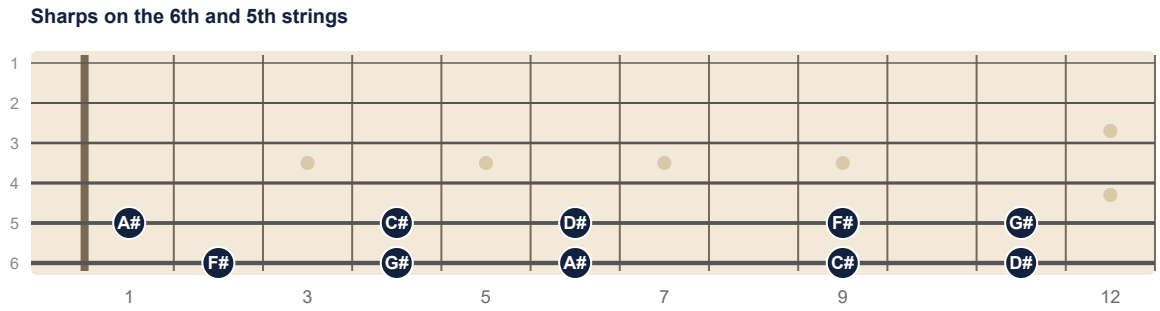
In practice, this halves your work. You never need to memorize frets 12 through 24 as a separate territory. If you meet a note on fret 15, subtract 12: it is fret 3. The high frets are just the low frets again in a higher voice. And when you play a riff near the 12th fret, you can think of it as if it were played near the nut.

> **Subtract twelve**
> Above fret 12, subtract 12 and read the low-fret answer. Fret 14 on the 5th string is B (fret 2 of the A string). Fret 17 on the 6th string is A (fret 5).

Sharps and flats on the neck

Sharps and flats are the notes between the naturals. On the guitar, they sit in the frets between the naturals you already know. A sharp is the fret right after (higher than) the natural. A flat is the fret right before (lower than) the natural.

- **Sharp (#)**: move one fret toward the body of the guitar (higher pitch).
- **Flat (b)**: move one fret toward the headstock (lower pitch).
- **Natural**: no change.



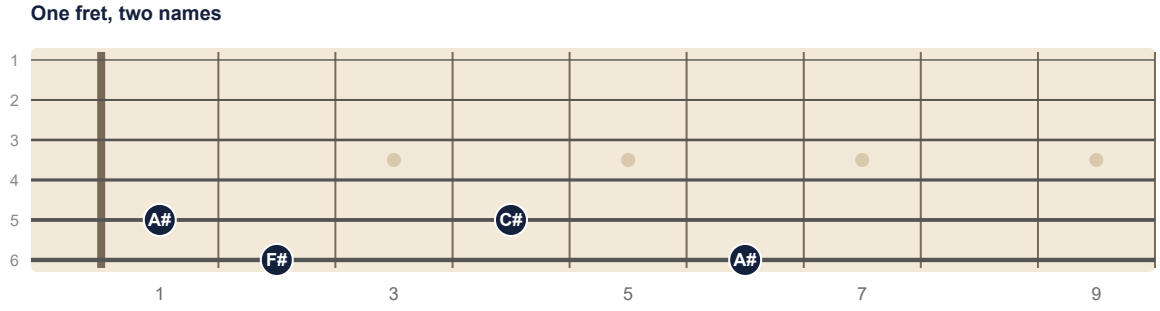
Every sharp on the 6th and 5th strings. F# is at fret 2 on the 6th string; A# at fret 1 on the 5th string.

Check three of them aloud. 6th string, fret 2: F is at fret 1, a sharp adds one, so F#. 5th string, fret 4: C is at fret 3, so fret 4 is C#. 6th string, fret 9: C is at fret 8, so fret 9 is C#. And do not forget that two letters have *no* sharp: E# and B# are not played as separate notes in everyday music (they are just F and C). That is why the sharps are not evenly spread: there is a missing one after E and after B.

Enharmonics: one note, two names

Every sharp or flat note has two names. The fret between F and G can be called F# (one above F) or Gb (one below G). Both names refer to the exact same fret and the exact same sound. This is called an **enharmonic** pair. Here are all five:

SHARP NAME	FLAT NAME	6TH STRING FRET	5TH STRING FRET
A#	Bb	6	1
C#	Db	9	4
D#	Eb	11	6
F#	Gb	2	9
G#	Ab	4	11



6th string fret 2 is F# or Gb. 5th string fret 1 is A# or Bb. 5th string fret 4 is C# or Db. 6th string fret 6 is A# or Bb.

Which name do you use? It depends on the key you are playing in, and you will learn that rule in Module 3. For now, accept both names and know they live on the same fret. If someone calls out Db and you were thinking C#, nothing changes on the neck.

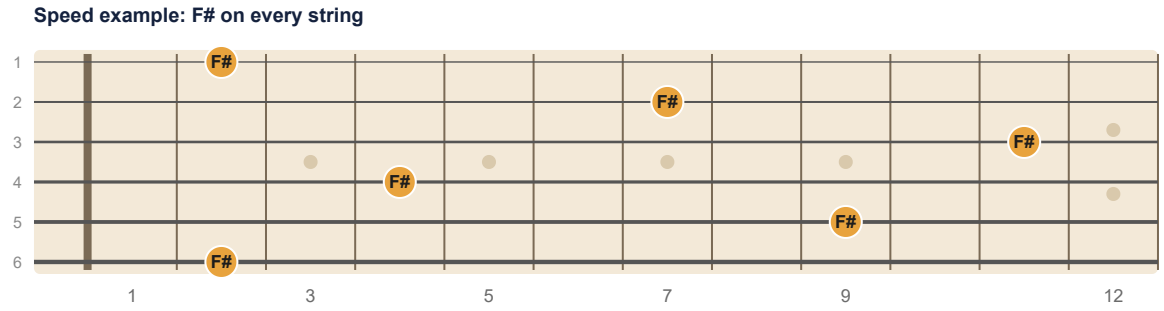
There are also a few rarely used enharmonics among the naturals. E# is the same as F, B# is the same as C, Fb is the same as E and Cb is the same as B. You will rarely meet them, but when you do, you will know they are just the natural note under another name.

A speed-finding method

You now have everything you need to find notes. The last step is turning it into a habit that takes a few seconds or less. Most players find that this routine gets noticeably faster within a week or two of short daily practice.

1. **Pick your string.** Start on the 6th or the 5th string. Both are anchors.
2. **Find the letter.** Use the naturals: E0 F1 G3 A5 B7 C8 D10 on the 6th string, A0 B2 C3 D5 E7 F8 G10 on the 5th string.
3. **Adjust for the sharp or flat.** One fret up for #, one fret down for b.
4. **Jump with octave shapes.** If you need the same note on another string, use +2 frets (strings 6-4 and 5-3) or +3 frets (strings 4-2 and 3-1).

Say the note aloud as you play it. Then go the other way: put a finger down at any random fret on any string and name the note. The skill you are building works in two directions: *name to fret* and *fret to name*. Both matter for the music you will learn later in this book.



F# on the 6th string at fret 2, the 5th string fret 9, the 4th string fret 4, the 3rd string fret 11, the 2nd string fret 7 and the 1st string fret 2.

Check them: F# on the 6th string is fret 2 (F at 1, plus one). On the 5th string, nine frets above A: A#, B, C, C#, D, D#, E, F, F#. On the 4th string, four frets above D: D#, E, F, F#. On the 3rd string, eleven frets above G, which is one fret below the next G at fret 12. On the 2nd string, seven frets above B: C, C#, D, D#, E, F, F#. And on the 1st string it is fret 2, the same as the 6th.

> **A realistic target**
>
> With a few minutes of practice each day, most players can name a fret in a second or two within a couple of weeks. Do not aim for perfection. Aim for 'quick enough that it does not interrupt the music'.

Common mistakes

- **Forgetting the gaps.** Assuming there is a sharp between B and C or between E and F. Fix: say 'no sharp between B-C and E-F' before every drill.
- **Using +2 for every octave.** Octaves that land on the 2nd or 1st string need +3. Fix: remember 'two ideas, not four'.
- **Counting from the nut every time.** Slow and error-prone. Fix: jump to the nearest anchor first.
- **Learning the map by reading only.** Eyes do not teach fingers. Fix: always play the note and say its name.
- **Practicing out of tune.** A flat string means wrong notes, so the method seems broken. Fix: tune before every session.
- **Mixing up the direction of sharps and flats.** A sharp goes up (toward the body). A flat goes down (toward the headstock). Fix: 'sharp is higher'.
- **Skipping the 1st and 2nd strings.** They matter for solos and chord tops. Fix: include them in your drills from day one.

Module 1 cheat sheet

WHAT TO REMEMBER	THE SHORT VERSION
The 12 notes	A A# B C C# D D# E F F# G G# (then A again)
The gaps	No sharp between B-C and between E-F
Open strings	6th E, 5th A, 4th D, 3rd G, 2nd B, 1st E
Distance	1 fret = half step, 2 frets = whole step
Anchor 1 (6th string)	E0 F1 G3 A5 B7 C8 D10 E12
Anchor 2 (5th string)	A0 B2 C3 D5 E7 F8 G10 A12
Anchor 3 (octaves)	Strings 6-4 and 5-3: +2 frets. Strings 4-2 and 3-1: +3 frets.
Sharps and flats	Sharp = one fret up. Flat = one fret down.
Fret 12	Same note as fret 0, one octave up. The neck repeats.
Fifth-fret shortcut	Fret 5 equals the next open string (on the 3rd string, fret 4).
Enharmonics	A#/Bb, C#/Db, D#/Eb, F#/Gb, G#/Ab: one fret, two names
The recipe	Pick an anchor string, find the letter, adjust for # or b, jump with octave shapes.

Glossary in plain words

WORD	PLAIN MEANING
Note	One specific pitch with a letter name, like C or F#.
Natural	A note with no sharp or flat: A B C D E F G.
Sharp (#)	One fret higher than the natural note.
Flat (b)	One fret lower than the natural note.
Half step	The smallest distance in music: one fret.
Whole step	Two half steps: two frets.
Octave	The same note, higher or lower. Twelve frets up on the same string.
Enharmonic	Two names for the same note, like F# and Gb.
Anchor	A note or string you know by heart, used as a starting point for counting.
Fretboard	The neck of the guitar with the frets on it, also called the fingerboard.
Open string	A string played without pressing any fret.
Nut	The small ridge at the top of the neck. Fret 0 is the open string at the nut.

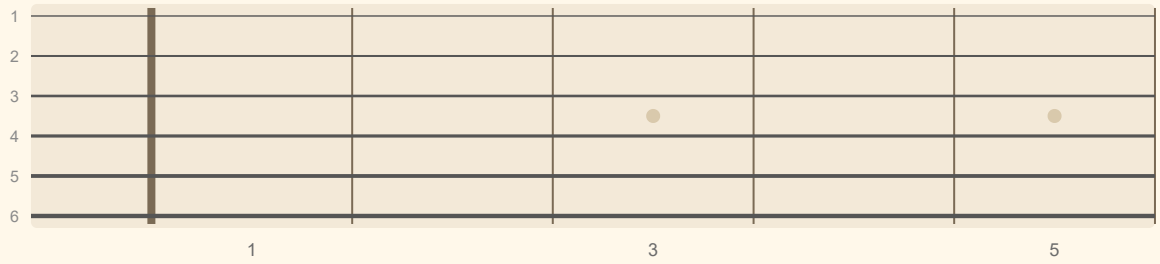
On the fretboard: five exercises

Each exercise takes about five minutes. Do them in order. After each one, check yourself with the self-check line before moving on. If you miss something, redo it before continuing; the next module builds on this one.

EXERCISE 1 5 MIN

Name the open strings and the twelve notes

1. Tune your guitar to standard tuning (E A D G B E).
2. Play each open string from the 6th to the 1st and say its name aloud.
3. On the 6th string, play frets 0 to 12 slowly, saying each note name with sharps (E, F, F#, G, G#, A, A#, B, C, C#, D, D#, E).
4. Repeat on the 5th string, starting from A.
5. Write the note names on the blank diagram for frets 0 to 5 on all six strings, then check them against the lists in this module.



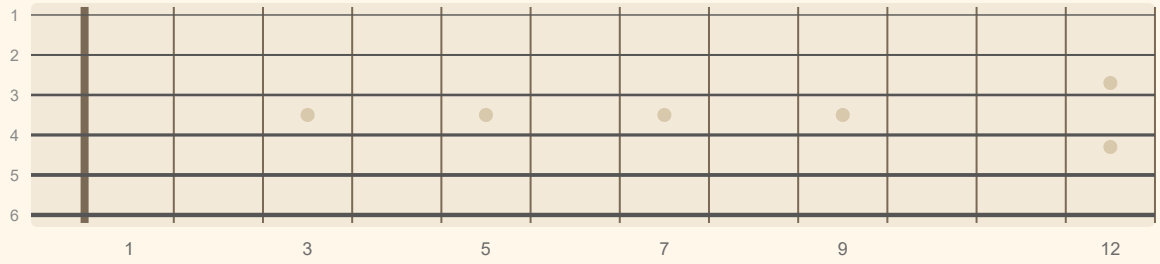
Write the note name on each position: strings 6 to 1, frets 0 to 5.

☐ Self-check: you can say E A D G B E without hesitation and name all 13 notes on the 6th string (frets 0-12) in order. If you stumbled at E-F or B-C, do it once more.

EXERCISE 2 5 MIN

Anchor and count: find the notes

1. Without looking at the lists, find these notes on the 6th string and play them: B, F, G#, D, C#.
2. Now find these on the 5th string: E, G, C, F#, Bb.
3. For each one, say the anchor you used and how many frets you counted.
4. Check your answers against the lists in this module.



Mark your answers here, then compare with the lists.

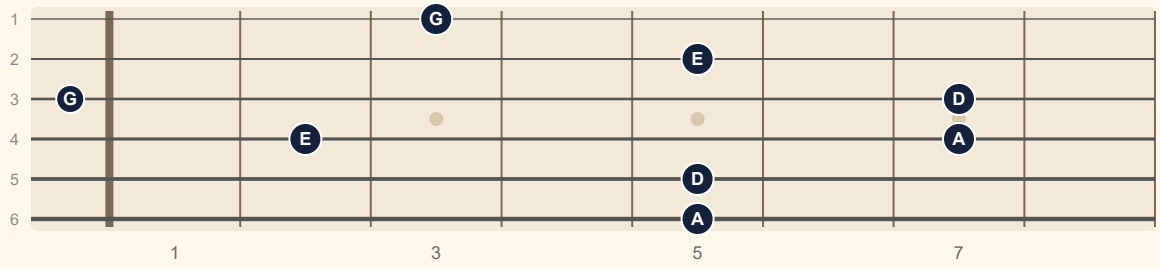
☐ Self-check: 6th string: B fret 7, F fret 1, G# fret 4, D fret 10, C# fret 9. 5th string: E fret 7, G fret 10, C fret 3, F# fret 9, Bb fret 1. Eight or more right, counted from an anchor and not from the nut, means you are on track.

EXERCISE 3 5 MIN

Use the octave shapes

1. Play A on the 6th string (fret 5). Without counting from the nut, find its octave on the 4th string.
2. Play D on the 5th string (fret 5). Find its octave on the 3rd string.
3. Play E on the 4th string (fret 2). Find its octave on the 2nd string (remember +3).
4. Play open G on the 3rd string. Find its octave on the 1st string (remember +3).
5. Play each pair one note after the other and listen: the two notes should sound like the same note in a lower and a higher voice.

Exercise 3 answers: octave pairs



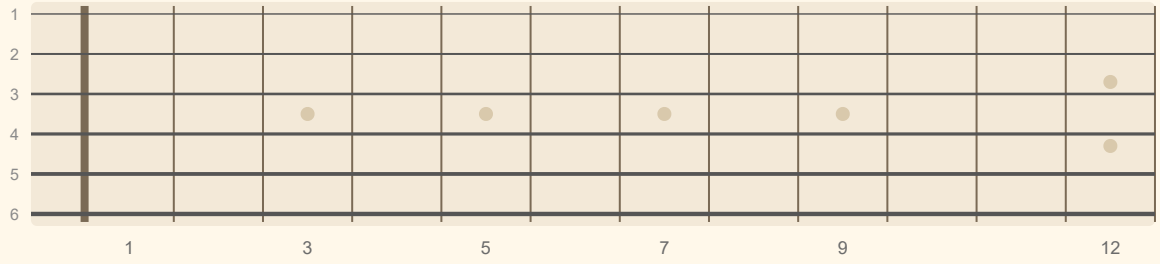
The four octave pairs from this exercise. Cover this diagram until you have tried them yourself.

☐ Self-check: A is fret 5 (6th) and fret 7 (4th). D is fret 5 (5th) and fret 7 (3rd). E is fret 2 (4th) and fret 5 (2nd). G is fret 0 (3rd) and fret 3 (1st). If a pair sounds like two different notes, check your tuning first and your fret counting second.

EXERCISE 4 6 MIN

Find every G on the neck

1. Start with G on the 6th string (fret 3) and G on the 5th string (fret 10).
2. Use octave shapes to find the other G's on the 4th, 3rd, 2nd and 1st strings.
3. Mark each one on the blank diagram.
4. Play them in order from the 6th string to the 1st, saying 'G' each time.
5. Then pick another note of your choice (say E or A) and repeat the whole process.



Mark every G on frets 0 to 12.

☐ Self-check: you should have six G's: 6th string fret 3, 5th fret 10, 4th fret 5, 3rd fret 0, 2nd fret 8, 1st fret 3. If one is off, find where the octave shape went wrong. Most mistakes are +2 versus +3.

EXERCISE 5 5 MIN

Speed drill: fret to name

1. Choose a random string and a random fret between 0 and 12, without looking.
2. Name the note aloud as fast as you can. If you do not know it, use an anchor and count.
3. Check the answer by playing the note and comparing it with your anchor.
4. Repeat for 20 positions. Try to get each answer under about three seconds by the end.
5. Warm up with these six: 4th string fret 7, 3rd string fret 9, 2nd string fret 6, 1st string fret 10, 5th string fret 11, 6th string fret 9.

☐ Self-check: the warm-up answers are A, E, F, D, G# (or Ab), C# (or Db). If you got five of six right, you are ready for Module 2. If not, repeat the drill on one string at a time for a few days.

Next step

You can now find any note on your guitar, and you know why the neck repeats. In Module 2 you will use that map to measure the distance between notes. Those distances, called intervals, are what give music its emotion: the reason one pair of notes sounds bright, another sad, another tense. Before you move on, spend five minutes with Exercise 5 on a different string each day for the next few days, so the map becomes automatic.

MODULE 2

Intervals: Distance Is Emotion

- An interval is just the distance between two notes, counted in half steps (frets).
- There are 12 intervals inside one octave, and each one has its own feel: tense, bright, sad, open, unresolved.
- Every interval is a shape on the fretboard, and that shape is the same everywhere, with one small exception at the B string.
- Once you know the shapes, you can find any note from any root without naming a single note.

By the end: Name any interval by sight and by ear, build the full interval map from any root, and use the shape that fits under your fingers instead of counting frets.

> *In Module 1 you learned to name the notes. Now comes the real shortcut: stop thinking about notes one at a time and start thinking about the distance between them. Distance is what you hear. It is also what your hand feels.*

What an Interval Is

An **interval** is the distance between two notes. That is the whole definition. On a guitar, distance has a very friendly unit: the **half step**, which is one fret. Two frets make a **whole step**. If you play fret 5 and then fret 8 on the same string, the interval is 3 half steps. Nothing else matters: not the note names, not the string, not the key.

This is why intervals are so useful. A note name tells you *where* you are. An interval tells you *how far* you moved, and how far is what creates the feeling. Play A and then E and you hear something open and strong. Play A and then A# and you hear something tight and nervous. The notes are different, but the real difference is the distance: 7 half steps versus 1.

Here is the second piece of good news. Because the guitar is built from regular steps, a given distance always makes the **same shape**, no matter where you play it. Learn the shape once and you own it in every key. That is the point of this module.

> **Counting Rule**
>
> Always count the **destination** in half steps from the starting note, and never count the starting note itself. From A, one fret up is A#: that is 1 half step. Two frets up is B: 2 half steps. Twelve frets up is A again: the octave.

The Starting Note Is the Root

In this module, the note we measure from is called the **root**. It is shown in a different color on every diagram. We will mostly use A as the example root, because its notes are easy to check: A, A#, B, C, C#, D, D#, E, F, F#, G, G#, and then A again. You can swap in any other note and the shapes stay exactly the same.

The 12 Intervals, One by One

Inside one octave there are 12 half steps, so there are 12 distances to learn, plus the octave itself. Each has a short name (like m3 or P5), a size in half steps, and a feel. The short names are simple: **m** is minor (the smaller one), **M** is major (the larger one), and **P** is perfect (a special group we will explain shortly).

HALF STEPS	SHORT NAME	FULL NAME	THE FEEL	FROM A YOU REACH
0	P1	Unison	Same note, nothing moved	A
1	m2	Minor second	Tense, crunchy, close to rubbing	A#
2	M2	Major second	Stepping, neutral, a gentle push forward	B
3	m3	Minor third	Sad, dark, tender	C
4	M3	Major third	Bright, happy, settled	C#
5	P4	Perfect fourth	Open, suspended, a little unfinished	D
6	b5 (tritone)	Flat fifth	Unstable, uneasy, wants to move	D#
7	P5	Perfect fifth	Strong, hollow, powerful	E
8	m6	Minor sixth	Bittersweet, aching	F
9	M6	Major sixth	Warm, hopeful, wide open	F#
10	m7	Minor seventh	Bluesy, relaxed, not finished	G
11	M7	Major seventh	Dreamy, yearning, almost home	G#
12	P8	Octave	The same note, higher, home again	A

Feelings are personal, and different players describe them differently. Treat the table as a starting map, not a law. What matters is that *you* attach a feel to each sound, so that you can recognize it later.

Famous Starting Points

A classic trick is to link each interval to the opening of a piece you already know. Use only the ones that click for you. Here are a few that are widely agreed on:

- **m2**: the creeping two-note motion of the shark-movie theme, one fret apart.

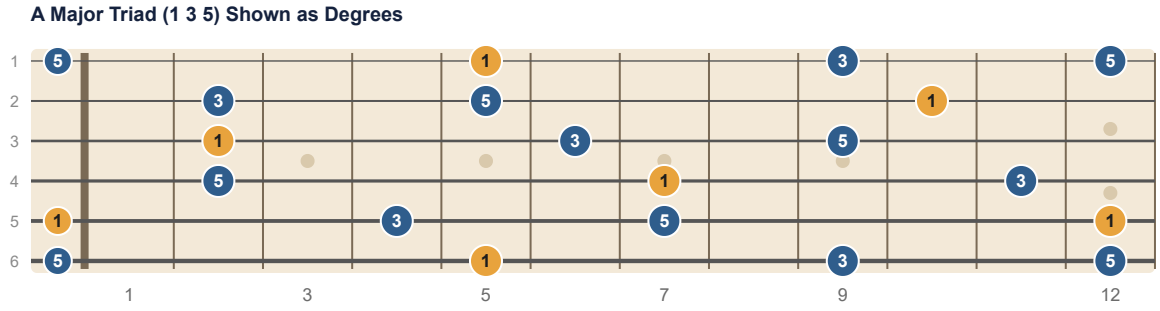
- **M2**: the first three notes of 'Frere Jacques' climb in major seconds, one after another.
- **m3**: the first two notes of 'Greensleeves' (an upward minor third).
- **M3**: the first two notes of 'When the Saints Go Marching In'.
- **P4**: the first two notes of the wedding march 'Here Comes the Bride'.
- **Tritone**: the first two notes of 'Maria' from West Side Story.
- **P5**: the jump in 'Twinkle, Twinkle, Little Star' between the second and third notes.
- **M6**: the first two notes of 'My Bonnie Lies Over the Ocean'.
- **m7**: the first two notes of 'Somewhere' from West Side Story.
- **Octave**: the first two notes of 'Somewhere Over the Rainbow'.

For the minor sixth and the major seventh there is no single famous opener that everyone agrees on, so rely on the feel words instead: aching for the m6, dreamy and unfinished for the M7.

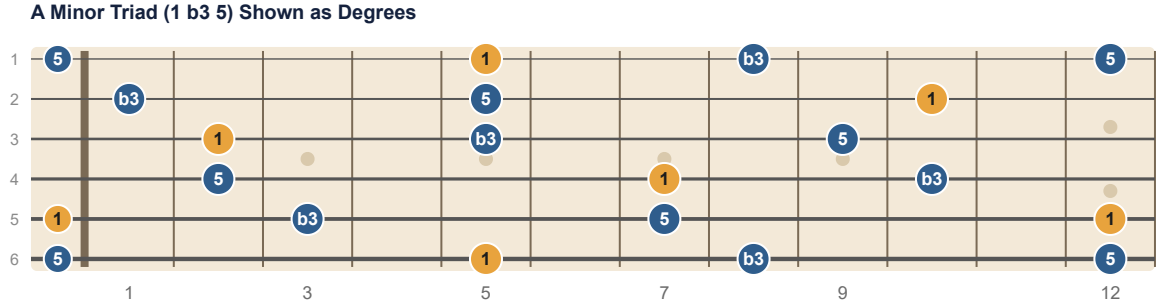
Major or Minor: Why Thirds Decide the Mood

Look at the table and notice the pairs: m2 and M2, m3 and M3, m6 and M6, m7 and M7. In each pair, the minor version is exactly one half step smaller than the major one. That one fret changes the whole mood. The pair that matters most is the third.

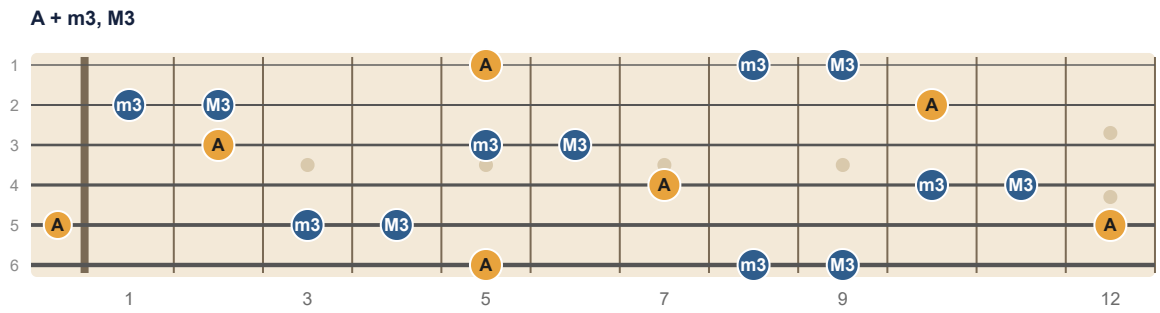
The **third** is the note that decides whether a chord sounds happy or sad. Play A with C# (a major third, 4 half steps) and it sounds bright. Play A with C (a minor third, 3 half steps) and it sounds darker. Everything else can stay the same: same root, same fifth, and the mood flips just because of that one fret.



A, C# and E across the neck. The 3 (C#) is 4 half steps above A. Find the 3 on each string.



Same root, same 5. Only the third moved down one fret (C# became C), and the mood changed.



Minor third (C) and major third (C#) above every A. They always sit one fret apart.

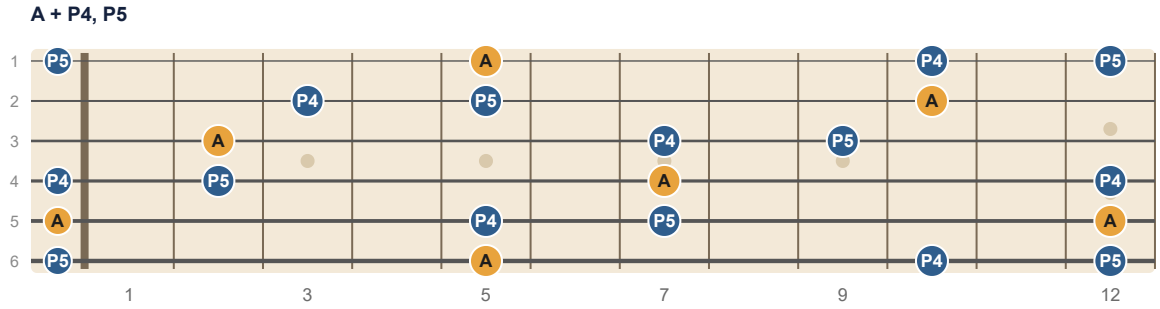
Hear It

Play the open A string (string 5), then C# on string 4, fret 4. That is a major third. Now move the second note one fret down to string 4, fret 3 (C). That is a minor third. Say 'bright' or 'sad' out loud as you play. Most players find that saying the word helps the sound stick.

Perfect Intervals Versus Major and Minor

Why are the fourth, the fifth and the octave called **perfect**, while seconds, thirds, sixths and sevenths are major or minor? Because the perfect ones keep the same size in the major scale and its mirror, and they sound stable and neutral, neither bright nor dark. A power chord is built from a root and a P5 for exactly this reason: it works over both major and minor music.

A useful quick sort: perfect intervals are the *skeleton* (P1, P4, P5, P8), and the major and minor intervals are the *color*, the thirds above all. If you can hear a P5, you can already hear the skeleton of almost every chord.

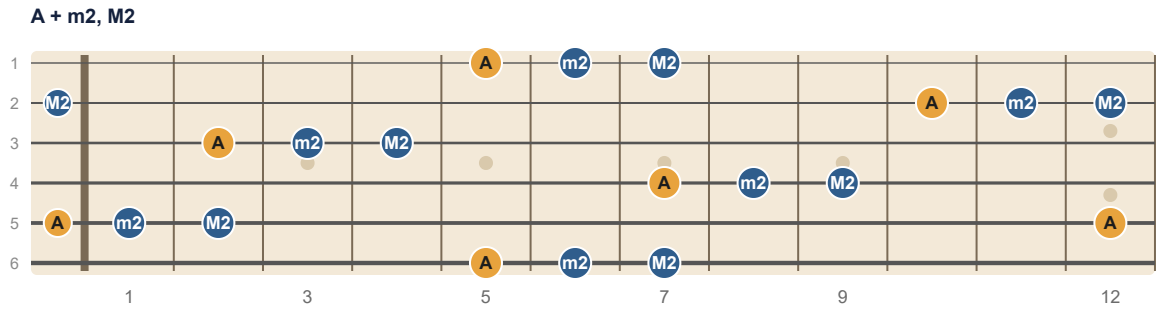


Perfect fourth (D) and perfect fifth (E) above every A. These two are your anchors.

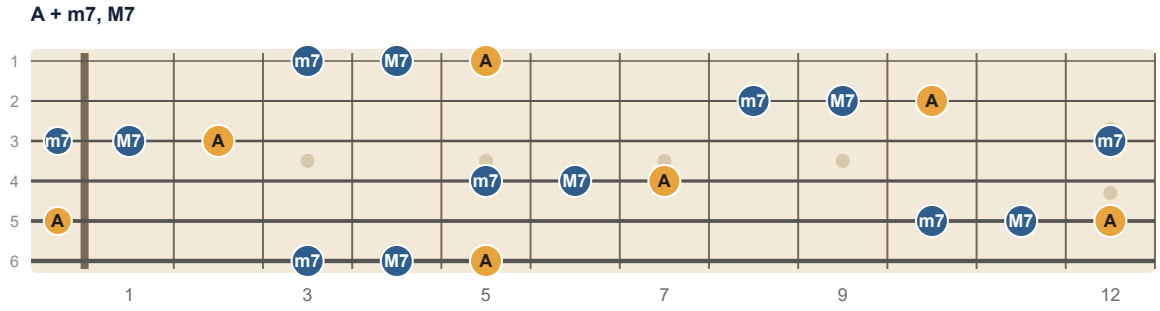
Fourth and Fifth Are Mirror Images

A P5 up from A is E. A P4 up from E is A again. A fifth and a fourth add up to an octave (7 + 5 = 12). The same is true for thirds and sixths: M3 + m6 = 12, and m3 + M6 = 12. Flip an interval upside down and the pair always totals 12.

Seconds and Sevenths



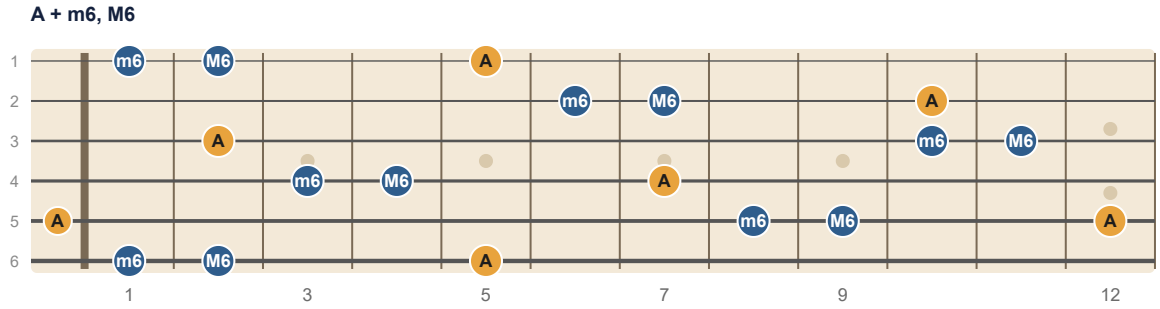
Minor second (A#) and major second (B). One and two frets away: the tight pair.



Minor seventh (G) and major seventh (G#). These sit two and one frets below the octave.

Notice something handy: a seventh is just a second turned upside down. The m7 (G) sits 2 frets *below* the next A, and the M7 (G#) sits 1 fret below it. When you are lost, find the octave and step down.

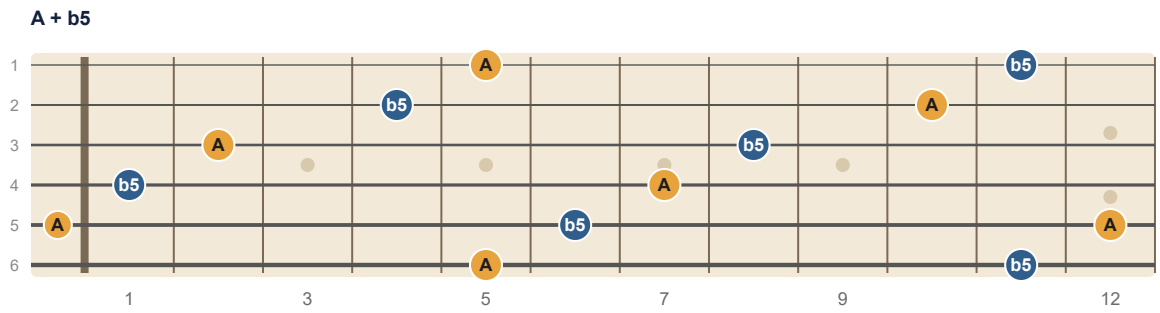
Sixths



Minor sixth (F) and major sixth (F#). Warm and wide, a third below the octave.

The Tritone: The Unsettled Middle

Exactly halfway through the octave, at 6 half steps, lives the **tritone**. It is also written b5 (flat five) or #4 (sharp four), depending on the situation. It does not belong to the perfect group and does not have a clean major or minor feel. It sounds restless, like a question with no answer, which is why blues and jazz players love to lean on it and then let it resolve.



The tritone from A is D# (also called Eb). It sits one fret above the P4 and one fret below the P5.

The tritone is the only interval that is the same size upside down: 6 up is 6 down. From A, the tritone is D#; from D#, the tritone is A. Keep that in mind when we reach chords, where it appears inside dominant sevenths and diminished chords.

Same Shape Anywhere

Now the part your hands will enjoy. On a guitar in standard tuning, neighboring strings are a perfect fourth apart (5 half steps), except for one pair. This regularity is what makes a shape slide up and down the neck without changing.

> **The Tuning Map**
>
> Moving to the next higher string adds 5 half steps on strings 6 to 5, 5 to 4 and 4 to 3 (E to A, A to D, D to G). Between strings 3 and 2 (G to B) it adds only 4. From string 2 to string 1 (B to E) it is 5 again.

So, from a root on one string, an interval on the next string up sits at a fret you can predict. Use this rule for any normal pair of strings: **fret on the next string = fret of the root + half steps in the interval - 5**. For the pair of strings 3 and 2, subtract only 4 instead of 5, so the note sits one fret higher than the normal rule says. You do not need to do this arithmetic every time: the table below turns it into a shape.

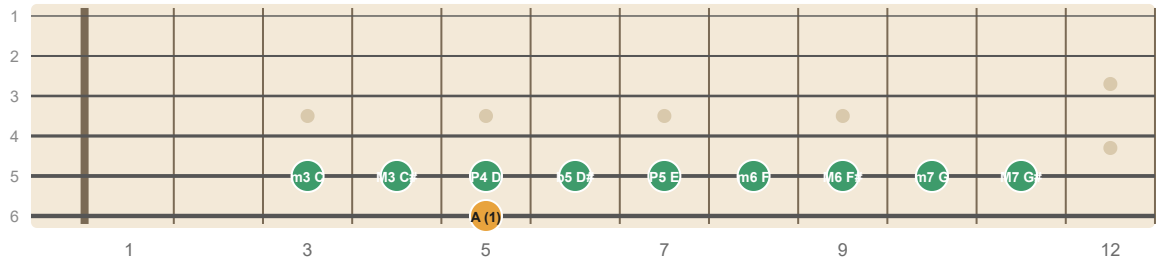
INTERVAL	NORMAL PAIRS (6/5, 5/4, 4/3, 2/1)	THE 3/2 PAIR (G TO B)
m3	2 frets back	1 fret back
M3	1 fret back	same fret
P4	same fret	1 fret up
b5	1 fret up	2 frets up
P5	2 frets up	3 frets up
m6	3 frets up	4 frets up
M6	4 frets up	5 frets up
m7	5 frets up	6 frets up

Read 'back' as toward the headstock (a lower fret number) and 'up' as toward the body (a higher fret number). All of these describe where the note sits on the *next higher string*, measured from the root's fret.

Strings 6 and 5

Take A on string 6, fret 5. On string 5, the intervals fall at the frets shown below. Every number follows from the rule: m3 (C) is at fret 5 + 3 - 5 = 3, M3 (C#) at fret 4, P4 (D) at fret 5, and so on up to M7 (G#) at fret 11.

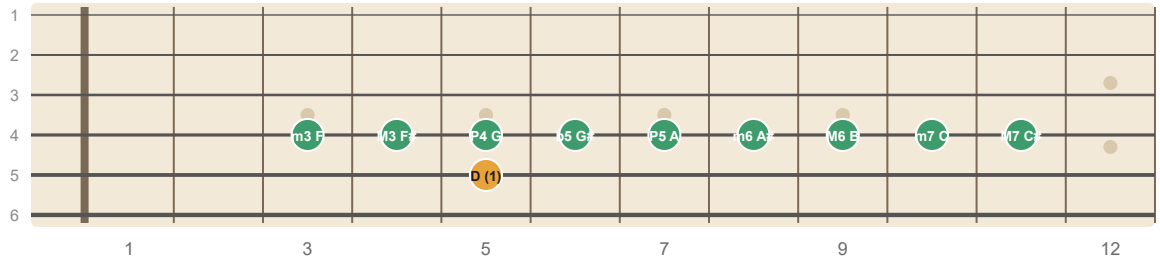
Strings 6 to 5: Root on the Low E String



Root A on string 6, fret 5. Targets appear on the A string at frets 3 to 11. The frets simply count up with the intervals.

Strings 5 and 4

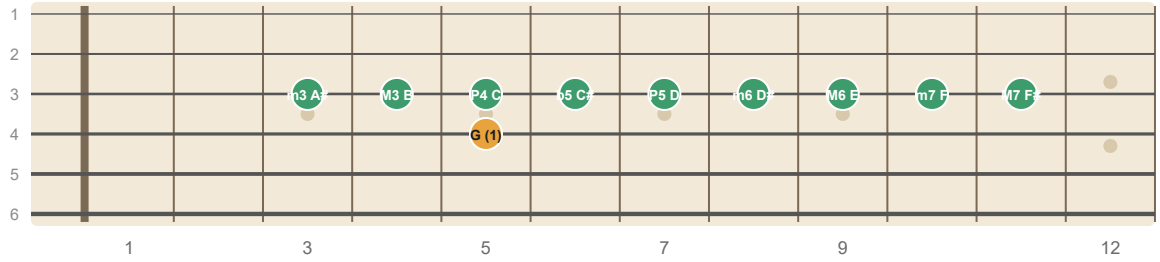
Strings 5 to 4: Root on the A String



Root D on string 5, fret 5. Same shape, same pattern on the D string. Only the note names change.

Strings 4 and 3

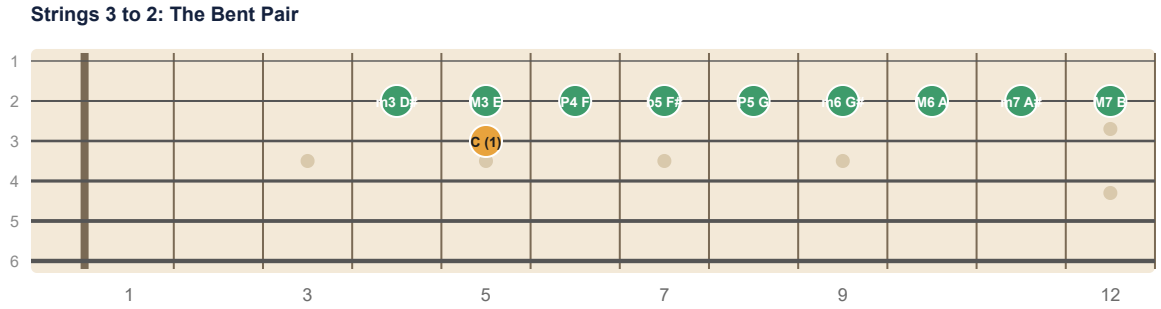
Strings 4 to 3: Root on the D String



Root G on string 4, fret 5. Once again the pattern is identical.

The B String Exception

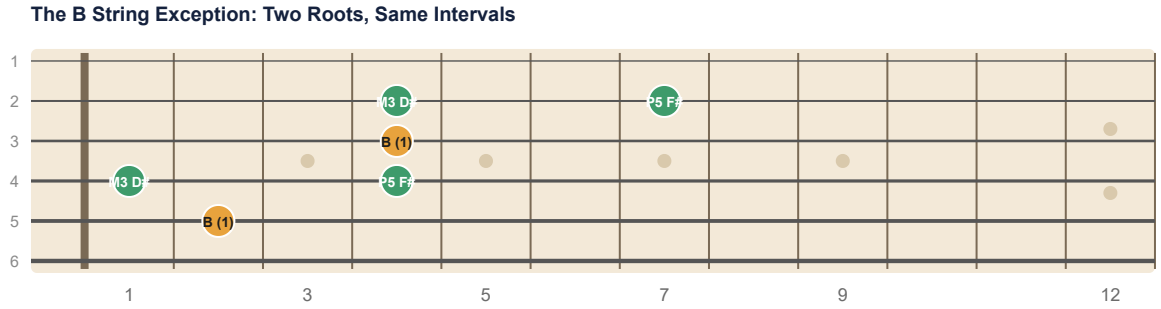
Here is the one place the shape bends. Between the G string and the B string, the gap is a major third instead of a fourth. Every interval that goes from string 3 to string 2 therefore lands **one fret higher** than the normal rule predicts. Compare the next diagram with the three above: the whole set of targets slides one fret to the right.



Root C on string 3, fret 5. Targets on the B string sit at frets 4 to 12, each one fret higher than the normal pattern would give.

> **A Memory Hook**
>
> Use this sentence: 'Crossing into the B string, add one.' Whenever your shape travels from the G string to the B string, move the upper note up by one fret. From the B string to the high E string everything is normal again, because B to E is a fourth.

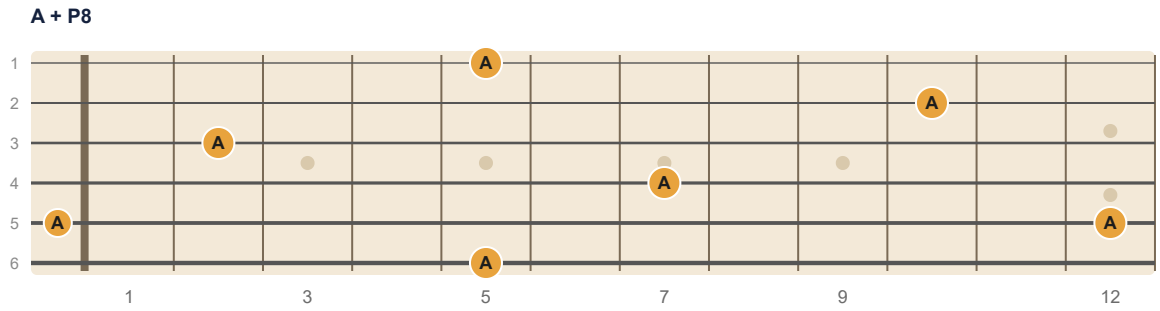
Here are two quick proofs. A major third from G (string 3, open) is B, which is string 2, open. Same fret, 0 and 0: that matches the table. A perfect fifth from G (string 3, open) is D, which is string 2, fret 3, and the table says 3 frets up. Everything checks out.



Root B on string 5 (fret 2): M3 is one fret back, P5 is two up. Root B on string 3 (fret 4): M3 is the same fret, P5 is three up.

What If the Shape Skips a String?

Intervals can also jump over a string. The most useful one is the octave. Going from the root up an octave, you skip one string and move 2 frets up: string 6 to string 4, string 5 to string 3. For the pairs that cross the bent gap (string 4 to string 2, and string 3 to string 1) the octave sits 3 frets up. Octaves are the quickest way to double a note and to check that you really know where a root is.

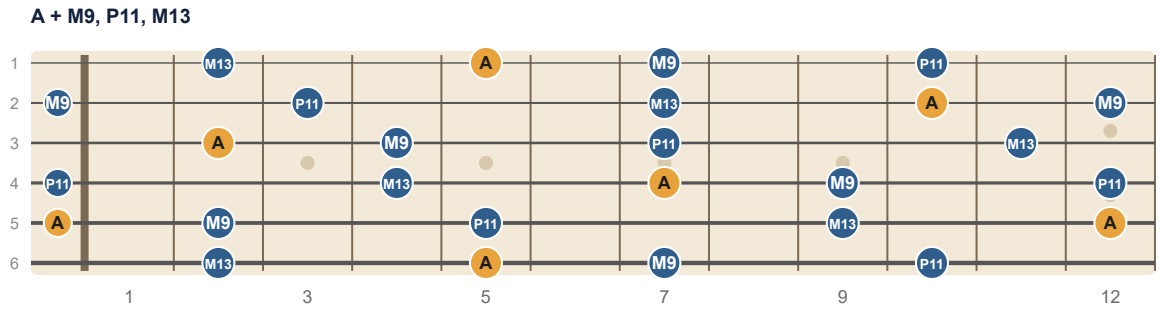


Every A and the A an octave above it. Look for the skip-a-string, two-frets-up pattern.

Beyond the Octave: 9ths, 11ths and 13ths

When you see names such as 9, 11 and 13 on a chord, they are not new notes. They are the same notes you already know, one octave higher. A **9th** is a 2nd plus an octave (2 + 12 = 14 half steps). An **11th** is a 4th plus an octave (5 + 12 = 17). A **13th** is a 6th plus an octave (9 + 12 = 21). Naming them this way matters later, because in a chord the 9 is used instead of the 2 when the note sounds above the root and has room to breathe.

NAME	SAME AS	HALF STEPS	FROM A YOU REACH
b9	m2 + octave	13	A#
9	M2 + octave	14	B
#9	m3 + octave (a bluesy sound)	15	C
11	P4 + octave	17	D
#11	tritone + octave	18	D#
b13	m6 + octave	20	F
13	M6 + octave	21	F#



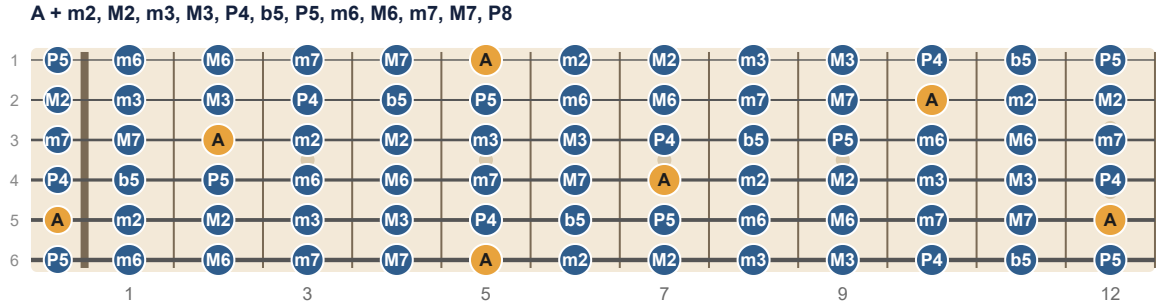
The 9 (B), 11 (D) and 13 (F#) above A. On the fretboard they land on the same spots as the 2, 4 and 6; the difference is only how you use them.

> **The Fast Conversion**
>
> To translate a number above 8 back to the one you already know, subtract 7: 9 is 2, 11 is 4, 13 is 6. This works for names only. The sound is the same note; the higher-octave version simply sits further above the root.

Building the Interval Map From Any Root

Now put it all together. This routine gives you every interval from any note, in about a minute. Try it first with A, then with any other root you like.

1. Pick a root and find it on string 6 and on string 5.
2. Mark the octave: skip a string and go 2 frets up (3 frets if you cross the bent gap).
3. Mark the P5 two frets up on the next string, and the P4 on the same fret on the next string.
4. Mark the M3 one fret back on the next string, and the m3 two frets back.
5. Mark the M6 four frets up on the next string, and the m6 three frets up.
6. Fill in the second, the sevenths and the tritone by moving one fret from the octave, the fifth or the fourth.



The complete map: all 12 intervals above every A in the first 12 frets. Notice how much of the neck is already covered.

You will not memorize all of this at once, and you do not have to. Start with the five that carry the most weight: the octave, the P5, the M3, the m3 and the P4. Together they cover almost every chord shape you will meet in the next modules.

Ear Training: Play Two Notes, Name the Feel

Seeing an interval is half the skill. Hearing it is the other half, and the two together are what make you feel at home on the neck. The goal is not perfect pitch. The goal is to attach a feeling to each sound, so that you can say 'that was a third' as easily as you recognize a friend's voice.

1. Choose one interval for the day (start with M3 or m3, or P5 and P4).
2. Play the root, then the interval note, one after the other. Let each ring.
3. Play them together as well, so you hear the blend, then apart again.
4. Say the feel word out loud before you say the name. 'Bright' first, 'major third' second.
5. Hum the top note while you play the root. If you can sing it, you know it.

6. Add a second interval the next day and flip between the two, eyes closed.

> **A Simple Order to Learn Them**
>
> Most players find this order works: P5 and P4 first (the skeleton), then M3 and m3 (the color), then the octave, then M6 and m7, and last the seconds, the m6, the M7 and the tritone. Keep each session short, about five minutes. A little each day beats an hour once a week.

Another useful habit: whenever you play a riff or a song you love, stop and ask 'what is the distance between these first two notes?' Look at the fretboard, apply the rule, and name it. Songs become free ear-training material.

Common Mistakes

- **Counting the starting note.** From A to C is 3 half steps, not 4. Count only the steps you move.
- **Mixing up m3 and M3.** The major third has 4 half steps and sounds bright. The minor third has 3 and sounds darker. If in doubt, ask 'happy or sad?'.
- **Forgetting the B string exception.** When a shape crosses from string 3 to string 2, the upper note sits one fret higher. Most wrong notes in shapes come from here.
- **Thinking a tritone is a fifth.** The b5 is one fret below the P5. They are neighbors, but they sound very different.
- **Calling every interval 'a note'.** Always name both the distance and the feel. Naming the feel makes your ear faster.
- **Memorizing shapes without listening.** A shape you cannot hear will not help you improvise. Play every shape you learn.
- **Trying to learn all 12 in one day.** Add one or two a week. Most players find that slow and steady works better than a big push.

Module 2 Cheat Sheet

INTERVAL	HALF STEPS	FEEL	FRETS ON THE NEXT STRING UP	FROM A
m2	1	Tense	4 back	A#
M2	2	Stepping	3 back	B
m3	3	Sad	2 back	C
M3	4	Bright	1 back	C#
P4	5	Open	same fret	D
b5	6	Unstable	1 up	D#
P5	7	Strong	2 up	E

INTERVAL	HALF STEPS	FEEL	FRETS ON THE NEXT STRING UP	FROM A
m6	8	Aching	3 up	F
M6	9	Warm	4 up	F#
m7	10	Bluesy	5 up	G
M7	11	Dreamy	6 up	G#
P8	12	Home again	skip a string, 2 up	A

The B String Rule

Crossing from the G string (3) to the B string (2), add one fret to everything in the column above.

Quick Reminders

Minor is one half step smaller than major. Perfect intervals are P1, P4, P5 and P8. 9 = 2, 11 = 4, 13 = 6 (one octave higher). Pairs that flip upside down add to 12.

Glossary in Plain English

- **Interval:** the distance between two notes.
- **Half step:** one fret, the smallest distance on the guitar.
- **Whole step:** two half steps (two frets).
- **Root:** the note you measure from.
- **Unison (P1):** the same note twice, zero half steps.
- **Minor (m):** the smaller version of an interval, one half step less than major.
- **Major (M):** the larger version of an interval.
- **Perfect (P):** a fourth, fifth or octave; stable and neutral.
- **Tritone:** the interval of 6 half steps, half an octave; also called b5 or #4.
- **Octave:** 12 half steps; the same note at a higher pitch.
- **Third:** the interval that decides major or minor feel.
- **Extension:** a 9, 11 or 13; a 2, 4 or 6 placed an octave higher.

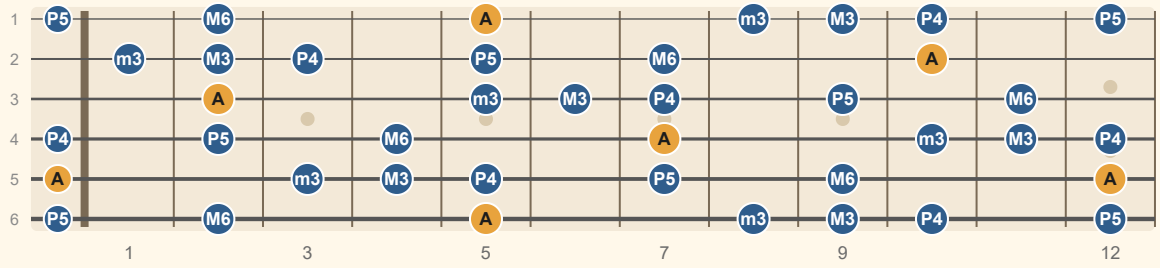
On the Fretboard: Exercises

EXERCISE 6 — 5 MIN

Name the Distance

1. Play the open A string (string 5), then the note on string 5, fret 7. Count the half steps (7) and name the interval.
2. Now play string 5, fret 4 after the open A. Count again and name it.
3. Do the same for frets 3, 5 and 9 on string 5.
4. For each one, say the feel word out loud before the name.

A + m3, M3, P4, P5, M6



Answer key on the A string: frets 3, 4, 5, 7 and 9.

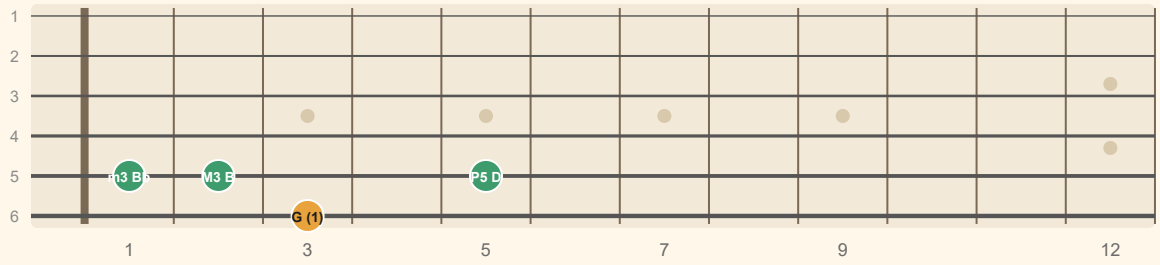
☐ Self-check: fret 7 is a P5 (E), fret 4 is a M3 (C#), fret 3 is a m3 (C), fret 5 is a P4 (D), fret 9 is a M6 (F#). If you missed any, count again from zero.

EXERCISE 7 — 6 MIN

Fifths and Thirds From G

1. Find G on string 6, fret 3. Play it.
2. Use the normal-pair rule to find the P5 on string 5. (Hint: 2 frets up.) Play both notes together.
3. Find the M3 (1 fret back) and the m3 (2 frets back) on string 5. Play each with the G.
4. Say which one sounds bright and which one sounds sad.

Root G on String 6



The three notes you are looking for, relative to the G on string 6.

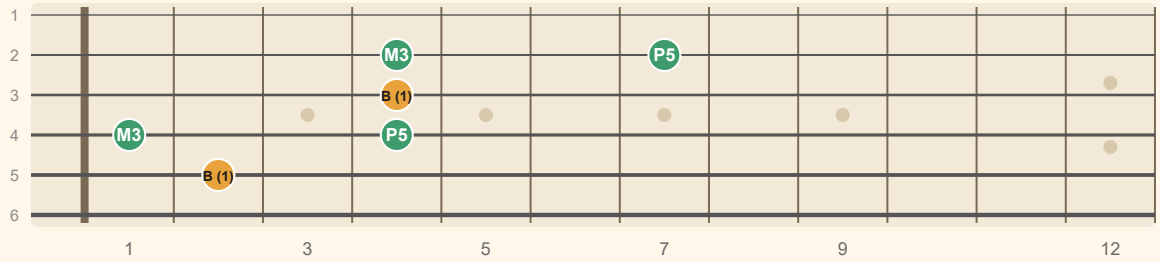
☐ Self-check: the P5 is D (string 5, fret 5), the M3 is B (string 5, fret 2) and the m3 is Bb (string 5, fret 1). The M3 should sound bright and the m3 darker.

EXERCISE 8 7 MIN

Cross the Bent Gap

1. Play B on string 5, fret 2. Find its M3 on string 4 (one fret back). That is D#.
2. Now play B on string 3, fret 4. Find its M3 on string 2. Use the 3/2 column of the table.
3. Play both pairs and compare how they sound and where your fingers land.
4. Repeat for the P5 in both places.

Same Intervals, Two Places



Compare the two shapes. The only difference is the bent gap.

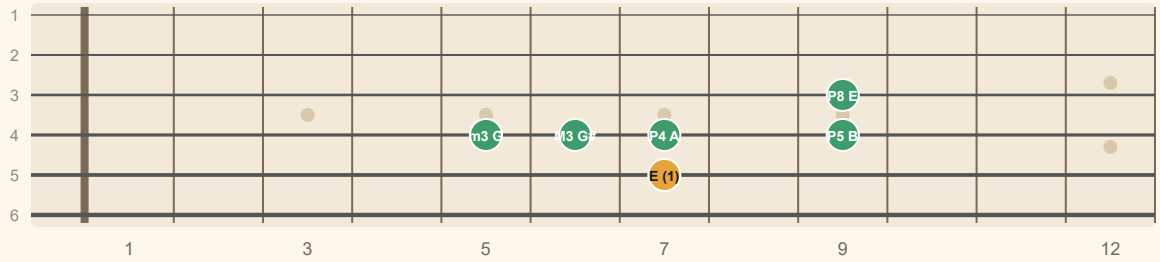
☐ Self-check: the first M3 is string 4, fret 1 (D#). The second M3 is string 2, fret 4 (D#, the same fret). The P5 is F# at string 4, fret 4 in the first place, and at string 2, fret 7 in the second place. The shape looks different but sounds the same.

EXERCISE 9 8 MIN

Build the Map From E

1. Find E on string 5, fret 7. This is your root.
2. Mark the octave: skip a string and go 2 frets up, string 3, fret 9.
3. On string 4, mark the P5 (B) and the P4 (A) with the rule: root fret + half steps - 5.
4. Mark the M3 (G#) and the m3 (G) on string 4 the same way.
5. Play each one and say its feel word.

Root E on String 5



Compare your marks with the answer.

☐ Self-check: P5 is B at string 4, fret 9; P4 is A at string 4, fret 7; M3 is G# at string 4, fret 6; m3 is G at string 4, fret 5; the octave E is at string 3, fret 9. If a number is off, recheck the rule: root fret + half steps - 5.

EXERCISE 10 8 MIN

Ear Check: Close Your Eyes

1. Pick two intervals you are still unsure about (for example M3 and P4).
2. With your eyes closed, play a root and then one of the two intervals. Do not look at which one.
3. Say your guess aloud: bright or open? Then open your eyes and check the frets.
4. Do ten rounds. Count how many you got right.
5. Write down which interval confused you most, and repeat it tomorrow.

☐ Self-check: after ten rounds, most players get six or seven right on the first day. If you are below five, slow down and play each note twice. Progress comes quickly with short daily practice.

Next Step

You can now measure the distance between any two notes and recognize its feel. In Module 3, **Scales and Keys: The Master Rule**, you will use these same distances to build the major scale from one pattern of steps. That single pattern unlocks every key on the guitar. Before you turn the page, take two minutes and play the A-string example from exercise 6 one more time, and say the feel word out loud.

THE FRETBOARD REVEALED

Part 2 - Build

MODULE 3

Scales and keys: the master rule

- A key is just seven notes. Almost every chord and melody in a song is built from them.
- The major scale follows one pattern, W W H W W W H, and that pattern works from any starting note.
- Every major key has a relative minor that uses exactly the same seven notes.
- The pentatonic scale trims those seven notes down to five and gives you five shapes that cover the whole neck.
- Because the fretboard is symmetrical, a shape you learn once moves to every other key.

By the end: Build the major scale in five keys, name the job of each degree, spell key signatures as note lists, find the relative minor, play all five pentatonic boxes, and work out the key of a song from its chords.

> *Here is the idea that makes the rest of the book easy: music does not use all twelve notes at once. A song picks seven of them, and almost everything you hear comes from that small group. This module shows you how to find those seven notes in any key, and where they live on the neck.*

The big picture: why scales matter

In Module 1 you learned to name any note on the neck. In Module 2 you learned that the distance between two notes has a sound and a feeling. Now we put those two skills to work. A scale is a set of notes arranged from low to high, starting and ending on a home note. That is all it is.

You do not need scales to impress anyone. You need them because they answer the question every self-taught player eventually asks: *why does this note sound right here and that one sound wrong?* The answer is almost always that one note belongs to the key and the other does not. Once you can see the seven notes of a key, you stop guessing and start choosing.

The master rule

> **The master rule**
>
> A key is a group of 7 notes. The chords in the song and the notes in the melody come from that group. Learn the 7 notes of a key and you know what is safe, what is strong, and what will sound like a surprise.

Hold on to that sentence. Every other idea in this module, from key signatures to pentatonic boxes, is a way of finding or using those seven notes faster. Think of a key as a small toolbox. You do not need

every tool in the shop; you need the right seven, close at hand.

The major scale: the pattern behind it

The major scale is the sound of do-re-mi. You know it by ear already. What we want now is the recipe, because the recipe works in every key.

Remember the two basic distances from Module 2. One fret is a **half step** (H). Two frets is a **whole step** (W). The major scale is built like this, starting from the root:

> **The major scale formula**
>
> W - W - H - W - W - W - H. Add up the frets: 2 + 2 + 1 + 2 + 2 + 2 + 1 = 12. Seven steps take you from the root to the same note one octave higher.

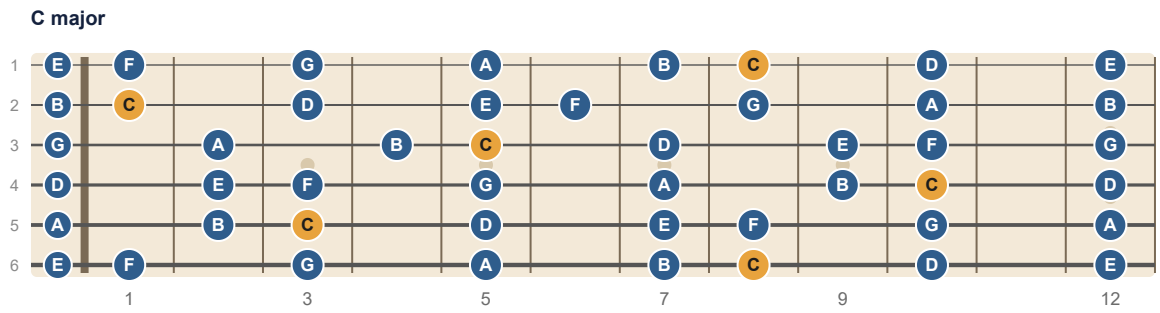
Start on C and follow the formula. C up a whole step is D. D up a whole step is E. E up a half step is F. Then whole steps to G, A and B, and a final half step back to C. No sharps or flats appear, which is why C is the easiest key to read.

STEP	MOVE	FROM	TO
1	W (2 frets)	C	D
2	W (2 frets)	D	E
3	H (1 fret)	E	F
4	W (2 frets)	F	G
5	W (2 frets)	G	A
6	W (2 frets)	A	B
7	H (1 fret)	B	C

Why this pattern?

The two half steps sit at positions 3 to 4 and 7 to 8. Those are the places where the scale feels pulled upward: the last note, B, leans hard toward the home note C. The whole steps in between spread the other notes out evenly. Most players find that this mix of mostly even steps and two small leaning steps is what gives the scale its settled, happy character.

You do not have to memorize why. Just remember that the half steps are in two places, and everything else is a whole step. In note names, the natural notes E to F and B to C already have a half step between them, which is why C major needs no sharps.



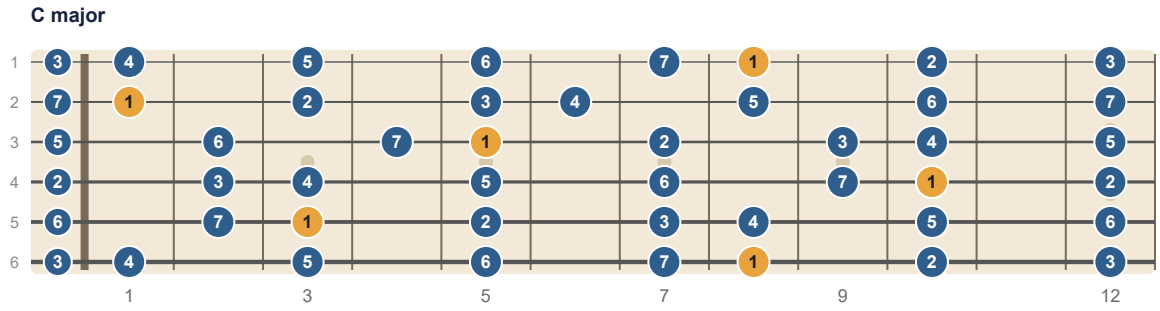
The C major scale across the neck: C D E F G A B. The notes follow the same pattern on every string.

The major scale in five keys, on the neck

Now we apply the formula to five keys that fit the guitar naturally. For each one, read the note list first, then find the notes on the diagram. Degree numbers show you the role of each note: 1 is the root, 3 is the third note of the scale, and so on.

C major

C D E F G A B. No sharps or flats. The diagram below shows the same scale as degree numbers, so you can see the pattern of 1 to 7 repeating along the neck.

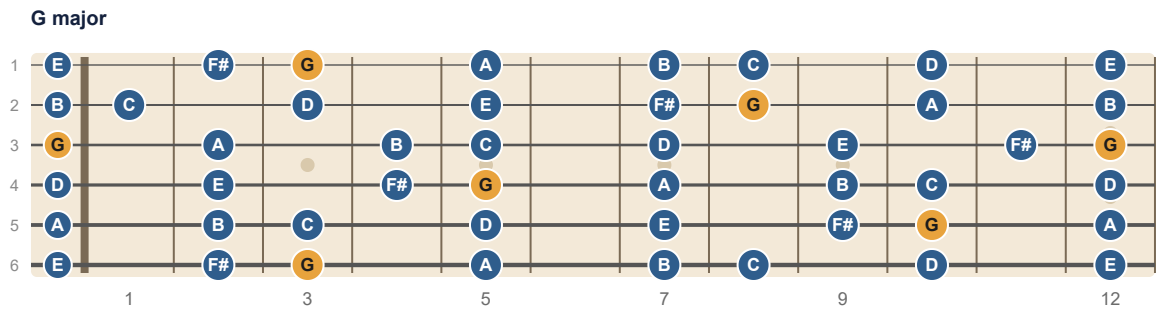


C major with degrees. Every 1 is a C.

G major

Follow the formula from G. G up a whole step is A. A up a whole step is B. B up a half step is C. C up a whole step is D, D up a whole step is E, E up a whole step is F#. That last note needs a sharp: from E, a whole step lands on F#, not F. Then a half step brings you back to G.

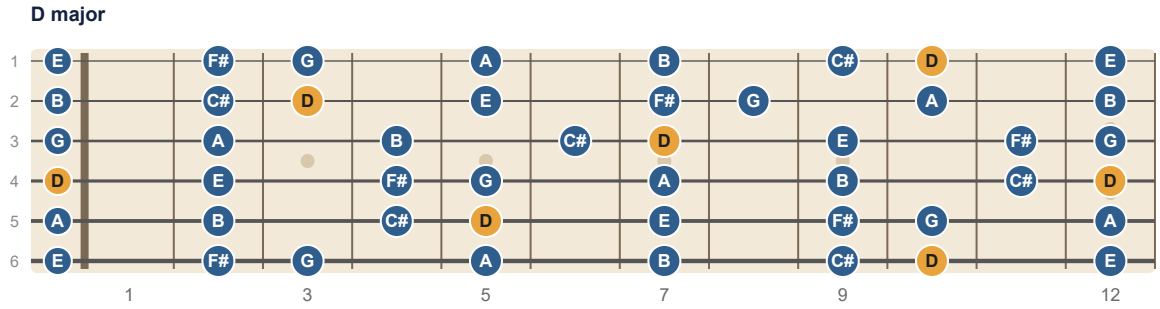
G A B C D E F#. One sharp.



G major: G A B C D E F#. Find the single sharp note, F#, on each string.

D major

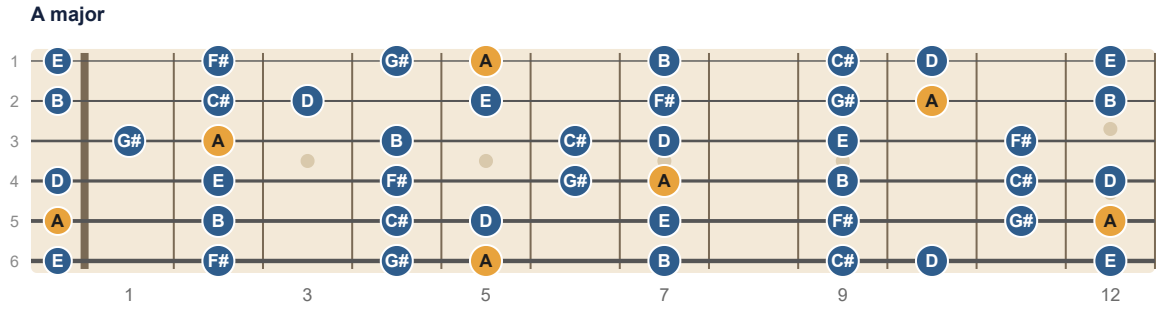
D E F# G A B C#. From D, a whole step is E. A whole step from E is F#. A half step from F# is G. Then whole steps to A, B and C#, and a half step back to D. Two sharps: F# and C#.



D major: D E F# G A B C#.

A major

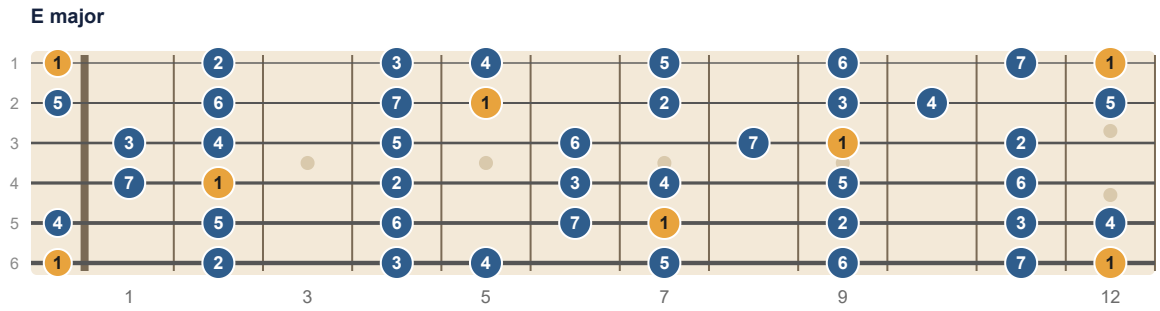
A B C# D E F# G#. Three sharps. Notice how the sharps keep being added in the same order: F#, then C#, then G#. This is no accident, and the next section explains it.



A major: A B C# D E F# G#.

E major

E F# G# A B C# D#. Four sharps: F#, C#, G#, D#. The low E string is the root, so this key sits comfortably in open position and is very common in rock, blues and folk.



E major with degrees. Compare it with the C major diagram: the same numbers, a different set of positions.

> **Check yourself**
>
> Count the notes in each key. It should always be seven different letters (C D E F G A B), with a sharp or flat added to some of them. If you ever see the same letter twice, such as F and F#, recheck the formula.

Scale degrees and their jobs

Each note of the scale has a number, called its **degree**, and each degree has a personality. Knowing the personalities lets you hear what a note is doing, not just what it is called.

DEGREE	NAME	JOB IN PLAIN ENGLISH	IN C MAJOR
1	Tonic	Home. Everything resolves here and feels finished.	C
2	Supertonic	A passing note. It wants to move to 1 or 3.	D
3	Mediant	The color note. It tells you major or minor.	E
4	Subdominant	A step away from home. It feels open and lifting.	F
5	Dominant	Strong tension. It pulls hard toward home.	G
6	Submediant	Warm and slightly sad. It leads to the relative minor.	A
7	Leading tone	Very tense. It leans up a half step into the tonic.	B

The most important three to remember are the **tonic** (1), the **subdominant** (4) and the **dominant** (5). When you meet chords in the next modules, the chords built on those three notes carry most of the music in most songs. For now, remember the roles: 1 is home, 5 is the pull toward home, and 4 is the step away from it.

> **Hear it**
>
> Play a C note, then play G (the dominant), then play C again. Feel the tension of G and the relief when you land back on C. That is what the dominant does. Then play B (the leading tone) and notice how badly it wants to climb to C.

Key signatures, without a staff

You may have heard that a key has a *key signature*: the number of sharps or flats in it. In sheet music this is drawn on a staff. We never need a staff here. A key signature is simply a list of which notes are altered. You already saw this: G major has F#, D major has F# and C#.

The sharps arrive in a fixed order: F#, C#, G#, D#, A#, E#, B#. Each new sharp key adds the next one to the list. The flat keys work the other way, with flats appearing in the order Bb, Eb, Ab, Db, Gb, Cb, Fb.

MAJOR KEY	SHARPS OR FLATS	ALTERED NOTES	ALL SEVEN NOTES
C	0	none	C D E F G A B
G	1 sharp	F#	G A B C D E F#
D	2 sharps	F# C#	D E F# G A B C#
A	3 sharps	F# C# G#	A B C# D E F# G#
E	4 sharps	F# C# G# D#	E F# G# A B C# D#
F	1 flat	Bb	F G A Bb C D E
Bb	2 flats	Bb Eb	Bb C D Eb F G A
Eb	3 flats	Bb Eb Ab	Eb F G Ab Bb C D

Notice that the keys G, D, A, E go up by a fifth each time (G to D is a fifth, D to A is a fifth, and so on). That is the beginning of the circle of fifths, which is the star of Module 9.

> **A shortcut for sharp keys**
>
> In a sharp key, the last sharp in the list is the note just below the root. In D major the last sharp is C#, one half step below D. In E major the last sharp is D#, one half step below E. If you know the order of sharps, you can name the key.

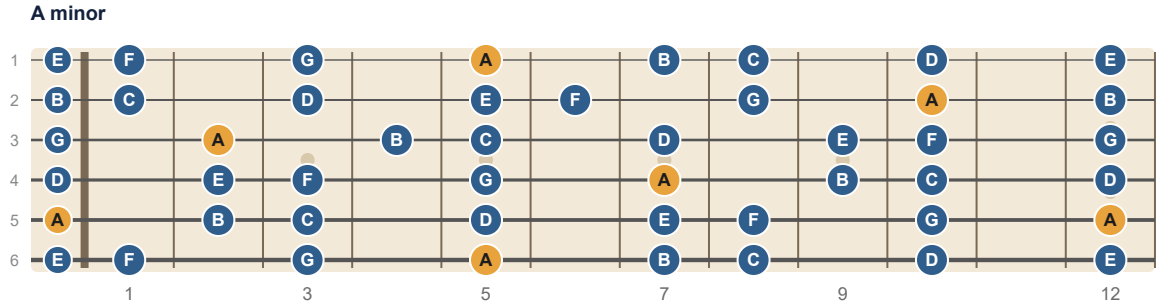
> **A shortcut for flat keys**
>
> In a key with two or more flats, the second-to-last flat in the list is the name of the key. Eb major has Bb, Eb and Ab; the second-to-last is Eb. Bb major has Bb and Eb; the second-to-last is Bb. The key of F, with one flat, is the one to memorize.

The relative minor

Every major key has a minor key that uses exactly the same seven notes. It is called the **relative minor**. You find it by starting on the sixth degree of the major scale.

In C major the sixth note is A. Play C D E F G A B but start and stop on A, and you get A B C D E F G: the A natural minor scale. Same notes, new home. A feels like the center now, and the music sounds darker and more serious.

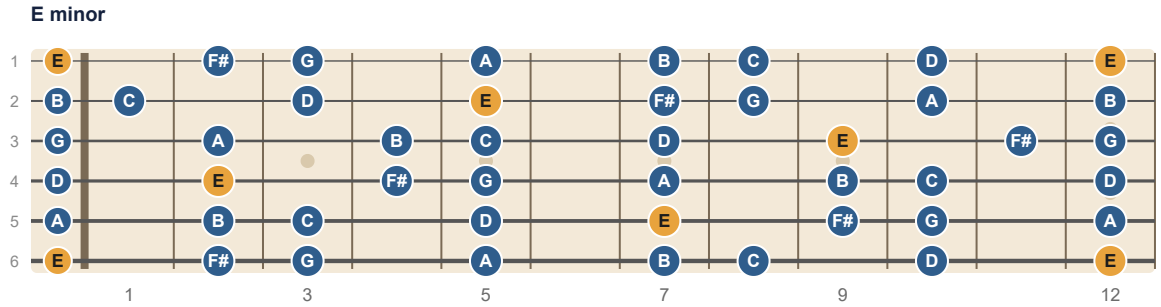
> **The natural minor formula**
>
> W - H - W - W - H - W - W. In degrees: 1 2 b3 4 5 b6 b7. Compared with the major scale, the third, sixth and seventh are each one half step lower.



A natural minor: A B C D E F G. These are the exact notes of C major, starting from A.

MAJOR KEY	RELATIVE MINOR	SHARED NOTES
C major	A minor	C D E F G A B
G major	E minor	G A B C D E F#
D major	B minor	D E F# G A B C#
A major	F# minor	A B C# D E F# G#
E major	C# minor	E F# G# A B C# D#
F major	D minor	F G A Bb C D E

To find the relative minor on the neck: from the major root, move down three frets (a minor third) on the same string. From C on fret 8 of string 6, go down to fret 5: that is A. From G on fret 3, go down three frets and you reach E on the open string. This shortcut will save you time again and again.



E natural minor: E F# G A B C D. The same seven notes as G major.

> **Same notes, different home**
>
> A major scale and its relative minor are not two scales to learn. They are one set of notes with two possible homes. Which one you hear depends on which note feels like home, and that depends on the chords and where the melody settles.

Pentatonic scales: five notes, five shapes

The word pentatonic means five notes. A pentatonic scale is a major or minor scale with two notes removed, and it is a favorite of guitarists because it sounds good almost everywhere and is easy to finger.

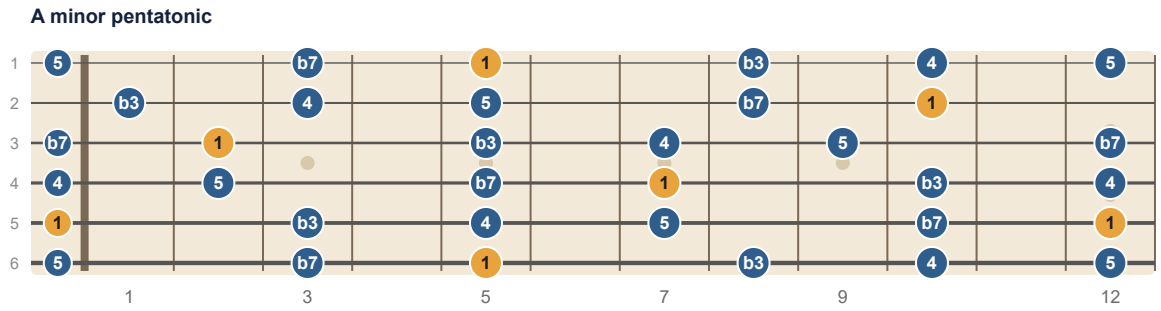
Minor pentatonic

Take the natural minor scale (1 2 b3 4 5 b6 b7) and remove the 2 and the b6. You are left with **1 b3 4 5 b7**. In A, that is **A C D E G**. Five notes, no half steps, no awkward spots.

Major pentatonic

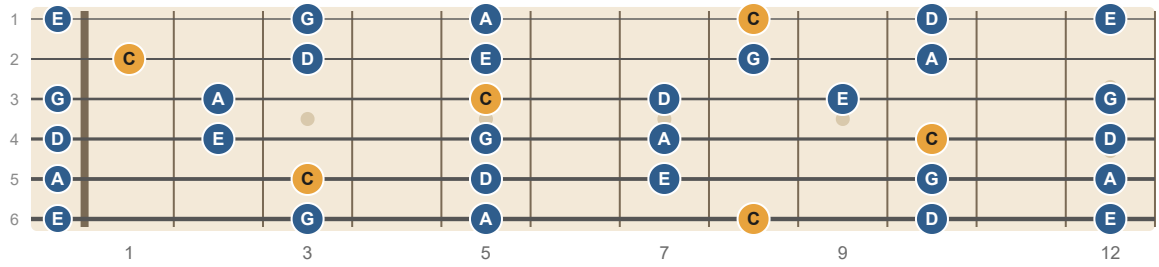
Take the major scale (1 2 3 4 5 6 7) and remove the 4 and the 7. You are left with **1 2 3 5 6**. In C, that is **C D E G A**. It has a bright, open sound often heard in country, pop and folk. The two notes it drops are the ones that clash most easily with chords, which is why it is so forgiving.

> **Two scales, one set of notes**
>
> C major pentatonic (C D E G A) and A minor pentatonic (A C D E G) contain exactly the same notes. Just as with the full scales, they are relatives.



A minor pentatonic, degrees: 1 b3 4 5 b7. Five notes repeated across the neck.

C major pentatonic



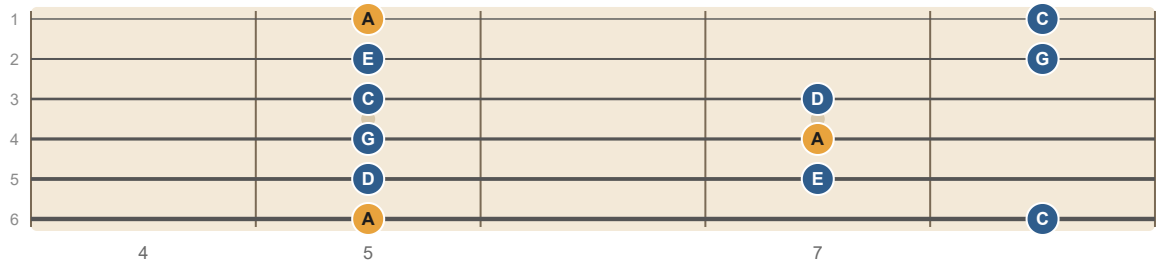
C major pentatonic: C D E G A. Compare it with the A minor pentatonic above.

The five positions of A minor pentatonic

The scale repeats along the neck, and it falls into five overlapping shapes, often called boxes. Each box covers about four frets and uses two notes per string. When you link all five, you cover the whole neck. Learn them one at a time, starting with the first.

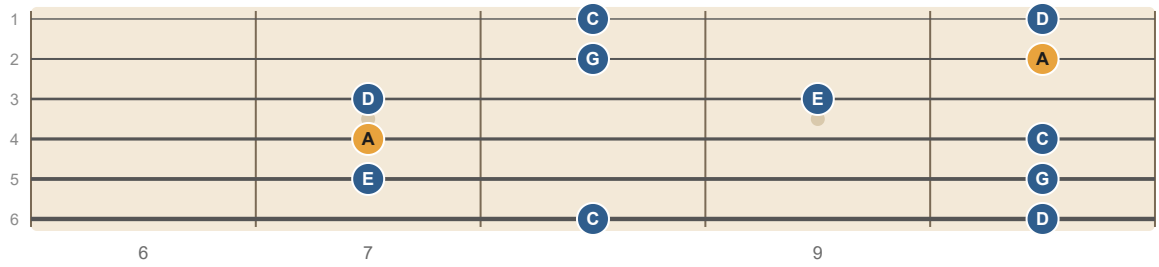
In the diagrams that follow, the root A is marked in its own color. The other dots are the remaining notes of the scale. Frets are shown by number so you can find them without counting.

Position 1 (frets 5-8)



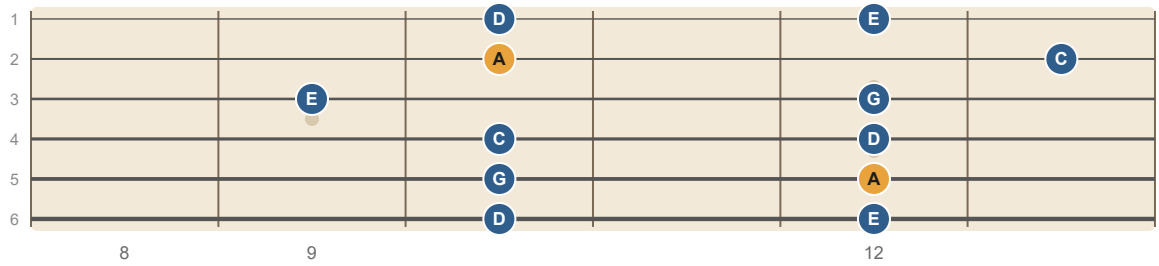
Position 1: the box most players learn first. The root A sits on string 6 fret 5, string 4 fret 7 and string 1 fret 5.

Position 2 (frets 7-10)

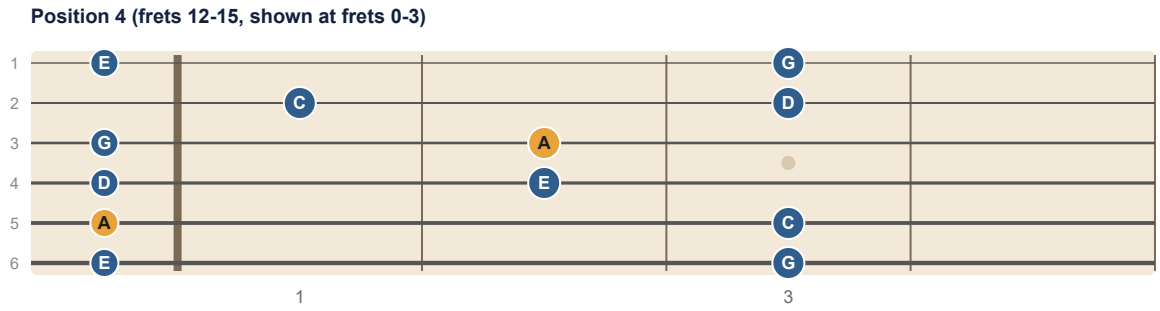


Position 2: it begins where Position 1 ends. The A roots are on string 4 fret 7 and string 2 fret 10.

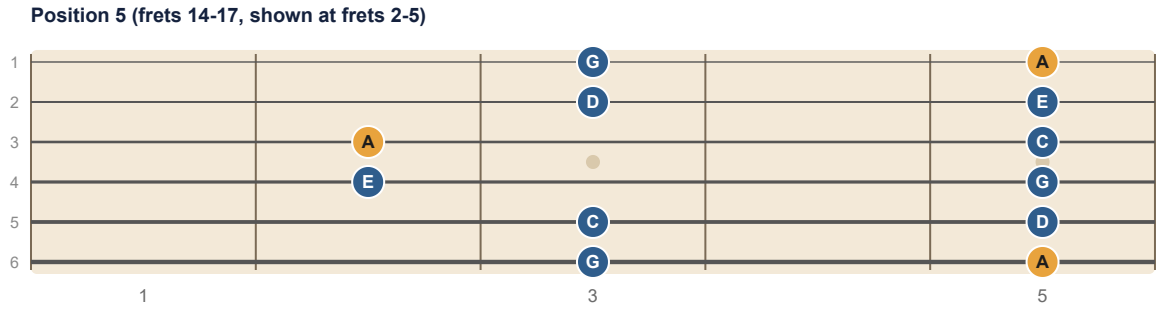
Position 3 (frets 9-13)



Position 3: the roots are on string 5 fret 12 and string 2 fret 10.



Position 4, shown in the low octave: the same shape twelve frets higher is the one that sits at frets 12-15. Roots: open string 5 and string 3 fret 2.

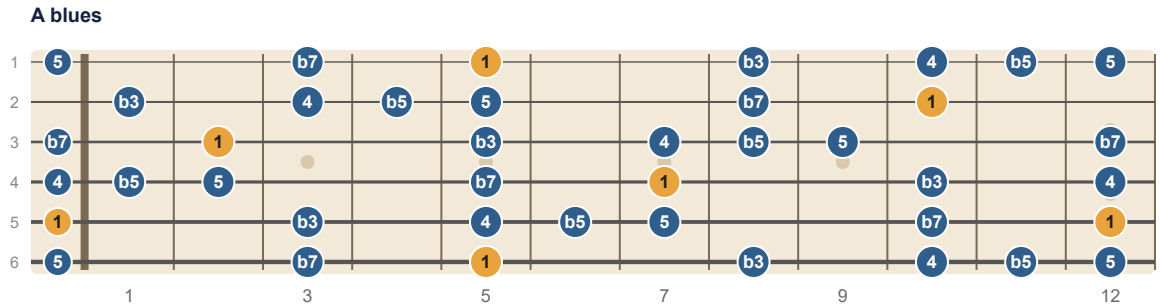


Position 5, shown in the low octave. It leads straight back into Position 1 at fret 5. Roots: string 6 fret 5, string 3 fret 2 and string 1 fret 5.

> **How the boxes fit together**
>
> Positions 1 to 5 overlap, like links in a chain. Position 5 ends where Position 1 begins, one octave lower. Once you know one position well, find a neighbor by sliding to the notes next to it. You never need to learn them all at once.

The blues scale

The blues scale is the minor pentatonic with one extra note: the b5, the so-called blue note. In A, the notes are **A C D Eb E G**, or in degrees **1 b3 4 b5 5 b7**.



A blues scale. The Eb (b5) is the added note that gives the scale its tension.

Use the b5 as a passing note: slide through it, bend into it, or touch it briefly on the way between D and E. Resting on it for a long time sounds out of tune, which is exactly why it adds tension. It is a color, not a home.

How to find the key of a song

The master rule gives you a method. If a song uses a group of seven notes, the chords in the song are made from the same group. So you can work backward from the chords to the key.

1. **Write down the chords.** Look at the chords used in the verse and the chorus.
2. **Check the first and last chord.** A song usually starts and ends on the home chord, which is the tonic. That is your strongest clue.
3. **Test the key.** Spell out the major scale you suspect. Do all the chord names use only those seven notes, with the correct sharps or flats?
4. **Look for the dominant.** If you suspect the key of G, listen for a D or D7 chord (a fifth above). A dominant leading back to your guess confirms it.
5. **Check the relative minor.** If the song seems to start on a minor chord, try the relative minor: Em can belong to the key of G major, and Am to the key of C major.

CHORDS IN THE SONG	LIKELY KEY	WHY
G C D Em	G major	All notes fit G A B C D E F#. G is first and D is the dominant.
Am F C G	C major (or A minor)	All notes are in C D E F G A B. It starts on Am, so A minor is also possible.
D G A Bm	D major	Notes fit D E F# G A B C#. D is home, A is the dominant.
E A B C#m	E major	Notes fit E F# G# A B C# D#.

> **Be flexible**
>
> Sometimes a song borrows one note from outside the key and still sounds like it belongs. That is normal, and Modules 8 and 9 will explain why. For now, ask: *do most of the chords come from one group of seven?* If yes, you have the key.

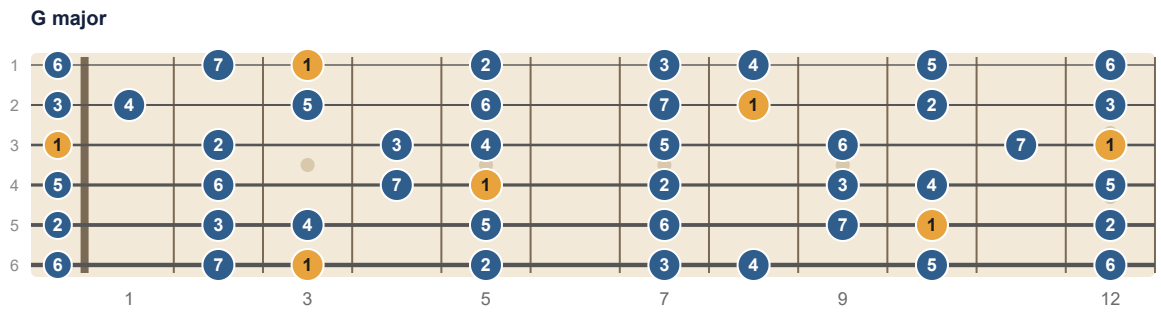
Move the shape, change the key

Here is the best news in this module: the guitar neck is symmetrical. Every string pair repeats the same relationships, so a scale shape is the same no matter where you play it. Only the root changes.

Take the C major shape with its root on string 6 at fret 8. Slide that exact shape down five frets, so the root lands on fret 3: you are now playing G major. Slide it to fret 5 and you are in A major. Slide it to fret 12 and you are in E major. The fingers do not change. Only the position does.

ROOT ON STRING 6	FRET	KEY
G	3	G major

ROOT ON STRING 6	FRET	KEY
A	5	A major
C	8	C major
D	10	D major
E	12 (or open)	E major



G major with degrees. Find the shape you learned in C and notice it has moved five frets down.

The same idea works for pentatonic boxes. The Position 1 shape of A minor pentatonic at fret 5 becomes G minor pentatonic at fret 3, B minor pentatonic at fret 7, and so on. Learn the shape once, and attach the key name by finding the root on string 6.

> **The one-shape habit**
>
> Always say two things when you play a shape: the shape's number, and the root note it is built on. This keeps you from treating every key as something new.

Common mistakes

- **Memorizing scales as finger patterns only.** The pattern is useful, but know the note names too. You will need them to change keys.
- **Forgetting that sharps and flats are part of the key.** If you play F in the key of G major, it will sound off. It must be F#.
- **Playing every scale from the lowest fret.** Practice from each root note you find on the neck, not just the open position.
- **Treating the relative minor as a different scale.** A minor and C major share the same seven notes. Learn them as one set.
- **Resting on the blue note.** The b5 in the blues scale is a passing note. Stay on it too long and it sounds like a mistake.
- **Trying to learn all five pentatonic boxes in one day.** Master Position 1, then add one neighbor at a time. Most players find that a week per box is a calm pace.

Module 3 cheat sheet

IDEA	THE RULE	EXAMPLE
Major scale formula	W W H W W W H	C D E F G A B
Natural minor formula	W H W W H W W	A B C D E F G
Minor from major	Go to the 6th degree	C major gives A minor
Degrees	1 tonic, 4 subdominant, 5 dominant	C, F, G in C major
Sharp key order	F# C# G# D# A# E# B#	G has 1, D has 2, A has 3, E has 4
Flat key order	Bb Eb Ab Db Gb Cb Fb	F has 1, Bb has 2, Eb has 3
Minor pentatonic	1 b3 4 5 b7	A C D E G
Major pentatonic	1 2 3 5 6	C D E G A
Blues scale	1 b3 4 b5 5 b7	A C D Eb E G
Move a shape	Same shape, new root	Fret 3 is G, fret 5 is A

Glossary

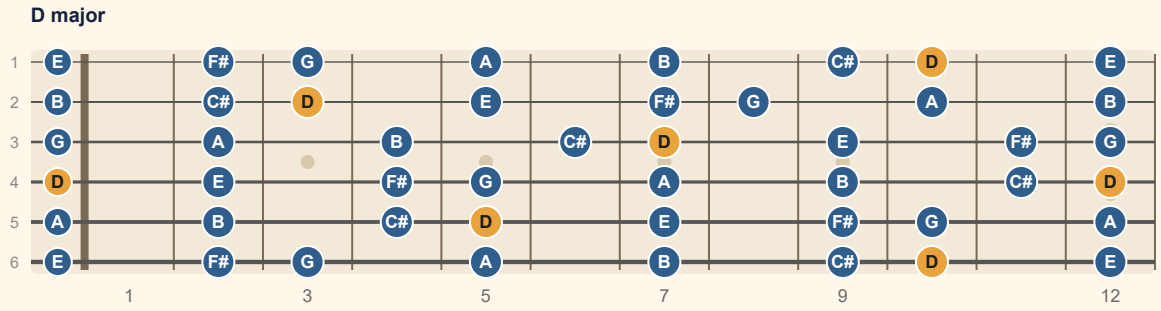
- **Scale:** a group of notes arranged from low to high, starting from a home note.
- **Key:** the seven-note group a song is built on.
- **Whole step (W):** a distance of two frets.
- **Half step (H):** a distance of one fret.
- **Degree:** a note's number inside the scale, from 1 to 7.
- **Tonic:** degree 1, the home note.
- **Dominant:** degree 5, the strongest pull back to home.
- **Key signature:** the list of sharps or flats that belong to a key.
- **Relative minor:** the minor key that shares all seven notes with a major key; it starts on degree 6.
- **Pentatonic:** a five-note scale, with no half steps.
- **Blue note:** the b5 added to the minor pentatonic to make the blues scale.

On the fretboard: exercises

EXERCISE 11 — 6 MIN

Spell major scales from the formula

1. Write the formula W W H W W W H on paper.
2. Starting on D, say each note out loud, applying the formula one step at a time.
3. Do the same for A, then for E.
4. Find each of the seven D major notes on string 3 (the G string), going up from the open string.



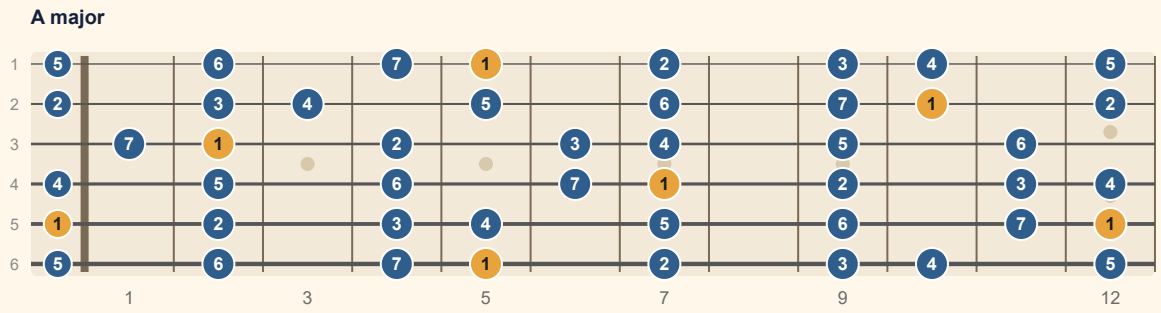
D major. Compare your answers with the diagram.

☐ Self-check: D major is D E F# G A B C#, A major is A B C# D E F# G#, and E major is E F# G# A B C# D#. If you named all notes correctly, with sharps where needed, you are done.

EXERCISE 12 — 7 MIN

Slide the major scale

1. Find C on string 6 at fret 8. Play C major in one position near fret 8, naming each note.
2. Slide the whole shape down to fret 3 so the root is G. Name the new notes aloud.
3. Slide to fret 5 for A major. Check that you hit F#, C# and G#.
4. Say the key name every time before you play.



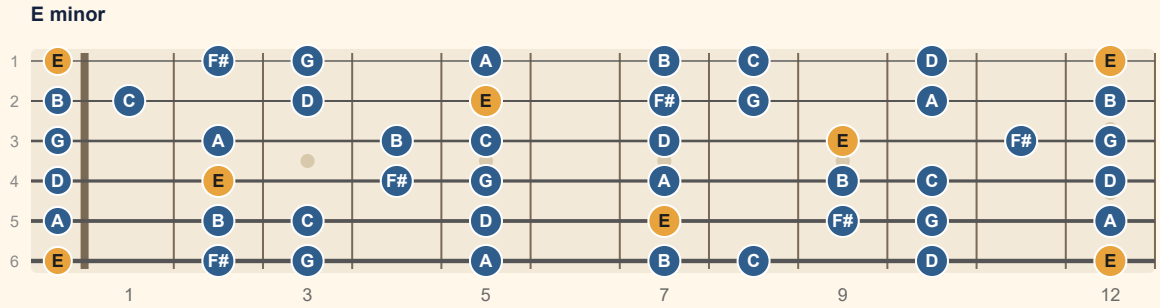
A major with degrees. Slide the shape to see the same pattern in a new place.

☐ Self-check: at fret 3 your notes should be G A B C D E F#. If you play an F natural, that is wrong; the key needs F#.

EXERCISE 13 5 MIN

Find the relative minor

1. Write out the seven notes of G major.
2. Count to the sixth note. That is the relative minor.
3. Play the seven notes from that note back up and down, and say its name.
4. Repeat for D major and F major.



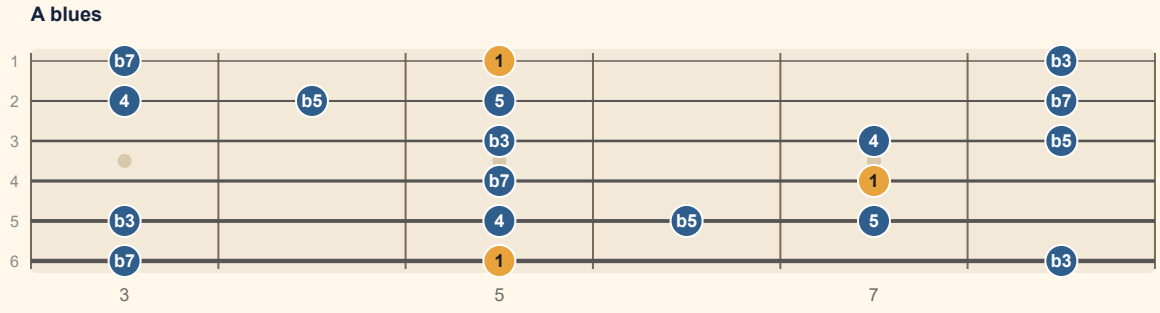
E natural minor: the same notes as G major.

☐ Self-check: G major has relative minor E (E F# G A B C D). D major has relative minor B. F major has relative minor D. If you started on the sixth degree each time, you got them all.

EXERCISE 14 10 MIN

Own Position 1

1. Play Position 1 of A minor pentatonic, from the low A on string 6 fret 5 up to the high C on string 1 fret 8, then back down.
2. Name each note as you play: A C D E G.
3. Play it again with only the root A, the five notes of the scale, and back to A.
4. Add the b5 (Eb) as a passing note between D and E to hear the blues scale.



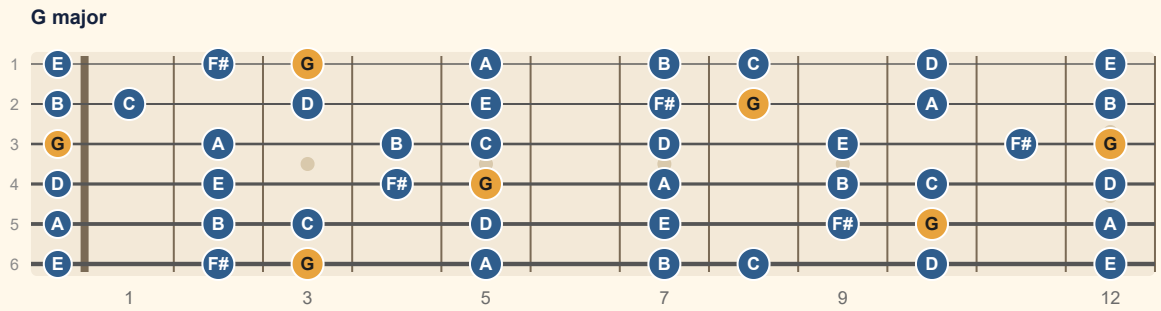
A blues scale in the area of Position 1. The b5 is the passing note.

☐ Self-check: you can play the box up and down without looking and name every note. Pick one slow note and bend or slide it just to feel the blues color.

EXERCISE 15 8 MIN

Find the key of three songs

1. Song A uses the chords: G, C, D, Em. Decide the key.
2. Song B uses: D, G, A, Bm. Decide the key.
3. Song C uses: Am, F, C, G. Decide the key, and name its relative major or minor.
4. For each song, write the seven notes of the key and check each chord's notes belong to it.



G major notes, for checking Song A.

☐ Self-check: Song A is G major (G A B C D E F#), Song B is D major (D E F# G A B C#), and Song C is C major, with relative minor A minor, because it starts on Am. Review the 'How to find the key' steps if any differ.

Next step

You now own the most useful tool in this book: a group of seven notes you can find anywhere on the neck. Take a few minutes today to play the major scale in two keys and name the notes aloud. In the next module, **The modes, without the mysticism**, you will start each of these scales from a different degree and discover seven new colors from the same notes.

MODULE 4

The modes, without the mysticism

- A mode is a major scale that starts on a different note. Same seven notes, new home.
- Seven modes means seven colors: Ionian, Dorian, Phrygian, Lydian, Mixolydian, Aeolian and Locrian.
- Every mode has one or two characteristic notes that give it its flavor.
- You can learn them as small changes to a scale you already know, not as seven new things.

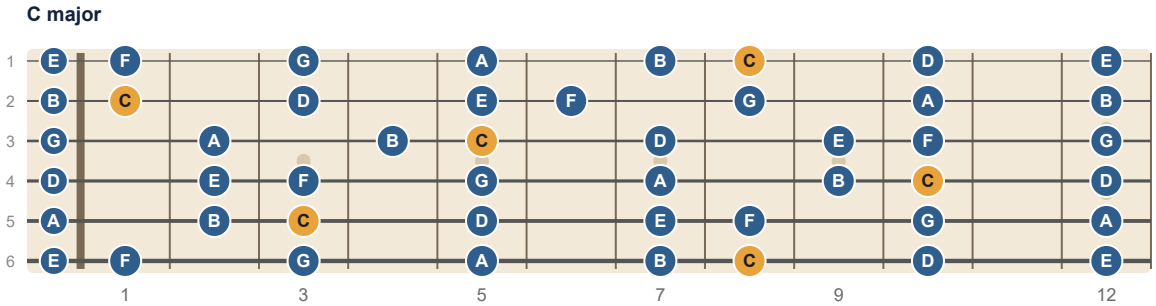
By the end: Play any of the seven modes from any root, hear the difference between them, and use a two-chord vamp to make each one sound like itself.

> *Modes have a reputation for being advanced, mystical, or only for jazz players with a degree. None of that is true. A mode is a scale you already know, played from a different starting note. By the end of this module you will see all seven on the fretboard, hear what makes each one different, and know which chord to put underneath it.*

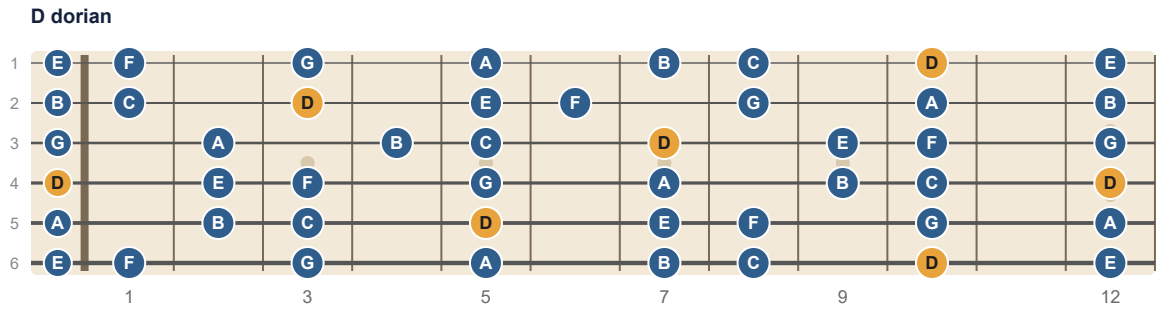
What a mode actually is

Take the C major scale: C D E F G A B. Now play the same seven notes, but start on D and end on the next D: D E F G A B C D. You did not change a single note. You changed the home note. That new scale is called D Dorian, and it sounds different from C major because your ear now hears D as home and every other note as a distance from D.

That is the whole idea. A mode is the same set of notes with a different note acting as the center. The seven notes of a major scale give you seven possible centers, so there are seven modes. Each one gets a name, and each one has its own mood.



C major (also called C Ionian): C D E F G A B, with C as home. The big dots are the C notes.



D Dorian: D E F G A B C. These are exactly the same notes as C major, but now D is home (big dots).

Look at the two diagrams. The dots sit in the same places on the neck. Only the big root dots move. That is why modes are less scary than they sound: you do not need seven new shapes, you need to move your sense of home.

> **The one sentence to remember**
>
> A mode is a scale with the same notes as a major scale, but a different note acting as home. Change the home note and the mood changes with it.

Two ways to think about modes

There are two useful ways to approach modes, and good players use both. They give the same notes in the end, so use whichever feels natural for the moment.

The relative approach: same notes, new home

This is what we just did. You start from a major key and slide the home note along its scale. C major has the notes C D E F G A B, so its seven modes are:

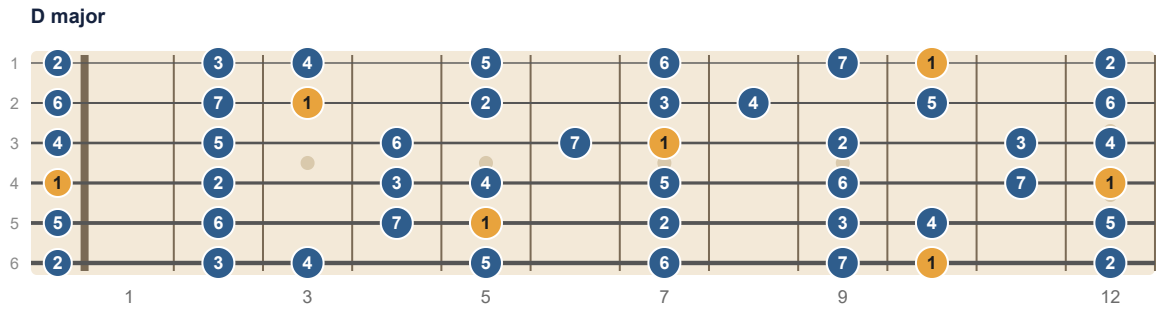
START ON	MODE NAME	NOTES (ALL FROM C MAJOR)
C	Ionian (plain major)	C D E F G A B
D	Dorian	D E F G A B C
E	Phrygian	E F G A B C D
F	Lydian	F G A B C D E
G	Mixolydian	G A B C D E F
A	Aeolian (natural minor)	A B C D E F G
B	Locrian	B C D E F G A

This approach is perfect when a song stays in one key and you want to know which notes are safe. If the chords all come from C major and the music is centered on D, think D Dorian and you already have the

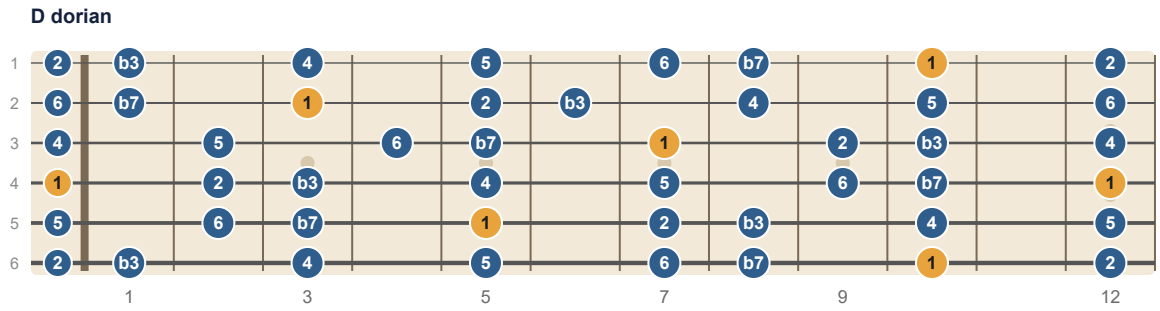
right notes. You can also run it backwards: if a song is centered on A and uses only the notes of G major, you are hearing A Dorian, because A is the second note of G major.

The parallel approach: same root, new notes

Now suppose you want to hear what Dorian sounds like next to major, with the same home note. Start with D major: D E F# G A B C#. To get D Dorian you lower two notes, the 3rd and the 7th: F# becomes F, and C# becomes C. The result is D E F G A B C, the same notes as before.



D major with degrees: 1 2 3 4 5 6 7. Find the 3 (F#) and the 7 (C#).



D Dorian with degrees: 1 2 b3 4 5 6 b7. Compared with D major, only the 3rd and 7th moved down a fret.

The parallel approach is how you hear the color. Same root, one or two notes changed, and your ear notices the difference at once. We will use it for most of this module, because the formulas make every mode easy to remember.

> **Which approach when?**
>
> Relative: you know the key of the song and want safe notes. Parallel: you know the sound you want and need to build it from a chosen root. Both land on the same notes.

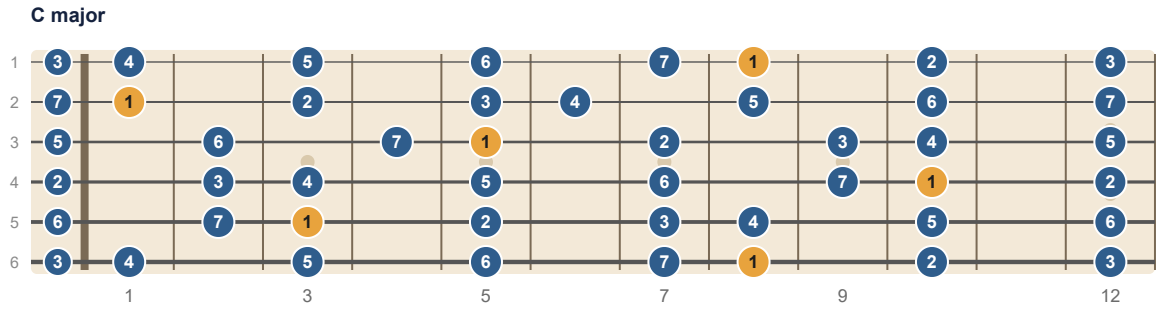
The seven modes as seven colors

Each mode is described below with its formula (the degrees compared with a major scale), its characteristic note (the one that gives it away), its feel, and where it fits. Every mode is shown twice: once rooted on C, and once rooted on A, so you can see that the pattern of degrees stays the same while the notes change. The C diagrams use degree labels like 1, b3 and 5; the A diagrams use note names.

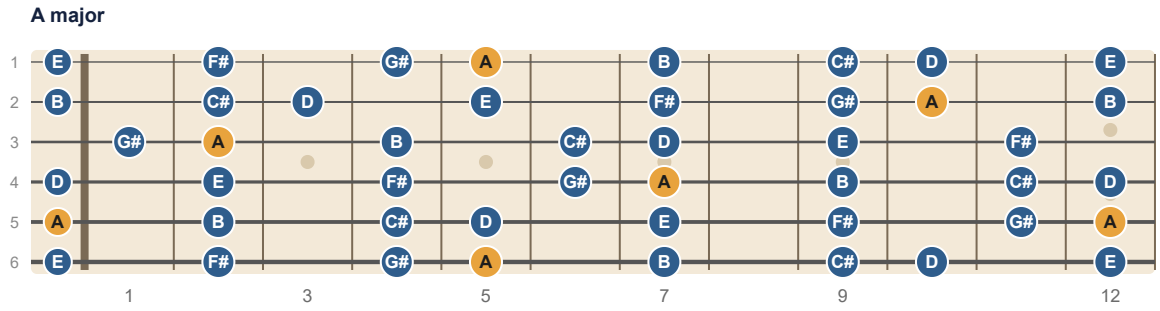
Words about feel are only a starting point. Two players can describe the same mode differently. Trust your ears more than the adjectives, and play each one against a held root or a simple chord before you decide.

1. Ionian: the plain major scale

Formula: 1 2 3 4 5 6 7. Characteristic note: the natural 7 (a half step under the root) together with the natural 3. Feel: bright, settled, happy, resolved. This is simply the major scale you learned in Module 3. Use it over major chords and major 7th chords when the music feels at rest. C Ionian: C D E F G A B. A Ionian: A B C# D E F# G#.



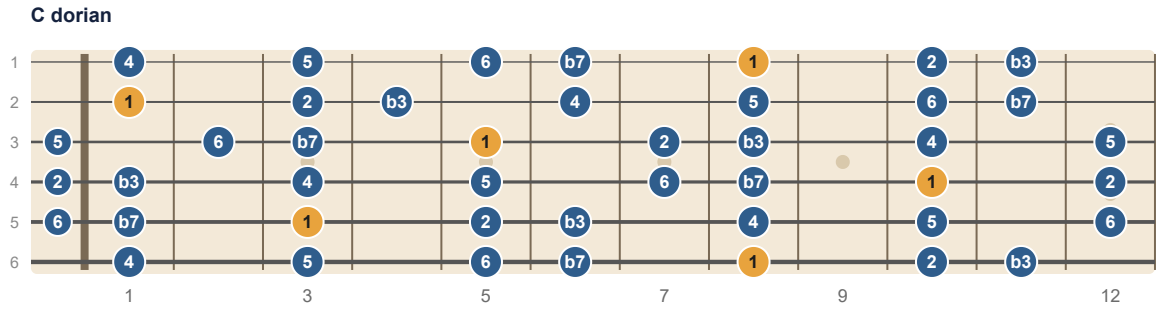
C Ionian (C major): C D E F G A B. Degrees 1 2 3 4 5 6 7.



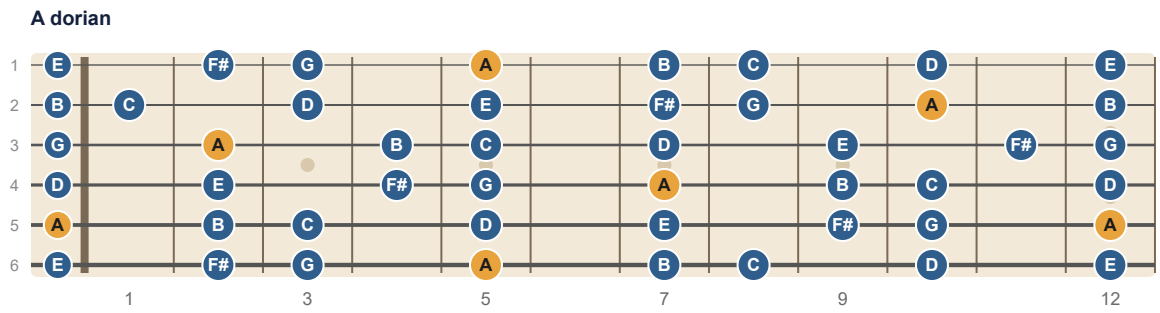
A Ionian (A major): A B C# D E F# G#.

2. Dorian: minor with a lifted sixth

Formula: 1 2 b3 4 5 6 b7. Characteristic note: the natural 6. It is a minor scale (it has the b3), but the 6 is a half step higher than in natural minor, which keeps it from sounding sad or dark. Feel: cool, soulful, a little hopeful. Use it over minor 7th chords, especially when the chord is a ii chord or when the music just sits on one minor chord for a long time. C Dorian: C D Eb F G A Bb. A Dorian: A B C D E F# G.



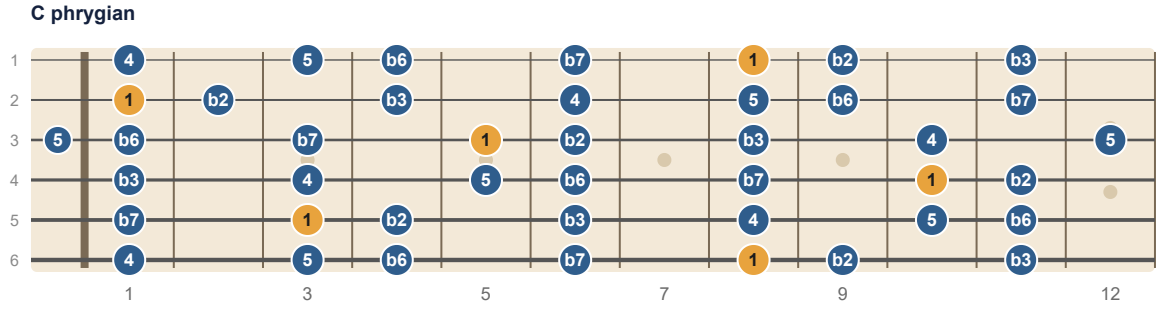
C Dorian: C D Eb F G A Bb. Spot the natural 6 (A).



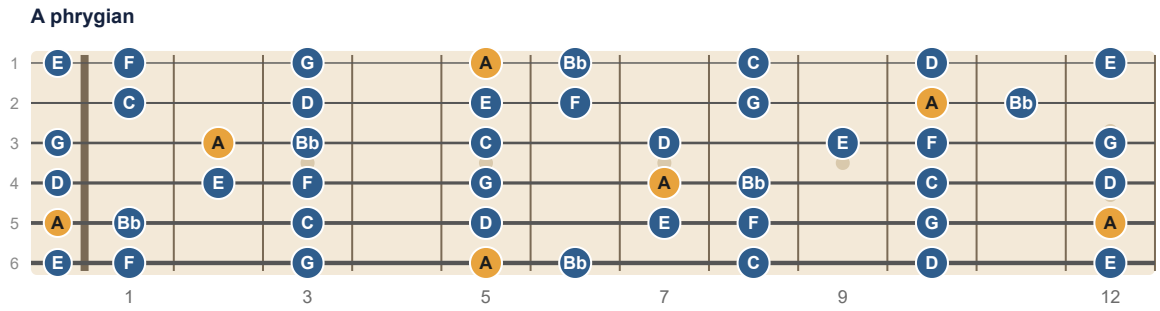
A Dorian: A B C D E F# G. The F# is the characteristic note.

3. Phrygian: minor with a flat second

Formula: 1 b2 b3 4 5 b6 b7. Characteristic note: the b2, a half step above the root. Feel: dark, tense, Spanish or heavy, depending on how you play it. That one close note pulls hard against the root. Use it over a minor chord that moves to a major chord a half step above (like Em to F). C Phrygian: C Db Eb F G Ab Bb. A Phrygian: A Bb C D E F G.



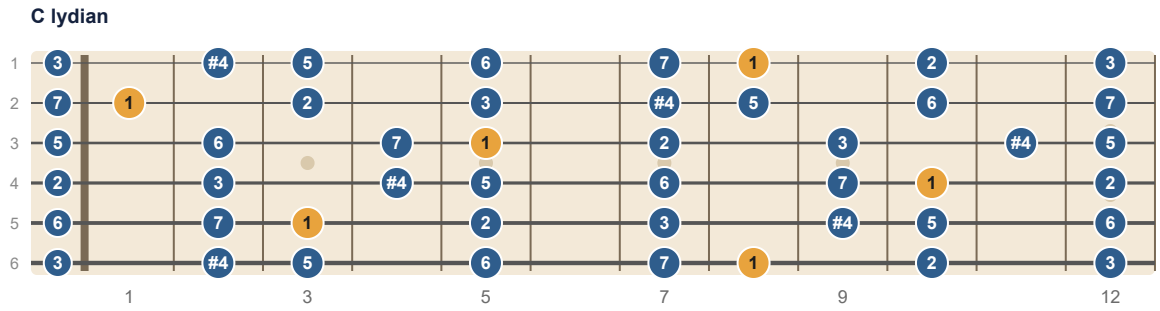
C Phrygian: C Db Eb F G Ab Bb. The Db (b2) is the giveaway.



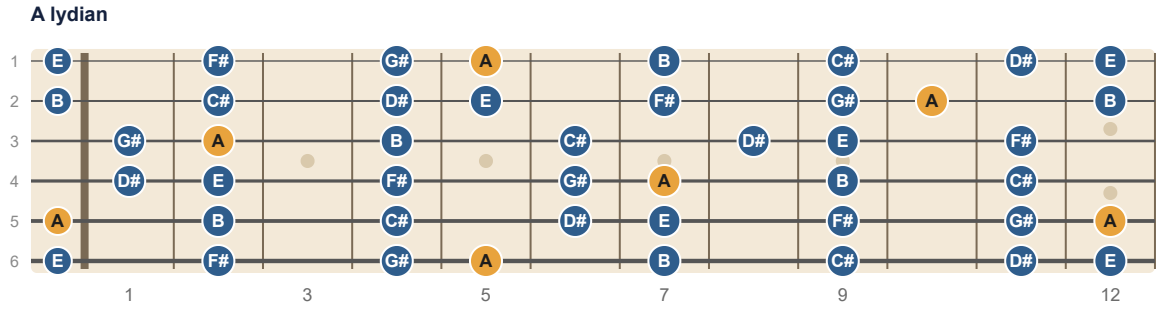
A Phrygian: A Bb C D E F G.

4. Lydian: major with a raised fourth

Formula: 1 2 3 #4 5 6 7. Characteristic note: the #4. Feel: bright, floating, dreamy, a little like wonder. It is the brightest of the seven. Use it over major 7th chords, particularly when the chord is not resolving anywhere and you want it to hover. C Lydian: C D E F# G A B. A Lydian: A B C# D# E F# G#.



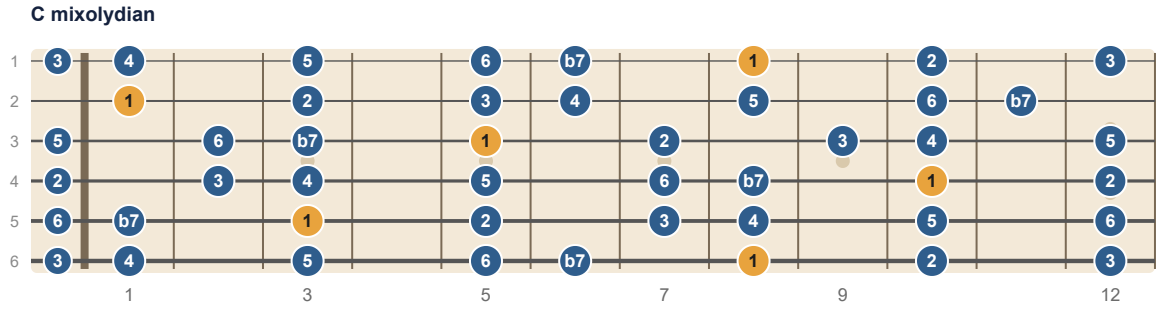
C Lydian: C D E F# G A B. The F# (#4) is the giveaway.



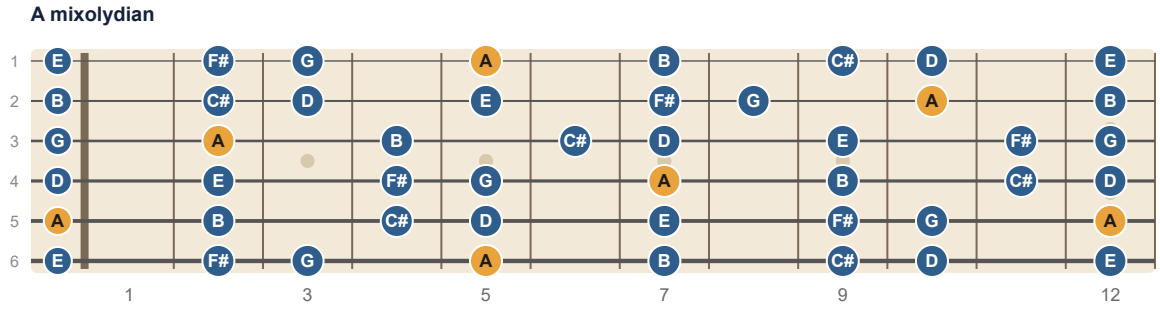
A Lydian: A B C# D# E F# G#.

5. Mixolydian: major with a flat seventh

Formula: 1 2 3 4 5 6 b7. Characteristic note: the b7. Feel: relaxed, bluesy, rootsy, a bit rock and roll. It is major with a loose, easy-going edge. Use it over dominant 7th chords (C7, A7 and so on) and over songs that live on a major chord with a b7 chord nearby. C Mixolydian: C D E F G A Bb. A Mixolydian: A B C# D E F# G.



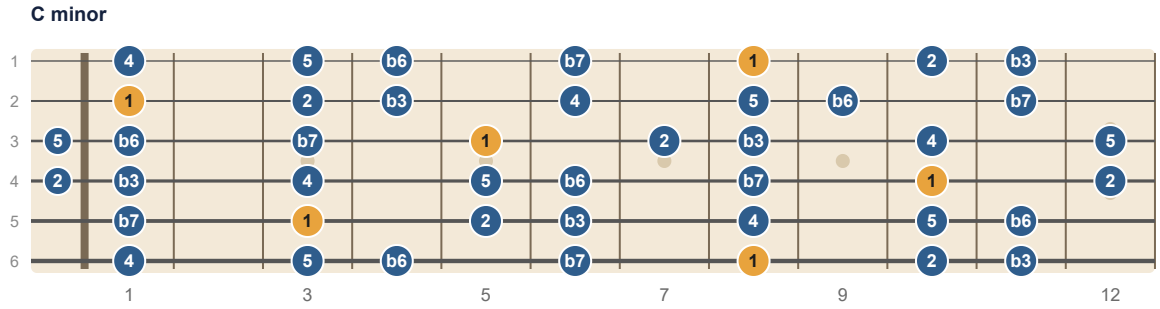
C Mixolydian: C D E F G A Bb. The Bb (b7) is the giveaway.



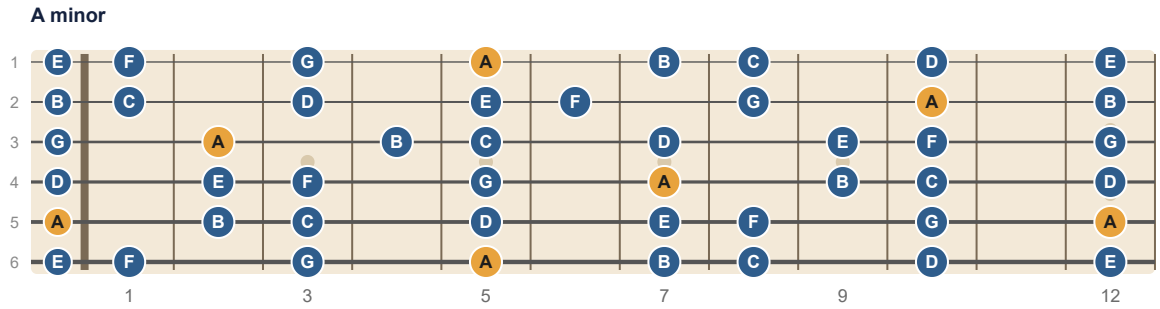
A Mixolydian: A B C# D E F# G.

6. Aeolian: the natural minor scale

Formula: 1 2 b3 4 5 b6 b7. Characteristic note: the b6. Feel: sad, serious, dramatic, the default minor sound. It is the minor scale from Module 3 under its modal name. Use it over minor chords when the music feels like it is in a minor key. C Aeolian: C D Eb F G Ab Bb. A Aeolian: A B C D E F G.



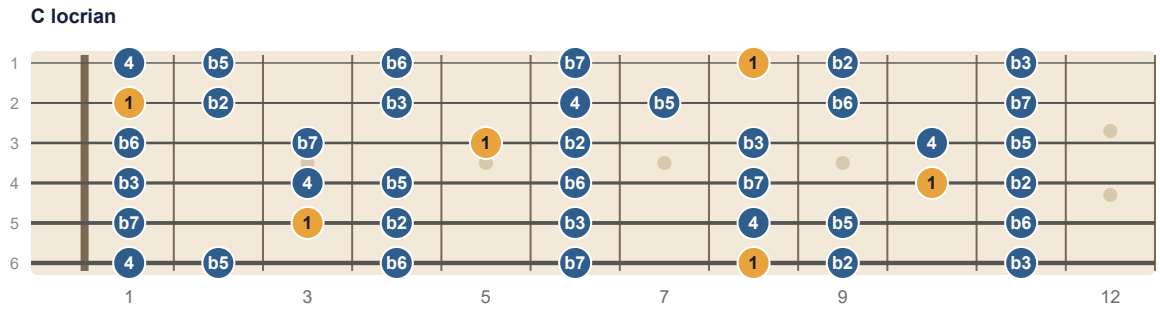
C Aeolian (C natural minor): C D Eb F G Ab Bb.



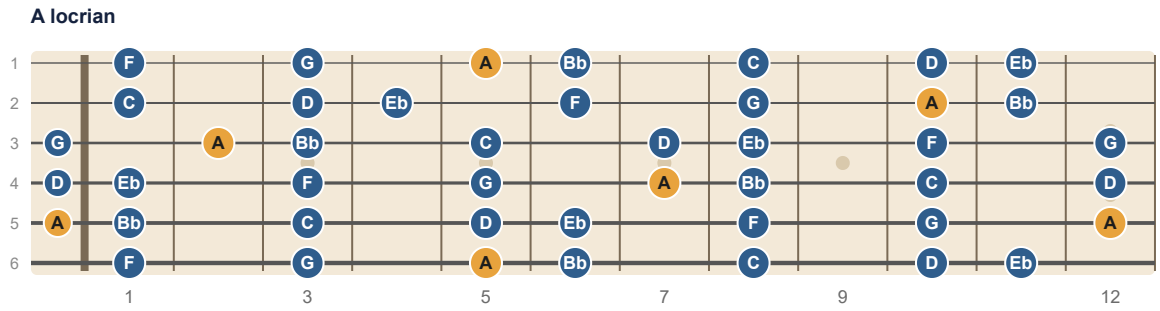
A Aeolian (A natural minor): A B C D E F G. No sharps and no flats.

7. Locrian: the unstable one

Formula: 1 b2 b3 4 b5 b6 b7. Characteristic notes: the b5 and b2. Feel: unstable, tense, restless. The root chord is a diminished sound that does not want to stay put. Use it sparingly, over a minor 7th flat 5 chord (a half-diminished chord, such as Bm7b5 in C major). C Locrian: C Db Eb F Gb Ab Bb. A Locrian: A Bb C D Eb F G.



C Locrian: C Db Eb F Gb Ab Bb. The Gb (b5) makes the root chord unstable.



A Locrian: A Bb C D Eb F G.

The brightness ladder

Here is a trick that makes the whole family easy to remember. Line the modes up from brightest to darkest by counting how many notes are lowered compared with the major scale. Lydian has one raised note (the #4), so it is brighter than major. Everything after major gets darker, one lowered note at a time.

RANK	MODE	COMPARED WITH MAJOR	MOOD
1 (brightest)	Lydian	#4	Dreamy, floating
2	Ionian (major)	nothing changed	Happy, settled
3	Mixolydian	b7	Relaxed, bluesy
4	Dorian	b3 b7	Cool, soulful
5	Aeolian (minor)	b3 b6 b7	Sad, serious
6	Phrygian	b2 b3 b6 b7	Dark, tense
7 (darkest)	Locrian	b2 b3 b5 b6 b7	Unstable

Read the third column from top to bottom and notice the pattern: each step down the ladder lowers one more note. Lydian to Ionian lowers the 4th. Ionian to Mixolydian lowers the 7th. Mixolydian to Dorian lowers the 3rd. Dorian to Aeolian lowers the 6th. Aeolian to Phrygian lowers the 2nd. Phrygian to Locrian lowers the 5th. If you remember the order 4, 7, 3, 6, 2, 5, you can rebuild the ladder from scratch.

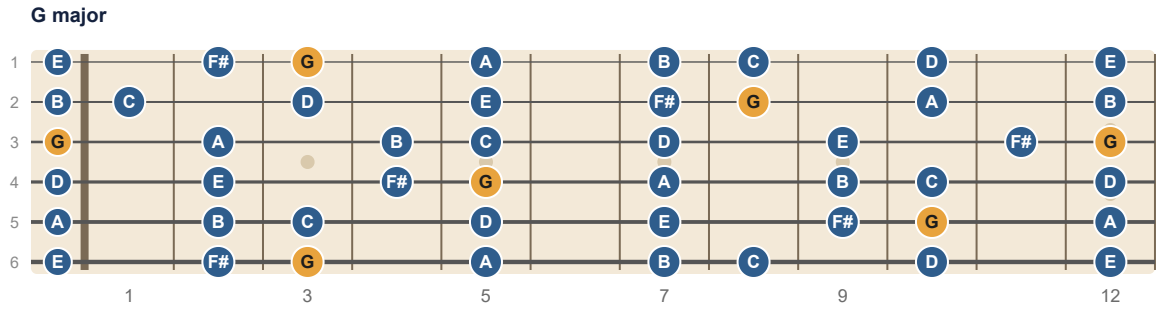
> **A handy memory hook**
>
> The lowered notes go 7, 3, 6, 2, 5 after major, and Lydian sits above major by raising the 4. Say it as a chant while you play and the whole ladder sticks.

Seeing modes as changes to the major scale

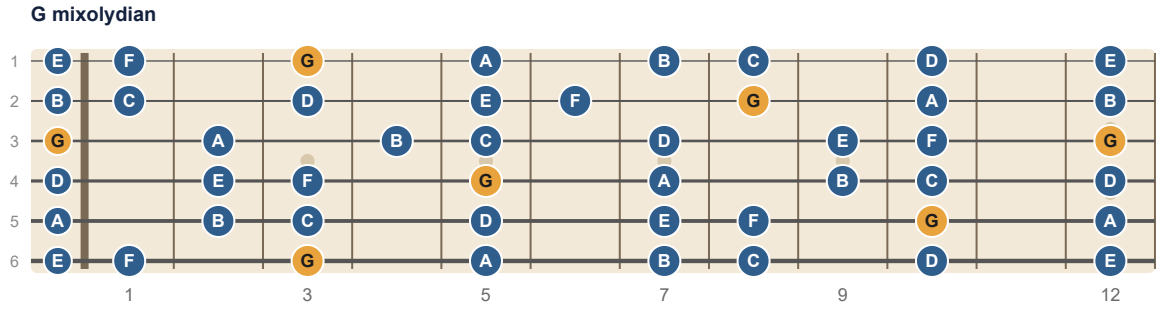
Since most players know the major scale best, the quickest way to learn a mode is to compare it with major. Here is every mode as a short list of changes, with a worked example on root G (G major is G A B C D E F#):

MODE	ALTERED DEGREES VS. MAJOR	EXAMPLE ON G
Ionian	none	G A B C D E F#
Dorian	b3, b7	G A Bb C D E F
Phrygian	b2, b3, b6, b7	G Ab Bb C D Eb F
Lydian	#4	G A B C# D E F#
Mixolydian	b7	G A B C D E F
Aeolian	b3, b6, b7	G A Bb C D Eb F
Locrian	b2, b3, b5, b6, b7	G Ab Bb C Db Eb F

Use this table on the fretboard. Play G major first. To get Mixolydian, move the F# down one fret to F. To get Dorian, also lower the B to Bb. To get Lydian, raise the C to C#. You never need to memorize seven separate scales; you only need one major scale and a list of tiny edits.



G major: G A B C D E F#. Start here and edit.



G Mixolydian: G A B C D E F. Only the F# moved down to F.

Check your work

Every edit is exactly one fret from where the major scale had the note. A flat 7 is one fret below the major 7. A sharp 4 is one fret above the perfect 4. If an edit feels bigger than one fret, something is off.

Choosing a mode for a chord

The simplest rule for choosing a mode is to match the quality of the chord. A major chord with a major 7th suggests Ionian or Lydian. A dominant 7th chord suggests Mixolydian. A minor 7th chord suggests Dorian or Aeolian (or Phrygian if the next chord is a half step up). A m7b5 chord suggests Locrian. The table below puts it in one place.

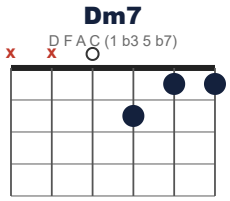
CHORD UNDERNEATH	GOOD MODE	WHY IT FITS
Cmaj7	Ionian or Lydian	Both have the 3 and the 7; Lydian adds the dreamy #4
C7	Mixolydian	It has the b7 that the chord has
Dm7	Dorian (or Aeolian)	Both have b3 and b7; Dorian's natural 6 is cooler
Em	Phrygian	Minor chord, with the b2 for tension
Am	Aeolian	Plain minor, no surprises
Bm7b5	Locrian	It has the b3, b5 and b7 the chord has

Modal vamps: the fastest way to hear a mode

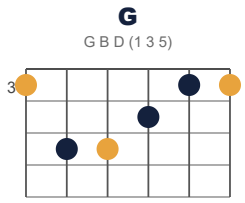
A vamp is two chords repeated over and over. Because there is no chord progression to pull your ear toward some other home, the repeating chords tell your ear which note is home, and the mode appears. Each of the vamps below puts the characteristic note into the chords themselves, so the mode comes through even when you just strum.

Dm7 to G: D Dorian

Play Dm7 for two beats or a whole bar, then G, and repeat. The G major chord contains a B natural, which is the natural 6 of D Dorian. If you used D Aeolian, the sixth would be Bb and it would clash with the G chord. So the vamp itself says: this is Dorian. Solo with D Dorian: D E F G A B C.



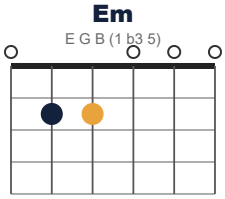
Dm7 (open): D F A C. The home chord.



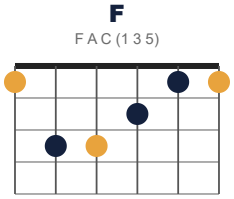
G major, E form (root on string 6, fret 3): G B D. The B is the natural 6 of D Dorian.

Em to F: E Phrygian

Play Em, then F, and repeat. F is a half step above E, which is exactly the b2 of E Phrygian. Hearing a major chord a half step above the root is the quickest way to land in Phrygian. Solo with E Phrygian: E F G A B C D. It is the same set of notes as C major, with E as home.



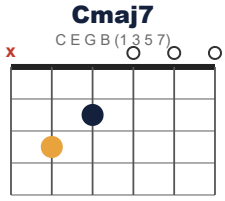
Em (open): E G B. The home chord.



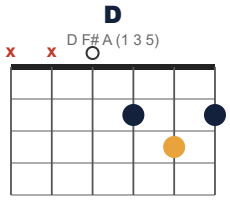
F major, E form (barre at fret 1): F A C. The F note is the b2 of E Phrygian.

Cmaj7 to D: C Lydian

Play Cmaj7, then D, and repeat. The D major chord contains F#, the #4 of C Lydian. Plain C major would want F natural. So the vamp gives you the Lydian note on a plate. Solo with C Lydian: C D E F# G A B.



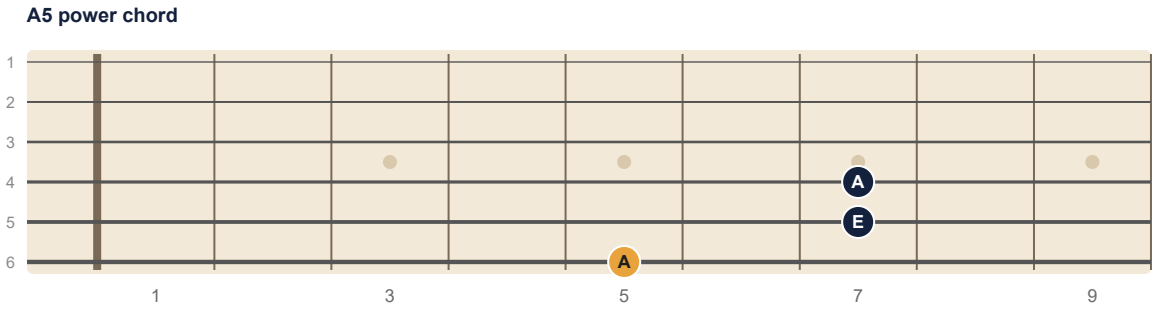
Cmaj7 (open): C E G B. The home chord.



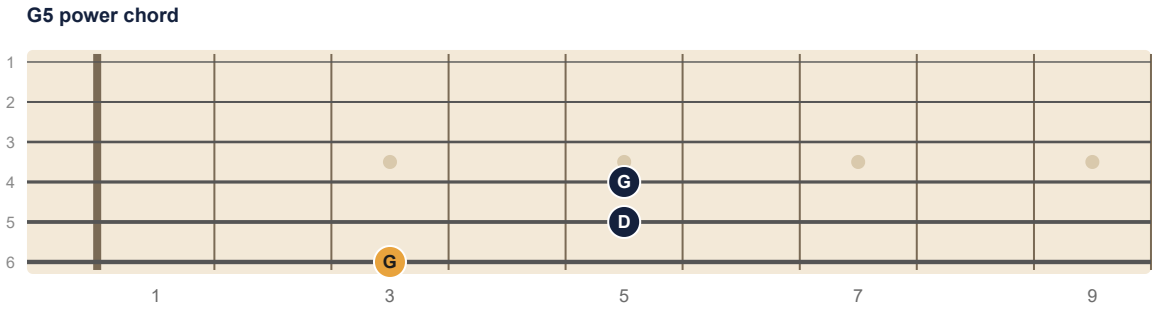
D major (open): D F# A. The F# is the #4 of C Lydian.

A5 to G: A Mixolydian

Play A5 (a power chord), then G, and repeat. G is a whole step below A, which makes it the b7 chord in A. The G note is the b7 of A Mixolydian. This is the sound of a lot of classic rock. Solo with A Mixolydian: A B C# D E F# G.



A5: A (string 6, fret 5), E (string 5, fret 7), A (string 4, fret 7).



G5: G (string 6, fret 3), D (string 5, fret 5), G (string 4, fret 5). The G note is the b7 of A Mixolydian.

VAMP	MODE	CHORD THAT GIVES IT AWAY
Dm7 - G	D Dorian	G major carries the natural 6 (B)
Em - F	E Phrygian	F major carries the b2 (F)
Cmaj7 - D	C Lydian	D major carries the #4 (F#)
A5 - G	A Mixolydian	G carries the b7 (G)

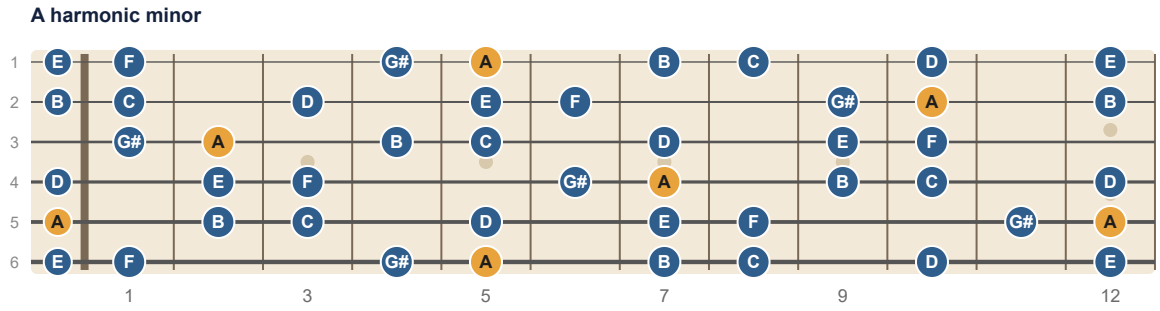
How to practice a vamp

Loop the two chords for a minute. Then play only the root of the mode over it, then the root plus the characteristic note. Hearing that one note against the chords teaches your ear more than any scale run.

Beyond: modes of the melodic and harmonic minor

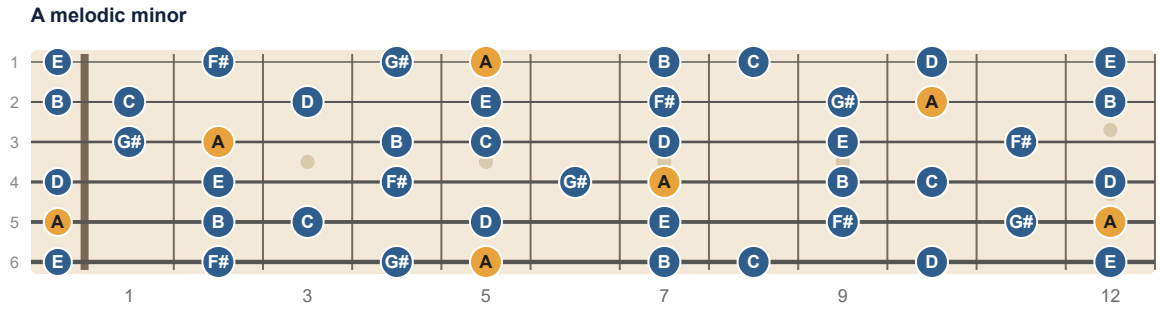
Once the seven major modes feel familiar, you may meet modes built on other scales. You do not need them to play great music, so treat this as a short peek. The idea is the same: take a scale, start on a different degree, and a new color appears.

The harmonic minor scale is a natural minor with a raised 7th. In A it is A B C D E F G#. Its fifth mode starts on E: E F G# A B C D. This is called Phrygian dominant, with the formula 1 b2 3 4 5 b6 b7. It sounds Spanish, dramatic, and a little Middle Eastern, and it fits over an E7 chord.



A harmonic minor: A B C D E F G#. The raised 7 (G#) is the signature.

The melodic minor scale (going up) is a natural minor with a raised 6th and 7th, or a major scale with a b3. In A it is A B C D E F# G#. Its fourth mode, starting on D, is D E F# G# A B C: Lydian dominant, with formula 1 2 3 #4 5 6 b7. It is Lydian with a b7 and gives a bright, jazzy, floating dominant sound. Its seventh mode is called altered, and it is the tense scale jazz players love on dominant chords.



A melodic minor: A B C D E F# G#. Note the natural 6 and 7 compared with natural minor.

> **Where to go next with these**
>
> Keep these as a curiosity for now. Master the seven major modes and the four vamps first. Most players find the others make sense quickly once that is solid.

Common mistakes

- **Treating each mode as a new set of notes to memorize.** It is the major scale with a different home. Learn one scale and a list of edits.
- **Playing the notes but never establishing home.** If the chords or the bass do not sit on the root, your ear will hear plain major or minor. Use a vamp or a drone.

- **Calling every minor sound Dorian.** Dorian has a natural 6. If your scale has a b6, it is Aeolian. If it has a b2, it is Phrygian.
- **Ignoring the chords.** A mode only works if it matches what is under it. Check that the mode's notes agree with the chord tones.
- **Landing on the characteristic note and staying there.** It is a flavor, not the main course. Use it as a passing or ending note, and let the root and chord tones do the heavy lifting.
- **Reaching for Locrian too early.** Because its root chord is unstable, it works best over a single m7b5 chord, briefly. Most songs do not stay in it.

Cheat sheet

MODE	FORMULA	CHARACTERISTIC NOTE	FEEL	USE OVER	ON C	ON A
Ionian	1 2 3 4 5 6 7	natural 3 and 7	Happy	maj, maj7	C D E F G A B	A B C# D E F# G#
Dorian	1 2 b3 4 5 6 b7	natural 6	Cool, soulful	m7, ii	C D Eb F G A Bb	A B C D E F# G
Phrygian	1 b2 b3 4 5 b6 b7	b2	Dark, Spanish	m, bII chord above	C Db Eb F G Ab Bb	A Bb C D E F G
Lydian	1 2 3 #4 5 6 7	#4	Dreamy	maj7	C D E F# G A B	A B C# D# E F# G#
Mixolydian	1 2 3 4 5 6 b7	b7	Bluesy	7, bVII chord	C D E F G A Bb	A B C# D E F# G
Aeolian	1 2 b3 4 5 b6 b7	b6	Sad	m, m7	C D Eb F G Ab Bb	A B C D E F G
Locrian	1 b2 b3 4 b5 b6 b7	b5	Unstable	m7b5	C Db Eb F Gb Ab Bb	A Bb C D Eb F G

MODULE 4 CHECKLIST

☐ I can say the difference between relative and parallel modes in one sentence.

☐ I can build any mode from a root using the altered-degrees table.

☐ I can play the four vamps and name the mode each one points to.

☐ I can order the modes from brightest to darkest.

☐ I know which chord type each mode likes.

Glossary

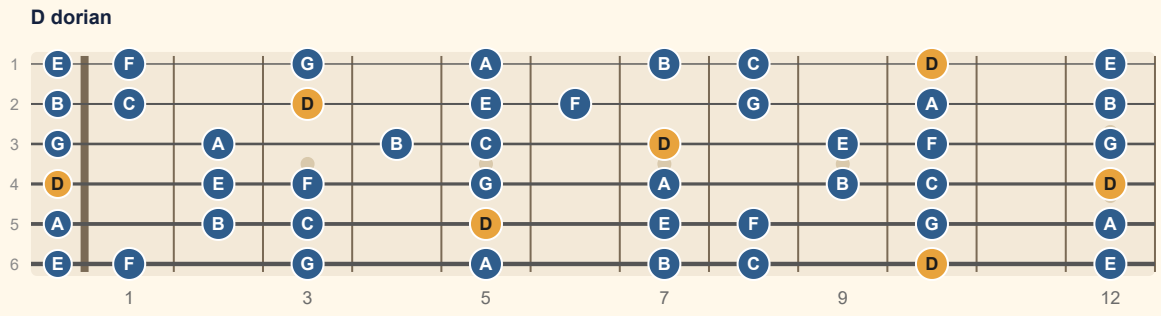
WORD	PLAIN-ENGLISH MEANING
Mode	A scale with the same notes as a parent scale, but a different note as home.
Home note (root)	The note that feels like the center; the note the mode is named after.
Relative modes	Modes that share the same notes but have different roots (D Dorian and C major).
Parallel modes	Modes that share the same root but have different notes (D major and D Dorian).
Characteristic note	The one note that most clearly gives a mode its flavor.
Vamp	Two (or a few) chords repeated to establish a sound.
Ionian / Aeolian	The modal names for the major scale and the natural minor scale.
Dominant 7th	A major chord with a b7 added, such as C7.
Half-diminished	The m7b5 chord, the home chord of Locrian.

On the fretboard: exercises

EXERCISE 16 — 6 MIN

Same notes, new home

1. Play the C major scale along the first string (open E string), using the diagram to find each note. Say each note name aloud.
2. Now start on D (string 2, fret 3, or string 1, fret 10) and play up to the next D using only the notes of C major.
3. Play a low D (string 4, open) as a drone, then play your D scale over it. Notice how D feels like home.
4. Repeat for E (string 1, open). Play E F G A B C D E and hold a low E underneath.



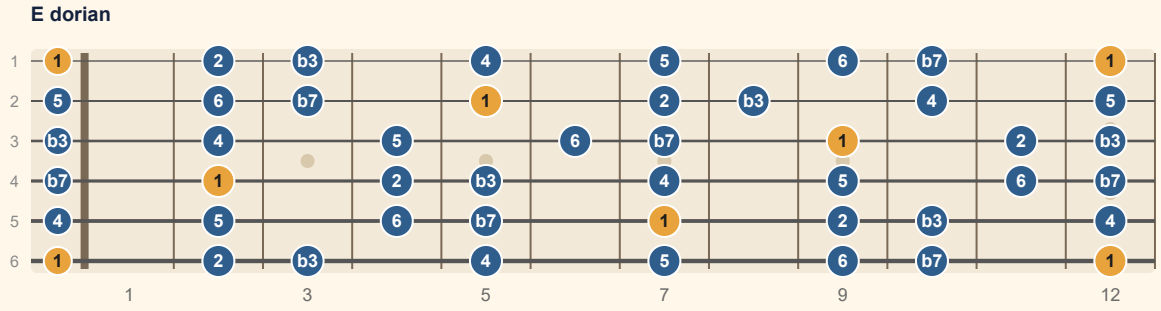
D Dorian: D E F G A B C. Use it to check your notes.

☐ Self-check: You can play D Dorian and E Phrygian and name all seven notes for each without looking. They use the same notes as C major, but D and E sound like home.

EXERCISE 17 8 MIN

Edit the major scale

1. Write down the notes of E major: E F# G# A B C# D#.
2. Make E Dorian by lowering the 3rd and 7th. Write it down.
3. Make A Mixolydian from A major (A B C# D E F# G#) by lowering the 7th.
4. Make G Lydian from G major (G A B C D E F#) by raising the 4th.
5. Make B Phrygian from B major (B C# D# E F# G# A#) by lowering the 2nd, 3rd, 6th and 7th.



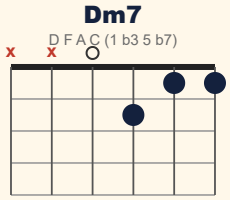
E Dorian with degrees: compare with E major.

☐ Self-check: E Dorian = E F# G A B C# D. A Mixolydian = A B C# D E F# G. G Lydian = G A B C# D E F#. B Phrygian = B C D E F# G A. If you got all four, you have the method.

EXERCISE 18 8 MIN

Hear the vamp

1. Play Dm7 for four beats, then G for four beats, and loop it for a full minute.
2. Over the loop, play only the note D, then add A, then add the B (the natural 6).
3. Now switch to D Aeolian (D E F G A Bb C) over the same chords. Hear how the Bb fights with the G chord.
4. Switch back to Dorian and notice how the B sits comfortably.



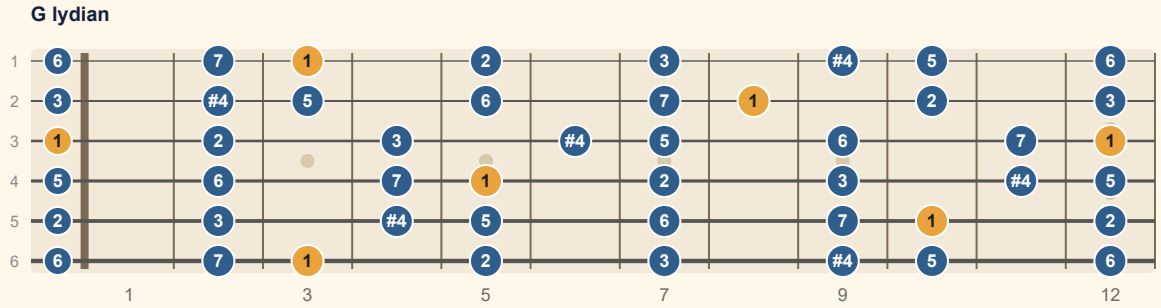
Dm7 (open) for the home chord.

☐ Self-check: You can hear the difference between B natural and Bb over the G chord, and you can explain why Dorian is the right choice for this vamp.

EXERCISE 19 10 MIN

Walk the brightness ladder on G

1. Pick the root G on string 6, fret 3 and play it between each scale as your home note.
2. Play G Lydian: G A B C# D E F#. Then G Ionian: G A B C D E F#. Then G Mixolydian: G A B C D E F.
3. Continue with G Dorian: G A Bb C D E F. Then G Aeolian: G A Bb C D Eb F. Then G Phrygian: G Ab Bb C D Eb F. Then G Locrian: G Ab Bb C Db Eb F.
4. After each scale, say aloud which note changed from the previous one.



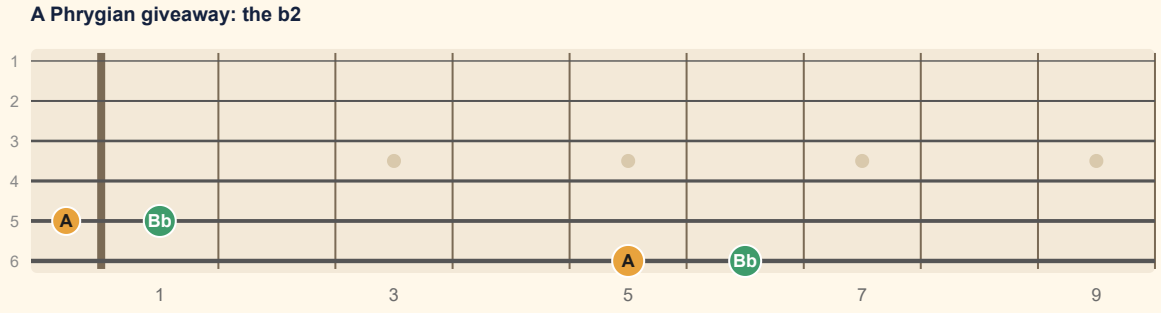
G Lydian, the brightest: G A B C# D E F#.

☐ Self-check: You named the changes in this order: C# to C (4th lowered), F# to F (7th), B to Bb (3rd), E to Eb (6th), A to Ab (2nd), D to Db (5th). That is the ladder: 4, 7, 3, 6, 2, 5.

EXERCISE 20 10 MIN

Write your own vamp

1. Choose a mode from the cheat sheet: Phrygian on A is a good start (A Bb C D E F G).
2. Pick two chords: Am and Bb (the Bb is a half step above A and carries the b2).
3. Loop them. Play only A, then add the Bb, then add the full A Phrygian scale.
4. Write down the chords, the mode, and the note that gives it away.



Home note A and its b2 (Bb) on strings 6 and 5.

☐ Self-check: Your vamp is Am - Bb, the mode is A Phrygian, and the giveaway is Bb (the b2). When you play it, the Bb should feel tense and want to fall back to A.

Next step

You now have seven colors and a way to hear each of them. In Module 5 we turn from scales to chords and ask where they really come from. You will see that every chord is built by stacking notes from a

scale, and the same major scale you just turned into seven modes also gives you the seven chords that fit together naturally.

MODULE 5

Chords: where they really come from

- A chord is just notes stacked in thirds, taken from a scale.
- Major is 1-3-5. Minor is 1-b3-5. Two small changes, two different moods.
- Every open chord you already play is a shape that holds those same notes, doubled across six strings.
- Barre chords are open chords you can slide, so one shape gives you twelve chords.

By the end: You will be able to build any major, minor, diminished, augmented, sus2 or sus4 chord from scratch, name every note in the shapes you already play, and find a chord on the neck without looking it up.

Why this module matters

> *You already know a handful of chords. Maybe a dozen. You know where the fingers go, and your hand knows it better than your head does. This module opens up the box and shows you what is inside. Once you see it, you stop memorizing shapes and start building them.*

Most self-taught players learn chords the same way: a chart, a finger pattern, repetition. That works until you meet a chord you have never seen, or want to play a song in a different key, or wonder why one shape sounds sad and another sounds bright. A chart cannot answer those questions. A formula can.

Here is the good news. Chords are not a separate subject from scales. They come straight out of the scales you built in Module 3. A chord is a small selection of notes from a scale, picked by a simple rule. If you can name notes on the neck (Module 1) and count intervals (Module 2), you already have everything you need.

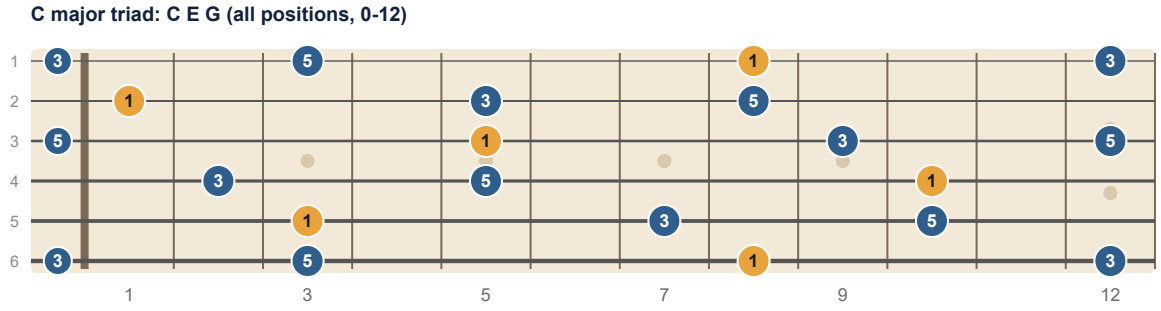
Everything here is done with note names, degrees (1, b3, 5) and fretboard diagrams. There is no sheet music anywhere in this book.

1. What a chord really is

A chord is three or more notes sounding together. The most common kind, the **triad**, has exactly three notes. The rule for building it is easy to say: **pick a note, skip one, take the next, skip one, take the next.** Each note you take is a third above the last.

Try it on the C major scale: C D E F G A B. Start on C. Skip D, take E. Skip F, take G. You now have C, E, G. That is the C major triad. The three notes have names: the first is the **root** (1), the second is the **third** (3), the third is the **fifth** (5).

Because you pulled the notes out of the C major scale, counting 1-3-5 of that scale gives you C-E-G. Counting 1-3-5 from the G major scale gives G-B-D. The formula never changes. Only the starting note does.



Every C, E and G on the neck up to the 12th fret. The root C shows in a different color. Labels show the degree: 1, 3 and 5.

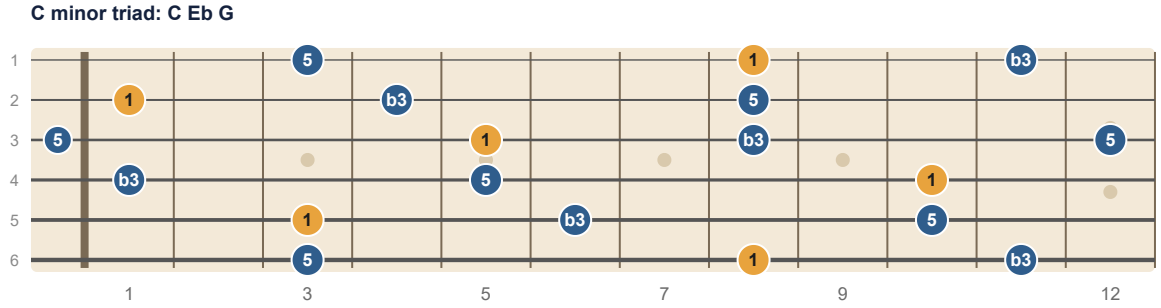
Look at the diagram. You are not looking at one chord. You are looking at the raw material of every C major chord that exists on the neck. A chord shape is just a choice of one note per string (or fewer) from these dots.

2. Major and minor by formula

The thirds you stack are not all the same size. A **major third** (M3) is 4 frets (4 semitones). A **minor third** (m3) is 3 frets. A fifth (P5) is 7 frets from the root. That is all you need.

TRIAD	FORMULA	INTERVALS FROM THE ROOT	MOOD
Major	1 - 3 - 5	M3 + P5 (4 and 7 frets up)	Bright, settled
Minor	1 - b3 - 5	m3 + P5 (3 and 7 frets up)	Darker, more inward
Diminished	1 - b3 - b5	m3 + b5 (3 and 6 frets up)	Tense, wants to move
Augmented	1 - 3 - #5	M3 + #5 (4 and 8 frets up)	Floating, unresolved

Major and minor share the same root and the same fifth. The **only** difference is the middle note. Lower the third by one fret and a major chord becomes minor. Every time. In any key.



Compare with the C major triad above. The E has moved down one fret to Eb. The root and the 5 have not moved. That single note is the whole difference.

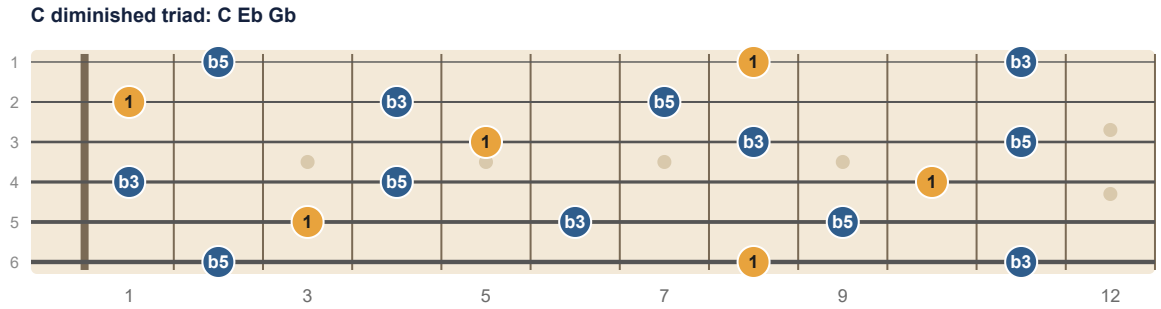
The one-fret rule

To turn any major chord into minor, drop the 3 by one fret. To turn minor into major, raise the b3 by one fret. Try it on an open chord right now: play E major, then lift the finger on string 3 (the G#) off and let it ring open. Your E has just become Em.

3. Diminished and augmented triads

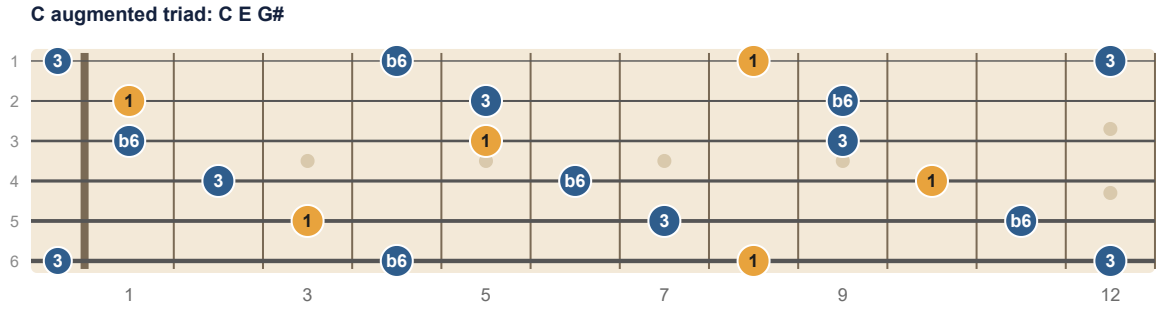
These two are less common than major and minor, but they matter, and building them with the formula takes seconds.

A **diminished** triad squeezes the fifth down by a fret: 1 - b3 - b5. Starting on C that gives C - Eb - Gb. It stacks two minor thirds, which is why it sounds tight and tense. You will meet it as the seventh chord of the major scale (next section).



Notice how close the three notes sit: every step is 3 frets. The tension you hear comes from that evenly stacked pair of minor thirds.

An **augmented** triad stretches the fifth up a fret: 1 - 3 - #5. Starting on C that gives C - E - G#. It stacks two major thirds. Because the steps are all equal (4 frets), the chord has no clear home, and it sounds like it is floating.



Every step here is 4 frets. Three equal steps nearly fill an octave, so the chord has no obvious center.

Do not worry about the odd ones yet

Most songs live on major and minor. Learn diminished and augmented as formulas now, and meet them in real songs later. You do not need to drill them.

4. Harmonizing the major scale

Here is where it all connects. Take a major scale and build a triad on **every** note, using only the notes of that scale. You get seven chords. Some come out major, some minor, and one diminished. The pattern is the same in every key.

SCALE DEGREE	1	2	3	4	5	6	7
Chord type	major	minor	minor	major	major	minor	diminished
Roman numeral	I	ii	iii	IV	V	vi	vii-dim

Capital numerals mean major. Small ones mean minor. The last one is the diminished chord. Memorize this one line and you can build the chords of any key.

In C major

The C major scale is C D E F G A B. Build a triad on each note, skipping every other scale note:

NUMERAL	CHORD	NOTES	TYPE
I	C	C - E - G	major
ii	Dm	D - F - A	minor
iii	Em	E - G - B	minor
IV	F	F - A - C	major
V	G	G - B - D	major
vi	Am	A - C - E	minor
vii-dim	Bdim	B - D - F	diminished

Check one: Dm is D - F - A. From D to F is 3 frets (a minor third), from D to A is 7 frets (a fifth). That is the minor formula 1 - b3 - 5. Check Bdim: B to D is 3 frets, B to F is 6 frets. That is 1 - b3 - b5, a diminished chord.

In G major

The G major scale is G A B C D E F#. Same pattern, new notes:

NUMERAL	CHORD	NOTES	TYPE
I	G	G - B - D	major
ii	Am	A - C - E	minor

NUMERAL	CHORD	NOTES	TYPE
iii	Bm	B - D - F#	minor
IV	C	C - E - G	major
V	D	D - F# - A	major
vi	Em	E - G - B	minor
vii-dim	F#dim	F# - A - C	diminished

Notice that G, C and D are all major, and Am, Bm and Em are all minor, just like Dm, Em and Am were in C. The numerals tell the story. If a song is in G and you hear the IV chord, you already know it is C.

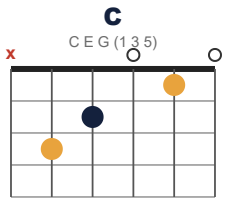
> **Why these chords sound like they belong together**
>
> Every note in all seven chords comes from one scale. That shared pool of notes is exactly why a song built from I, IV, V and vi sounds coherent. Module 8 shows you how to use that.

5. How your open chords are built

Now the payoff. The open chords you have played for years are not random finger patterns. Each one is the triad formula, spread across the strings. The same three notes appear several times, in different octaves. That doubling is what makes a six-string chord sound full.

Read each table like this: **string** is the string number (6 is the low E, 1 is the high E), **fret** is where you press, **note** is what you hear, **degree** is its job in the chord. A dash means the string is not played.

C major (x32010)



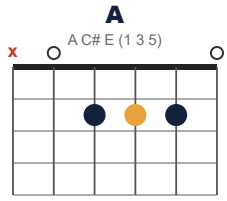
C major open. Notes: C E G, with the C and E doubled.

STRING	FRET	NOTE	DEGREE
6	-	muted	-
5	3	C	1
4	2	E	3
3	0	G	5

STRING	FRET	NOTE	DEGREE
2	1	C	1
1	0	E	3

Five strings, three different notes: C, E, G. Nothing more.

A major (x02220)

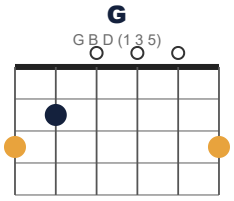


A major open. Notes: A E A C# E.

STRING	FRET	NOTE	DEGREE
6	-	muted	-
5	0	A	1
4	2	E	5
3	2	A	1
2	2	C#	3
1	0	E	5

The notes are A, C#, E. The C# on string 2 is the third. This is why A major sounds bright: it carries that major third.

G major (320003)

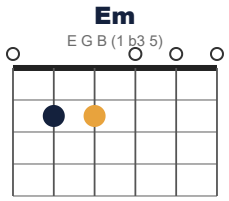


G major open. Notes: G B D G B G.

STRING	FRET	NOTE	DEGREE
6	3	G	1

STRING	FRET	NOTE	DEGREE
5	2	B	3
4	0	D	5
3	0	G	1
2	0	B	3
1	3	G	1

E minor (022000)



E minor open. Notes: E B E G B E.

STRING	FRET	NOTE	DEGREE
6	0	E	1
5	2	B	5
4	2	E	1
3	0	G	b3
2	0	B	5
1	0	E	1

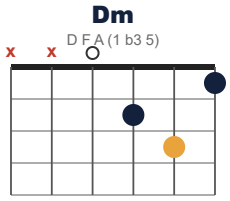
Compare this with E major (022100). The only difference is string 3: G# (the 3) in E major, G (the b3) in E minor. One fret. The one-fret rule at work.

The others at a glance

Same method for the rest. Here are the notes by string, from the lowest sounding string up:

CHORD	SHAPE	NOTES BY STRING (LOW TO HIGH)	DEGREES
E	022100	E - B - E - G# - B - E	1 - 5 - 1 - 3 - 5 - 1
D	xx0232	D - A - D - F#	1 - 5 - 1 - 3
Am	x02210	A - E - A - C - E	1 - 5 - 1 - b3 - 5

CHORD	SHAPE	NOTES BY STRING (LOW TO HIGH)	DEGREES
Dm	xx0231	D - A - D - F	1 - 5 - 1 - b3



D minor open. Notes: D A D F.

Every chord uses only 1, 3 (or b3) and 5. That is the whole secret. The shapes look different because the guitar's tuning scatters the same three notes in different places.

> **Try this tonight**
>
> Play each open chord one string at a time, and say the note name out loud as you go. Then say the degree. Slow, a few minutes. Most players find that this one habit changes how the neck feels.

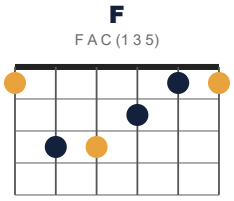
6. Barre chords: the shapes you can slide

An open chord uses open strings, so it is stuck in one place. A barre chord has no open strings. Your index finger acts as a movable nut, and the rest of your hand holds an open-chord shape in front of it. Now the shape can slide anywhere, and the root note tells you which chord you are playing.

There are two forms you will use constantly, named after the open chord they come from.

The E-form: root on string 6

Take the open E major shape (022100). Slide it up one fret and barre with your index finger: you get F major. The index finger now plays the role of the open strings. The root is the lowest note, on string 6.

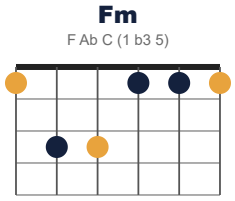


F major, E-form barre at fret 1. Root on string 6. Notes: F C F A C F.

STRING	FRET	NOTE	DEGREE
6	1	F	1
5	3	C	5

STRING	FRET	NOTE	DEGREE
4	3	F	1
3	2	A	3
2	1	C	5
1	1	F	1

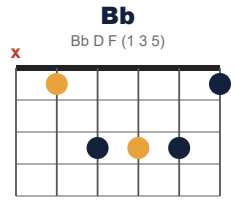
To play any major chord in this form, find the root on string 6 and put the barre there. G is fret 3, A is fret 5, B is fret 7. Make the shape minor by lifting the 3: that is the Em shape slid up.



F minor, E-form. Same barre, with the third (A) lowered to Ab.

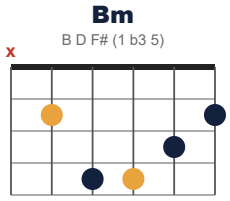
The A-form: root on string 5

Take the open A major shape (x02220) and slide it up. The root is now on string 5. One fret up from the open A gives Bb.



Bb major, A-form barre at fret 1. Root on string 5. Notes: Bb F Bb D F.

STRING	FRET	NOTE	DEGREE
6	-	muted	-
5	1	Bb	1
4	3	F	5
3	3	Bb	1
2	3	D	3
1	1	F	5



B minor, A-form at fret 2. Root on string 5 (B). Notes: B F# B D F#.

> **Finding the right fret**
>
> For an E-form chord, name the fret from string 6. For an A-form, name it from string 5. If you can name notes on those two strings (Module 1), you can play any major or minor barre chord in the key you need.

7. Build any chord without looking it up

This is the skill the module has been heading toward. Say a chord name, such as Eb minor or A sus4, and you can play it. Use four steps:

1. **Pick the root.** The letter in the name is the root.
2. **Pick the formula.** Major is 1-3-5. Minor is 1-b3-5. Others are in the cheat sheet at the end.
3. **Spell the notes.** Count frets from the root: M3 is 4 frets, m3 is 3 frets, P5 is 7 frets, then name each note.
4. **Find them on the neck.** Use a shape you know, or pick one note per string from the dots on the diagram.

Example: **D minor**. Root D. Formula 1-b3-5. D plus 3 frets is F. D plus 7 frets is A. The notes are D, F, A. Your open Dm shape (xx0231) holds exactly those: D, A, D, F. Done.

Example: **Ab major**. Root Ab. Formula 1-3-5. Ab plus 4 frets is C. Ab plus 7 frets is Eb. The notes are Ab, C, Eb. No open shape exists, so use the E-form barre with the root on string 6 at fret 4 (string 6, fret 4 is G#, the same note as Ab). Done.

Example: **F# minor**. Root F#. F# plus 3 frets is A. F# plus 7 frets is C#. The notes are F#, A, C#. Use the E-form minor shape with the barre at fret 2.

> **Quick check: does a chord make sense?**
>
> Name the notes you are playing, string by string. If every note belongs to the 1, 3 (or b3) and 5 of the chord, it is right. If one note is something else, you have found a mistake before your ears did.

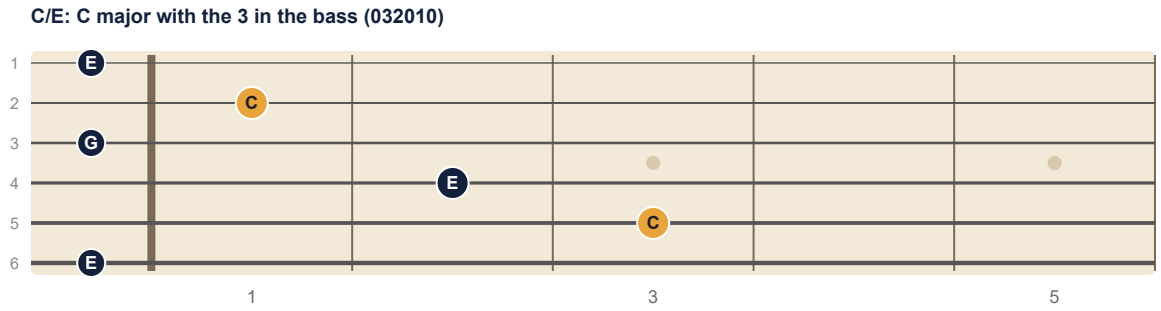
8. Inversions and voicings

So far the root has been the lowest note. It does not have to be. The order of the notes from lowest to highest is the **voicing**, and changing which chord tone sits on the bottom is called an **inversion**.

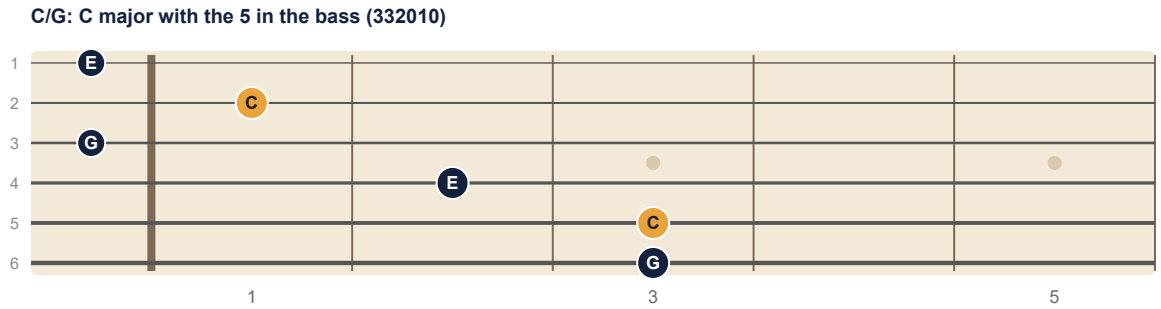
POSITION	LOWEST NOTE	C MAJOR EXAMPLE
Root position	1 (root)	C in the bass: C - E - G
First inversion	3	E in the bass: E - G - C
Second inversion	5	G in the bass: G - C - E

The chord is still C major in all three cases. The same three notes are present. Only the bass note changes, and the feel changes with it. Root position is firm and grounded. First inversion is lighter and flows toward the next chord. Second inversion is open and a little unsettled.

On the guitar you already play inversions. Take the open C and add the low E on string 6 (open). The bass is now E, the third. That is C with E in the bass, written C/E.



All six strings sound. The notes are E C E G C E. Same three notes as C major, but the lowest is E, so this is first inversion.



Put your fourth finger on string 6, fret 3. The bass is G, the fifth. Second inversion.

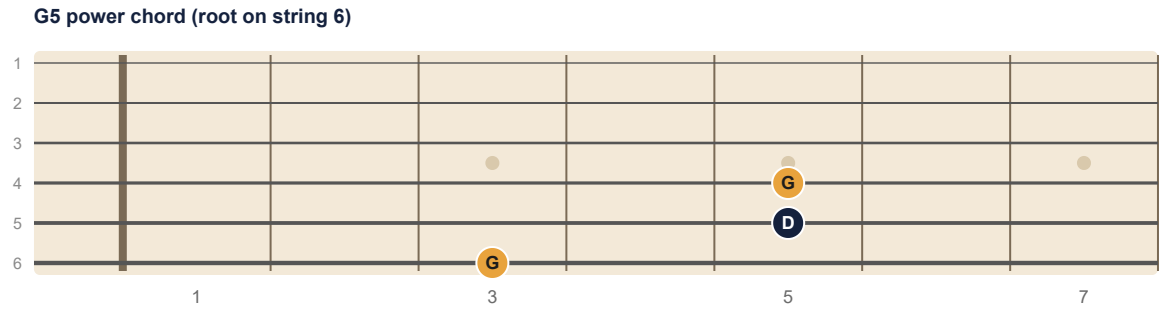
Inversions let you move between chords with less hand travel, which is one of the reasons a bass line can walk smoothly under a progression. You do not need to master them today. Just know that when a chord chart says C/E, the letter after the slash is the bass note.

Voicing is the broader idea. A C major chord can be spread across the neck in dozens of ways: different strings, different octaves, some notes doubled and some omitted. Each is a different voicing of the same chord. Module 7 (CAGED) will show you five systematic ones.

9. Power chords

A power chord has only two notes: the root and the fifth. There is no third, so it is neither major nor minor. It just says the root and a solid fifth above it. Usually the root is doubled one octave up, which makes a three-string shape.

The formula is **1 - 5 (- 8)**. On the guitar it is a shape you can slide: root on string 6 or string 5, the fifth one string over and two frets up from the root, and the octave on the next string, level with the fifth.



G on string 6 (fret 3), D on string 5 (fret 5), G on string 4 (fret 5). The notes are G - D - G: root, fifth, root an octave up.

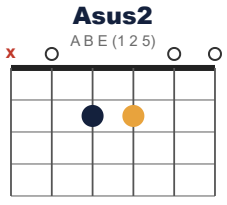
Because there is no third, a power chord fits over both major and minor passages. That is why it is the default in loud rock and heavy distortion: the thirds muddy up under gain, and the 1-5 skeleton stays clear. Slide the same shape to fret 5 on string 6 and you get A5. To fret 7: B5.

> **Same shape, same sound**
>
> Any 1-5 shape works the same way. It is the root that changes the name. Name the note on string 6 and you have named the power chord.

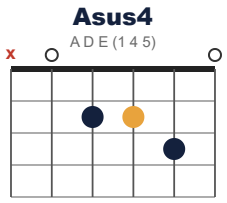
10. Sus2 and sus4: replacing the third

A **suspended** chord drops the 3 and replaces it with a neighbor. Without the third, the chord is neither major nor minor, which gives it an open, unfinished quality.

CHORD	FORMULA	WHAT REPLACES THE 3	EXAMPLE ON A
sus2	1 - 2 - 5	The 2nd (one whole step above the root)	A - B - E
sus4	1 - 4 - 5	The 4th (a half-step above the 3)	A - D - E



A sus2 open. Notes: A E A B E. The C# (3) has gone; B (the 2) took its place.



A sus4 open. Notes: A E A D E. The C# has been replaced by D (the 4).

Compare both with the open A major (x02220). The third on string 2 is at fret 2 in A major. In Asus2 you open that string (B), in Asus4 you reach to fret 3 (D). That is the whole difference.

A sus chord usually wants to move. Sus4 leans back toward the major third (D wants to fall to C#). Sus2 is calmer, more floating. Most players find that a quick sus4 to major move adds a lot of life to a plain chord.

Common mistakes

- **Learning shapes without the notes.** If you only know where the fingers go, a new key means starting over. Know what each string is playing.
- **Mixing up root and chord name.** In an E-form barre, the root is the lowest note on string 6, not wherever your barre finger lands on string 1.
- **Forgetting the b3 makes it minor.** A single fret. If a chord sounds sadder than you expected, check that note.
- **Thinking sus chords are 'fancier' major chords.** They have no third. They are neither major nor minor.
- **Calling a power chord 'major'.** It has no third, so it is neither. It works over both.
- **Treating Bdim like any other chord in the key.** It is the odd one in the major scale. In most songs it appears rarely and briefly.
- **Muffled barre chords.** Press just behind the fret with the bony side of the finger, not the flat pad. If one string buzzes, move the finger down a little. With practice the barre gets easier.

Module 5 cheat sheet

CHORD	FORMULA	INTERVALS UP FROM THE ROOT	NOTES IN C
Major	1 - 3 - 5	M3, P5	C - E - G
Minor	1 - b3 - 5	m3, P5	C - Eb - G
Diminished	1 - b3 - b5	m3, b5	C - Eb - Gb
Augmented	1 - 3 - #5	M3, #5	C - E - G#
sus2	1 - 2 - 5	M2, P5	C - D - G

CHORD	FORMULA	INTERVALS UP FROM THE ROOT	NOTES IN C
sus4	1 - 4 - 5	P4, P5	C - F - G
Power chord (5)	1 - 5	P5	C - G

KEY	I	II	III	IV	V	VI	VII-DIM
C	C	Dm	Em	F	G	Am	Bdim
G	G	Am	Bm	C	D	Em	F#dim

- Major scale chord pattern: **major, minor, minor, major, major, minor, diminished** (I ii iii IV V vi vii-dim).
- Major to minor: lower the 3 by one fret.
- E-form: root on string 6. A-form: root on string 5.
- Inversion: same notes, different bass note (first = 3 in bass, second = 5 in bass).

Module 5 glossary

WORD	PLAIN MEANING
Triad	A three-note chord: root, third, fifth.
Root	The note the chord is named after (1).
Third	The note that makes a chord major (3) or minor (b3).
Fifth	The supporting note, 7 frets above the root (5).
Diminished	A chord with a b3 and a b5. Tense.
Augmented	A chord with a 3 and a #5. Floating.
Harmonizing	Building a chord on each note of a scale.
Barre	Using one finger across several strings as a movable nut.
Inversion	Playing a chord with a note other than the root in the bass.
Voicing	The particular order and spread of a chord's notes.
Power chord	A root and a fifth, with no third.
Sus (suspended)	A chord where the 3 is replaced by the 2 or the 4.

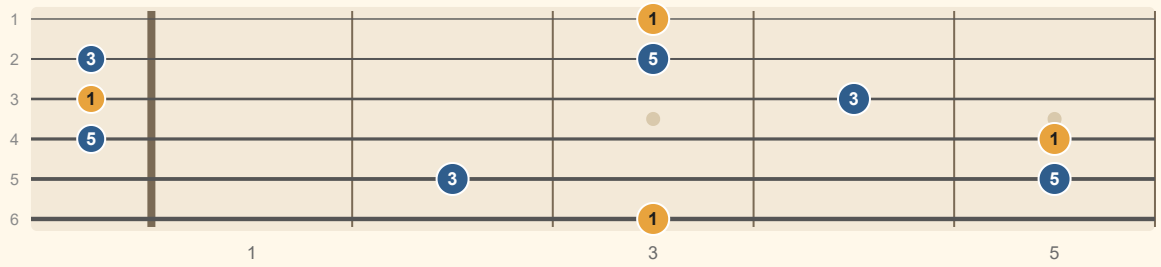
On the fretboard: exercises 21 to 25

EXERCISE 21 — 6 MIN

Spell the triads

1. Write the notes of these chords using the formulas: G major, D minor, A major, E minor, Bb major.
2. For each, find all three notes on the neck within the first five frets.
3. Play the three notes and check they sound like the chord.

Check: G major triad notes



All G, B and D positions in the first five frets.

☐ Self-check: G = G B D, Dm = D F A, A = A C# E, Em = E G B, Bb = Bb D F. If any of yours differ, count the frets again (M3 = 4, m3 = 3, P5 = 7).

EXERCISE 22 — 8 MIN

Name your open chords

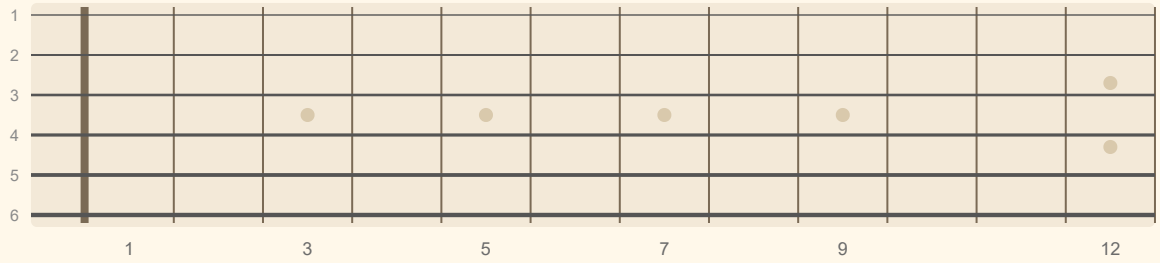
1. Play open C, G, D, A and E, one at a time.
2. On each, say the note on every string, then say its degree.
3. Find the string that carries the 3 in each chord.

☐ Self-check: C has its E (the 3) on strings 4 and 1. G has B on strings 5 and 2. D has F# on string 1. A has C# on string 2. E has G# on string 3. If you can name those five without looking, you know the chords from the inside.

EXERCISE 23 — 8 MIN

Harmonize in a new key

1. Pick the key of D major: D E F# G A B C#.
2. Build a triad on each note using only those scale notes.
3. Write chord name, notes and type (major, minor or diminished) for all seven.



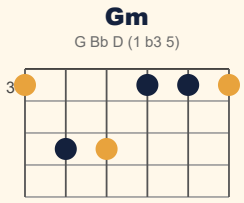
Use this neck to mark the D major scale notes, then circle the notes of your three favorite chords.

☐ Self-check: D (D F# A), Em (E G B), F#m (F# A C#), G (G B D), A (A C# E), Bm (B D F#), C#dim (C# E G). The type pattern must read major, minor, minor, major, major, minor, diminished.

EXERCISE 24 — 8 MIN

Slide the barre

1. Play an E-form major barre at fret 1 (F), then at 3 (G), then at 5 (A).
2. Say the chord name before you play it, using the string-6 note.
3. Repeat with the A-form: fret 1 (Bb), fret 3 (C), fret 5 (D).
4. Make each chord minor by lowering the 3.



G minor, E-form barre at fret 3.

☐ Self-check: E-form roots are F (fret 1), G (3), A (5). A-form roots on string 5 are Bb (1), C (3), D (5). If you named the chord from the wrong string, the pattern will not line up.

EXERCISE 25 — 10 MIN

Build it, then play it

1. Without a chart, spell and play: Eb major, C# minor, F sus2, B sus4, Gm.
2. For each: root, formula, notes, then find them on the neck.
3. Play a power chord on the root of each one, and name its two notes.

☐ Self-check: Eb = Eb G Bb. C#m = C# E G#. Fsus2 = F G C. Bsus4 = B E F#. Gm = G Bb D. Power chords: Eb5 = Eb Bb, C#5 = C# G#, F5 = F C, B5 = B F#, G5 = G D.

Next step

You can now build any triad and see inside every shape you already play. Module 6 adds one more note on top, the seventh, and with it a whole new palette: maj7, m7, dominant 7 and the tensions that give chords color. Keep this module's formulas handy: the sevenths are just triads with one more stacked third.

MODULE 6

Seventhes, tensions and color

- A seventh chord is a triad plus one more stacked third. That one extra note changes the whole mood.
- Four main sevenths cover almost everything: maj7, m7, dominant 7 and m7b5.
- Stack sevenths on every degree of the major scale and you get Imaj7, ii7, iii7, IVmaj7, V7, vi7 and viiø7.
- Tensions (9, 11, 13) are just the 2, 4 and 6 sitting an octave higher. Used well, they are color, not trouble.

By the end: Build, name and play seventh chords from memory, explain why V7 pulls home, and swap triads for sevenths in a progression without losing the song.

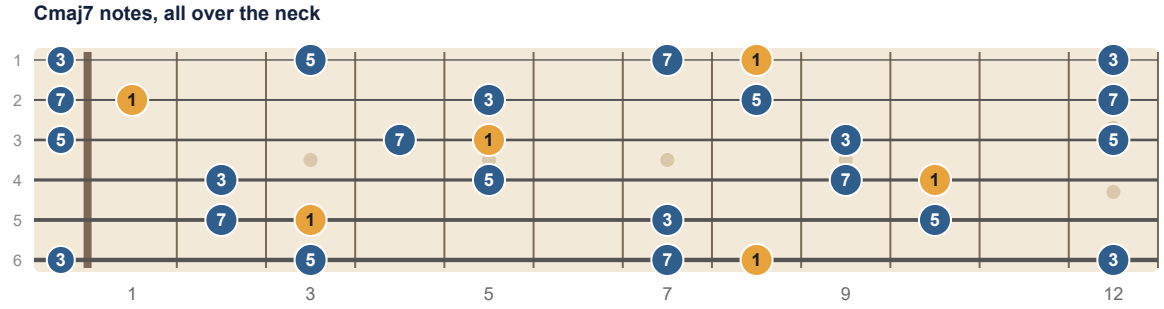
> *In Module 5 you learned that a chord is a stack of thirds. Three notes stacked that way give you a triad. Stack one more third on top and you get a seventh chord. That is the whole idea of this module, and it opens the door to the sound of jazz, soul, R&B, bossa and a lot of modern pop.*

You already know some of these chords by sight. The open Am7, Dm7 and G7 shapes you may have strummed for years are seventh chords. What has probably been missing is the why: which note is the seventh, why one chord sounds sweet and another sounds tense, and how to swap one for another on purpose. By the end of the module you will have that map.

What a seventh chord is

Take a C major triad: C, E, G. These are the notes 1, 3 and 5 of the C major scale. A triad is built by skipping every other note of the scale: C (skip D) E (skip F) G. To add a fourth note, keep going the same way: skip A and land on B. Now you have C, E, G, B. The new note, B, is the 7th degree of the scale.

Count it in letters: C to E is a third, E to G is a third, G to B is a third. Four notes, three stacked thirds. Any chord built this way is a seventh chord, because the top note sits a 7th above the root (C up to B).



Every C, E, G and B from the open strings to fret 12. The root C is marked in a different color. Labels show the degree: 1, 3, 5 and 7. Together these four notes are Cmaj7.

Why bother with the extra note? Because the triad is a clear, plain statement and the seventh adds a layer of feeling. Triads sound solid and finished. Sevenths sound richer, a little more grown-up, and some of them sound unfinished on purpose, as if the music is asking a question.

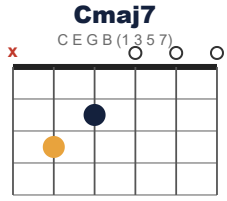
The four main seventh chords

The seventh can sit in two spots: a half step below the octave (a major 7th, written 7) or a whole step below the octave (a minor 7th, written b7). Combine the two kinds of 7th with the different triads and you get four chords that do nearly all the work.

CHORD	FORMULA	EXAMPLE IN C	FEEL
maj7	1 3 5 7	Cmaj7: C E G B	Dreamy, warm, peaceful
m7	1 b3 5 b7	Cm7: C Eb G Bb	Mellow, smooth, a bit moody
7 (dominant)	1 3 5 b7	C7: C E G Bb	Bluesy, restless, wants to move
m7b5 (half-diminished)	1 b3 b5 b7	Cm7b5: C Eb Gb Bb	Dark, tense, a sigh before resolving

maj7: major triad plus a major 7th

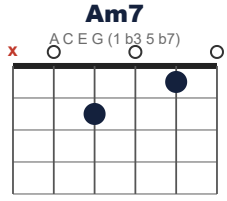
Cmaj7 is C E G B. The B sits just one fret below the next C, so it rubs gently against the root. That small rub is what gives maj7 its floating, dreamy quality. It sounds like a happy chord that has exhaled.



Open Cmaj7. Notice the open E and B strings ringing: the 3 and the 7 are right next to the root C. Strum from string 5 down.

m7: minor triad plus a minor 7th

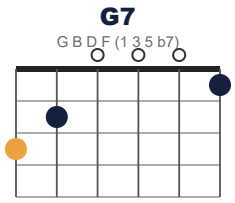
Am7 is A C E G. Take the A minor triad (A, C, E) and add G, a whole step below the next A. The result is soft and relaxed. If minor triads can sound sad, m7 chords usually sound more cool than sad.



Open Am7 (A E G C E). Lift the finger from string 3 of open Am and the open G string rings: that G is the b7.

Dominant 7: major triad plus a minor 7th

G7 is G B D F. This is the odd one out, a major triad with a minor 7th on top. It is called dominant because it is the chord built on the 5th degree of the major scale (G is the 5th of C), and that position gives it a special job we will meet in a moment. For now, hear it as bluesy and unsettled.

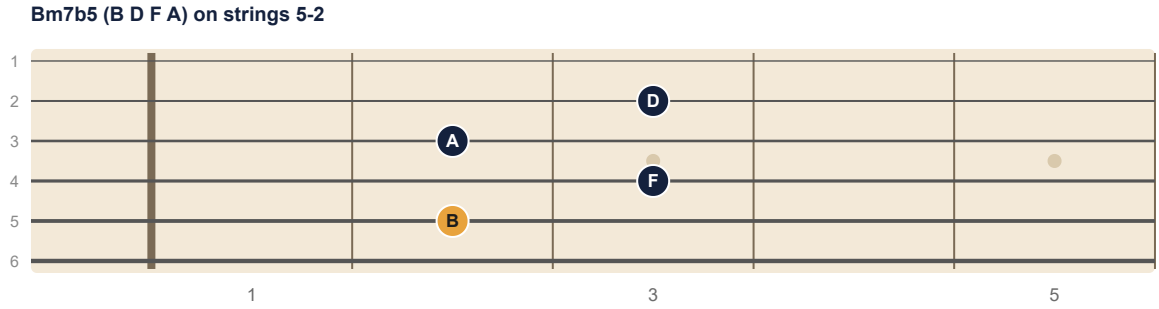


Open G7: G B D G B F. The F on string 1 (fret 1) is the b7 and the note that makes the chord restless.

m7b5 and dim7

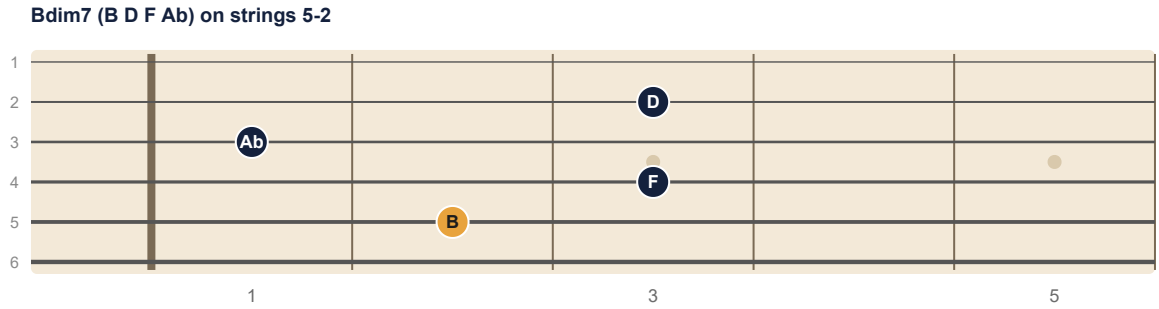
Take a minor triad and flatten the 5th: you get a diminished triad (1 b3 b5). Add a minor 7th (b7) and you get m7b5, also called half-diminished. Add a double-flat 7th (bb7, which is the same pitch as the 6th) and you get the fully diminished 7th, or dim7: 1 b3 b5 bb7.

The fretboard tool in this book draws diminished triads, so for these two seventh chords we draw the notes ourselves. Here is Bm7b5 on strings 5 to 2: B (root), F (b5), A (b7) and D (b3).



Fret 2 on string 5 is B, fret 3 on string 4 is F, fret 2 on string 3 is A, fret 3 on string 2 is D. The shape is a compact 'two-three-two-three' pattern. Mute strings 6 and 1.

Bdim7 is B D F Ab. Compared with Bm7b5, only one note moves: the A drops a half step to Ab. On the fretboard that is one finger sliding down a fret.



Same as Bm7b5 except string 3 drops from fret 2 to fret 1 (A becomes Ab). Dim7 chords are made of stacked minor thirds, so the shape repeats every three frets.

Quick naming tip

A plain '7' on a chord name always means the flat 7th (dominant). The major 7th must say 'maj7' (or 'M7'). So C7 is C E G Bb, but Cmaj7 is C E G B. Mixing them up is the number one seventh-chord mistake.

Harmonizing the major scale into sevenths

In Module 5 you stacked thirds on each degree of the major scale and got triads: I ii iii IV V vi vii°. Do the same with one more third on each and you get seventh chords. The pattern is always the same in every major key.

DEGREE	NAME	CHORD TYPE
I	Imaj7	maj7
ii	ii7	m7
iii	iii7	m7
IV	IVmaj7	maj7
V	V7	dominant 7
vi	vi7	m7
vii	viiø7	m7b5

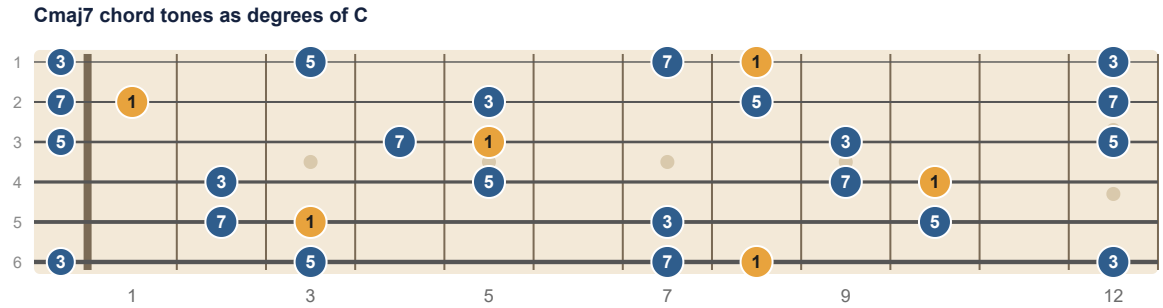
Here it is in C major. The scale is C D E F G A B, and each chord uses only those notes:

DEGREE	CHORD	NOTES
I	Cmaj7	C E G B
ii	Dm7	D F A C
iii	Em7	E G B D
IV	Fmaj7	F A C E
V	G7	G B D F
vi	Am7	A C E G
vii	Bm7b5	B D F A

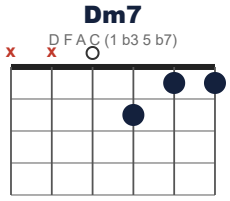
And in G major (G A B C D E F#):

DEGREE	CHORD	NOTES
I	Gmaj7	G B D F#
ii	Am7	A C E G
iii	Bm7	B D F# A
IV	Cmaj7	C E G B
V	D7	D F# A C
vi	Em7	E G B D
vii	F#m7b5	F# A C E

Notice something useful: every seventh chord in the key uses only notes from the key, so you can switch between them freely. Also notice the count: two maj7 chords (I and IV), three m7 chords (ii, iii, vi), one dominant 7 (the V) and one m7b5 (the vii).



The chord tones of Cmaj7 again, this time to compare with the next chord: Dm7 (D F A C) is the same stack of thirds started one scale step higher.



Open Dm7: D A C F. The ii chord of C. It uses the 1 b3 5 b7 formula with D as the root.

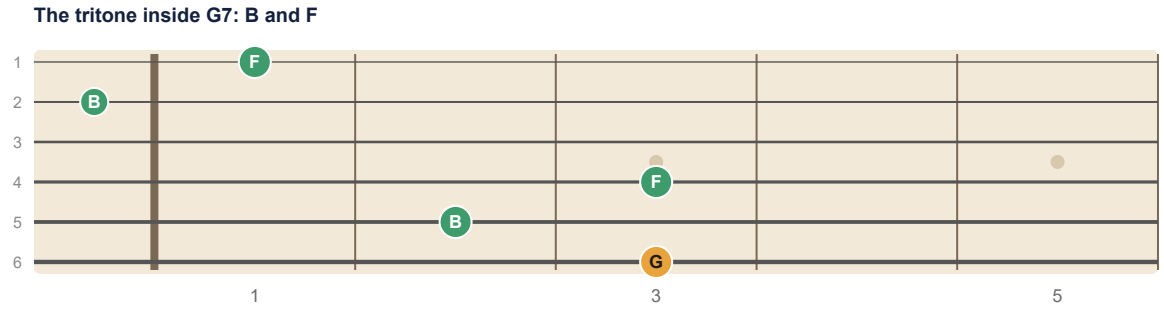
> **Memory trick**
>
> Lowercase roman numerals are minor, uppercase are major. For sevenths, remember the family: IVmaj7 and Imaj7 are the 'maj' pair, ii7 iii7 vi7 are the 'm7' trio, V7 stands alone, and vii is the odd half-diminished one.

Why V7 pulls toward I

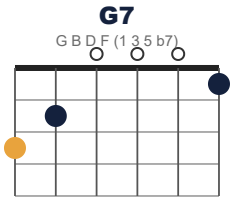
Of all the seventh chords, the dominant 7 has the strongest personality. Look inside G7: G, B, D, F. The notes B and F are three whole steps apart, which is an interval called a tritone (six half steps, the b5/#4).

A tritone is the most unstable sound in the scale. It wants to move.

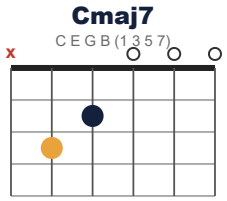
Watch how it resolves in C major. The B (the 7th of the scale) is only a half step below C, so it leans up into C. The F (the 4th) is only a half step above E, so it leans down into E. Both notes slide a half step into notes of the C major chord. The tension of the tritone releases and the chord feels like it has come home.



The two notes that make G7 want to move are B and F. They appear at strings 5/4 (frets 2 and 3) and strings 2/1 (frets 0 and 1). Play each pair together and listen to the tight, tense sound. G is shown as the root.



Resolve it: play G7, then play open C. B moves up to C, F moves down to E. Listen for the feeling of 'arriving'.



The arrival chord. Compare how G7 to C sounds versus G to C. With the seventh the pull is stronger.

This V7-to-I move is the heartbeat of most western songs, from folk to jazz to pop. In Module 8 you will see it again as the most common way a progression closes. Remember: it is the tritone (the 3 and b7 of the dominant) that does the pulling.

> **Your ear is the proof**
>
> Play G, then C, then G7, then C. Most players find the second pair feels more finished. Trust what you hear. The tritone is a real physical reason, not a rule invented to confuse you.

Tensions: the 9, 11 and 13

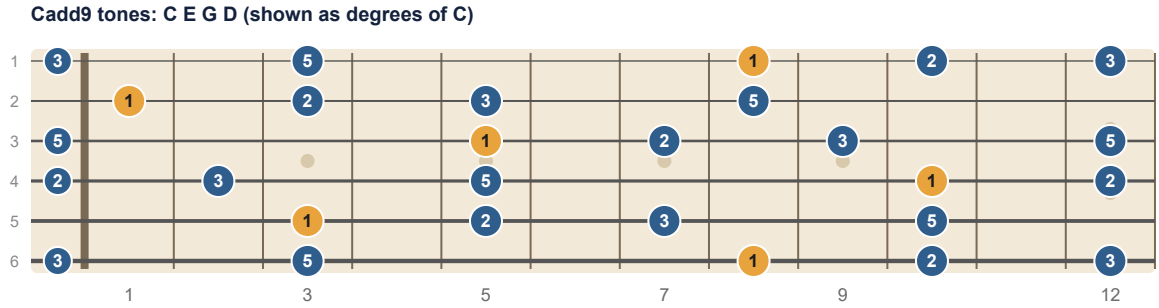
So far we have built chords by stacking thirds only up to the seventh. Keep stacking and you get a 9th, an 11th and a 13th. These are called tensions or extensions. They sound mysterious, but here is the simple truth: they are the same notes as 2, 4 and 6, an octave higher.

TENSION	SAME NOTE AS	IN C	INTERVAL FROM ROOT
9	2	D	M9 (a 2nd plus an octave)
11	4	F	P11 (a 4th plus an octave)
13	6	A	M13 (a 6th plus an octave)

Why name them as 9, 11 and 13 if they are the same notes as 2, 4 and 6? Because chord names tell you how the chord was stacked and what other notes it has. A 2 means the note replaces the 3rd. A 9 means the note is added on top and the 3rd stays. You can hear the difference.

add9 and 6 chords: the easy entry

The simplest tension chords are the ones you can play with a triad plus one note. Cadd9 is C E G plus D (the 9th): C E G D. Add the 9th to a triad and you get a shimmering, open sound used everywhere in modern acoustic playing. The 6 chord is a triad plus the 6th: C6 is C E G A.



The D appears labelled '2' because the tool shows the degree inside one octave. In the chord name it is the 9th because it is placed above the 3rd and 5th.

A nice detail: C6 (C E G A) uses exactly the same notes as Am7 (A C E G). Only the bass note decides which name you use. If the lowest note is C, call it C6. If the lowest note is A, call it Am7.

9 chords and 13 chords

A dominant 9 chord is a dominant 7 plus the 9th. C9 is C E G Bb D. You get the same bluesy drive of C7 with extra shine. A dominant 13 adds the 6th as well: C13 contains C E G Bb D A, usually with a note or two left out because six notes will not fit under four or five fingers. Most players drop the 5th and the 11th and keep the 3rd, 7th and the color note.

The 11th: a careful one

The 11th (same as the 4th) clashes with the major 3rd. In a C chord the F sits a half step above the E, so the two notes rub against each other. That is why you rarely add the 11th to a major chord. It works well on minor chords: Am11 is A C E G D, and the D simply sits above the minor 3rd without fighting it. On major chords, players usually replace the 3rd with the 4th and call it a sus chord. That brings us to the next section.

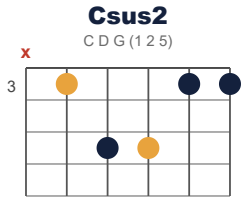
When tension is color, and when it is trouble

- **Color:** a tension that sits a whole step or more away from the chord notes around it, like the 9th on a major chord or the 13th on a dominant.
- **Trouble:** a tension that lands a half step above a chord tone that stays in the chord, like the 11th over a major 3rd, or the b9 over the root unless you mean it.
- **Rule of thumb:** if a note sounds sour, ask whether it is a half step above another note in the chord. If yes, lower the tension, change the voicing, or drop that chord tone.

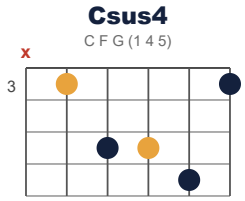
Sus chords: suspended color

A sus chord replaces the 3rd with a neighbor. There is no 3rd, so the chord is neither major nor minor. It sounds open, floating and waiting. 'Sus' is short for suspended: the note is held up from where it would normally land.

- **sus2:** 1 2 5. Csus2 is C D G.
- **sus4:** 1 4 5. Csus4 is C F G.



Csus2 (C D G) using the A form. The 3rd (E) is gone and the D takes its place.

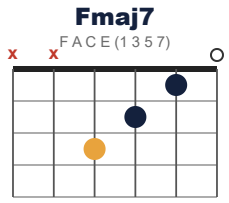


Csus4 (C F G) in the A form. Tension: the F wants to fall a half step to E, which is the 3rd.

Sus4 is the older classic: it wants to resolve down into the major 3rd, so Csus4 to C sounds like a small exhale. Sus2 does not need to resolve; players often leave it ringing. Folk, rock and pop use sus chords constantly because they add color without committing to major or minor.

Replacing triads with sevenths in a progression

Here is a simple, powerful trick. Take any progression of triads and swap chords for their seventh versions. In C major, a common pop loop is C, Am, F, G. Replace each chord with its diatonic seventh and you get Cmaj7, Am7, Fmaj7, G7.



Open Fmaj7: F A C E. It replaces F. Easier than the barre and it sounds lovely.

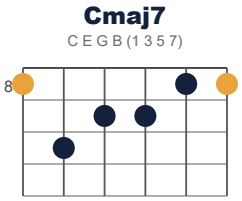
Try it slowly. Strum two beats of each chord. The progression keeps the same bass motion and the same song, but each chord now has a more polished, relaxed color.

TRIAD PROGRESSION	SEVENTH VERSION	WHAT CHANGES
C - Am - F - G	Cmaj7 - Am7 - Fmaj7 - G7	Softer, more open; G7 pulls harder to C
Am - Dm - G - C	Am7 - Dm7 - G7 - Cmaj7	A classic jazz-style loop (vi - ii - V - I)
G - Em - C - D	Gmaj7 - Em7 - Cmaj7 - D7	Same shape of song, but D7 now really wants to return to G

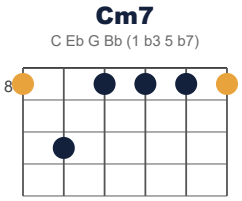
> **Do not seventh everything**
>
> Sevenths are seasoning. In a loud strummed rock tune, plain triads are often right. Use sevenths where you want warmth or movement, and keep triads when you want punch. Most players find that changing only the V and the vi chords already makes a big difference.

Common voicings on strings 6-5-4-3 and 5-4-3-2

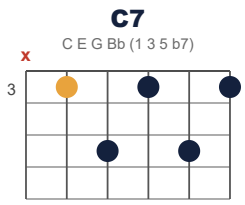
You already know the E form and the A form from Module 5: the E form has its root on string 6, the A form has its root on string 5. Those shapes slide up the neck to change the chord. Seventh chords work the same way. You just change one note in the shape.



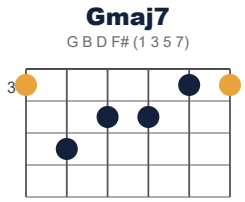
Cmaj7 in the E form: root on string 6 (fret 8). Slide this shape and the root is whichever note sits on string 6.



Cm7 in the E form. Compare with the maj7 above: the 3rd drops to the b3 and the 7th drops to the b7.



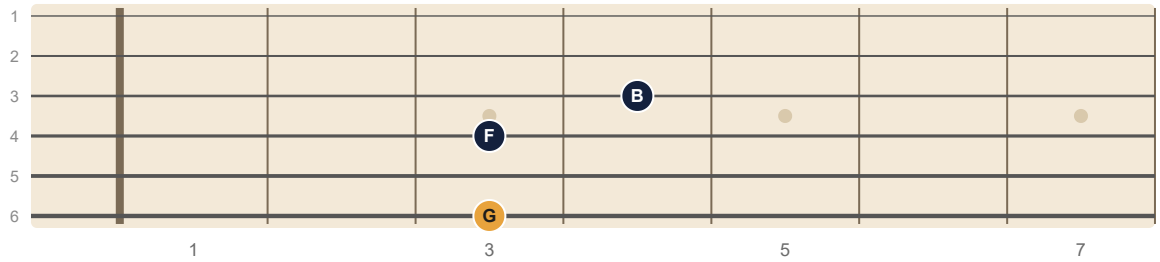
C7 in the A form: root on string 5 (fret 3). The Bb is the b7 that makes it dominant.



Gmaj7 in the E form with its root on string 6, fret 3.

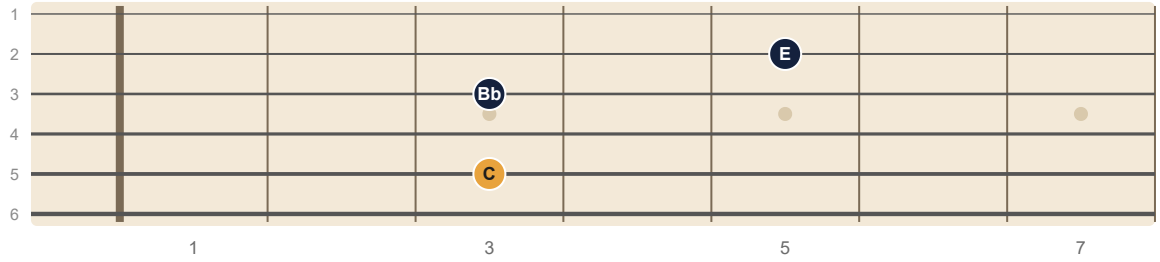
Not every note needs to be played. Most seventh-chord work in jazz and pop is done with just three or four strings, often keeping only the root, 3rd and 7th. These small voicings are called shell voicings, because they are the shell (outline) of the chord. They are easy to move and leave space for other instruments.

G7 shell on strings 6-4-3: 1, b7, 3



Root G on string 6 fret 3, the b7 (F) on string 4 fret 3, and the 3rd (B) on string 3 fret 4. Skip strings 5, 2 and 1.

C7 shell on strings 5-3-2: 1, b7, 3



Root C on string 5 fret 3, the b7 (Bb) on string 3 fret 3, and the 3rd (E) on string 2 fret 5. Leave strings 4 and 1 silent.

Why shells work

The 3rd tells you major or minor. The 7th tells you which kind of seventh. The 5th is the least important note, because it adds the least information. That is why pros often leave it out.

Color chords for jazz and pop, in plain English

People say 'color chord' to mean any chord that sounds richer than a basic triad. Here is a short list of the ones you will meet most, with the feel of each:

- **maj7:** the 'sunset' chord. Great for the I or IV when you want calm.
- **m7:** the 'late night' chord. Works anywhere a minor triad would, and is usually smoother.
- **7:** the 'pull' chord. Place it on the V, or use it anywhere you want blues flavor.
- **m7b5:** the 'sigh' chord. It usually sits on the vii or just before a minor ii-V.
- **sus2 / sus4:** the 'open window' chords. Use them where you want air.
- **add9:** the 'shimmer' chord. Brilliant on acoustic guitar, especially with open strings ringing.
- **6:** the 'vintage' chord. Warm and a little old-fashioned in the best way.
- **9:** the 'funk' chord. A dominant with a little extra gloss.

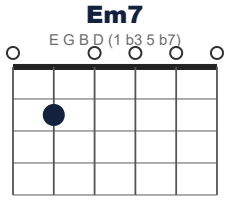
You do not need to learn all of them at once. Learn the open versions of four: Cmaj7, Am7, Dm7 and G7. Then learn how to find those same types with E-form and A-form shapes, and you can play maj7, m7 and dominant 7 in any key.

A closer look at the ii - V - I

Look back at the table of the C major sevenths and find three chords: Dm7, G7 and Cmaj7. This trio, ii - V - I, is the most important chord movement in jazz and shows up constantly in pop, soul and bossa. Each chord leads naturally to the next: the ii sets the stage, the V creates the pull, and the I releases it.

Spell them in notes and watch how little changes from one chord to the next. Dm7 is D F A C. G7 is G B D F. Cmaj7 is C E G B. Between Dm7 and G7, the D and F stay put as common tones (both chords contain D and F). Between G7 and Cmaj7, the F falls to E and the B rises to C. Smooth movement is the reason these chords feel so natural together.

Try it in G major too: Am7, D7, Gmaj7. Here the common tones between Am7 (A C E G) and D7 (D F# A C) are A and C, and in the final step the C falls to B while the F# rises to G. The same logic works in every key. Only the names change.

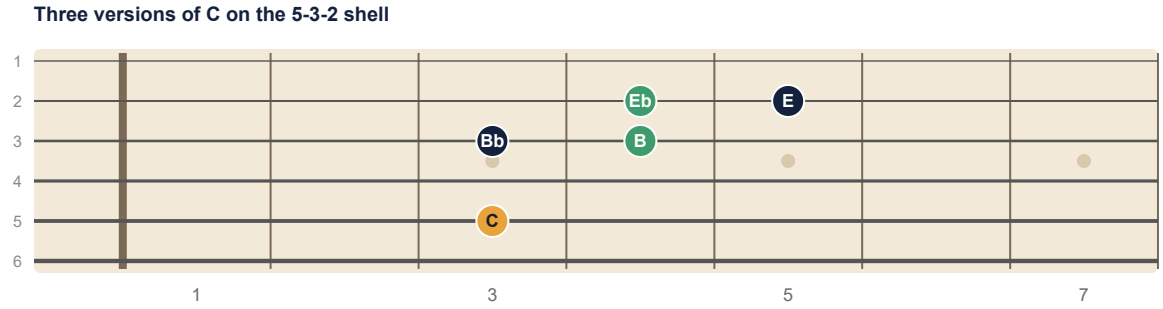


Open Em7 (E B D G B E): the vi chord of G and the iii chord of C. It is the easiest seventh chord on the neck, and it is open Em with one finger lifted.

More on voicings: the 5-4-3-2 set

So far the shapes were rooted on string 6 or string 5. A third useful family keeps the bass on string 5 and uses the middle strings 4, 3 and 2 as the top of the chord. Because you only play four strings, the sound is clear and the chord sits in the middle of the mix, which is perfect when another instrument holds the low notes.

Here is how to think about it. Pick a chord. Find its root on string 5. Then ask where the 3rd and the 7th are, close to the root, on strings 4, 3 and 2. Maj7 and m7 and dominant 7 differ only by one or two notes, so you can build all of them from one starting position. Start with C on string 5, fret 3, and look at the C7 shell you already learned: C on string 5, Bb on string 3, E on string 2. To turn it into Cmaj7, raise the Bb a half step to B (string 3 fret 4). To turn it into Cm7, lower the E a half step to Eb (string 2 fret 4).



Root C (string 5 fret 3). The b7 Bb (string 3 fret 3) and the 3rd E (string 2 fret 5) make C7. For Cmaj7 use B at string 3 fret 4 with the E. For Cm7 use Bb with the Eb at string 2 fret 4. Only one shape moves at a time: that is the point.

This is a big idea: you do not need to memorize dozens of separate shapes. You need to know where the root, 3rd and 7th live, and how a half step turns one chord type into another. Five minutes of that kind of thinking is worth an hour of copying chord charts.

A short practice routine for sevenths

1. Pick a key. Say its seven seventh chords out loud, in order, with notes. (For C: Cmaj7, Dm7, Em7, Fmaj7, G7, Am7, Bm7b5.)
2. Play the I, IV, V and vi in open shapes. Slowly. Four beats each. Use only the strings that sound clean.
3. Choose one shell voicing and slide it up and down the neck so the root lands on C, then D, then E, F, G, A and B. Name the chord at each stop.
4. Finish by playing a progression of your choice with the seventh version of every chord, then again with triads. Notice which one you prefer.

Keep it short. Ten focused minutes beats an hour of aimless strumming. Most players find that after a week of this, seventh chords stop feeling like a different language and start feeling like a different color of the same crayon.

Reading a chord name like a map

Chord names look cryptic until you see that they are built in a fixed order. First comes the root letter. Then the quality (m for minor, nothing for major). Then the number (7, 9, 13) and any alteration (b5, #9). Finally, any added or suspended notes. Take Am7b5 apart: A is the root, m means minor third, 7 means a seventh is included, b5 means the fifth is lowered. That gives A C Eb G, which is a half-diminished chord.

- **Cmaj9:** a Cmaj7 with the 9th added (C E G B D).
- **Cm9:** a Cm7 with the 9th added (C Eb G Bb D).
- **C7sus4:** a dominant 7 with the 3rd replaced by the 4th (C F G Bb).
- **C6/9:** a C6 chord with the 9th added (C E G A D). A favorite ending chord.

If a name ever confuses you, break it into pieces and write the notes in letters. A chord name is only a recipe, and you now know the ingredients.

Training your ear to tell them apart

Knowing the formulas is half the work. The other half is hearing the difference. Do this small experiment with an open C chord and its neighbors. Play C, then Cmaj7, then C7. The first sounds plain, the second sounds dreamy, the third sounds bluesy and unfinished. Your fingers moved by only one note each time.

Next, play Am, then Am7, then Am7 followed by Dm7. Notice that the minor seventh makes the chord softer, not sadder. A minor triad has a clear mood. The m7 version blurs the edges, which is why it fits so naturally into relaxed, groovy music.

Finally, play Dm7, G7, Cmaj7 and stop on the G7. Hold it. Your ear will ask for the next chord. That reaction is the tritone doing its job. Training your ear is simply paying attention to these feelings and naming them: calm, mellow, restless, tense. Later, when you hear a song, those labels will help you guess the chord type before you ever touch the guitar.

- **Calm and floating** usually means maj7 or sus2.
- **Soft and mellow** usually means m7.
- **Restless and bluesy** usually means a dominant 7 or 9.
- **Dark and about to resolve** usually means m7b5 or dim7.

Common mistakes

- **Calling a plain 7 a major 7.** C7 has a Bb. Cmaj7 has a B. They are neighbors in name and very different in sound.
- **Forgetting the key.** In G major the ii chord is Am7, not Am. Build every chord from the scale, not from habit.
- **Using maj7 on every major chord.** In C major only the I (Cmaj7) and IV (Fmaj7) are naturally maj7. The V is a dominant 7: G7, not Gmaj7.

- **Adding the 11th to a major chord.** It fights the 3rd. Use sus4 or choose a minor chord.
- **Strumming all six strings.** Many seventh voicings sound best with only four or five strings. Mute the rest with a fingertip or by aiming your pick.
- **Leaving every chord as a seventh.** If everything is colored, nothing stands out. Keep contrast.

Module 6 cheat sheet

TYPE	FORMULA	EXAMPLE (C)	NOTES	FEEL
maj7	1 3 5 7	Cmaj7	C E G B	Dreamy
m7	1 b3 5 b7	Cm7	C Eb G Bb	Mellow
7	1 3 5 b7	C7	C E G Bb	Restless, bluesy
m7b5	1 b3 b5 b7	Cm7b5	C Eb Gb Bb	Dark, tense
dim7	1 b3 b5 bb7	Cdim7	C Eb Gb Bbb (A)	Spooky, symmetrical
sus2	1 2 5	Csus2	C D G	Open
sus4	1 4 5	Csus4	C F G	Waiting
add9	1 3 5 9	Cadd9	C E G D	Shimmering
6	1 3 5 6	C6	C E G A	Vintage
9	1 3 5 b7 9	C9	C E G Bb D	Funky, rich

DEGREE IN A MAJOR KEY	I	II	III	IV	V	VI	VII
Seventh chord	maj7	m7	m7	maj7	7	m7	m7b5
In C	Cmaj7	Dm7	Em7	Fmaj7	G7	Am7	Bm7b5
In G	Gmaj7	Am7	Bm7	Cmaj7	D7	Em7	F#m7b5

BEFORE YOU MOVE ON, CAN YOU...

☐ Name the notes of Cmaj7, Dm7, Em7, Fmaj7, G7, Am7 and Bm7b5 without looking?

☐ Say why G7 wants to go to C (the tritone B-F)?

☐ Turn a triad into maj7, m7 and 7 by changing one note?

☐ Play open Cmaj7, Am7, Dm7 and G7 cleanly?

☐ Find a shell voicing (1, 3, 7) on strings 6-4-3 or 5-3-2?

Glossary

- **Seventh chord:** a triad with a fourth note, the 7th, stacked on top.
- **maj7:** a major triad plus the major 7th (1 3 5 7).
- **m7:** a minor triad plus the minor 7th (1 b3 5 b7).
- **Dominant 7:** a major triad plus the minor 7th (1 3 5 b7), built on the 5th degree.
- **m7b5 (half-diminished):** a diminished triad plus the minor 7th (1 b3 b5 b7).
- **dim7:** a diminished triad plus the double-flat 7th (1 b3 b5 bb7).
- **Tritone:** an interval of three whole steps (six half steps), the b5/#4. Very unstable.
- **Tension (extension):** a note above the 7th, named 9, 11 or 13.
- **Sus chord:** a chord where the 3rd is replaced by the 2nd (sus2) or 4th (sus4).
- **Shell voicing:** a stripped-down chord with just the root, 3rd and 7th.
- **Diatonic:** made only of notes that belong to the key.

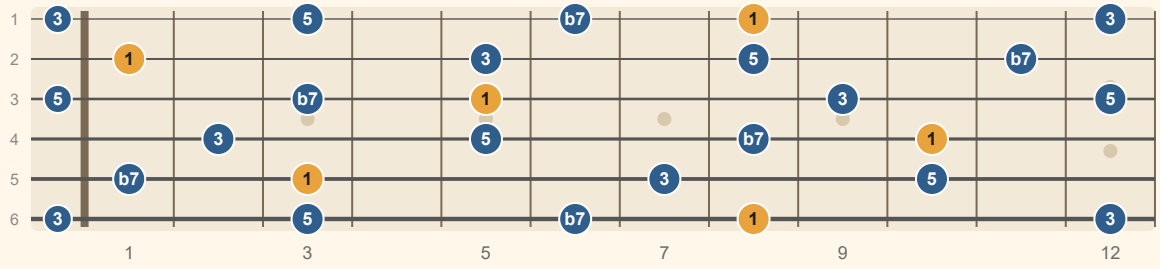
On the fretboard: exercises 26 to 30

EXERCISE 26 — 5 MIN

Build the four sevenths on C

1. Write C as your root. Using the formulas, spell out Cmaj7, Cm7, C7 and Cm7b5 in letter names.
2. Check that each chord has exactly four notes and that only the 3rd, 5th and 7th change.
3. Find all the C notes on strings 6 and 5. Pick one and build each chord around it.

Notes of C7: C E G Bb



All C7 tones. Find the Bb: it is the only note that is not in the C major scale.

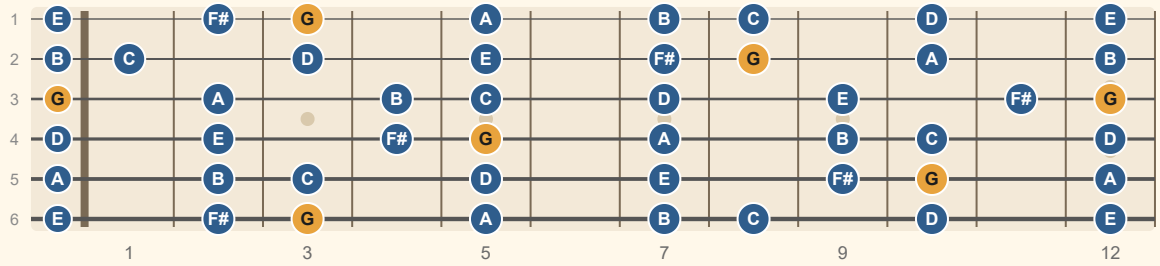
☐ Self-check: Cmaj7 = C E G B, Cm7 = C Eb G Bb, C7 = C E G Bb, Cm7b5 = C Eb Gb Bb. If any chord has more than one different note from the next, recount.

EXERCISE 27 8 MIN

Harmonize G major into sevenths

1. Write the G major scale: G A B C D E F#.
2. For each degree, stack thirds by skipping every other scale note until you have four notes.
3. Name each chord type using the pattern: maj7, m7, m7, maj7, 7, m7, m7b5.

G major



Use these notes to build each chord: skip one, take one, skip one, take one.

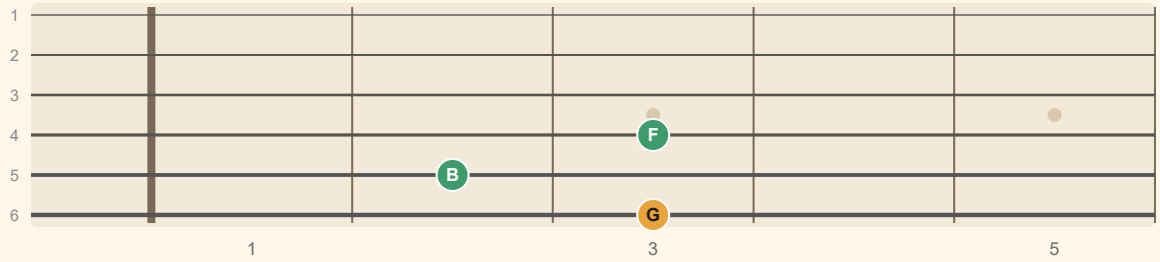
☐ Self-check: Gmaj7, Am7, Bm7, Cmaj7, D7, Em7, F#m7b5. The V chord D7 should contain F# and C (the tritone).

EXERCISE 28 6 MIN

Hear the pull of V7

1. Play open G major, then open C. Listen to how it ends.
2. Now play open G7, then open C. Listen again.
3. Find the B and F inside G7. Play only those two notes together (string 5 fret 2 and string 4 fret 3), then play the open C chord and listen for each note stepping into place.
4. Repeat with D7 to G (use the open D7 shape).

The tritone B-F



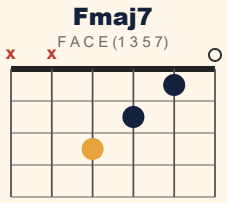
B (string 5 fret 2) and F (string 4 fret 3). They resolve to C and E.

☐ Self-check: you should be able to point to the two notes of the tritone (B and F in G7) and say where each one wants to go (B up to C, F down to E).

EXERCISE 29 10 MIN

Swap triads for sevenths

1. Play C - Am - F - G for four bars with plain triads.
2. Replace every chord with Cmaj7 - Am7 - Fmaj7 - G7 using the open shapes.
3. Now mix: keep C and F as triads, but use Am7 and G7. Which version do you like for the verse? For the chorus?
4. Repeat in G major: G - Em - C - D versus Gmaj7 - Em7 - Cmaj7 - D7.



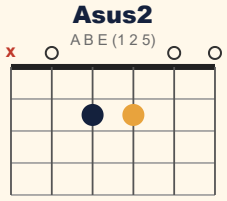
Open Fmaj7 (F A C E) replaces F in the loop.

☐ Self-check: you can describe in one sentence how the sevenths changed the mood, and you can name the one chord that adds the most pull (V7).

EXERCISE 30 8 MIN

Color a chord with a tension

1. Play an open C (x32010). Now move your first finger off string 2 and put your third finger on fret 3 of string 2 instead: x32030. That D is the 9th, and the chord is Cadd9.
2. Play open Asus2 and Asus4 and compare. Which one wants to move?
3. Pick one chord in a song you know and replace it with a sus or add9 version. Decide whether it is color or trouble.
4. Write down one tension you will use again.



Open Asus2 (A B E): the 3rd is replaced by the 2nd.

☐ Self-check: Cadd9 contains C E G D. The D on string 2 sits a whole step below the open E on string 1. If you hear a clash, make sure string 1 rings open and that no finger touches it.

Next step

Play the seventh chords you built today inside a real song idea. Pick one of the progressions from this module, loop it for two minutes, and notice which chord makes you lean forward. That chord is your new

favorite color. When you are ready, move to Module 7, where CAGED shows you how every chord shape connects across the whole fretboard.

THE FRETBOARD REVEALED

Part 3 - Connect

MODULE 7

CAGED: The Whole Fretboard Opens Up

- Five open chords you already know (C, A, G, E, D) are five puzzle pieces that tile the entire neck.
- Every piece slides up the neck, and the pieces always link in the same order: C - A - G - E - D, then back to C twelve frets higher.
- Inside each piece lives a scale position, a pentatonic box and an arpeggio, so one map gives you chords, scales and licks.
- You will play one chord progression in three different places and hear why it is the same music with a different feel.

By the end: By the end of this module you will be able to pick any major key, find all five CAGED shapes for it, name the one you are in, and move smoothly from one to the next.

> *For years you may have lived in the same four or five frets. This module is where that stops. You already own the key: five chord shapes you learned in your first months on the guitar.*

The same four frets for years

Be honest with yourself for a moment. When you pick up the guitar, where do your hands go? For most self-taught players the answer is the same small neighborhood: the open chords near the nut, a minor pentatonic box somewhere around the fifth fret, and a handful of licks that were learned note by note. That is not a failure. It is how almost everyone starts. You learned what you needed to play the songs you loved, and you stopped when the songs stopped asking for more.

The trouble is that the guitar is a very strange instrument. On a piano, every note has one home, and moving up the keyboard always means the same thing. On a guitar, the same note lives in several places, and the same chord can be played in many different spots. A beginner sees that as chaos. A player who understands the pattern sees it as freedom: the whole neck becomes one connected map, and any idea you have can be played in the place where it sounds best.

The map is called **CAGED**, named after the five open chords it is built from: C, A, G, E and D. Most players find that once they see it, the neck stops being a wall of dozens of positions and starts looking like five neighboring rooms. You will not have to memorize hundreds of fingerings. You will learn five shapes, one order, and a handful of simple rules.

There is also a practical payoff that has nothing to do with theory. When you only know one position for each chord, a song in an awkward key forces you into awkward fingerings, and a solo is stuck wherever your favorite box happens to be. When you know the chain, you can ask a better question: where on the

neck do I want to be for this part of the song? Near the nut for a warm, ringing feel, up in the middle for body and punch, or high for a thin, bright color. The notes are the same. The choice is yours.

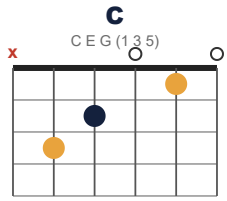
> **Before you start**
> This module builds on the earlier ones. You should be able to name the notes on the three anchor strings (Module 1) and know the major scale formula (Module 3). If a note name makes you hesitate, flip back and refresh. Every diagram below is labeled, so you can still follow along.

Here is the promise of this module in one sentence: **there is no new chord to learn.** You already know all five pieces. All we do is slide them, link them, and look inside them.

Five open chords, five puzzle pieces

Think of the neck as a floor that needs tiling. You have five tiles. Each has a different outline, but they fit together edge to edge with no gaps. Let us look at each tile in its original open position first, because that is where your fingers already know the way. As you look, pay attention to two things for every shape: which strings are played, and **which string holds the root note** (the note that gives the chord its name). The root is the anchor that lets you slide the shape anywhere.

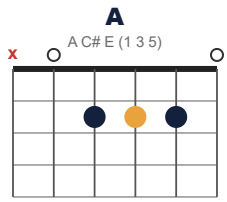
The C shape



The open C shape: x 3 2 0 1 0. Notes from string 5 to string 1: C E G C E. The root C sits on string 5, fret 3.

The C shape is a compact, arched shape. Your ring finger sits on string 5, fret 3, which is the root C. Index and middle fingers sit above it. String 6 is not played. Because the main root is on string 5, this is a **string-5 root shape**. It also has a second C on string 2, fret 1.

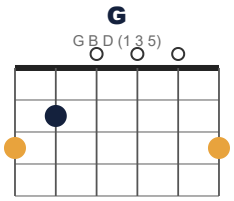
The A shape



The open A shape: x 0 2 2 2 0. Notes: A E A C# E. The root A is the open string 5.

The A shape stacks three fingers side by side on strings 4, 3 and 2, all at fret 2. The root is the open A string (string 5). When you slide this shape up the neck, a barre replaces the open string and takes over the job of holding the root, still on string 5. This is another string-5 root shape, but with a very different outline from C.

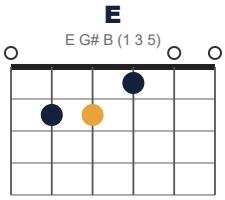
The G shape



The open G shape: 3 2 0 0 0 3. Notes: G B D G B G. The root G is on string 6, fret 3.

The G shape is the widest one. Fingers sit on string 6 (fret 3), string 5 (fret 2) and string 1 (fret 3). The main root is on string 6, and the high string gives you another G. This is a **string-6 root shape**.

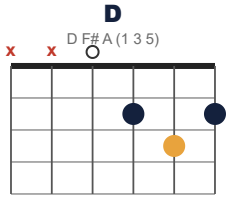
The E shape



The open E shape: 0 2 2 1 0 0. Notes: E B E G# B E. The root E is the open string 6.

The E shape is the famous one. It is the shape behind almost every barre chord you have ever played. The root is on string 6, and the same shape also has roots on strings 4 and 1.

The D shape



The open D shape: x x 0 2 3 2. Notes: D A D F#. The root D is the open string 4.

The D shape is the small one: only the top four strings, a tight triangle of fingers on strings 3, 2 and 1. The root is on string 4. It is the hardest shape to see on the neck at first, because it is so compact, and that is exactly why we will give it extra attention later.

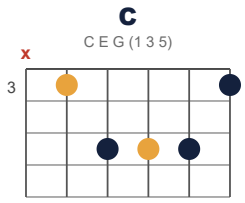
A pattern worth noticing

Two shapes use string 5 as the main root string (C and A), two use string 6 (G and E), and one uses string 4 (D). This tiny fact will save you a lot of searching later.

The capo idea: how a shape moves up the neck

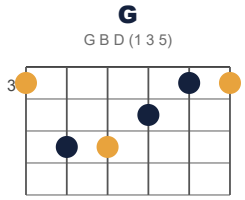
Here is the idea that unlocks everything. Put a capo on the second fret of a guitar and play an open G shape. What do you hear? An A chord. The shape did not change. Every note simply went up two frets, which is two semitones, which moved G to A. A capo turns the nut into a movable fret, and every open shape into a movable shape.

Now think of your index finger as the capo. Lay it flat across all six strings at fret 3 and play the A shape with your other fingers. The shape is still an A shape, but everything is three semitones higher: A plus three semitones is C. You are now playing a **C chord with an A shape**. The open strings of the original shape are now held down by the barre, so they behave like the nut did before.



The A shape moved up to fret 3: x 3 5 5 5 3. Notes: C G C E G. This is a C major chord with the A-shape outline.

Check the notes. String 5 at fret 3 is C, string 4 at fret 5 is G, string 3 at fret 5 is C, string 2 at fret 5 is E, and string 1 at fret 3 is G. That is C, G, C, E, G: a C major chord (C E G) in a different arrangement than the open C. Same chord, different neighborhood.



The E shape moved up to fret 3: 3 5 5 4 3 3. Notes: G D G B D G. This is a G major chord with the E-shape outline.

The same logic works with the E shape. Slide it up until the root on string 6 lands on G (fret 3) and you get G major. This is the barre chord you already know. You were already using CAGED. You just never called it that.

The one rule

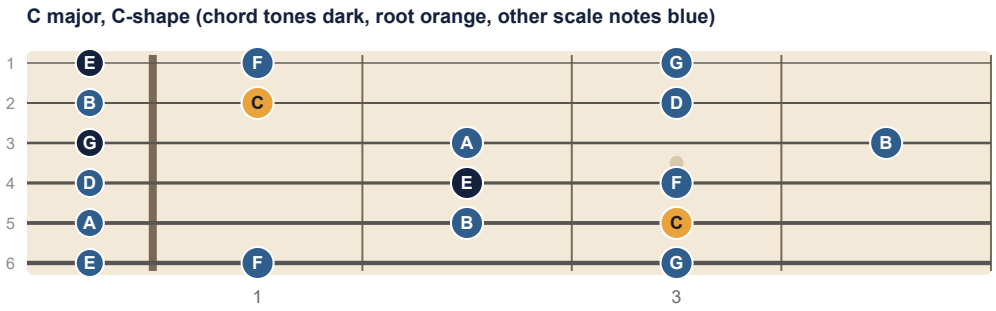
To move any shape, find its root note in the new place. If the root of the shape is on string 5, look for the key name on string 5. If it is on string 6, look on string 6. The shape then falls into place around it. The fret number of the root is also the capo number: a root on fret 3 means a capo at fret 3.

SHAPE	STARTS AS	MAIN ROOT STRING	CAPO EXAMPLES (LETTER PLUS FRETS)
C shape	Open C	string 5	C + 2 = D, C + 5 = F
A shape	Open A	string 5	A + 3 = C, A + 5 = D
G shape	Open G	string 6	G + 2 = A, G + 5 = C
E shape	Open E	string 6	E + 3 = G, E + 5 = A
D shape	Open D	string 4	D + 2 = E, D + 5 = G

The chain: C - A - G - E - D

If you take the key of C and slide each of the five shapes until it makes a C chord, something beautiful happens: the five C chords line up along the neck in a fixed order, each one starting about where the last one ends. Let us watch it happen one piece at a time.

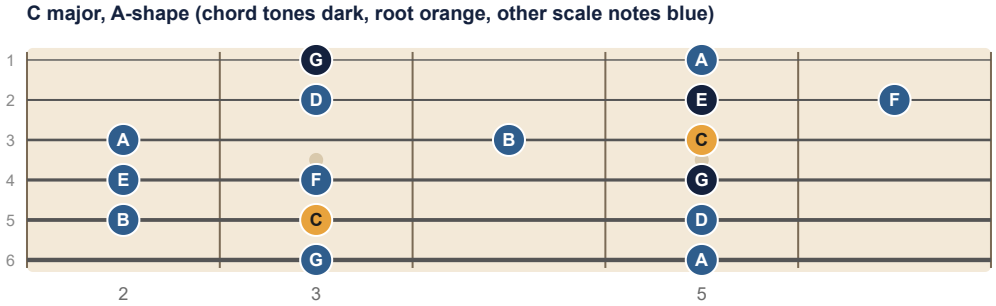
Piece 1: the C shape, at the nut



The C shape in the key of C: frets 0 to 3, the open C chord.

The C shape is the first tile. In the key of C it sits at the very start of the neck, in the open position. Its neighborhood covers frets 0 to 3.

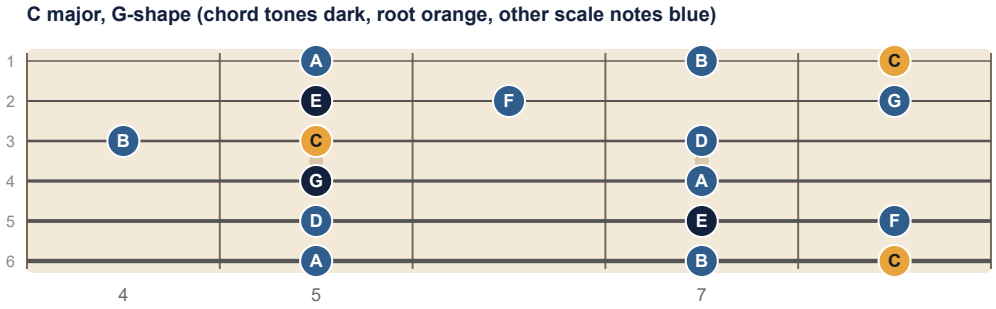
Piece 2: the A shape, around fret 3



The A shape in the key of C: barre at fret 3, covering frets 3 to 5.

The next tile begins where the C shape stops. The A shape with the root on string 5, fret 3, covers frets 3 to 5. Your hand is still next to where it was.

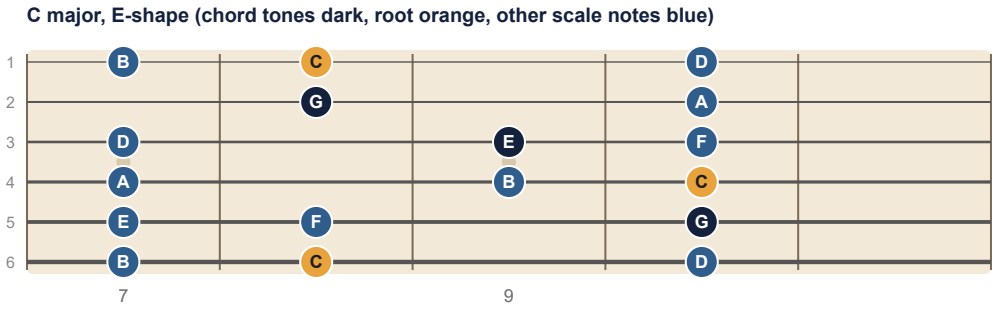
Piece 3: the G shape, around fret 5



The G shape in the key of C: root C on string 6, fret 8, with the shape reaching down to fret 5.

Now the G shape. The root is on string 6, so look for C on string 6: that is fret 8. But the shape reaches down below the root, so the chord spans frets 5 to 8 (8 7 5 5 5 8). This is the shape that surprises many players, because the root sits at the top of the shape, not the bottom.

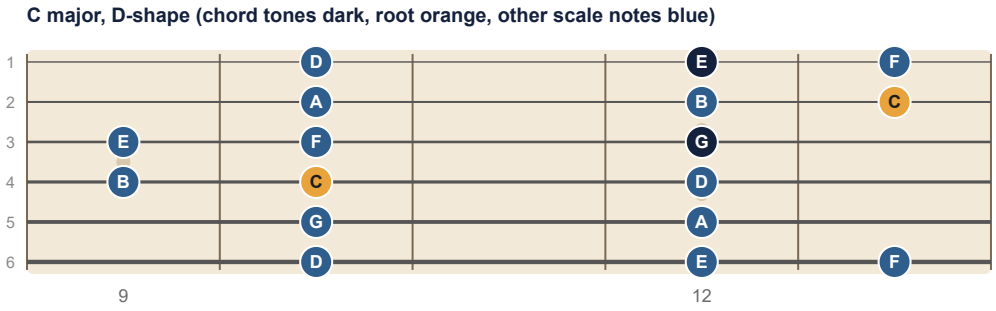
Piece 4: the E shape, around fret 8



The E shape in the key of C: barre at fret 8, covering frets 8 to 10.

The E shape is the barre chord you know best, with the root on string 6, fret 8 (8 10 10 9 8 8). It covers frets 8 to 10, starting exactly where the G shape ended.

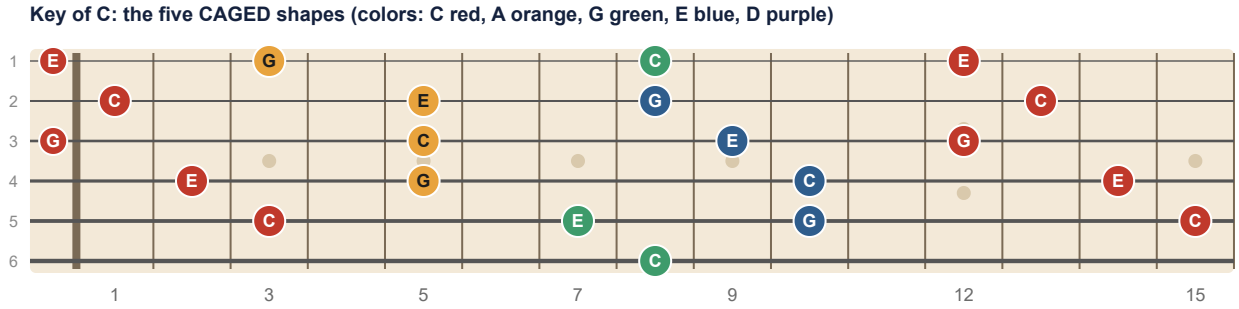
Piece 5: the D shape, around fret 10



The D shape in the key of C: root C on string 4, fret 10, covering frets 10 to 13.

Last, the D shape. The root is on string 4, so find C on string 4: fret 10. The shape reaches up to fret 13. After that, the chain starts over: the C shape again at fret 12, exactly one octave above where we began.

All five together



The five CAGED shapes for the key of C, linked in order: C - A - G - E - D, then C again at fret 12.

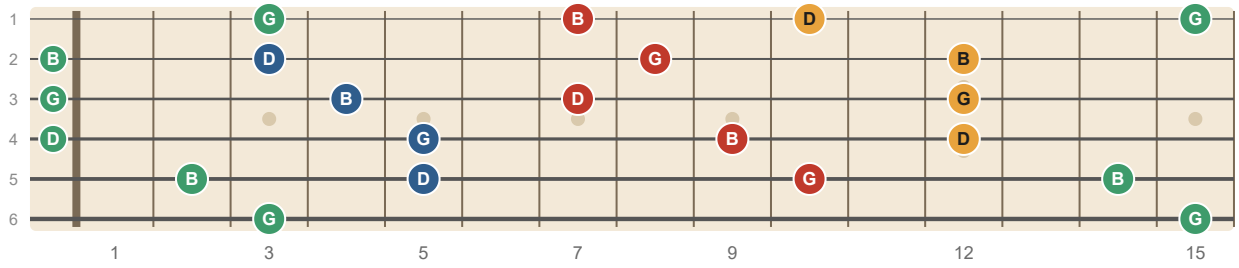
This is the picture to carry in your head. Read it left to right: C, A, G, E, D. The letters of the word are the order of the shapes, in every key. The neck is no longer a maze. It is a row of five rooms, and each room has a door into the next one.

> **Why the chain repeats at fret 12**
>
> The neck is a repeating pattern. Twelve frets equal twelve semitones, and twelve semitones is one octave, so at fret 12 every string plays the same note name as its open string. Everything you learned from fret 0 to fret 12 happens again from fret 12 upward.

Same chain, other keys

The best part is that the order never changes. What changes from key to key is **which shape you meet first at the nut**. In the key of C the chain starts with the C shape. In other keys it starts with a different shape, but it always continues in the same circular order: C, A, G, E, D, C, A, G, E, D, and so on.

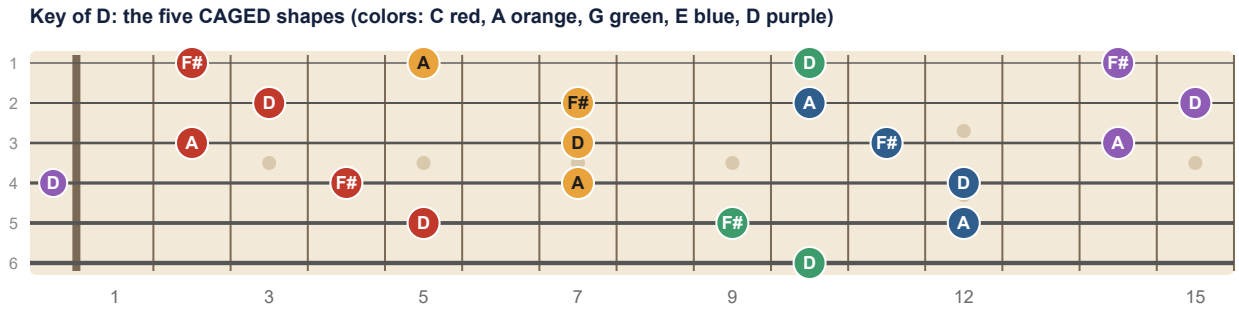
Key of G



The five CAGED shapes for the key of G. The chain starts with the G shape at the nut: G - E - D - C - A.

In the key of G, the open G chord is the G shape, so the chain starts there. Next in the circle comes E (the E shape, a G barre at fret 3), then D (around fret 5), then C (around fret 7), then A (the A shape with the root G at fret 10), and then the G shape returns at fret 12.

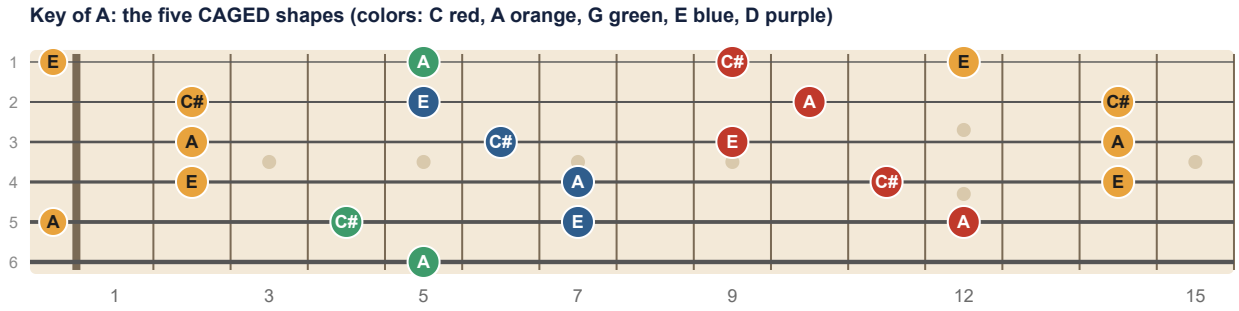
Key of D



The five CAGED shapes for the key of D: D - C - A - G - E, then D again at fret 12.

In the key of D, the chain starts with the D shape. After D comes C (the C shape, the small arched chord around frets 2 to 5), then A (around fret 5), then G (the G shape with the root D on string 6, fret 10, reaching down to fret 7), then E (the E shape with the root D at fret 10), and the D shape comes back at fret 12.

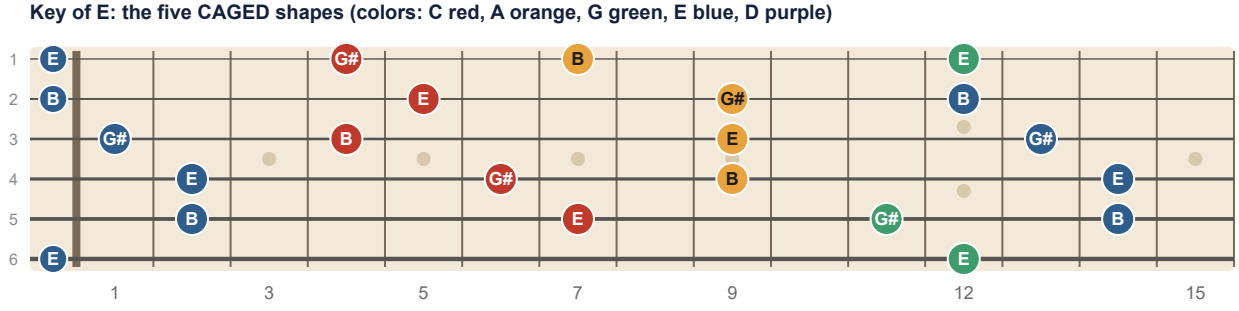
Key of A



The five CAGED shapes for the key of A: A - G - E - D - C, then A again at fret 12.

In the key of A the chain starts with the A shape at the nut, then G (around frets 2 to 5), E (the barre at fret 5), D (around fret 7), C (around frets 9 to 12), and then the A shape again at fret 12.

Key of E

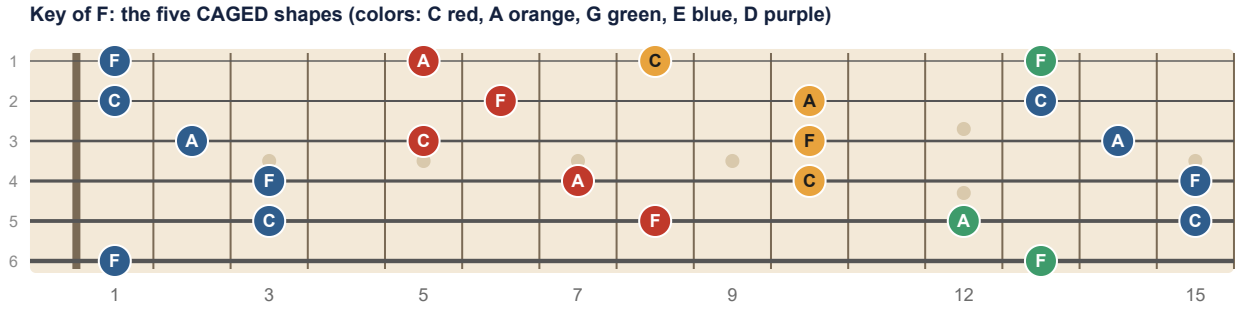


The five CAGED shapes for the key of E: E - D - C - A - G, then E again at fret 12.

In the key of E, the E shape starts at the nut, then D (around fret 2), C (around fret 4), A (the A shape with the root E at fret 7), G (around frets 9 to 12), and the E shape returns at fret 12.

Keys that are not in the word CAGED

What about keys like F, B-flat or F-sharp? The same rule applies, with a small twist. Use the capo idea: pick one shape, find where its root lands on the key, and the rest of the chain follows in order. For the key of F, the E shape is the natural starting point: put the root of the E shape on string 6, fret 1 and you have F (1 3 3 2 1 1). Then the chain continues D (around fret 3), C (around fret 5), A (the A shape with the root F at fret 8), G (around fret 10), and E again at fret 13.



The five CAGED shapes for the key of F. The E shape sits at fret 1, then D, C, A, G, and E again at fret 13.

KEY	ORDER OF THE SHAPES	WHERE EACH SHAPE SITS (FRETS, APPROXIMATE)
C	C - A - G - E - D	C at 0, A at 3, G at 5-8, E at 8, D at 10
G	G - E - D - C - A	G at 0, E at 3, D at 5, C at 7, A at 10
D	D - C - A - G - E	D at 0, C at 2, A at 5, G at 7-10, E at 10
A	A - G - E - D - C	A at 0, G at 2-5, E at 5, D at 7, C at 9
E	E - D - C - A - G	E at 0, D at 2, C at 4, A at 7, G at 9-12
F	E - D - C - A - G	E at 1, D at 3, C at 5, A at 8, G at 10

These frets are rough guides to where each shape begins. The diagrams show the exact drawings. What matters is the pattern: the chain always follows C - A - G - E - D, and the key tells you which shape you meet first.

How to find where a shape lives in any key

You do not need a table for every key. You need a method. Here it is, in five steps. It works for any major key, and it is the single most useful tool in this module.

1. **Name the key.** For example, the key of A.
2. **Choose the shape you want.** For example, the E shape, because you like barre chords in the middle of the neck.
3. **Find that shape's root string.** The E shape has its root on string 6.
4. **Find the key name on that string.** A on string 6 is fret 5.

5. **Lay the shape down around that root.** You now have an A chord at fret 5 with the E shape: 5 7 7 6 5 5.

Try it again for the D shape in the key of C. Root string of the D shape: string 4. C on string 4: fret 10. The shape goes around it: x x 10 12 13 12. Check the notes: C on string 4, G on string 3 (fret 12), C on string 2 (fret 13), and E on string 1 (fret 12). C, G, C, E. It is a C major chord.

And once more for the C shape in the key of D. The C shape has its root on string 5. D on string 5 is fret 5. The shape goes around it, and since the open C shape reaches down toward the nut, the chord is x 5 4 2 3 2. Every note is two frets higher than in the open C, which makes sense: D is two semitones above C.

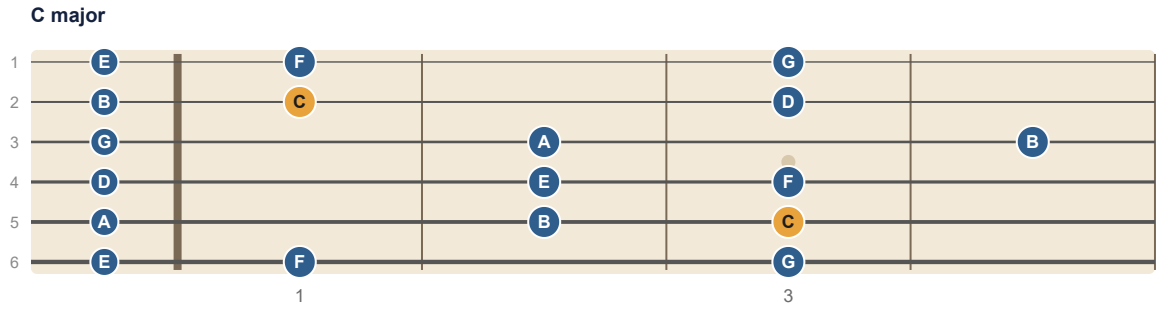
> **Fast check**
>
> If your chord does not sound right, the most common reason is that the root is on the wrong string. Re-read the root string for your shape (C: 5, A: 5, G: 6, E: 6, D: 4) and count again from the open string note.

Scale positions inside each shape

So far we have used the shapes for chords. Now the same shapes become a map for scales. Here is the key idea: **every chord shape has a scale position wrapped around it.** The notes of the chord are the strongest notes of the scale, and the other scale notes fill the gaps in the neighborhood. If you know where the chord is, you know where the scale lives.

We will use the key of C so that all notes are natural, and the diagrams show the notes of the C major scale (C D E F G A B) in a window of frets around each shape. Notice the chord tones inside each window: C, E and G. They are the target notes you can aim for when you play a melody or a solo.

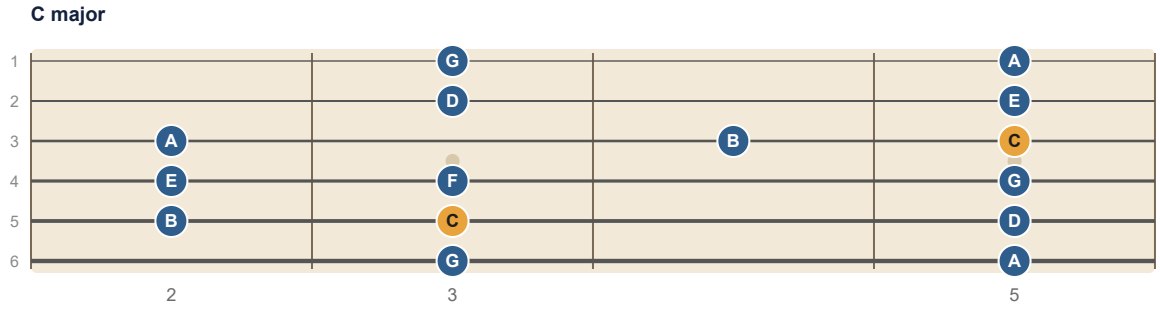
Around the C shape (frets 0 to 4)



C major scale around the C shape, frets 0 to 4. Chord tones C, E, G are the landmarks.

Start with the position you probably already know from the nut. On the low strings you have E, F and G on the sixth string (frets 0, 1, 3), A, B and C on the fifth (frets 0, 2, 3), and so on. Find the C chord inside it: C on string 5, fret 3, E on string 4, fret 2, G on string 3, open.

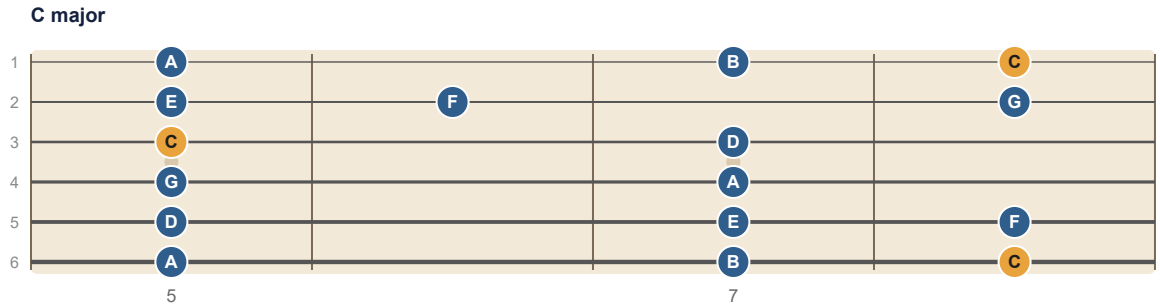
Around the A shape (frets 2 to 6)



C major scale around the A shape, frets 2 to 6. The A-shape C chord is x 3 5 5 5 3.

This position overlaps with the one before it, which is exactly what we want. Your hand moves up only a little. The A-shape C chord at fret 3 sits right in the middle of this window, and the root C appears on strings 5, 3 and 1 (frets 3, 5 and 8 in the neighboring windows).

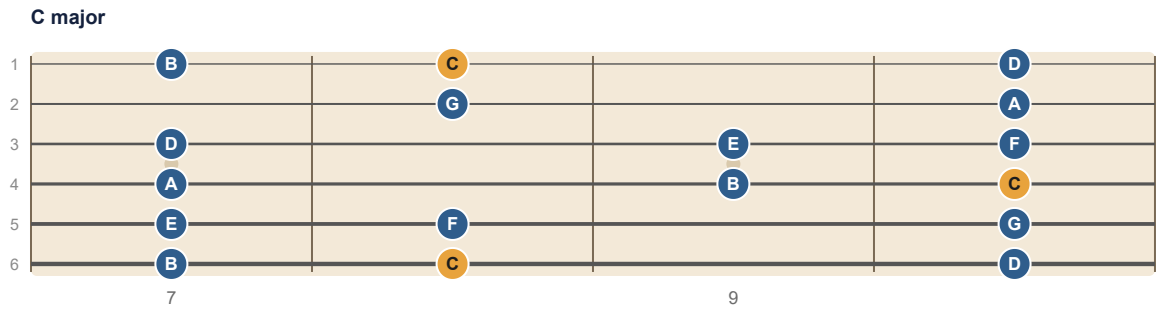
Around the G shape (frets 5 to 9)



C major scale around the G shape, frets 5 to 9. The root C is on string 6, fret 8.

Here the scale appears with the root C at the top of the low string, fret 8. Many guitarists learn the next position first and never see this one. It is the missing link that makes the chain whole.

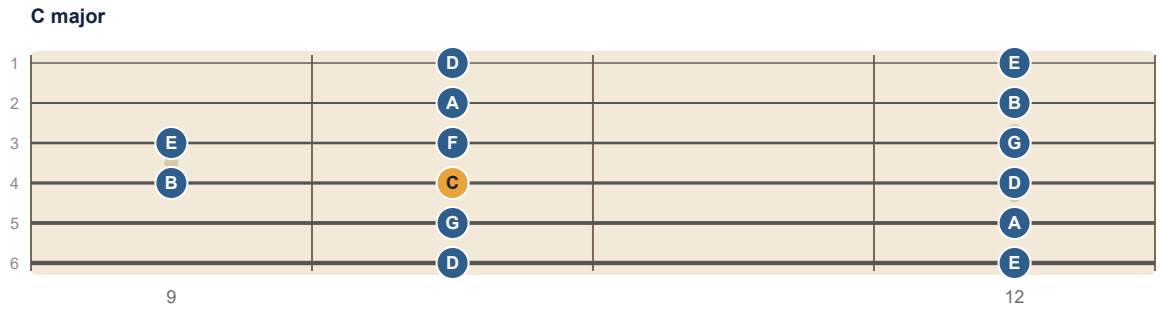
Around the E shape (frets 7 to 11)



C major scale around the E shape, frets 7 to 11. The barre chord at fret 8 is the landmark.

The E shape is the best-known scale position for most players. Put your index finger on fret 8 and the major scale falls into a clear pattern across the strings. Because the chord and the scale share the same home, you can switch between strumming and playing single notes without moving your hand.

Around the D shape (frets 9 to 13)



C major scale around the D shape, frets 9 to 13. The D-shape C chord is x x 10 12 13 12.

Last, the D position, which wraps around the small chord at fret 10. After this, the cycle begins again at fret 12 with the C position. Every note of the major scale is covered by one of these five windows, and each window overlaps its neighbors.

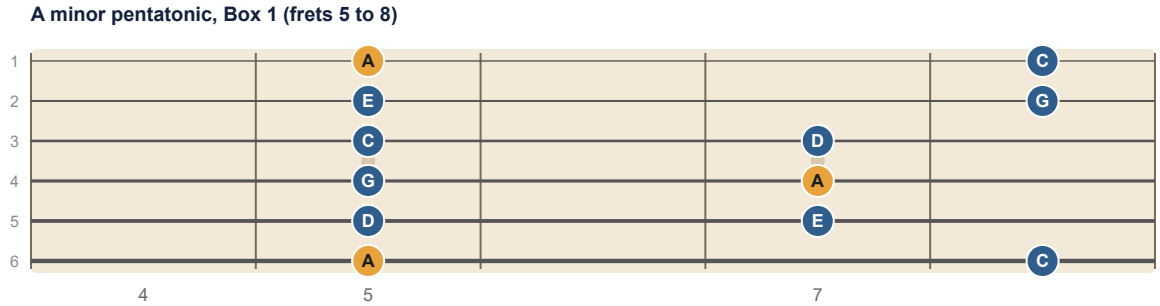
> **What to practice**
>
> Play each position slowly, up and then down, saying the note names out loud. Then play the chord for that position and find the chord tones inside the scale. Most players find that after a few days the shapes start to feel connected instead of five separate drawings.

The five pentatonic boxes

You probably know one pentatonic box already: the minor pentatonic shape at the fifth fret. It is the first scale most players learn and the one that sounds good over nearly everything. What most self-taught players do not realize is that there are **five boxes**, and they correspond one to one with the five CAGED shapes.

Pentatonic means five notes. The major pentatonic is the major scale with two notes removed: for C, that is C D E G A (the 4th and 7th, F and B, are gone). The minor pentatonic of A has exactly the same notes: A C D E G. The two scales are relatives: same notes, different home base. That means the five boxes of A minor pentatonic live in the same places as the five CAGED shapes of C major.

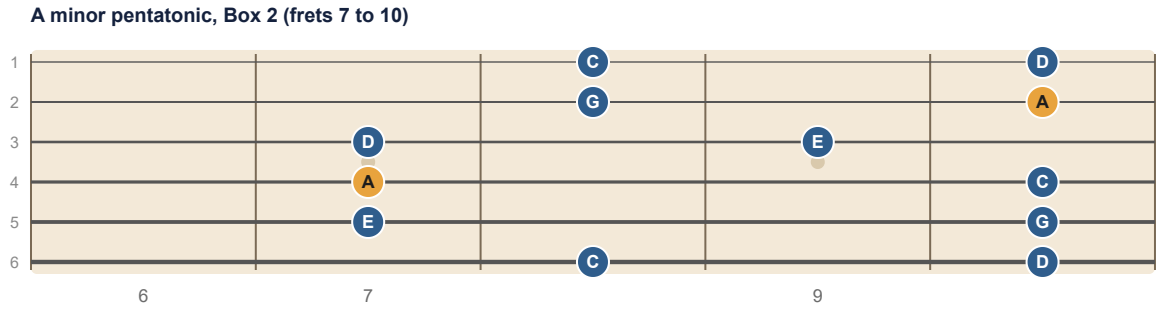
Here is Box 1 of A minor pentatonic, the one you probably know. Notice the notes: A, C, D, E, G, repeated across the strings, with two notes on each string.



Box 1 of A minor pentatonic. The A roots are on strings 6, 4 and 1. In the key of C, this box lives around the G shape.

In the key of C major, this box lives in the same region as the G shape (frets 5 to 8). That is why this box is so useful: it holds the chord tones of both C and Am, so it fits over either chord.

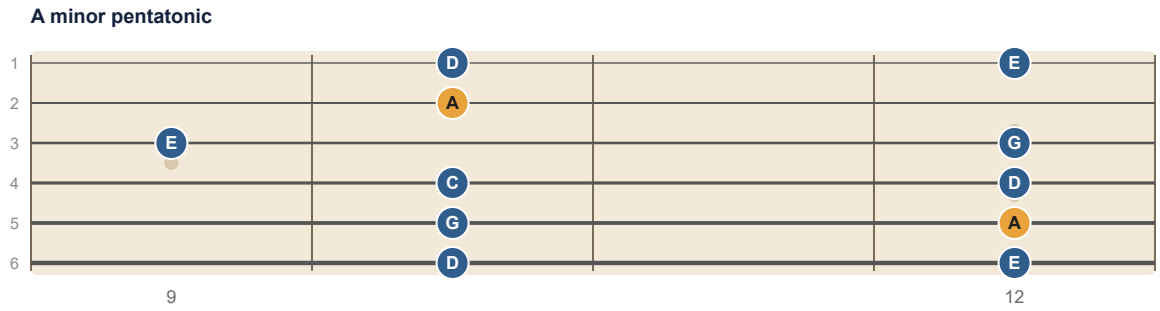
The next box starts a little higher. It overlaps Box 1 and uses the same notes in a new arrangement.



Box 2 of A minor pentatonic, around the E shape of C major. Notes: A C D E G in a new order.

BOX	FRETS (A MINOR PENTATONIC)	CAGED SHAPE (KEY OF C)	LANDMARK
Box 1	5 to 8	G shape	The box most players know
Box 2	7 to 10	E shape	The barre chord region
Box 3	9 to 13	D shape	Around the small D shape
Box 4	12 to 15	C shape	The chain repeats here
Box 5	14 to 17 (or 2 to 5)	A shape	Also available one octave lower

Box 3, Box 4 and Box 5 follow the same pattern: each box starts about where the last one ends, until Box 5 connects back to Box 1 an octave higher. Here is Box 3 so that you can see it with your own eyes.



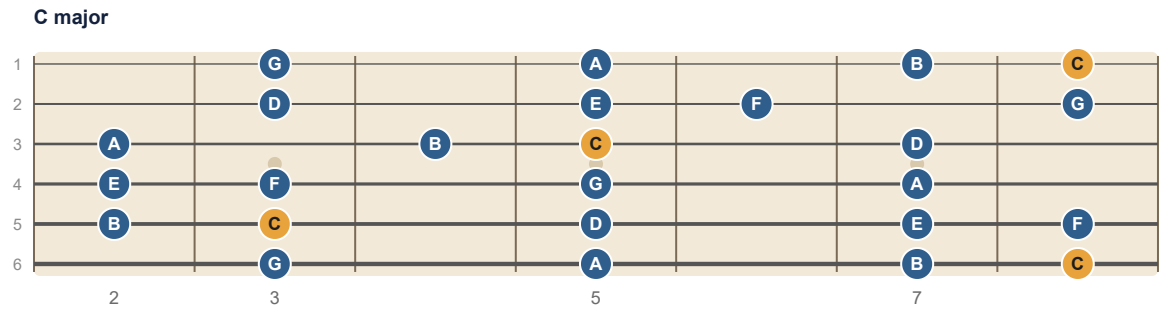
Box 3 of A minor pentatonic, frets 9 to 13, around the D shape of C major.

> **One scale, five faces**
>
> The five boxes are the same five notes. When you move from Box 1 to Box 2 you are not learning a new scale. You are looking at the same scale from the next room over. This is why players who know all five boxes can play the same lick in any part of the neck.

If you do one thing after reading this section, do this: play Box 1, then play the Am chord in the same area, then play Box 2 and the chord that belongs there. You are linking scales to chords to places.

Connecting two neighboring shapes: the overlap

Knowing five positions is good. Moving smoothly between them is the real skill. The good news is that neighbors always overlap. The last frets of one shape area are the first frets of the next. That shared area is your bridge.



C major scale across the A shape and G shape (frets 2 to 9). The shared area around fret 5 is the bridge.

Look at the diagram above. The A shape is on the left (frets 3 to 5) and the G shape is on the right (frets 5 to 8). The notes around fret 5 belong to both. If you play a line in the A position and want to continue up, you do not have to jump. You keep going, and the notes you land on are already part of the next shape.

Here is a practical way to practice this. Choose a pair of neighbors, for example the E shape and the D shape in the key of C. Play the scale in the E position, ascending, until you reach the strings that overlap. Then continue in the D position. Come back down the same way. Do it slowly, naming the notes. The goal is not speed; it is to feel that the join is invisible.

- **Overlap on a string:** the same two or three notes appear in both shapes. Slide your finger up the string to change position.
- **Shift with a common note:** land on a note that belongs to both positions, then shift your hand using that finger.
- **Mix:** play the low strings in one position and the high strings in the next. Many great solos live in these in-between spaces.

Naming shapes and keys

One source of confusion is that CAGED uses letters twice: once for the chord shape, and once for the key. Players say "the C shape in the key of G" and the whole thing becomes a puzzle. The trick is to always say the key first and the shape second.

- **Key** = the home note of the music you are playing. Example: the key of G.
- **Shape** = the outline your hand makes. Example: the D shape.
- Say it as: **"the D shape in G"**. This means a G chord (or a G scale) played with the D outline: root on string 4, fret 5, with the chord notes x x 5 7 8 7.

Check that last one: string 4, fret 5 is G. String 3, fret 7 is D. String 2, fret 8 is G. String 1, fret 7 is B. G, D, G, B: a G major chord. Naming also helps you talk to other musicians. If a friend says "play it with the E

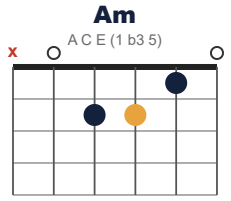
shape at the eighth fret", you know to look for C on string 6 and make the barre. If a teacher says "the open position", you know it is the first shape of the chain.

> **A simple habit**
> Whenever you play a chord or a scale, say two things out loud: the key and the shape. For example, "C major, A shape." It takes two seconds and builds the habit of seeing the neck as a map.

CAGED in chord changes: one progression, three places

Now let us use all of this in a real situation. Take one of the most common progressions in popular music: I - vi - IV - V. In the key of C, that is C, Am, F, G. Most players know it as a first-position progression. We are going to play it in three different places, and you will find that each place has its own character.

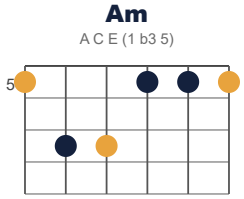
Place 1: the open position. C (open), Am (open), F (the E shape at fret 1, 1 3 3 2 1 1), G (open). This is the familiar one, ringing and full of open strings.



Open Am: x 0 2 2 1 0. Notes: A E A C E.

Place 2: the middle of the neck. The chords sit between frets 5 and 10, and every chord is a barre chord.

- C: the E shape at fret 8 (8 10 10 9 8 8).
- Am: the E shape at fret 5 (5 7 7 5 5 5).
- F: the A shape at fret 8 (x 8 10 10 10 8).
- G: the A shape at fret 10 (x 10 12 12 12 10).



Am with the E shape at fret 5: 5 7 7 5 5 5. Notes: A E A C E A.

Place 3: the high neck. The same chords, one octave up from the open position. C is the C shape at fret 12 (x 15 14 12 13 12), Am is the A shape at fret 12 (x 12 14 14 13 12), F is the E shape at fret 13 and G is the E shape at fret 15. This place has a bright, thin quality that works well when a bass or a low keyboard part is already covering the low end.

CHORD	PLACE 1 (OPEN)	PLACE 2 (MIDDLE)	PLACE 3 (HIGH)
C	C shape, open	E shape, fret 8	C shape, fret 12
Am	A shape (minor), open	E shape, fret 5	A shape (minor), fret 12
F	E shape, fret 1	A shape, fret 8	E shape, fret 13
G	G shape, open	A shape, fret 10	E shape, fret 15

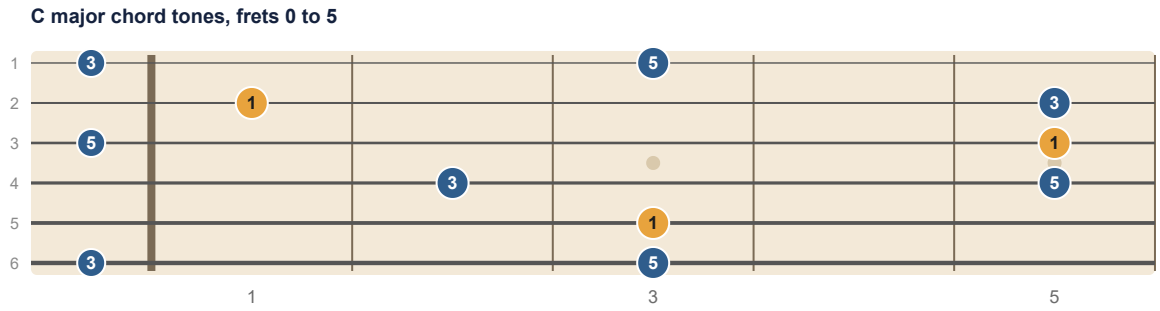
The chords have the same names in all three places. What changes is the voicing, meaning which notes are on top and which are on the bottom. Voicings change the feel of a song: lower is heavier, higher is lighter. A player who knows CAGED can choose the feel on purpose.

> **Try this**
>
> Play the progression in Place 1 for four bars, then Place 2 for four bars, then Place 3 for four bars. Do it with a slow, steady strum. Listen to how the same chords feel different. Most players find the middle place the most useful for playing along with a recording.

Arpeggios from the shapes

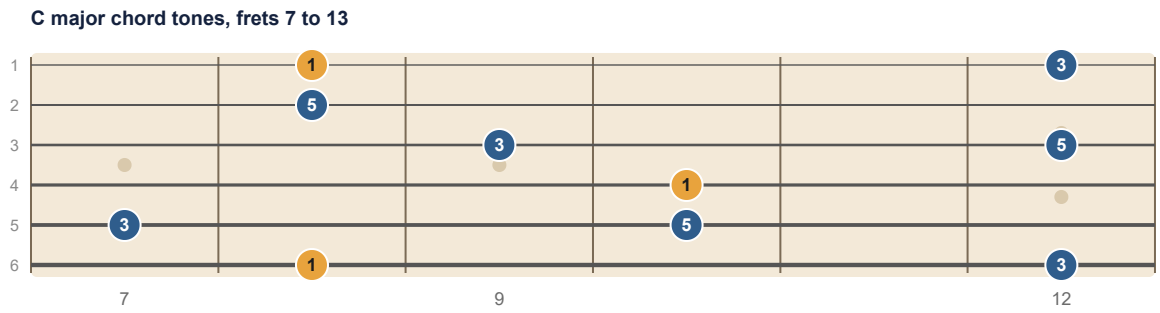
An **arpeggio** is a chord played one note at a time. In CAGED, every shape contains an arpeggio: just play the notes of the chord in order, from low to high. The arpeggio is the best way to hear the chord inside the scale.

In the key of C, the chord tones are C, E and G (the 1, 3 and 5). Here they are in the first region of the neck.



C, E and G near the nut. The root C is highlighted; the labels show 1, 3 and 5.

Look at the diagram and find the C shape and the A shape in it. They both contain the same three notes. The root is marked, so you can see where the chord is anchored.



C, E and G in the middle of the neck. The E shape and D shape chords live here.

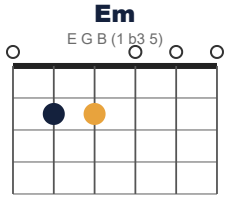
Playing the arpeggio is a powerful exercise. Take the E shape at fret 8. Play the notes slowly from the lowest string to the highest: C, G, C, E, G, C. Then go back down. You are playing a C major arpeggio, and every note is a chord tone. Over a C chord, it will sound smooth and clear. When the music moves to F, you move to the F chord in the nearest shape and play its tones (F, A and C). This is the secret of soloing that follows the changes: the arpeggio leads the ear from one chord to the next.

- **Step 1:** play the chord shape.
- **Step 2:** play the same notes one at a time, up and then down.
- **Step 3:** add one scale note between two chord tones.
- **Step 4:** move to the neighboring shape and repeat.

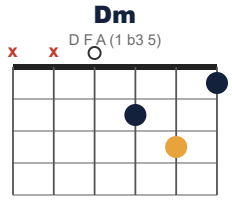
CAGED for minor chords and the minor pentatonic

Everything so far was major, and that is deliberate: the CAGED name comes from major chords, and the diagrams of the full system are drawn for major keys. But your songs are full of minor chords too. How do they fit in? There are two good answers.

Answer 1: minor versions of the shapes. Three of the five shapes have friendly open minor cousins: the E shape becomes Em, the A shape becomes Am, and the D shape becomes Dm. These shapes move up the neck exactly like the major ones, with the same root strings. For example, Am with the E-shape outline at fret 5 is 5 7 7 5 5 5, as you saw above. The C and G shapes have no comfortable open minor version, so most players do not use a minor C shape or a minor G shape on the open strings.



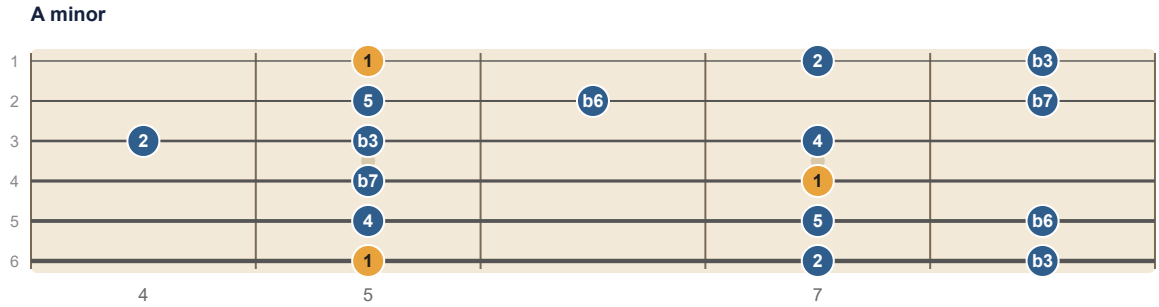
Open Em: 0 2 2 0 0 0. Notes: E B E G B E.



Open Dm: x x 0 2 3 1. Notes: D A D F.

Answer 2: the relative major. Every minor key shares its notes with a major key three semitones higher. A minor shares its notes with C major. So all the maps you learned for C major are also maps for A minor: the five scale positions, the five arpeggios and the five pentatonic boxes. You just start and end on A instead of C. This is the shortcut that most players use to play in minor keys.

Here is the A natural minor scale (A B C D E F G) around the root A on string 6, fret 5, with the labels showing the degree from A. Compare it with the C major picture of the G shape: the notes are the same, only the home base moves.



A natural minor around frets 4 to 9. Same notes as C major, with A as the root.

In the key of A minor, the shapes you see in C major are the shapes you use, but you focus on the notes A, C and E (the notes of the Am chord) instead of C, E and G. This is the whole trick, and it works for every minor key: find the relative major (three semitones up) and use its CAGED map.

> **Quick reference**
> Relative pairs: A minor and C major, E minor and G major, B minor and D major, F-sharp minor and A major, C-sharp minor and E major. To find a relative major, count up three semitones from the minor key.

Common mistakes

- **Memorizing shapes as drawings instead of notes.** A shape is a set of notes with a root. If you only memorize the picture, you cannot move it. Always know where the root is.
- **Forgetting the order.** The chain is C - A - G - E - D. If you reverse it, you will end up in the wrong place. Say the word out loud as you go up the neck.
- **Learning all five at once.** Most players find it easier to start with two neighbors, for example E and D in the key of C. Add one more shape each week.
- **Confusing the key and the shape.** Always say the key first: "the D shape in G".

- **Ignoring the repeat at fret 12.** The chain does not stop. After the D shape comes the C shape again, one octave higher.
- **Playing the shapes without hearing them.** Play each chord, then its scale, then its arpeggio. Always listen for the root.
- **Skipping the overlap.** If you learn the five positions as separate islands, you will be stuck inside one. Practice the bridges.

> **A word of encouragement**
>
> No one learns the whole system in a weekend. With practice, most players find that the first two or three shapes click within a couple of weeks, and the rest follow. It is normal to have to stop and think. That thinking is the learning.

Module 7 cheat sheet

SHAPE	OPEN CHORD	ROOT STRING	NOTES OF THE OPEN CHORD	NEXT IN THE CHAIN
C	x 3 2 0 1 0	5	C E G C E	A
A	x 0 2 2 2 0	5	A E A C# E	G
G	3 2 0 0 0 3	6	G B D G B G	E
E	0 2 2 1 0 0	6	E B E G# B E	D
D	x x 0 2 3 2	4	D A D F#	C (12 frets later)

KEY	SHAPES IN ORDER FROM THE NUT
C	C - A - G - E - D
G	G - E - D - C - A
D	D - C - A - G - E
A	A - G - E - D - C
E	E - D - C - A - G

- The chain is always C - A - G - E - D, then back to C at fret 12.
- To move a shape: find the root string, find the key name on that string, and lay the shape around it.
- Each shape has a scale position around it, an arpeggio inside it and a pentatonic box that matches it.
- Neighbors overlap: use the shared notes to move without jumping.
- Minor keys use the relative major map (A minor uses C major, E minor uses G major).
- Say the key first, then the shape: "the A shape in C".

Glossary

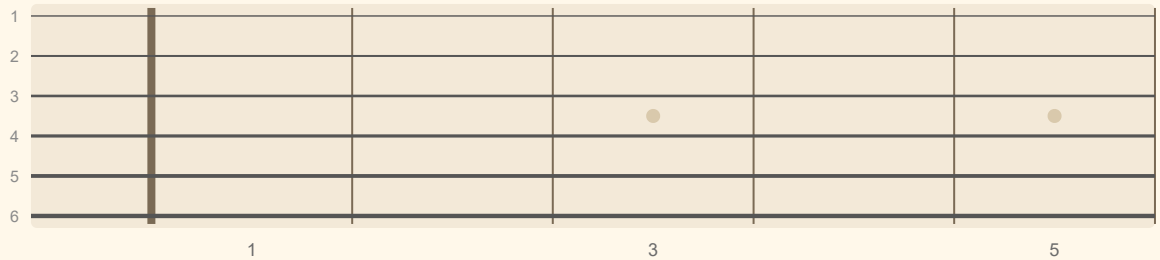
TERM	PLAIN-ENGLISH MEANING
CAGED	A system that names the five open chord shapes (C, A, G, E, D) that tile the whole neck.
Shape	The outline your fingers make for a chord, no matter where on the neck it is.
Root	The note that names the chord or the scale. It is the anchor of each shape.
Capo idea	Moving a shape up the neck is like putting on a capo: all the notes go up by the same number of frets.
Barre	Pressing one finger across several strings so it works like a movable nut.
Position	A group of frets where your hand stays, with one scale pattern.
Pentatonic box	A five-note scale pattern, with two notes per string, in one position.
Arpeggio	The notes of a chord played one at a time.
Overlap	The area where two neighboring shapes share notes. It works as a bridge.
Voicing	The way the notes of a chord are arranged, from low to high.
Relative major/minor	A major and a minor key that share the same seven notes.

On the fretboard: exercises

EXERCISE 31 — 10 MIN

Name the five puzzle pieces

1. Play the open C, A, G, E and D chords one after the other, slowly and cleanly.
2. For each chord, say out loud its main root string (C: 5, A: 5, G: 6, E: 6, D: 4).
3. Look at the blank neck below. On paper or in your head, mark the root of each open shape.
4. Close your eyes and play each shape again, naming the root string before you press a single finger.



Blank neck, frets 0 to 5. Mark the roots of the five open shapes.

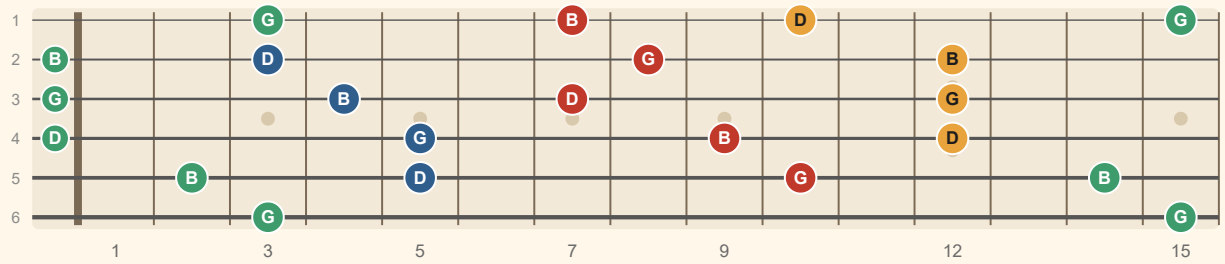
☐ Self-check: you can say the five root strings (5, 5, 6, 6, 4) without looking at the module, and you can play each open chord with every string ringing clearly.

EXERCISE 32 12 MIN

Five Gs on the neck

1. Look at the diagram below. It shows the five CAGED shapes in the key of G.
2. Play the G chord in the first shape (the open G).
3. Then play G in the next shape (the E shape at fret 3), then the D shape (around fret 5), then the C shape (around fret 7), then the A shape (the root at fret 10).
4. Say the shape name each time. After the A shape, play the G shape again at fret 12.

Key of G: the five CAGED shapes (colors: C red, A orange, G green, E blue, D purple)



The five CAGED shapes in the key of G: G - E - D - C - A.

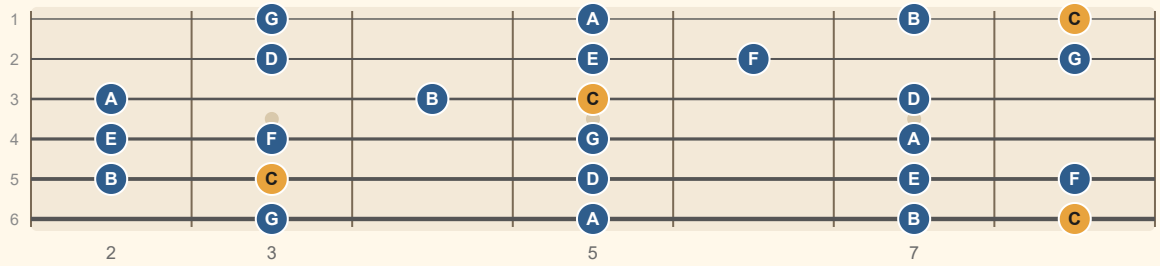
☐ Self-check: you can play five different G chords in order up the neck, name the shape of each one, and return to the open shape at fret 12.

EXERCISE 33 10 MIN

Scale and bridge

1. Play the C major scale in the A position (frets 2 to 6), up and down, saying the note names.
2. Do the same in the G position (frets 5 to 9).
3. Now play the scale in the A position, and continue it into the G position without stopping, using the notes around fret 5 as the bridge.
4. Go back down. Repeat with the E position and the D position.

C major



C major across the A and G positions.

☐ Self-check: you can move from one position to the next without a pause, and you can name the note you land on at the bridge.

EXERCISE 34 15 MIN

One progression, three places

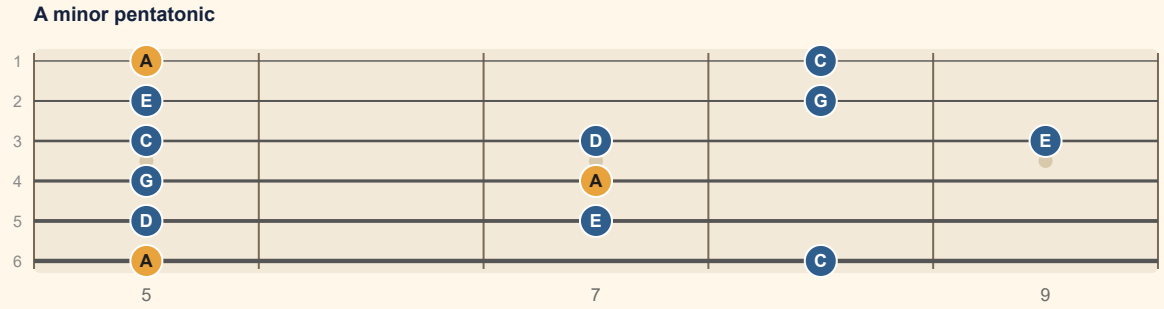
1. Choose the progression C - Am - F - G (I - vi - IV - V in C).
2. Play it in Place 1 (open chords) for four bars.
3. Play it in Place 2: C (E shape, fret 8), Am (E shape, fret 5), F (A shape, fret 8), G (A shape, fret 10).
4. Play it in Place 3: C (C shape, fret 12), Am (A shape, fret 12), F (E shape, fret 13), G (E shape, fret 15).
5. Repeat the whole idea in the key of G (G - Em - C - D), starting from the open shapes.

☐ Self-check: you can play the progression in three places, you can name the shape of each chord, and you can hear that the chords are the same but the feel is not.

EXERCISE 35 15 MIN

Five boxes, five shapes

1. Play Box 1 of A minor pentatonic (frets 5 to 8), up and down, saying the notes.
2. Play Box 2 (frets 7 to 10) and then Box 3 (frets 9 to 13).
3. For each box, name the CAGED shape it matches in the key of C: G shape, E shape and D shape.
4. Play an Am chord, then improvise a short phrase in the matching box. Land on the root A to end the phrase.
5. Repeat with Box 4 (frets 12 to 15) and Box 5 (frets 14 to 17).



A minor pentatonic, frets 5 to 10, covering Box 1 and Box 2.

☐ Self-check: you can play all five boxes, you can say which CAGED shape each one matches (G, E, D, C, A), and you can end a phrase on the root A in each.

Next step

You now have a map of the whole neck: five shapes, one order, and the knowledge of where the scales and arpeggios live inside them. The next module puts the map to work. In Module 8 you will see why chord progressions sound the way they do, and you will use CAGED to play them anywhere you like. Before you move on, go back to the cheat sheet, pick one key, and play the five shapes in order, up and down the neck. Keep it slow and let it be a little boring. That calm repetition is what turns a map on paper into a map in your hands.

MODULE 8

Progressions: why songs work

- Every chord in a key has a job: home, away, tension, or color.
- Roman numerals (I, IV, V, vi) describe jobs, so one pattern fits all 12 keys.
- A handful of progressions power a huge share of the songs you know.
- Once you can read the numbers, you can analyze, transpose and write songs.

By the end: Name the job of every chord in a key, recognize the 8 most common progressions, analyze any song in 4 steps, transpose in seconds, and write your first progression.

> *You already know chords and you already know keys. This module connects them. A song is not a random pile of chords: it is a small story told in a few steps, home, away, tension, and back home. Once you hear that story, learning songs gets faster and writing them stops feeling like magic.*

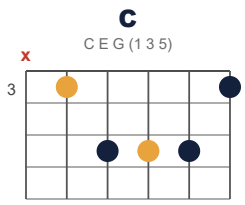
Chord function: every chord has a job

Take the key of C major. Its seven chords come from the C major scale, one chord built on each note: C, Dm, Em, F, G, Am and Bdim. Play them one after another and you will notice they do not feel equal. Some feel settled. Some feel like they are leaning forward. That feeling is called **function**, and it is the single most useful idea in this whole module.

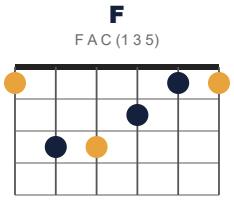
Here are the jobs, in plain words:

- **Home (I):** C in the key of C. This is where the song feels finished. Songs usually start here and end here.
- **Away (IV):** F in the key of C. It lifts you off home in a calm, open way. It does not demand to go anywhere in particular.
- **Tension (V):** G in the key of C. It pulls hard back toward home. Add a seventh (G7 = G B D F) and the pull gets even stronger.
- **Relative minor (vi):** Am in the key of C. It shares all its notes with C major but sounds sadder, like home with the lights turned down.
- **Predominant (ii):** Dm in the key of C. It sets up the V chord, the way a deep breath comes before a long sentence.

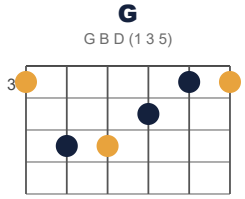
The remaining two chords, iii (Em) and vii° (Bdim), appear less often. Em can stand in for C because it shares two notes with it (E and G). Bdim is a tight, unstable chord that leans toward C. For most songs you will rely on I, ii, IV, V and vi.



C major (I), home. Root on string 5, fret 3. Notes C E G.



F major (IV), away. Barre at fret 1, root on string 6. Notes F A C.



G major (V), tension. Barre at fret 3, root on string 6. Notes G B D.

> **Try it by ear**
>
> Strum C, then F, then G, then C again, leaving a couple of beats on each. Close your eyes on the last C. Most players feel a small 'ahh, I am back' moment. That feeling is the whole point of this module.

Roman numerals, explained simply

Roman numerals are just a way to say which note of the scale a chord is built on. I means the chord built on the 1st note, IV on the 4th, V on the 5th, and so on. You do not need to read music for this. You only need to count the scale.

Capital numerals mean a major chord (I, IV, V). Lowercase numerals mean a minor chord (ii, iii, vi). The small circle (vii°) means a diminished chord. In a major key the pattern is always the same: **I ii iii IV V vi vii°**. Here it is in six keys. Every note and chord below follows from the major scale of that key.

KEY	I	II	III	IV	V	VI	VII°
C	C	Dm	Em	F	G	Am	Bdim
G	G	Am	Bm	C	D	Em	F#dim
D	D	Em	F#m	G	A	Bm	C#dim

KEY	I	II	III	IV	V	VI	VII°
A	A	Bm	C#m	D	E	F#m	G#dim
E	E	F#m	G#m	A	B	C#m	D#dim
F	F	Gm	Am	Bb	C	Dm	Edim

Why bother? Because the numbers travel. When someone says 'it is a I-IV-V', you instantly know the chords in any key. That is far easier than memorizing every song chord by chord.

The same four chords in all 12 keys

The most popular group of chords in pop is I, IV, V and vi. Here it is in every key. Notice that the pattern of major and minor never changes, only the starting note moves.

KEY	I	IV	V	VI
C	C	F	G	Am
Db	Db	Gb	Ab	Bbm
D	D	G	A	Bm
Eb	Eb	Ab	Bb	Cm
E	E	A	B	C#m
F	F	Bb	C	Dm
F#	F#	B	C#	D#m
G	G	C	D	Em
Ab	Ab	Db	Eb	Fm
A	A	D	E	F#m
Bb	Bb	Eb	F	Gm
B	B	E	F#	G#m

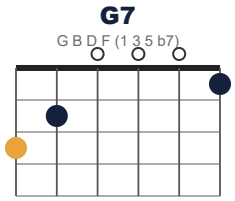
> **A shortcut for the guitar**
>
> Because of the way chord shapes move, the IV chord is always 5 frets above the I chord (same string, same shape) and the V chord is 7 frets above. Example: a barre chord with its root on fret 3 of string 6 is G. Five frets up (fret 8) is C, and seven frets up (fret 10) is D. That is I, IV and V of G.

The big picture: tension and release

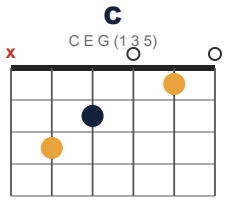
Music works like a conversation between pulling and settling. A chord that creates tension makes you want something to happen next. A chord that releases tension gives you that something. Good progressions move through both on purpose.

A simple map: **home** (I and vi) is rest. **Away** (IV and ii) is movement. **Tension** (V and vii°) is the strongest pull. The classic route goes home, away, tension, home. The V chord is the loudest question and the I chord is the answer.

Why does V pull so hard? In C major, the V chord is G (G B D). Its middle note B is the 7th note of the C scale, only one fret below C. Your ear wants that B to step up to C. When the next chord is C, the wish comes true. With G7 (G B D F) there is also the note F, which wants to step down to E, the third of C. Two small steps resolve at once, and that is why a G7 to C move sounds so decisive.



G7, the classic tension chord in C. Notes G B D F. The B wants to rise to C, the F wants to fall to E.



C major, the release. Open position: notes C E G.

The 8 most common progressions

A progression is a short loop of chords, usually 4 bars. These eight appear in a huge number of songs across rock, pop, country, folk and blues. For each one you get the numbers, the chords in C and G, and the feel. Learn the feel first, the chords second.

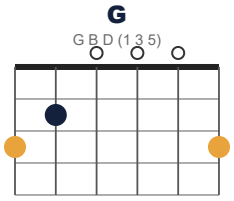
#	NUMBERS	IN C	IN G	THE FEEL
1	I - V - vi - IV	C G Am F	G D Em C	Bright, big, singable. The anthem loop.
2	I - vi - IV - V	C Am F G	G Em C D	Warm and nostalgic, with a gentle build.
3	ii - V - I	Dm7 G7 Cmaj7	Am7 D7 Gmaj7	Smooth and sophisticated. The jazz basic.

#	NUMBERS	IN C	IN G	THE FEEL
4	I - IV - V	C F G	G C D	Simple, strong, bluesy-rock. Three chords, endless songs.
5	vi - IV - I - V	Am F C G	Em C G D	Same chords as #1 but starting on the sad one: more serious.
6	12-bar blues	C7 F7 G7	G7 C7 D7	Rooted, swaggering, built on three dominant 7th chords.
7	I - bVII - IV	C Bb F	G F C	Rock and open road. A little rebellious.
8	i - bVI - bIII - bVII	Cm Ab Eb Bb	Gm Eb Bb F	Dark, epic, cinematic. A minor-key anthem.

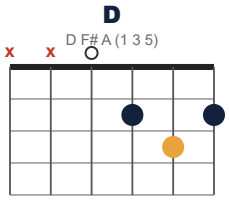
Check any of these against the scale. Progression 1 in G: G is I, D is V (the 5th note of G major is D), Em is vi (the 6th note is E), C is IV (the 4th note). Progression 7 in C: Bb is not in the C major scale, which is why it is written bVII, the 7th note lowered by one fret. More on that soon.

1. I - V - vi - IV

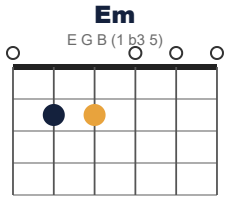
Four chords that sound like a chorus. C to G feels like a lift, G to Am a small dip into emotion, Am to F a soft fall, and F leads back to C. Because it loops cleanly, you can repeat it all night. Play it in G with an open-chord shape set:



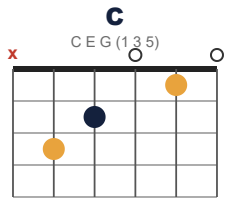
G major, I of G.



D major, V of G.



E minor, vi of G.



C major, IV of G.

2. I - vi - IV - V

An old favorite. It is the same four chords as number 1, reordered so the tension chord comes last. Ending on V means the loop begs to start again, which is why it sounds so classic in slow dances and early rock and roll.

3. ii - V - I

The most important small move in jazz. In C: Dm7 (D F A C), G7 (G B D F), Cmaj7 (C E G B). Watch what the notes do: C in Dm7 slides down to B in G7, and F in G7 slides down to E in Cmaj7. Each chord changes by only a step or two, so it flows. In G the chords are Am7 (A C E G), D7 (D F# A C) and Gmaj7 (G B D F#).

4. I - IV - V

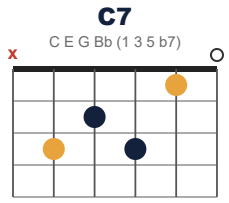
The great three-chord engine. With only home, away and tension you can play thousands of songs. It is also the skeleton of the blues below.

5. vi - IV - I - V

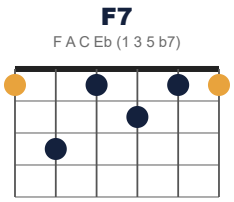
Take progression 1 and begin on vi. Same four chords, but starting on the minor chord changes the mood completely: the loop now feels like it is rising out of sadness rather than starting happy. This is a good example of the idea that *order matters, not just the chords*.

6. The 12-bar blues

Three chords, usually all dominant 7ths, spread over 12 bars. In C: C7 (C E G Bb), F7 (F A C Eb) and G7 (G B D F). In G: G7, C7, D7. We look at the exact bar layout in the section on forms.



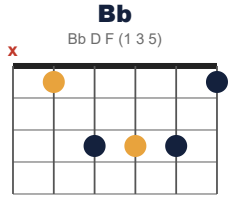
C7 in open position. Notes C E G Bb. The blues I chord in C.



F7, E-shape barre at fret 1. Notes F A C Eb. The blues IV chord in C.

7. I - bVII - IV

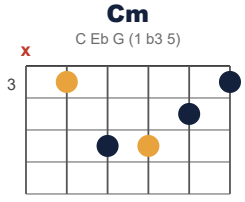
A rock favorite. The bVII chord (Bb in C) is borrowed from outside the key and has a bold, open sound. Going from Bb to F and back to C feels like stepping off a ledge and landing safely.



Bb major, A-shape barre at fret 1. Notes Bb D F. The bVII of C.

8. i - bVI - bIII - bVII

A minor-key loop. Lowercase i means the home chord is minor. In C minor the chords are Cm (C Eb G), Ab (Ab C Eb), Eb (Eb G Bb) and Bb (Bb D F). All four chords come from the C natural minor scale (C D Eb F G Ab Bb), so this is not 'borrowed', it is simply the minor key. In A minor (a very common guitar key) the same loop is Am, F, C, G, which is easy to play on open strings.



C minor, A-shape barre at fret 3. Notes C Eb G.

Cadences: how phrases end

A cadence is the two-chord ending of a musical phrase, like punctuation in a sentence. There are four you will hear all the time. All examples are in C.

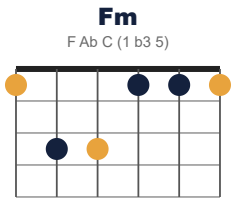
- **Authentic cadence (V to I):** G to C. The full stop. Strong and final. This is the most conclusive ending there is.
- **Plagal cadence (IV to I):** F to C. The 'amen' ending. Softer than the authentic one, like a warm goodbye.
- **Half cadence (ending on V):** Often something like C to G, and stop. It sounds like a question mark or a comma: the phrase pauses but is not finished.
- **Deceptive cadence (V to vi):** G to Am. Your ear expects C, but you land on Am instead, a pleasant surprise. It extends the phrase when a song does not want to end yet.

> **Listen for them**
>
> Next time a song's verse ends, try to hear which of the four it uses. A half cadence usually means more is coming; an authentic cadence usually means the section is done.

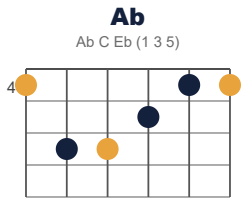
Borrowed chords: a little color from next door

So far every chord has come from the major scale of the key. A borrowed chord comes from a close relative: the parallel minor, which shares the same home note. For C major the parallel minor is C minor (C D Eb F G Ab Bb). Borrow a chord from it and you add a dark splash of color without leaving the key.

- **bVII (Bb in the key of C):** From C minor and also from the Mixolydian sound (C D E F G A Bb). Open, confident, rock.
- **bVI (Ab in the key of C):** From C minor. Dramatic and wistful, like a movie turning point.
- **iv (Fm in the key of C):** The minor version of IV. It is the 'sad IV' and often leads smoothly back to C. Fm contains Ab, which is the note that gives it that tearful feel.



F minor (the borrowed iv in C), E-shape barre at fret 1. Notes F Ab C.



Ab major (the bVI in C), E-shape barre at fret 4. Notes Ab C Eb.

A good test of a borrowed chord: play C, then F, then Fm, then C. The F to Fm change swaps A for Ab and the whole mood shifts from bright to bittersweet in one fret.

How to analyze any song in 4 steps

Analysis sounds fancy but it is a recipe. You will use it for songs you want to learn, songs you want to steal ideas from, and songs you want to transpose.

1. **Find the key.** The chord a song begins and ends on is usually home. If a song starts on G and ends on G, try G. Check that the other chords fit the G major scale (G A B C D E F#).
2. **Find the chords.** Write down each chord in order, from a chord chart or by ear. Do not worry about rhythm yet.
3. **Write the numbers.** Replace each chord with its Roman numeral in your key. Use the table earlier in this module if you need it.
4. **Spot the pattern.** Compare your numbers with the eight common progressions. Most songs match one, or a small variation.

Worked example. A song uses the chords D, Bm, G and A, looping, and it begins and ends on D. Step 1: the key is D. The D major scale is D E F# G A B C#, and all four chords fit. Step 2: D, Bm, G, A. Step 3: D is I, Bm is vi (the 6th note is B), G is IV, A is V. Step 4: I - vi - IV - V, progression 2. Done in under a minute.

STEP	WHAT YOU DO	EXAMPLE
1	Find the key	Starts and ends on D, so the key is D
2	Find the chords	D Bm G A
3	Write the numbers	I vi IV V
4	Spot the pattern	Progression 2, the classic loop

Transposing any song in seconds

To transpose means to move a song to a new key. With letter names it is slow and error-prone. With numbers it takes seconds, because the numbers never change.

The number method

1. Write the song as numbers (the 4-step recipe above).
2. Pick the new key and list its seven chords (use the table).
3. Swap each number for the new chord.

Example: the song is G, D, Em, C (I - V - vi - IV in G). You want it in E. The chords of E major are E, F#m, G#m, A, B, C#m, D#dim. So I is E, V is B, vi is C#m and IV is A. The song in E is E, B, C#m, A. Every relationship is the same, only the pitch is different.

The capo method

A capo clamps across the neck and raises every open string by one semitone (one fret) per fret. If you play G-shape chords with the capo on fret 2, the chords sound one whole step higher, as A. So you keep the easy shapes while the key changes.

SHAPES YOU PLAY	CAPO FRET	KEY YOU HEAR
C G Am F (key of C)	2	D
C G Am F (key of C)	5	F
C G Am F (key of C)	7	G
G D Em C (key of G)	2	A
G D Em C (key of G)	4	B
G D Em C (key of G)	5	C

Check one: G shapes with the capo on fret 2 become A, D shapes become E, Em shapes become F#m and C shapes become D. The progression G - D - Em - C sounds like A - E - F#m - D. To find the capo fret you need, count semitones from your shape key to the key you want. G up to B: G to G# is 1, A is 2, A# is 3, B is 4, so capo 4.

> **Tip for singers**
> If a song sits too high or too low for your voice, move it by a few semitones with a capo or new shapes. A singer-friendly key is more useful than a 'correct' key.

Song forms: 8 bars and 12 bars

A bar is one group of beats, usually 4. Most progressions in this module are 4 bars long and repeat. Two longer shapes show up all the time.

The 8-bar form

Two copies of a 4-bar progression make an 8-bar section. The second copy often ends differently, creating a question-and-answer feel. Example in C with I - V - vi - IV: bars 1 to 4: C G Am F (question, ends away from home). Bars 5 to 8: C G F C (answer, ends at home).

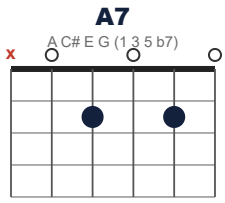
The 12-bar blues form

The blues takes three chords and sets them in a fixed pattern of 12 bars. Here it is in C, using dominant 7th chords:

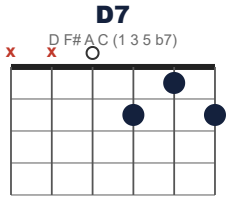
BARS 1-4	BARS 5-8	BARS 9-12
C7 | C7 | C7 | C7	F7 | F7 | C7 | C7	G7 | F7 | C7 | G7

In numbers: I I I I, IV IV I I, V IV I V. The last bar (V) is called the turnaround and sets up the next lap. The same form in other keys:

KEY	I	IV	V
C	C7	F7	G7
G	G7	C7	D7
A	A7	D7	E7
E	E7	A7	B7



A7 open. Notes A E G C#. I of the blues in A.



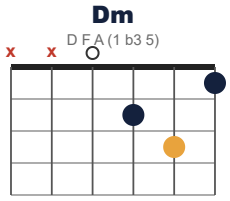
D7 open. Notes D A C F#. IV of the blues in A.

Writing your first progression

You do not need an idea, a lyric or a genre. You need a key and a plan. Use this 4-step template.

1. **Start at home.** Pick a key and open with I. Example: G.
2. **Move away.** Choose IV or vi (or ii) for the second and third chords. Example: Em, then C.
3. **Build tension.** Choose V for the fourth chord. Example: D.
4. **Decide the ending.** Return to I for a full stop, or loop back to the start, or land on vi for a surprise.

That gives G - Em - C - D, which is progression 2. Try changing one thing: swap Em for Bm or end on Em instead of D. Change only one chord at a time so you can hear what each choice does. Hum a tune over the top, or just strum a rhythm you like, and you have the seed of a song.



D minor open. Notes D F A. ii of C, a good predominant chord.

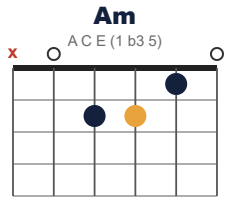
Common mistakes

- **Guessing the key from the first chord only.** Songs sometimes start on IV or vi. Check the final chord and the chord that feels like 'home'.
- **Mixing up capital and lowercase.** vi is minor, VI is major. They are different chords.
- **Forgetting that numbers follow the key, not the song.** If you change the key, change the chords but keep the numbers.
- **Overusing the V chord.** If every phrase ends with tension, nothing is ever resolved. Vary your endings.
- **Confusing the capo fret with the key.** The key you hear is the shape key plus the capo fret in semitones.
- **Adding too many borrowed chords.** One bVII or one iv is color. Four of them is a different key.

Module cheat sheet

IDEA	WHAT TO REMEMBER
Key chords	I ii iii IV V vi vii° (major, minor, minor, major, major, minor, diminished)
Jobs	I home, IV away, V tension, vi sad relative, ii setup
Strongest pull	V (or V7) to I
Cadences	Authentic V-I, plagal IV-I, half ends on V, deceptive V-vi
Borrowed chords	bVII, bVI, iv from the parallel minor
Top loops	I-V-vi-IV, I-vi-IV-V, ii-V-I, I-IV-V, vi-IV-I-V

IDEA	WHAT TO REMEMBER
Blues	I I I I, IV IV I I, V IV I V
Analyze	Key, chords, numbers, pattern
Transpose	Keep numbers, change the key table, or use a capo (fret = semitones)



Am open: the vi of C. Notes A C E.

Glossary

- **Progression:** a series of chords played in order.
- **Function:** the job a chord does in a key: home, away, tension.
- **Roman numeral:** a number that says which scale note a chord is built on.
- **Tonic (I):** the home chord.
- **Dominant (V):** the chord that pulls strongest toward home.
- **Subdominant (IV):** the 'away' chord.
- **Relative minor (vi):** the minor chord sharing all notes with the major key.
- **Cadence:** a two-chord ending to a phrase.
- **Borrowed chord:** a chord taken from the parallel minor key.
- **Turnaround:** the last bar or two that sets up the next lap.
- **Capo:** a clamp that raises the pitch of all open strings.
- **Bar:** one group of beats, usually 4 in this module.

On the fretboard: exercises

EXERCISE 36 — 8 MIN

Numbers to chords

1. Write the seven chords of D major as I ii iii IV V vi vii°.
2. Write the I, IV, V and vi chords for E and for A.
3. Check each one against the table earlier in this module.

☐ Self-check: D major is D, Em, F#m, G, A, Bm, C#dim. E gives E, A, B, C#m. A gives A, D, E, F#m. If any differ, recount the scale notes.

EXERCISE 37 10 MIN

Analyze three songs

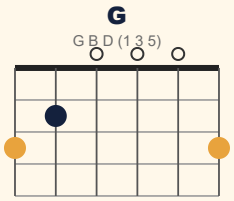
1. Take the loops Em, C, G, D and Am, F, C, G and D, A, Bm, G.
2. For each one, find the key, write the numbers and name the pattern.
3. Say which chord feels like home in each case.

☐ Self-check: Em C G D is in G (vi IV I V). Am F C G is in C (vi IV I V). D A Bm G is in D (I V vi IV). Two of the three use the same numbers, in different keys.

EXERCISE 38 10 MIN

Transpose with numbers and a capo

1. Write G, D, Em, C as numbers.
2. Transpose the loop to E using the numbers.
3. Using G-shape chords, find the capo fret that sounds in the key of B.



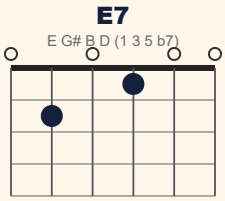
G open shape: the shape stays the same while the capo changes the key.

☐ Self-check: the numbers are I V vi IV. In E the loop is E, B, C#m, A. For B you need capo 4, because G up 4 semitones is B.

EXERCISE 39 12 MIN

Play a 12-bar blues in A

1. Use A7, D7 and E7.
2. Play the layout I I I I, IV IV I I, V IV I V using four beats per bar.
3. Count out loud so you stay on form. Repeat three laps without stopping.



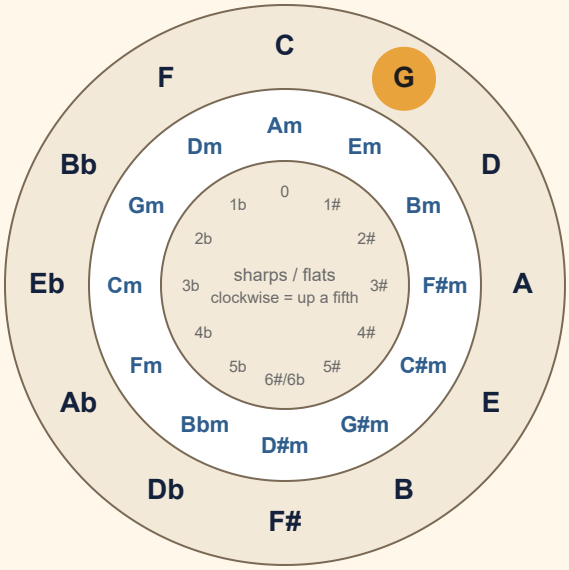
E7 open: notes E B D G#. The V chord of A.

☐ Self-check: the bars are A7 A7 A7 A7, D7 D7 A7 A7, E7 D7 A7 E7. You should feel the change to D7 at bar 5 and the pull of E7 at bar 9 before returning to A7.

EXERCISE 40 12 MIN

Write your own progression

1. Pick a key and write its seven chords.
2. Use the 4-step template: home, away, tension, ending.
3. Play it for two minutes and write the chords and numbers on paper.
4. Change exactly one chord and note how the mood shifts.



Neighbors of G on the circle of fifths (C and D) are the IV and V chords.

☐ Self-check: your first chord is I, your last chord is I, vi or V, and every chord is in the key table. If one is not, either rewrite it or label it a borrowed chord (bVII, bVI or iv).

Next step

You can now hear what chords are doing and describe it with numbers. Next, Module 9 shows how the circle of fifths organizes all 12 keys on one page and how to change key in the middle of a song. Before you move on, pick one song you love, run it through the 4-step analysis and play it in two different keys.

MODULE 9

The circle of fifths and modulation

- One picture that holds all 12 keys, how many sharps or flats each one has, and which keys are close relatives.
- Any key plus its two neighbors on the circle gives you the key's chords, with the relative minors inside.
- Chord progressions love to move down a fifth: that is the pull behind ii-V-I and behind a lot of songs you already know.
- Changing key (modulation) stops being mysterious when you see which chords two keys share, or when you borrow a dominant.

By the end: By the end of this module you will be able to read the circle, build the six main chords of any key in seconds, spot a key change coming, and move a song to a new key without writing anything down.

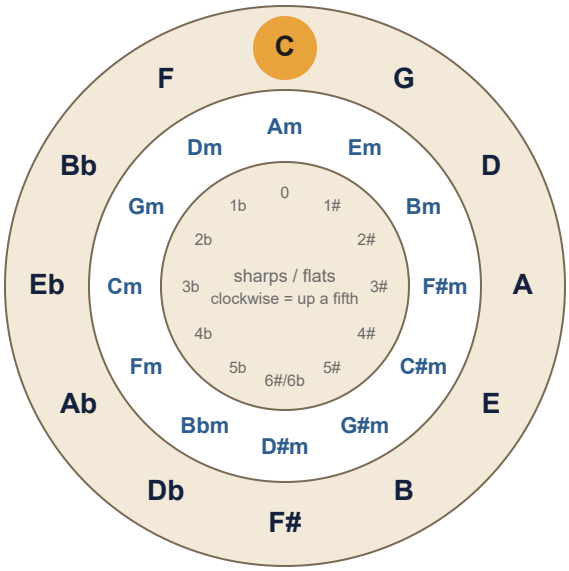
> *You already know that a key is a home base and that its chords come from its scale. The problem is that there are twelve keys, and nobody wants to rebuild the whole map each time. The circle of fifths is the shortcut: a single round diagram that organizes all twelve keys by how closely they are related. Once you can read it, you stop memorizing and start seeing.*

This module sits at the end of Part 3, Connect, because it ties together everything before it: the major scale from Module 3, the chords from Module 5, the sevenths from Module 6, and the progressions from Module 8. Take your time with the first section. Everything else is built on it.

1. Building the circle, one fifth at a time

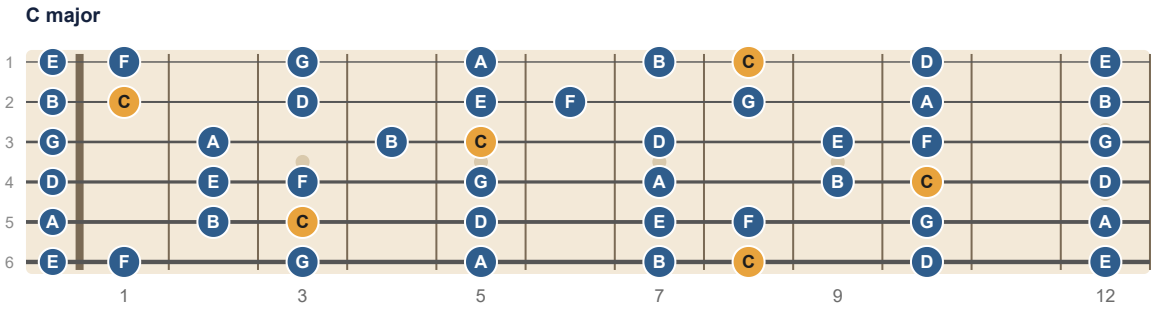
A fifth is the interval from 1 to 5 of a scale (Module 2). From C, the fifth is G. From G, the fifth is D. If you keep stacking fifths upward you walk through every note name and finally land back where you started. That walk, drawn as a circle, is the circle of fifths.

Here is the rule that makes the whole thing work. **Each step up a fifth adds one sharp to the key signature.** C major has no sharps or flats. G major, one fifth up, has one sharp (F#). D major has two (F# and C#). And so on. Going the other direction, **each step down a fifth adds one flat**. F major, a fifth below C, has one flat (Bb). Bb major has two (Bb and Eb).

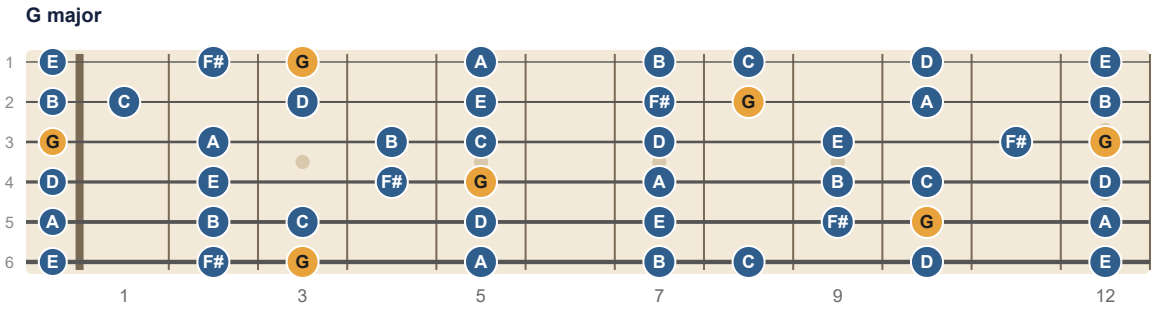


The circle of fifths with C at the top. Clockwise is up a fifth (one more sharp each step). Counterclockwise is down a fifth (one more flat each step).

Why does adding a sharp work out so neatly? Look at G major: G A B C D E F#. Compared with C major, only one note changed. The 7th degree moved up a half step from F to F#. When you build a new major scale a fifth higher, the new scale keeps six notes from the old one and replaces exactly one. That is why neighboring keys sound so close to each other: they share six of seven notes.



C major: no sharps, no flats. Every note is a natural note.



G major, one fifth up: the same notes as C major except F becomes F#.

The order never changes

Sharps are added in this order: F# C# G# D# A# E# B#. Flats are added in the reverse order: Bb Eb Ab Db Gb Cb Fb. In D major (two sharps) the sharps are always F# and C#. In Eb major (three flats) the flats are always Bb, Eb and Ab.

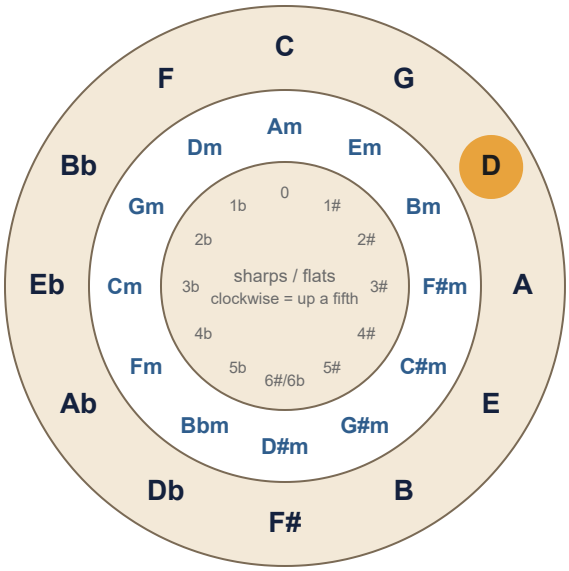
2. How to read the circle

Think of the circle like a clock. C sits at 12 o'clock. Every step clockwise is up a fifth and adds a sharp. Every step counterclockwise is down a fifth and adds a flat. At the bottom the two directions meet: the key with six sharps (F#) and the key with six flats (Gb) are the same key on the guitar, because F# and Gb are the same fret. Db also has a twin, C#, but it is easier to think of it with flats.

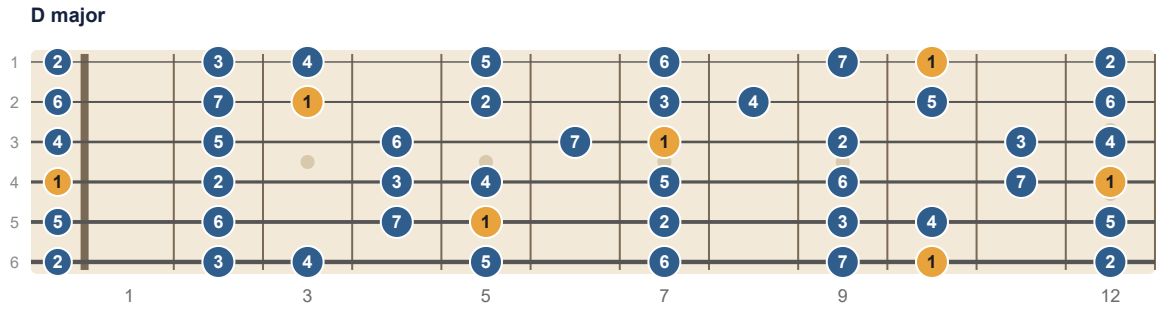
KEY (MAJOR)	SHARPS OR FLATS	WHICH NOTES ARE ALTERED
C	none	none
G	1 sharp	F#
D	2 sharps	F#, C#
A	3 sharps	F#, C#, G#
E	4 sharps	F#, C#, G#, D#
B	5 sharps	F#, C#, G#, D#, A#
F# / Gb	6 sharps / 6 flats	F#, C#, G#, D#, A#, E# / Bb, Eb, Ab, Db, Gb, Cb
Db	5 flats	Bb, Eb, Ab, Db, Gb
Ab	4 flats	Bb, Eb, Ab, Db
Eb	3 flats	Bb, Eb, Ab
Bb	2 flats	Bb, Eb
F	1 flat	Bb

Read the table from the top: C, then G, D, A, E, B, F#/Gb, Db, Ab, Eb, Bb, F, and back to C. That is the circle, in order. If you only memorize one thing from this module, memorize that sequence.

On a guitar you do not need the key signature to play, because you are looking at the fretboard, not at a page. The count of sharps and flats still matters for two reasons: it tells you at a glance how related two keys are (fewer steps apart means more notes in common), and it tells you which notes are outside the key when you improvise.



D major: two steps clockwise from C, so two sharps (F# and C#). Its neighbors are G (one step back) and A (one step forward).

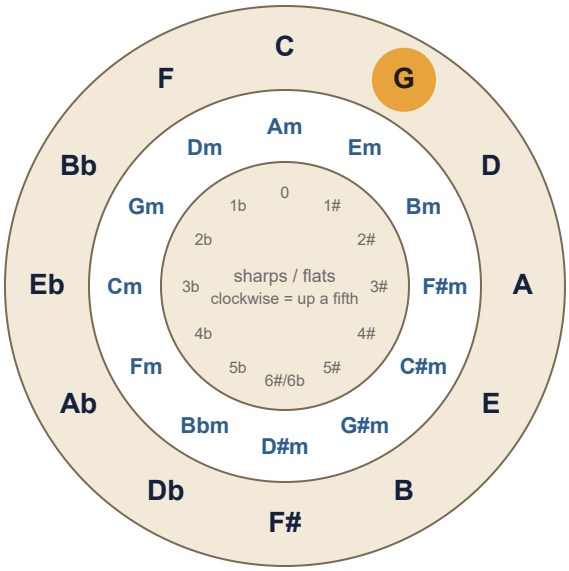


D major with scale degrees. The 3rd (F#) and the 7th (C#) are the two sharps.

> **Quick check**
>
> A song uses Bb and no sharps at all. Which key is it most likely in? Count counterclockwise from C: F has exactly one flat (Bb). So F major (or its relative, D minor) is the safest guess.

3. Neighbors are related keys

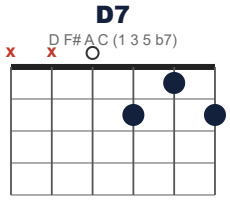
Pick any key on the circle and look at the keys on either side of it. Those two neighbors are its closest relatives. Take C as the example. To its right is G, to its left is F. In Module 8 you saw that a key's three main chords are I, IV and V. Here they are on the circle: **the key in the middle is I, the neighbor on the left (counterclockwise) is IV, and the neighbor on the right (clockwise) is V.**



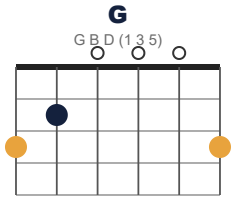
Key of G: its left neighbor C is the IV chord, its right neighbor D is the V chord. So G, C and D are the three main chords.

KEY (I)	IV (COUNTERCLOCKWISE)	V (CLOCKWISE)
C	F	G
G	C	D
D	G	A
A	D	E
E	A	B
F	Bb	C
Bb	Eb	F

This is a very practical thing to know on the neck. Open-position players often jump between G, C and D, or between A, D and E. Those are three neighbors in a row on the circle. When a friend says 'it is a three-chord song in A', you can say 'A, D, E' without thinking, because those three sit side by side.



D7 is the dominant of G. A fifth above G is D, so D (or D7) pulls toward G like a magnet.



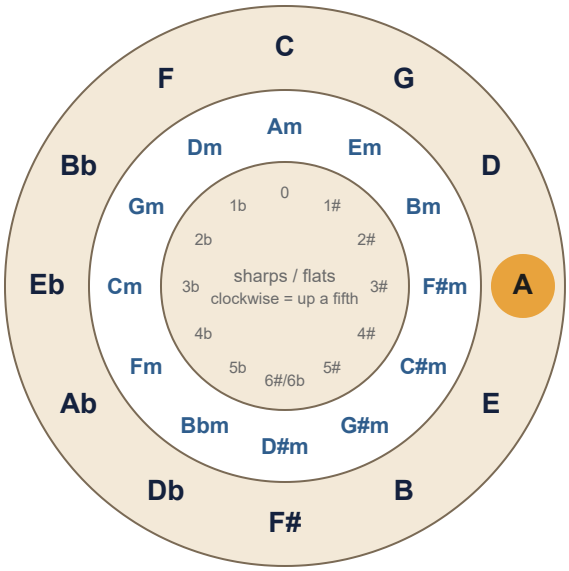
G major, the I chord in the key of G. C is its left neighbor and D its right neighbor.

4. The inner ring: relative minors

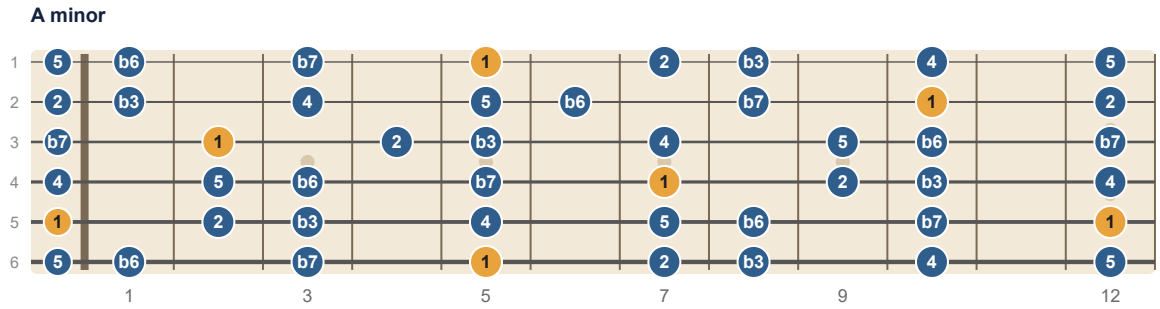
Most printed circles have a second ring inside. That ring holds the minor keys. Each major key shares its seven notes with exactly one minor key, called its **relative minor**. The relative minor starts on the 6th degree of the major scale. C major's 6th degree is A, so A minor is the relative minor of C. They use the same seven notes: C D E F G A B. Only the home note changes.

MAJOR KEY	RELATIVE MINOR	SAME NOTES, DIFFERENT HOME
C	Am	no sharps or flats
G	Em	1 sharp (F#)
D	Bm	2 sharps
A	F#m	3 sharps
E	C#m	4 sharps
B	G#m	5 sharps
F# / Gb	D#m / Ebm	6 sharps / 6 flats
Db	Bbm	5 flats
Ab	Fm	4 flats
Eb	Cm	3 flats
Bb	Gm	2 flats
F	Dm	1 flat

Put it together in one picture. The outer ring is the major keys: C G D A E B F#/Gb Db Ab Eb Bb F. The inner ring, directly underneath each major key, is its relative minor: Am Em Bm F#m C#m G#m D#m/Ebm Bbm Fm Cm Gm Dm. A major key and its relative minor sit in the same slot, so they always have the same number of sharps or flats.



A major (outer ring) and its relative minor F#m (inner ring) share the same slot: three sharps (F#, C#, G#).



A minor with degrees: the same notes as C major, but now A is the home note (1).

> **Minor and major share a slot**
>
> If someone tells you a song is in E minor, look at the slot of G major. It has one sharp. So E minor has one sharp too (F#). You never have to learn minor key signatures separately.

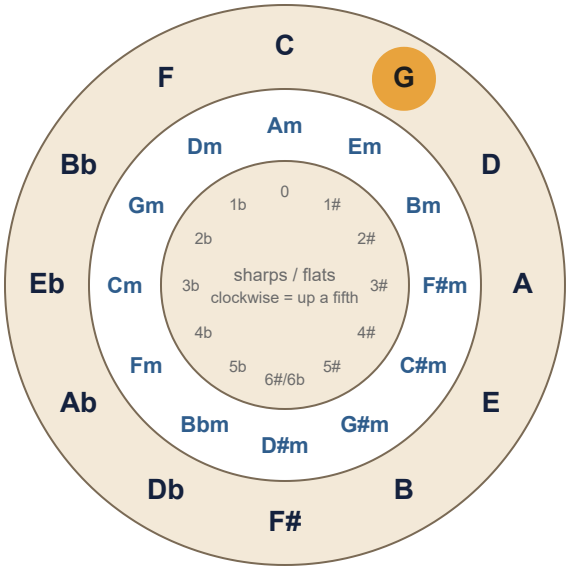
5. A fast way to find the chords of any key

Here is the payoff. In Module 5 you harmonized the major scale and got seven chords: I ii iii IV V vi vii°. Three of them are major (I IV V), three are minor (ii iii vi), and one is diminished (vii°). The circle holds the first six in a tidy cluster. To find them:

1. Find your key on the outer ring. That is the I chord.
2. Take the key to its left (IV) and the key to its right (V) on the outer ring.
3. Drop to the inner ring. The relative minor of each of those three majors gives you the three minor chords.

That is it. Three major chords and the three minor chords underneath them. Let's test it on G. The key is G. Its neighbors are C (left) and D (right). Underneath them on the inner ring: Em (under G), Am (under

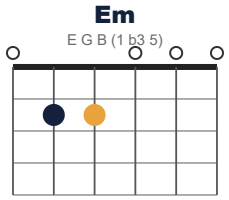
C), Bm (under D). So the six main chords of G major are **G, C, D, Em, Am, Bm**. The seventh chord is the diminished one built on F#. You will rarely need it for songs at this stage.



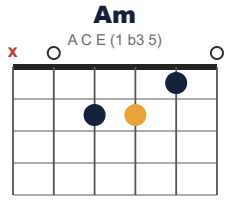
The six main chords of G major: G, C and D on the outer ring, with Em, Am and Bm directly underneath.

KEY	I	IV	V	VI	II	III	VII° (THE ODD ONE OUT)
C	C	F	G	Am	Dm	Em	Bdim
G	G	C	D	Em	Am	Bm	F#dim
D	D	G	A	Bm	Em	F#m	C#dim
A	A	D	E	F#m	Bm	C#m	G#dim
E	E	A	B	C#m	F#m	G#m	D#dim
F	F	Bb	C	Dm	Gm	Am	Edim

Spend a minute with that table and notice the pattern: the six main chords of any key always form a little block of three over three on the circle. If you can picture the block, you can name the chords. The vi chord is directly under the I, the ii chord is under the IV, and the iii chord is under the V.



Em is the vi chord of G, and it sits directly under G on the inner ring.



Am: the ii chord in the key of G, under the C on the outer ring.

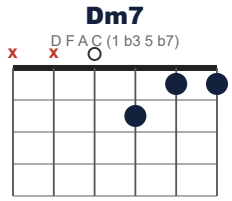
> **Keep it honest**
>
> The circle gives you the chords that belong to a key. It does not tell you what order to play them in. That is Module 8's job. Use the circle to decide which chords are available, then use your ear to pick the order.

6. The circle in chord movement: down a fifth

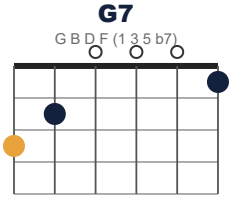
The circle is not only a map of keys. It also describes how chords like to move. In a lot of music, a chord resolves to another chord whose root is a fifth lower (or, the same thing, a fourth higher). G goes to C. D goes to G. A goes to D. Each of those moves is one step counterclockwise on the circle. It feels like gravity, a settling feeling, a question that gets an answer.

The strongest version uses a dominant 7 chord. G7 resolves to C, because G is a fifth above C, and because the 7th of G7 (F) wants to fall to E, and the 3rd (B) wants to rise to C. That pull is why V7 to I is the strongest cadence in this style of music, and it is why ii-V-I feels so complete.

Now look at ii-V-I in C: Dm7, G7, Cmaj7. The roots are D, G, C. Each is a fifth below the last. That is two steps counterclockwise around the circle. You can extend it backward: vi-ii-V-I, which is Am7, Dm7, G7, Cmaj7. The roots A, D, G, C are three steps going down in fifths. Add iii as well and you get iii-vi-ii-V-I: Em7, Am7, Dm7, G7, Cmaj7. Every chord resolves to the next one by a fifth, and the whole thing is just walking counterclockwise on the circle.



Dm7: the ii chord of C. Its root D is a fifth above G.



G7: the V7 chord of C. Its root G is a fifth above C, and the pull to C is strong.

PROGRESSION IN C	CHORDS	ROOT MOTION ON THE CIRCLE
ii - V - I	Dm7 - G7 - Cmaj7	D to G to C: two fifths down
vi - ii - V - I	Am7 - Dm7 - G7 - Cmaj7	A to D to G to C: three fifths down
iii - vi - ii - V - I	Em7 - Am7 - Dm7 - G7 - Cmaj7	E to A to D to G to C: four fifths down
I - IV - V - I	C - F - G - C	one step left, two steps right, one step back

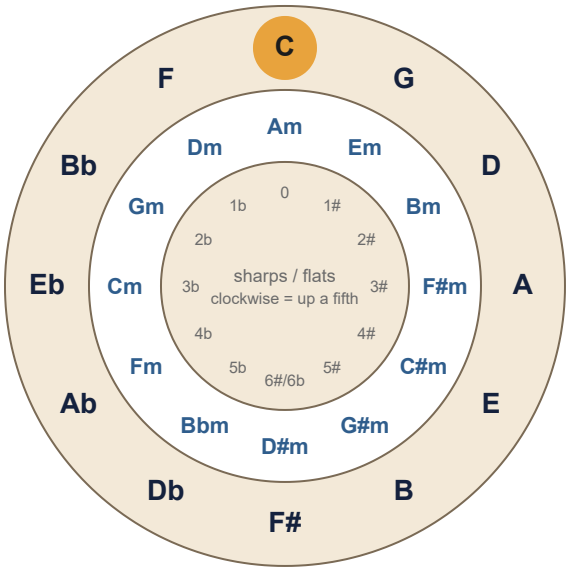
Why it works

Moving a fifth down is the same as moving a fourth up. Both give you the strongest root motion in this style of music. When you hear a progression and it feels like it is falling gently toward home, the roots are probably going around the circle counterclockwise.

7. Modulation: changing keys

A **modulation** is a change of key in the middle of a song. The home note changes. The old key stops feeling like home, and a new one takes over. Songs do it for lift: the last chorus goes up and everything feels brighter. They also do it to surprise: a bridge visits a different key and then comes back.

The circle shows you how far apart two keys are. Keys that are one step apart (like C and G) share six of seven notes, so moving between them feels smooth. Keys that are far apart (like C and F#) share very few notes, so moving between them feels dramatic. The distance on the circle tells you how big the jolt is.



Start in C. Neighboring keys F and G are one step away, so a modulation there is smooth. Keys on the far side of the circle are a bigger leap.

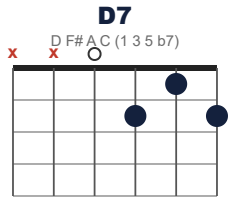
Pivot chord modulation

The smoothest way to change key is to use a **pivot chord**: a chord that belongs to both the old key and the new key. You play it in the old key, then treat it as a chord in the new key. Your ear never feels the seam.

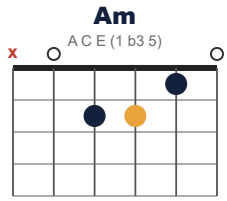
Example, C to G. C major has the chords C Dm Em F G Am Bdim. G major has G Am Bm C D Em F#dim. The shared chords are C, G, Am and Em. Pick Am. It is vi in C and it is ii in G. From there, D7 (V7 of G) leads you to G major, and you have arrived in a new key.

STEP	CHORD	ROLE IN C	ROLE IN G
1	C	I	IV
2	F	IV	(not in G)
3	Am	vi	ii (the pivot)
4	D7	(not in C)	V7
5	G	V	I (new home)

Play it: C, F, Am, D7, G. After D7 your ear expects G, not C. The key has changed. Notice that you needed only one new note: the F# inside D7. That one note is the whole difference between the two keys.



D7 contains F#, the one note that belongs to G major but not to C major.



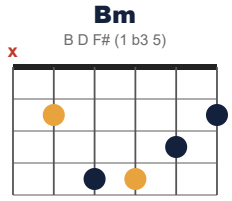
Am: the pivot chord. It is vi in C and ii in G.

Direct modulation (the truck driver)

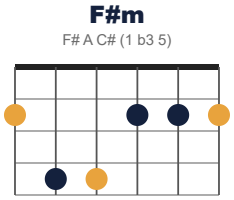
The simplest key change has no preparation at all. The song just jumps. Most often the last chorus goes up a half step or a whole step, and everyone grins. People call it the **truck driver modulation** for the way it shifts gears. It is not subtle, but it works.

A simple example: a song plays its verse in G, then the next chorus starts a whole step higher, in A. Every chord shifts up two frets: G becomes A, C becomes D, D becomes E, Em becomes F#m. Because you

move every chord by the same distance, the shapes on your fretboard are the same, only slid two frets up. Barre chords are perfect for this, because you can slide the whole shape.



Barre shape with its root on string 5. Slide the whole shape up two frets and the key moves up a whole step.



Barre shape with its root on string 6 at fret 2. Moving the same shape keeps the sound and changes the key.

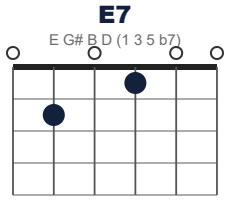
Secondary dominants: borrowing a V

Here is the other great tool. Any chord in the key can be treated, for a moment, as a temporary home. To make it feel like home, you play *its* dominant 7 right before it. That dominant 7 chord is called a **secondary dominant**. 'Secondary' just means it is the dominant of something other than the main key.

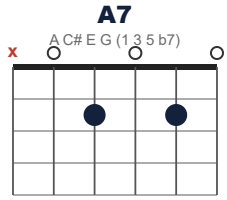
You name them with a slash: V/V means 'the five of five'. In the key of C, V is G. The dominant of G is D. So V/V is D7, and it leads to G. Likewise V/vi means 'the five of six'. In C, vi is Am. The dominant of A is E. So V/vi is E7, and it leads to Am.

NAME	READS AS	IN THE KEY OF C	RESOLVES TO
V/V	five of five	D7	G
V/vi	five of six	E7	Am
V/ii	five of two	A7	Dm
V/IV	five of four	C7	F
V/iii	five of three	B7	Em

Each of these has at least one note that is not in C major. D7 has F#. E7 has G#. A7 has C#. C7 has Bb. B7 has D# and F#. That outside note is what gives the progression its color: a sudden brightness or a little tension, resolved a beat later. Use them as spice. One secondary dominant in a verse is plenty for most songs.



E7 is V/vi in C: the dominant of Am. The G# is the note that does not belong to C major.



A7 is V/ii in C: the dominant of Dm. The C# is the outside note.

Try this progression: C, E7, Am, D7, G, C. E7 pulls toward Am. D7 pulls toward G. Each chord is led to by its own dominant, and the result sounds like a complete little journey. Secondary dominants are also an easy way to start a modulation: land on D7 and then settle on G as the new home instead of treating G as V, and you have just moved into G.

> **Three tools, three jobs**
>
> A pivot chord is a chord that two keys share, and it gives the smoothest change. A direct modulation just jumps, and it gives the biggest lift. A secondary dominant borrows a V to lean on one chord, and it gives color without leaving the key.

8. Dominant 7 chains

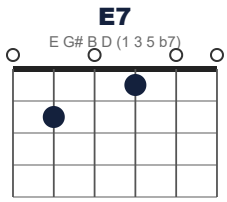
Secondary dominants can link up. If every chord in a row is a dominant 7 whose root is a fifth below the last, you get a **dominant 7 chain**. It is the circle of fifths in motion, and it shows up in blues turnarounds, swing, and a lot of older songs. A classic one in the key of C is: **E7 - A7 - D7 - G7 - C**.

Walk through it. E7 is V/vi, but instead of resolving to Am, it goes on to A7. A7 is V/ii, but instead of resolving to Dm, it goes to D7. D7 is V/V, which goes to G7. G7 is V7 of C, which goes to C. Each chord is a fifth below the one before it: E to A to D to G to C. Four steps counterclockwise on the circle.

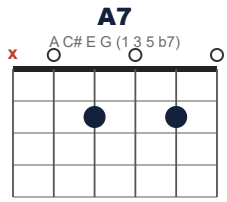
CHORD	NOTES	ROLE IN C	PULLS TO
E7	E G# B D	V/vi	A7
A7	A C# E G	V/ii	D7
D7	D F# A C	V/V	G7
G7	G B D F	V7	C

CHORD	NOTES	ROLE IN C	PULLS TO
C	C E G	I	home

Notice the little trick that makes the chain flow. The 7th of each chord falls a half step to the 3rd of the next. The D in E7 falls to the C# in A7. The G in A7 falls to the F# in D7. The C in D7 falls to the B in G7. The F in G7 falls to the E in C. Those half-step slides are what your ear is following.



E7: the start of the chain.



A7: the second link.

In open position this chain is comfortable under the fingers, because E7, A7, D7 and G7 are all open shapes. Strum each one for two beats and then land on C for a full bar. Most players find it takes only a few minutes to hear the circle as a long slide downhill.

9. Using the circle to transpose a song

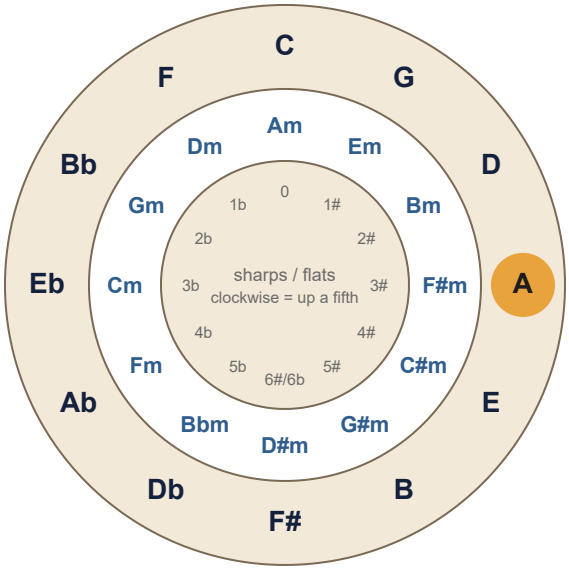
Transposing means playing a song in a different key. You might do it to fit your singing voice, to match a friend, or to swap open chords for barre chords. The circle makes it easy: **count the steps between the two keys, then move every chord the same number of steps in the same direction.**

Take a simple four-chord song in C: C, Am, F, G. Move it one step clockwise to G. C becomes G, Am becomes Em, F becomes C, G becomes D. The new song is G, Em, C, D. Same relationships (I vi IV V), different home. Go two steps clockwise and you have D, Bm, G, A. Go one step counterclockwise from C and you land in F: F, Dm, Bb, C.

KEY	I	VI	IV	V
C	C	Am	F	G
G	G	Em	C	D
D	D	Bm	G	A

KEY	I	VI	IV	V
A	A	F#m	D	E
F	F	Dm	Bb	C
Bb	Bb	Gm	Eb	F

Notice you do not need to think note by note. You only need to know where the key lives on the circle, and the chords come from that neighborhood. Many players find that G, D, A and E feel good for strumming in open position because the shapes are comfortable, which is one reason a song is often moved to one of those keys.



Transposing C to A is three steps clockwise: C to G to D to A. Every chord moves three steps too.

> **Capo as a transposing tool**
>
> A capo does the same job as moving around the circle: each fret is a half step up. G shapes with a capo on fret 2 sound in A. C shapes with a capo on fret 2 sound in D. Same shapes, new key. The circle tells you which shapes are easy, and the capo gets you to the key you want.

10. Common mistakes

- **Reading the circle backward.** Clockwise adds sharps. Counterclockwise adds flats. If you mix up the direction, check with C: G (one step clockwise) has an F#, so clockwise means sharps.
- **Forgetting the inner ring.** The circle is not only for major keys. If you only use the outer ring, you will miss half the chords.
- **Thinking the circle gives the order of chords.** It gives you the available chords and the strongest pull (down a fifth). The sequence is still your choice.

- **Using every secondary dominant at once.** A progression with five outside notes in four bars sounds busy. Pick one and let it breathe.
- **Modulating without a reason.** A key change lifts a song when the song is ready for it. If you move up a step just because you can, it can sound like a trick.
- **Mixing up F# and Gb.** They are the same fret but different names. Use sharps in keys with sharps and flats in keys with flats, so the names stay consistent.

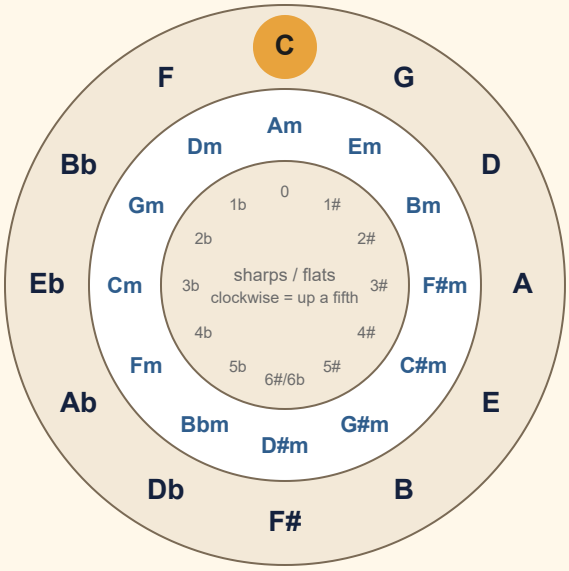
On the fretboard

Five short exercises. Do them with the guitar in your hands. The self-check sits right under each one.

EXERCISE 41 — 6 MIN

Walk the circle and count

1. Start at C. Say the next five keys clockwise out loud, and say how many sharps each has.
2. Go back to C. Say the next three keys counterclockwise, and say how many flats each has.
3. For each key, name the sharps or flats in its signature (use the order F C G D A E B for sharps, B E A D G C for flats).



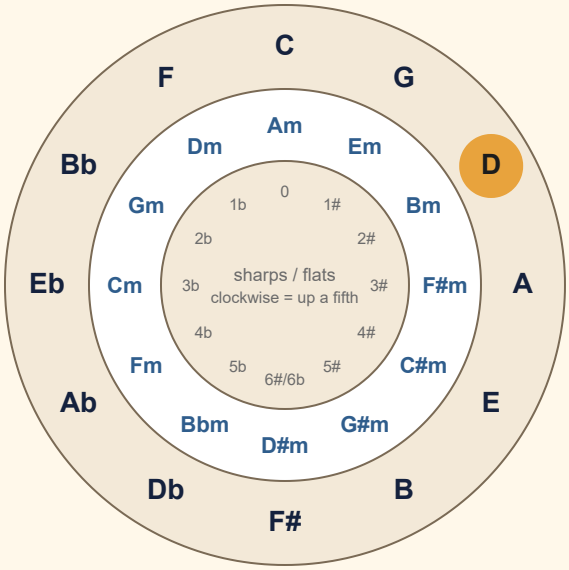
Use this circle: start at C and walk both ways.

☐ Self-check: Clockwise you should have said G (1 sharp), D (2), A (3), E (4), B (5). Counterclockwise: F (1 flat), Bb (2), Eb (3). If the first sharp you named for any sharp key was not F#, or the first flat was not Bb, go back to Section 1.

EXERCISE 42 6 MIN

Six chords of a key in thirty seconds

1. Choose the key of D. Find D on the outer ring of the circle and name its two outer neighbors.
2. Name the three relative minors directly under D and its two neighbors.
3. Play the six chords one after another, strumming each for four beats.
4. Now do the same for the key of A, without looking at the table.



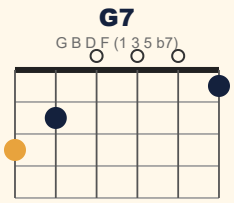
Key of D: neighbors G and A, with minors Bm, Em, F#m inside.

☐ Self-check: Key of D: D, G, A, Bm, Em, F#m. Key of A: A, D, E, F#m, Bm, C#m. If you named a G# in the key of D or a Bb anywhere, check the neighbors again: neighbors are one step left and one step right, nothing more.

EXERCISE 43 8 MIN

Ride ii-V-I down the circle

1. In the key of C, play Dm7 for four beats, G7 for four beats and C for a full bar.
2. Say the roots out loud: D, G, C. Ask yourself: is each a fifth below the last?
3. Move it to the key of G: Am7, D7, G. Say the roots: A, D, G.
4. Extend it to vi-ii-V-I in C: Am7, Dm7, G7, C. Say the roots: A, D, G, C.



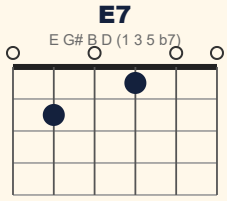
G7: the V7 chord in the key of C.

☐ Self-check: In every case the roots fall by a fifth, which is one step counterclockwise on the circle each time. If you cannot hear the pull from G7 to C, play G7 and stop. Your ear should want C.

EXERCISE 44 8 MIN

Find the outside note

1. Play this progression in C: C, E7, Am, D7, G, C. Strum each chord for two beats.
2. Listen for the two chords that feel like they have a new color: E7 and D7.
3. Name the note in each of those chords that is not in the C major scale.
4. Say what each one is: 'the five of ...'.



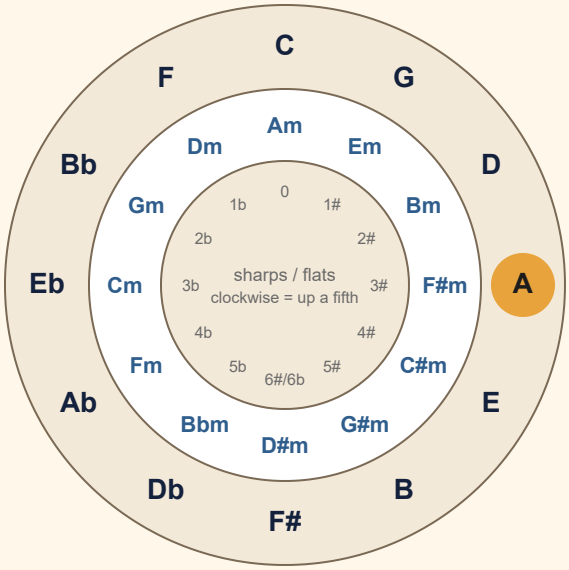
E7: V/vi in the key of C.

☐ Self-check: E7 contains G#, which is not in C major, and it leads to Am (so it is V/vi). D7 contains F#, which is not in C major, and it leads to G (so it is V/V). C major has no sharps, so any sharp you hear is an outside note.

EXERCISE 45 8 MIN

Transpose by the circle

1. Take this song: C, Am, F, G. Play it once in C.
2. Count three steps clockwise on the circle from C.
3. Write the new chords by moving each chord three steps.
4. Play the new version, using the open shapes where you can.



Three steps clockwise from C: the key of A.

☐ Self-check: C to G to D to A, so the new key is A. The chords are A, F#m, D, E. If you wrote Am instead of F#m, remember that the vi chord is the relative minor of the new key: F#m for A major. The progression should feel exactly like the original, only brighter.

Module 9 cheat sheet

IDEA	THE SHORT VERSION
Up a fifth	One more sharp (C, G, D, A, E, B, F#)
Down a fifth	One more flat (C, F, Bb, Eb, Ab, Db, Gb)
Order of sharps	F C G D A E B
Order of flats	B E A D G C F
Neighbors	Left of a key is IV, right is V
Relative minor	Inner ring under the major: Am Em Bm F#m C#m G#m D#m Bbm Fm Cm Gm Dm
Six main chords	The key, its two neighbors, and the three relative minors beneath them
Strong root motion	Down a fifth: ii - V - I, vi - ii - V - I

IDEA	THE SHORT VERSION
Pivot chord	A chord both keys share, used to glide to a new key
Direct modulation	Jump up a half or whole step; slide the same shapes
Secondary dominant	The dominant 7 of a chord other than I: V/V is D7 in C, V/vi is E7 in C
Dominant 7 chain	E7 - A7 - D7 - G7 - C, each root a fifth below the last
Transposing	Count the steps between keys and move every chord the same number

Module 9 glossary

- **Circle of fifths:** a round diagram of the 12 keys in order of fifths, with sharps added clockwise and flats counterclockwise.
- **Key signature:** the list of sharps or flats that belong to a key.
- **Relative minor:** the minor key that uses the same notes as a given major key, starting on its 6th degree.
- **Neighbor key:** a key one step away on the circle, sharing six of seven notes.
- **Modulation:** a change of key during a song.
- **Pivot chord:** a chord that belongs to both the old key and the new one.
- **Direct modulation:** a key change with no preparation, often up a half or whole step.
- **Secondary dominant:** the dominant 7 of a chord other than the home chord, written like V/V.
- **Dominant 7 chain:** a string of dominant 7 chords, each a fifth below the previous.
- **Transpose:** to play a song in a different key by moving every chord the same distance.

Next step

You now have the map that holds every key together. The last part of the book, Create, puts it to work. In Module 10 you will use the scales, chords and circle you already know to improvise with intention, choosing notes on purpose instead of hoping. Before you move on, play the chain E7 - A7 - D7 - G7 - C three times and listen to how each chord leans into the next. That feeling is the circle of fifths, and it will help you in every module that follows.

THE FRETBOARD REVEALED

Part 4 - Create

MODULE 10

Improvising with intention

- Improvising is choosing notes on purpose, not hoping the right ones show up.
- Three questions guide every note: what key, what chord is happening now, what do I want to say.
- Chord tones, especially the 3rd, are your landing spots; the other scale notes are the road between them.
- Space, repetition and rhythm shape a solo as much as the notes do.
- Bends, slides, the blue note and modal color are tools for expression, each with a clear pitch target.

By the end: Play over a chord loop and make phrases that start somewhere, say something and land on a note you chose.

> *Part 4 - Create. Everything you have learned so far (the fretboard, intervals, scales, modes, chords, CAGED, progressions) now goes to work. This module is about turning that knowledge into music you invent on the spot.*

What improvising really is

Many players think improvising is a gift: some people have it and some do not. Most players find that it is a skill built from a few simple choices, repeated until they become habit. Improvising is choosing notes on purpose, in real time, over a chord loop. The word that matters is **purpose**. A random note from the right scale may sound fine. A note you chose because you knew where it was going to land sounds like a musician.

Think of a conversation. You do not recite every word of the dictionary. You pick a few words, say them in a certain rhythm, pause, and let the other person answer. A solo works the same way. It has short ideas, silence between them, and a point where each idea finishes. Scales are your vocabulary. This module teaches you grammar.

You do not need to be fast. You do not need to know every scale. A player with one scale shape and a clear plan will usually sound better than a player with seven shapes and no plan. So we begin small, with one shape, and ask the right questions.

The three questions

Before and during every solo, ask yourself three questions. They take two seconds and they keep you out of trouble.

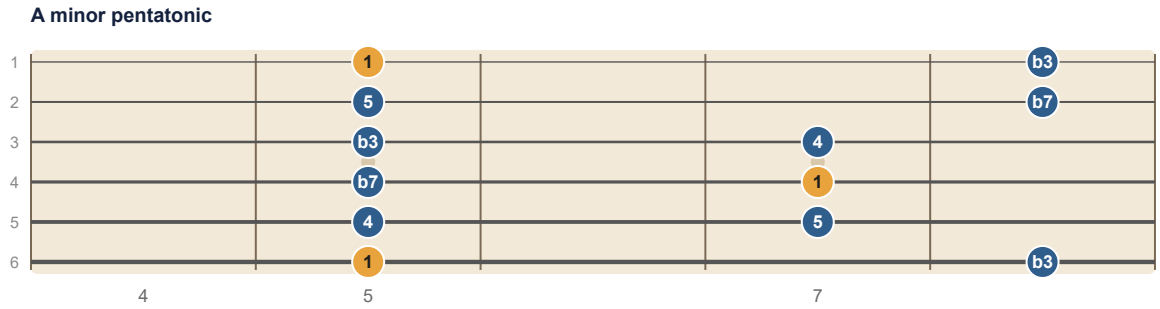
1. **What key am I in?** This tells you which notes belong. If the song is in A minor, your home scale is A minor or A minor pentatonic.
2. **What chord is happening right now?** The chord tells you which notes are the safest landing spots at this exact moment. The key is the map; the chord is where you are standing.
3. **What do I want to say?** Calm or tense? A question or an answer? A long sigh or a quick shout? Decide the feeling first, then pick notes that match it.

Notice that question 3 is not about notes at all. When you answer it, the notes get easier. A calm phrase uses long notes and low-tension tones. A tense phrase uses short notes and the notes that want to move. We will make that precise in a moment.

> **Try this**
>
> Put on any chord loop. Before you play a single note, say out loud: key, chord, feeling. Then play one phrase. If you can answer all three, you are already improvising with intention.

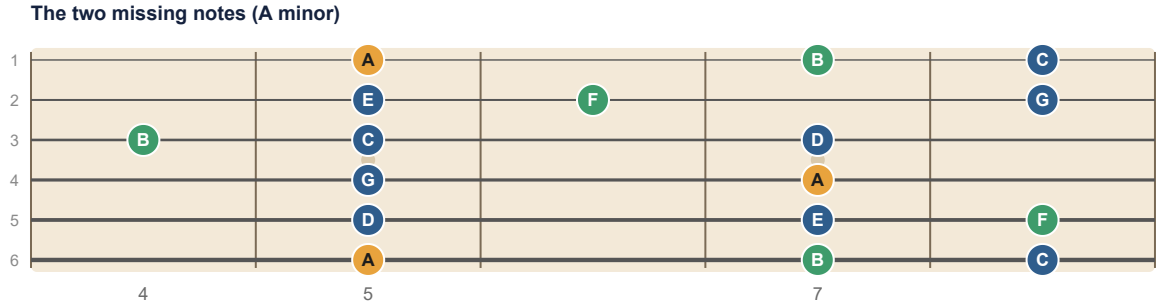
Start with the pentatonic and add the missing notes

The minor pentatonic is the friendliest scale on the guitar. It has five notes, no awkward clashes, and it fits most songs in a minor key. In A minor it is A C D E G, or 1 b3 4 5 b7. Here is the most famous shape, anchored on the 5th fret.



A minor pentatonic, box 1 (frets 5 to 8). Every dot is a safe note over an Am chord.

The full A natural minor scale has seven notes: A B C D E F G, or 1 2 b3 4 5 b6 b7. Compare it with the pentatonic. The pentatonic left out two notes: **B** (the 2) and **F** (the b6). Those two missing notes give the full minor scale its more melodic, slightly sadder flavor.



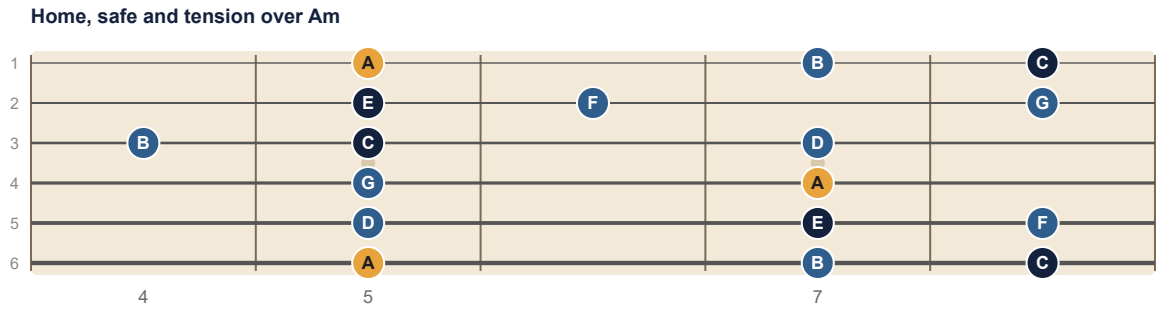
Pentatonic notes (A C D E G) plus the two missing notes, B and F, shown as targets.

Here is how to use this. Play your pentatonic as your home base. When you want a richer color, pass through B or F briefly, usually on the way to a pentatonic note. B walks naturally up a half step to C, or down a whole step to A. F walks naturally down a half step to E. Used as passing notes, they sound smooth. Held for a long time over the wrong chord, they can sound sour. That leads to the next idea.

Safe, tension, home

Every note in your scale has a job relative to the chord under it. The simplest way to remember this is three words.

- **Home**: the root of the chord. It feels finished. Over an Am chord, home is A.
- **Safe**: the other chord tones. Over Am these are C (the b3) and E (the 5). They sound stable and agree with the chord.
- **Tension**: scale notes that are not in the chord. Over Am: D, G, B and F. They lean toward a nearby chord tone and ask to be resolved.



Root dots are home (A). Chord dots are safe (C and E). Scale dots are tension (D, G, B, F).

Tension is not bad. Tension is what makes a phrase interesting. A phrase that only uses safe notes sounds like a lullaby. A phrase that only uses tension never feels settled. Good solos travel: they leave home, build a little tension, and come back to a safe note or home. Think of it as breathing out and in.

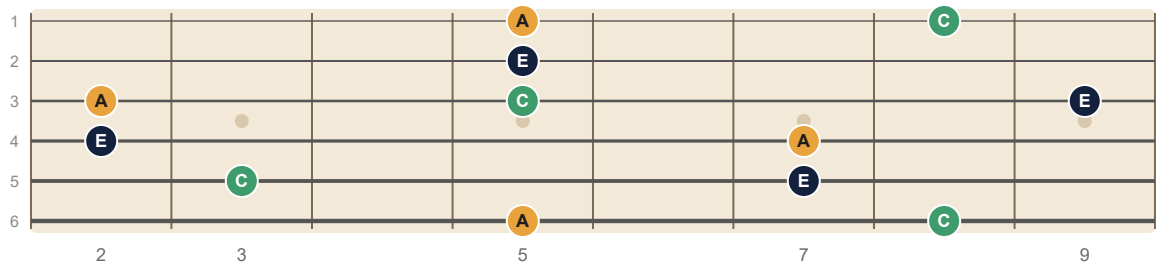
Chord tones are your targets

Here is the single most useful trick in this module: **when the chord changes, land on a chord tone of the new chord.** You do not need to play only chord tones. You only need to arrive on one at the right moment, usually the first beat of the new chord. The notes in between can be any scale notes you like.

Of the three notes in a triad, the **3rd** is the best target. The root says where you are. The 5th is neutral. The 3rd tells the listener whether the chord is major or minor, so landing on it makes your line sound like it follows the harmony. Let us try it over the loop Am, F, C, G (in the key of C major that is vi, IV, I, V; in A minor it is i, VI, III, VII).

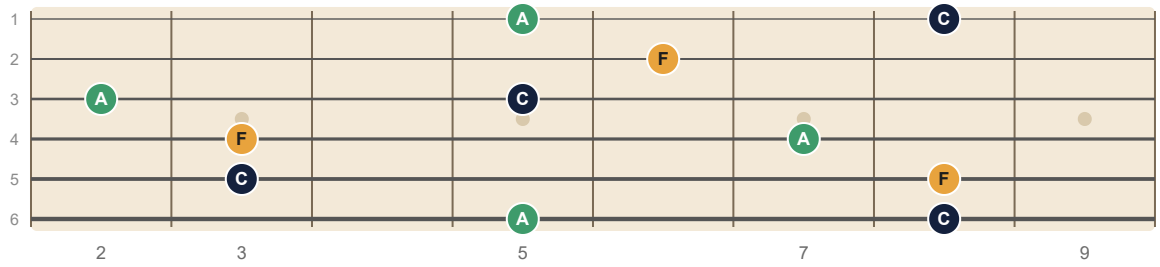
The 3rds are: Am has C, F has A, C has E, G has B. Each diagram below marks the 3rd as a target, the root as home, and the 5th as a chord tone.

Am: aim for C



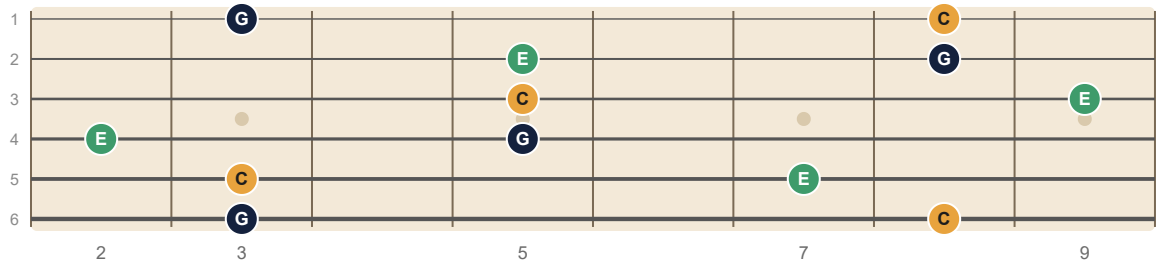
Am = A C E. The 3rd is C. Home is A, the 5th is E.

F: aim for A



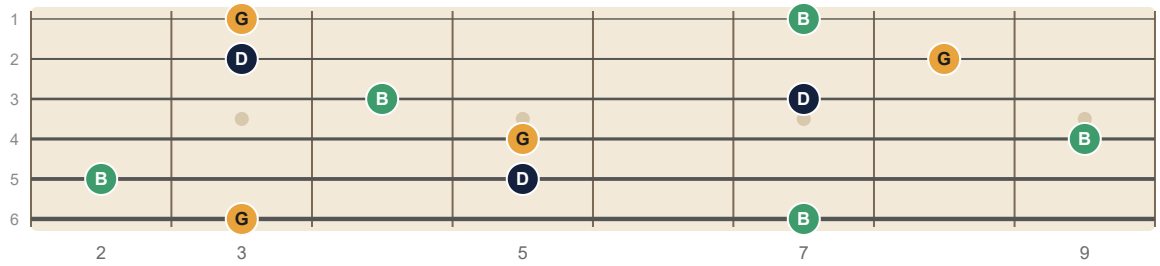
F = F A C. The 3rd is A. Home is F, the 5th is C.

C: aim for E



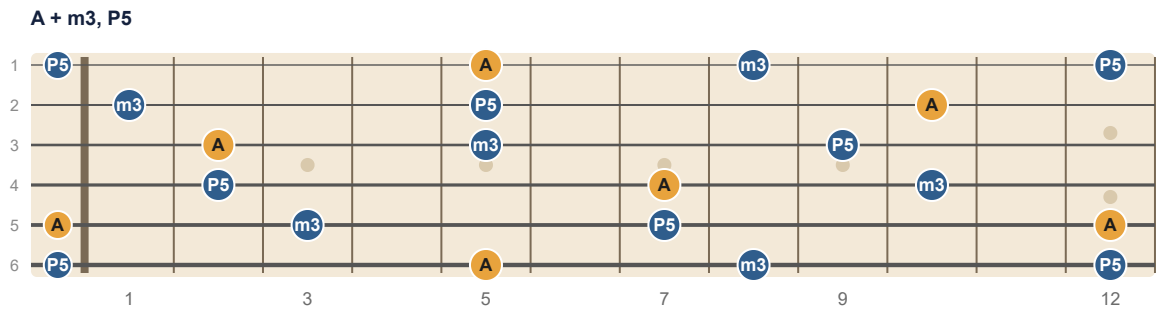
C = C E G. The 3rd is E. Home is C, the 5th is G.

G: aim for B



G = G B D. The 3rd is B. Home is G, the 5th is D.

Notice something clever: A is the 3rd of F and the root of Am. C is the 3rd of Am and the root of C. Your pentatonic notes keep showing up as targets in different chords. That is why one scale can cover a whole progression, and why changing your target note as the chord changes makes the solo sound like it is listening.



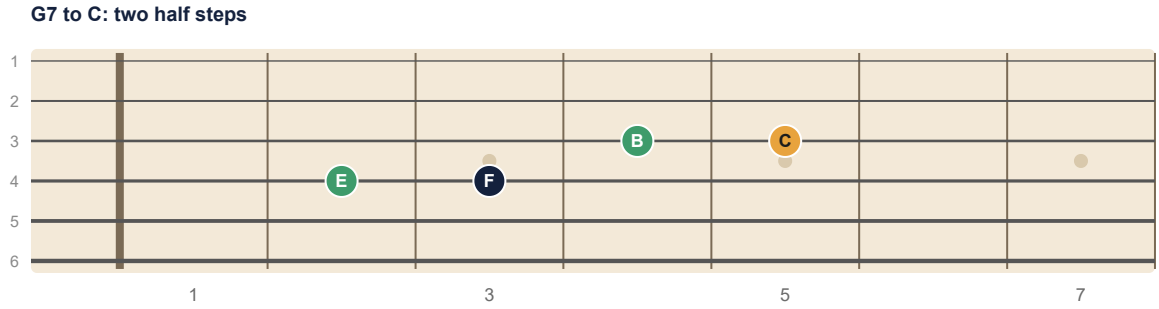
From A: a minor 3rd is C and a perfect 5th is E. Over Am these are your two safe landing notes.

The interval from the root is easy to hear. For a minor chord the 3rd is a minor 3rd (3 half steps). For a major chord it is a major 3rd (4 half steps). If you can hear that distance, you can find the 3rd without a diagram.

Voice leading by half steps

Here is the next level. The best landing notes are often one half step away from the note you just played. A half step is the smallest move on the guitar: one fret. Moving by half step between chords gives the strongest sense of connection.

Take the move from G7 to C. G7 is G B D F. C is C E G. The B in G7 wants to rise one half step to C. The F in G7 wants to fall one half step to E. Those two small moves make the chord change sound inevitable.



B (fret 4, string 3) rises to C (fret 5). F (fret 3, string 4) falls to E (fret 2). The 3rd and the 7th of G7 both resolve to chord tones of C.

A similar idea works for Dm7 to G7 to Cmaj7. The 3rd of Dm7 is F. Hold it, because F is also the 7th of G7. Then drop it one half step to E, the 3rd of Cmaj7. One note, three chords, and a smooth path. Players who practice this often find their solos start sounding like the song instead of sitting on top of it.

Phrasing: the part that makes it music

Notes are only half of a solo. Phrasing is the other half. Phrasing is how you group notes, how you pause, and how you repeat. Here are four habits that most players find make the biggest difference.

1. Leave space

A phrase needs room to breathe. After you play an idea, stop for a beat or two. Silence makes the next phrase sound important. If you feel the urge to fill every gap, play half as many notes and see how it feels.

2. Repeat on purpose

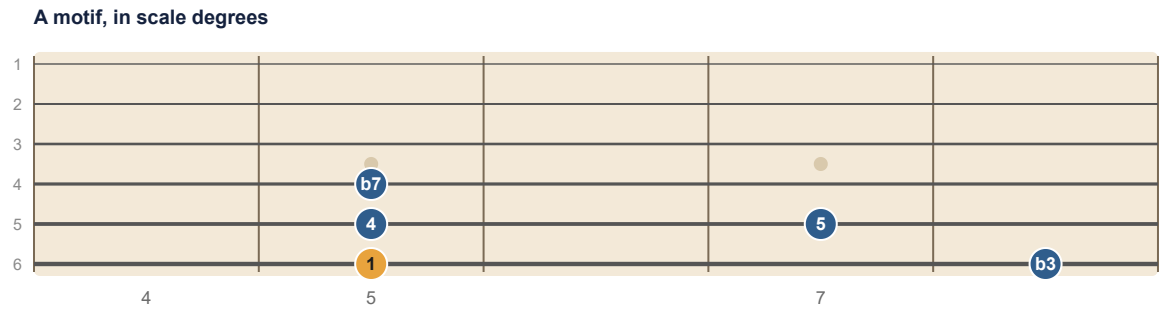
Repetition is not boring; it is the way listeners recognize your idea. Play a short phrase of two to four notes. Then play it again. Then change one thing. Many memorable guitar parts are a short idea repeated with small changes.

3. Call and response

Split your phrases into two halves. The first half asks a question: it ends on a tension note or a note that feels unfinished. The second half answers: it ends on a safe note or home. The question and answer can both be short, and the second half can reuse the rhythm of the first.

4. Develop a motif

A **motif** is a tiny idea, often two to four notes with a rhythm. To develop it, you transform it. Move it to a new starting note. Play it one string higher. Stretch the rhythm. Reverse the order. Keep the rhythm and change the pitches. Each version sounds new but connected, so your solo has a story.



The sequence 1, b3, 4, 5, b7 in A minor pentatonic (A, C, D, E, G). Play it low to high, then start it from a different note of the scale.

Rhythm is half of the solo

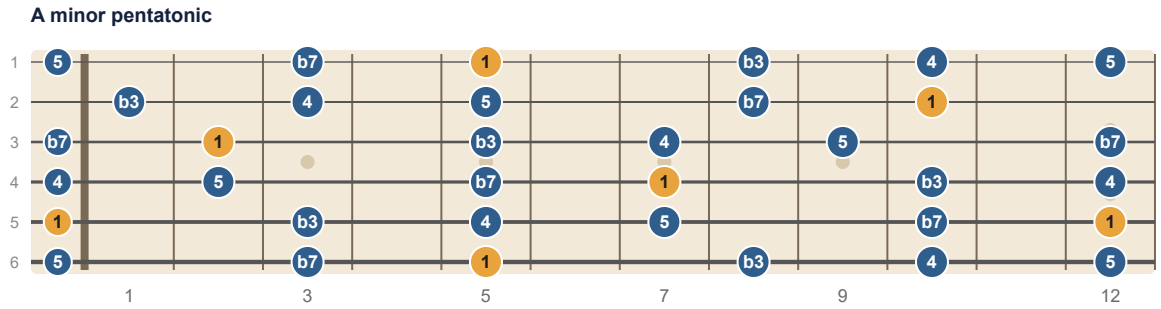
If you play the same notes with a different rhythm, you get a different phrase. Before you worry about new notes, try this: choose two notes and make five different phrases with them just by changing when you play and when you stop. Long-short-short. Short-short-long. Start on the offbeat. Wait two beats and then play three quick notes.

Listen to singers. They do not sing on every beat. They stretch some words, rush others, and leave gaps. Copy that. A solo with strong rhythm and only three notes often beats a solo with twenty notes and no groove. Tap your foot, count out loud, and place each phrase on purpose.

> **The two-note rule**
>
> Whenever you feel lost, drop to two notes and play with the rhythm only. Most players find that confidence returns quickly and their ear starts offering new notes.

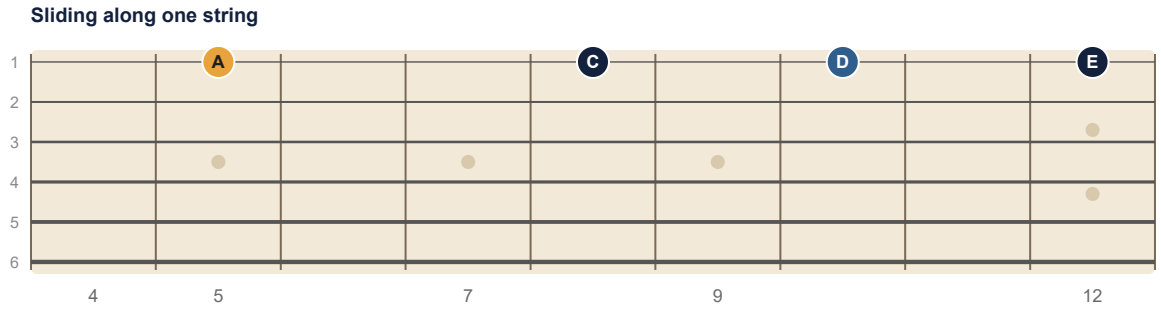
Connecting boxes along the neck

A single box traps you in four frets. In Module 7 you saw that the fretboard is one pattern repeating, and CAGED links the pieces. Pentatonic shapes work the same way: five boxes that cover the neck and overlap their neighbors.



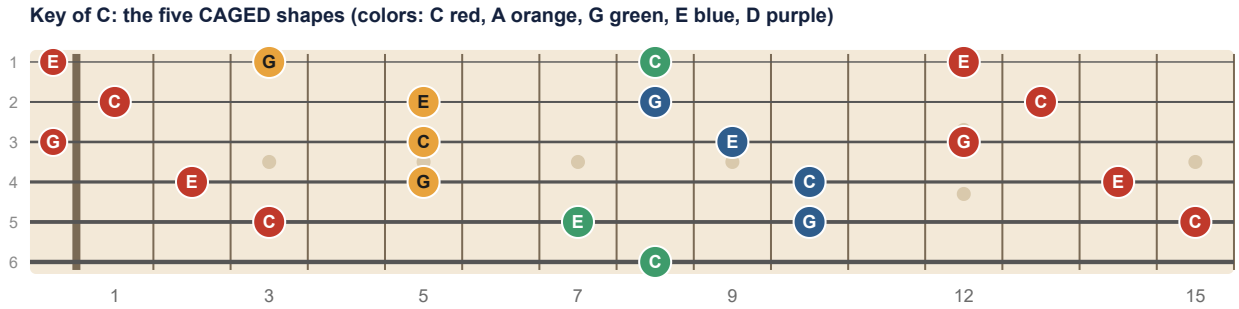
A minor pentatonic across the first 12 frets. Look for the same clusters of dots repeating as you move up.

To move from one box to the next without stopping, use a **slide**. Choose a note that belongs to both boxes, slide to the next note on the same string, and keep playing. On the high E string the A minor pentatonic notes are A (fret 5), C (fret 8), D (fret 10), E (fret 12) and G (fret 15). Sliding along this string shows you the path from one box to the next.



High E string: A (5), C (8), D (10), E (12). Slide between neighbors to travel from one box to the next.

The same idea works for chords. The CAGED shapes you learned map directly to areas of the neck, so wherever the chord is played, the pentatonic and major scale notes around it are close at hand. When the chord moves to a new shape, your phrase moves with it.

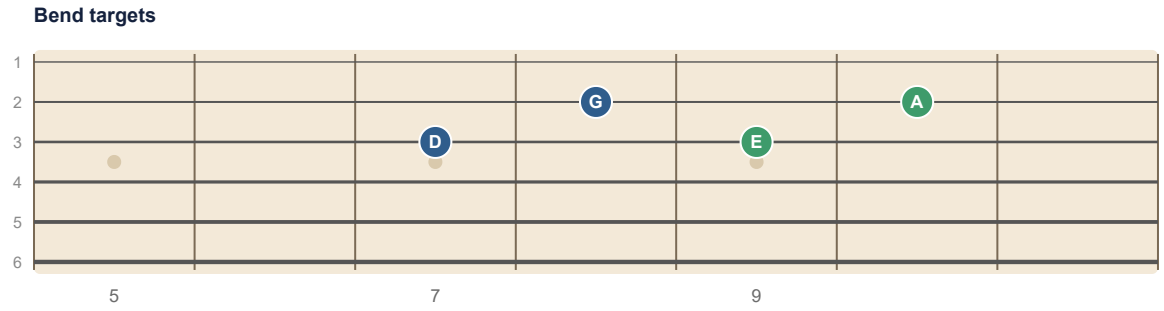


The five CAGED shapes for C major. Each shape sits inside a region of the neck where you can also find the C major scale and its pentatonic.

Bends and slides as expression

A bend and a slide are not decoration. They are the guitar's way of singing. The key is to treat each one as a path to a specific target note. Every bend has a pitch target. Every slide has a landing fret.

In a **whole-step bend**, you push the string until it sounds two frets higher without moving your fretting finger. On the G string, the D at fret 7 bent a whole step reaches the pitch of the E at fret 9. On the B string, the G at fret 8 bent a whole step reaches the pitch of the A at fret 10. Both targets are notes of the A minor pentatonic, so both sound right.
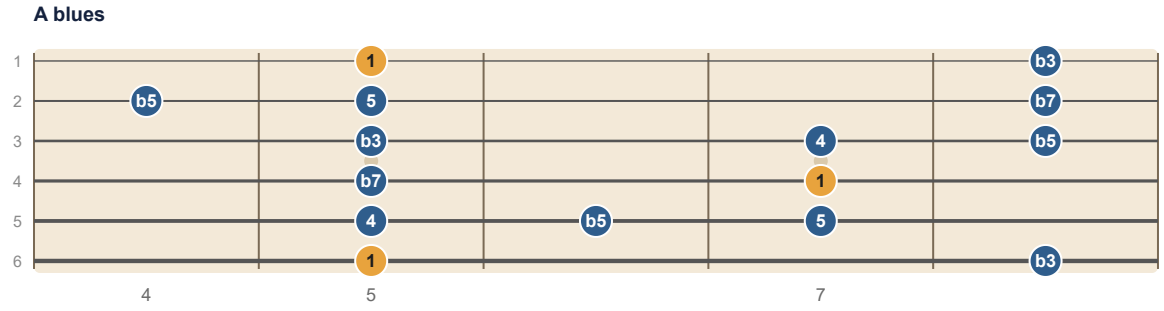


Bend D (string 3, fret 7) a whole step to the pitch of E (fret 9). Bend G (string 2, fret 8) a whole step to the pitch of A (fret 10).

How to practice a bend: first play the target note at its normal fret and listen. Then go back to the lower note and bend until it matches. Your ear is the judge. If it sounds flat, push farther; if it sounds sharp, ease back. A **half-step bend** targets the note one fret higher. A slide works the same way with no pushing: pick the first note and glide along the string to the target fret.

Use expression sparingly. Bend the note you want to be the center of a phrase, usually the end of a call or the peak of the line. If every note bends, none of them stand out.

The blues scale and the blue note

Add one note to the minor pentatonic and you get the blues scale. In A it is A C D Eb E G, or 1 b3 4 b5 5 b7. The new note is the b5 (Eb). It is the **blue note**: the tense, bendy note that sits between the 4 (D) and the 5 (E).



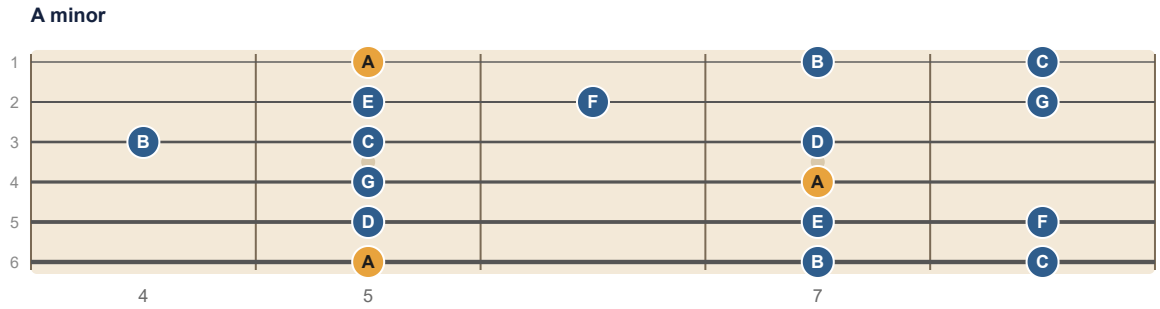
A blues scale around box 1. The b5 is the new note, a half step above D and a half step below E.

The blue note works best as a passing tone. Play D, step to Eb, and keep moving to E. Do not stop and rest on Eb unless you want that sour, tense effect on purpose. Because the blue note is a half step from each neighbor, it is a perfect way to practice half-step connections.

Soloing over a minor key

Minor keys are where the pentatonic feels most natural. Over a loop in A minor such as Am, G, F, E7 or Am, F, C, G, your home is A and your pentatonic is A C D E G. Use these steps.

1. Begin on A, C or E (home, safe, safe) on the first chord.
2. Use pentatonic notes to travel.
3. Add B or F as passing tones when the line needs more color.
4. When the chord changes, aim at the 3rd of the new chord.
5. End the solo on A, the root, for a clear finish.

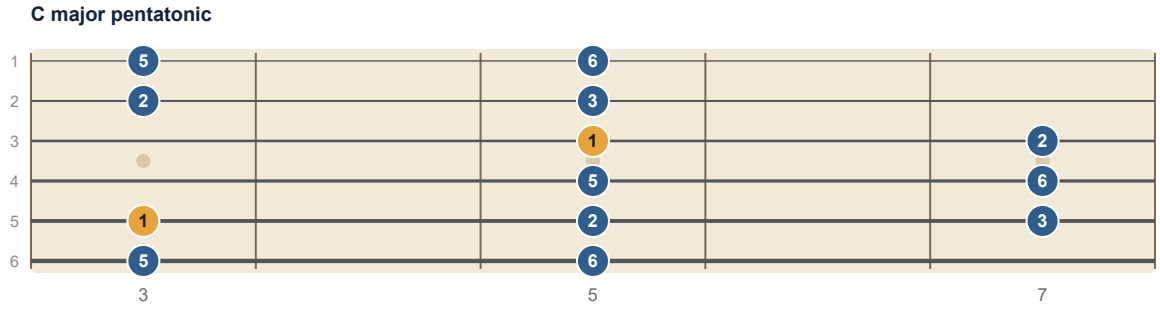


A natural minor around box 1: A B C D E F G. Pentatonic notes plus B and F.

If you play over an E or E7 chord, the harmonic minor (A B C D E F G#) gives you one special note: G# is the 3rd of E. This is the one place where a note outside the natural minor sounds excellent, because it is the chord's 3rd. You met this scale in Module 3; keep it as a special color for the V chord.

Soloing over a major key

Major keys work with the major pentatonic: 1 2 3 5 6. In C that is C D E G A. These are the same notes as A minor pentatonic; only the home note moves. That is the idea of **relative** keys from Module 3: A minor and C major share the same seven notes.



C major pentatonic: 1 2 3 5 6 = C D E G A. Same notes as A minor pentatonic, with a different home.

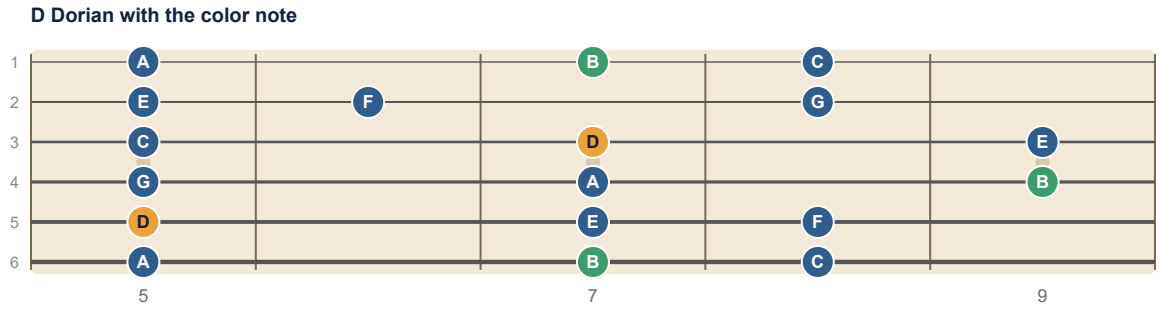
Over a loop such as C, Am, F, G, emphasize the 3rd of each chord: E over C, C over Am, A over F, B over G. Notice that B is not in C major pentatonic, so when the G chord arrives, step onto B briefly as a target. This single note is the difference between playing over the chords and playing with them.

The mood is different, too. Minor pentatonic sounds darker and more driving. Major pentatonic sounds sunnier and more open. A good way to learn the difference is to play the same shape over a minor loop and a major loop, changing only which note you treat as home.

Modal soloing over a vamp

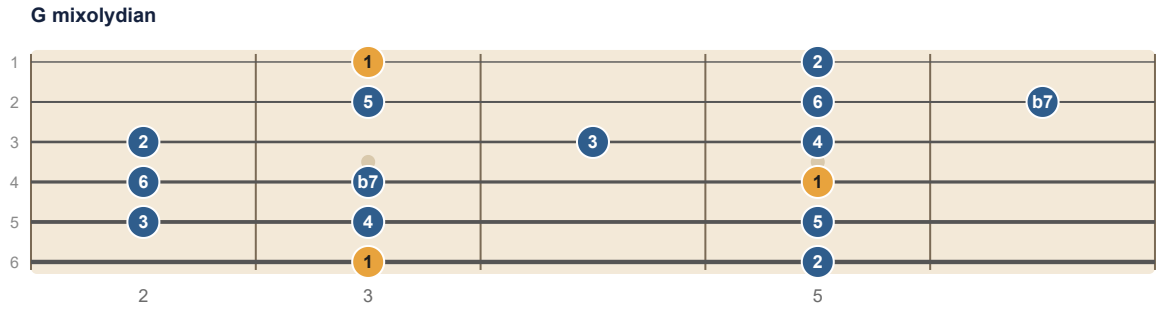
A **vamp** is a short chord loop that stays in one place, often just one or two chords. Vamps are where modes (Module 4) become useful. When the harmony does not travel through many keys, you can choose one mode and let its color define the solo.

Try a vamp of Dm7 and G7, two bars each. The notes D E F G A B C make up D Dorian (1 2 b3 4 5 6 b7). Dorian is minor with a bright twist: the natural 6, here B. That B is the characteristic note. It is also the 3rd of G7, so it lands naturally on the second chord.



D E F G A B C. The natural 6 (B) is the Dorian color note. Land on it over G7, and land on F (the b3) over Dm7.

Another example: a one-chord vamp on G7 invites G Mixolydian (G A B C D E F, or 1 2 3 4 5 6 b7). It is a major sound with a b7. The b7 (F) gives the dominant flavor; the 3rd (B) gives the major quality. Landing on B and F alternately is a good pattern.



G Mixolydian: 1 2 3 4 5 6 b7. Land on B (3) for sweetness and F (b7) for dominant color.

The rule for modal playing is simple: **pick one mode, find its color note, and keep returning to it.** Do not try to prove you know all seven modes in one solo.

Phrases to practice

Here are patterns described as scale degrees, so you can use them in any key. Play each one in A minor pentatonic first, then move your home note and play it again. The degree numbers are relative to home: in A minor pentatonic, 1 is A, b3 is C, 4 is D, 5 is E and b7 is G.

PATTERN (SCALE DEGREES)	IN A MINOR PENT.	WHAT IT DOES
1 - b3 - 4 - 5 - b7	A C D E G	Climbs the whole scale: a plain run
5 - b7 - 5 - 4 - b3	E G E D C	Turns around at the top and settles on the b3
1 - b3 - 1 - b3 - 4	A C A C D	A motif that repeats, then steps up
b3 - 4 - 5 - 4 - b3 - 1	C D E D C A	An arch: goes up, comes back, rests at home
5 - 5 - b7 - 5	E E G E	Repeated note with a leap, mostly rhythm

Play each one in three rhythms. Then end the phrase on a different chord tone each time. You are building a small toolbox of ideas that you can recognize and reuse in the middle of a solo.

A practice routine with backing tracks

You do not need a band. A backing track or a chord loop gives your phrasing something to react to. Better still, **record your own loop**. Strum Am, F, C, G, four beats each, for a minute. Use a phone or computer, play it back, and solo over yourself. When you hear your own chords, you learn to match them. A recording also stays at the same tempo every time, which a live strum may not.

1. **Warm up (3 minutes).** Play your pentatonic box slowly, naming each note out loud.
2. **Targets (5 minutes).** Over the loop, play any notes, but land on the 3rd of each chord on beat one.
3. **Phrase work (5 minutes).** Make one motif and develop it four ways.
4. **Free play (5 minutes).** Ask the three questions, then play freely with space.
5. **Listen back (2 minutes).** Record and listen. Choose one thing you liked and one you would change.

Twenty minutes a day is enough. With practice, most players find that landing on chord tones moves from effort to habit in a few weeks, and the rest of their playing opens up.

Common mistakes

- **Playing nonstop.** No space means no phrasing. Count two beats of silence between ideas.
- **Ignoring the chord changes.** If you stay on the same notes when the chord changes, you are playing over the chords, not with them. Aim for the new 3rd.
- **Resting on a tension note.** Leaving B or F hanging over Am sounds unfinished. Resolve to a chord tone.
- **Using too many scales.** One scale with a plan beats three scales with none.
- **Fast and sloppy.** Slow, clean phrases with good rhythm are almost always more convincing.
- **Bending without a target.** Know the pitch you want before you push the string.
- **Never repeating.** A good idea deserves a second hearing.

Module 10 cheat sheet

IDEA	RULE	EXAMPLE (AM, F, C, G)
Three questions	Key, chord, feeling	A minor or C major; chord now; calm or tense
Home / safe / tension	Root, other chord tones, other scale notes	Am: A / C E / D G B F
Landing notes	Aim for the 3rd of the new chord	C, A, E, B
Half-step resolve	Move 1 fret to a chord tone	B to C, F to E (G7 to C)
Minor pentatonic	1 b3 4 5 b7	A C D E G
Blues scale	Add b5 to minor pentatonic	A C D Eb E G
Major pentatonic	1 2 3 5 6	C D E G A
Dorian vamp	Highlight the natural 6	D Dorian over Dm7 to G7; land on B
Mixolydian vamp	Highlight the b7 and the 3	G Mixolydian over G7; B and F
Whole-step bend	Two frets up in pitch	D to E, G to A
Phrasing	Space, repeat, call and response, motif	End with a rest, then answer

Glossary

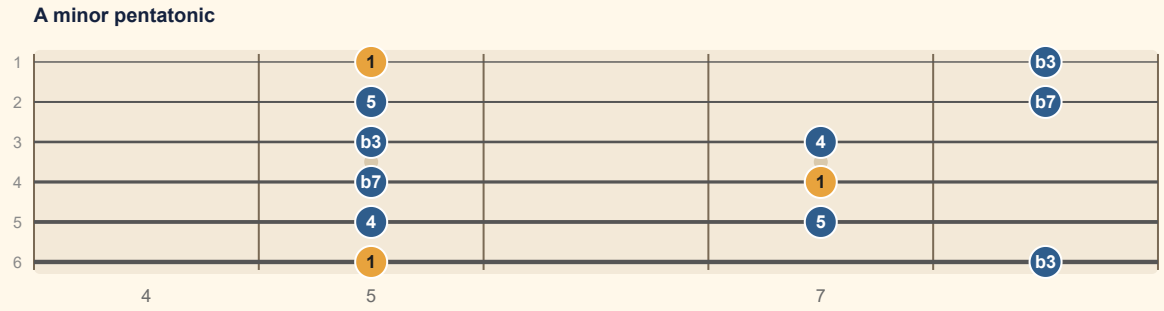
- **Improvising**: making up music in real time, over a chord loop or vamp.
- **Landing note**: a note you aim to arrive on, usually on the first beat of a chord.
- **Chord tone**: a note that is part of the chord (root, 3rd, 5th, and so on).
- **Tension note**: a scale note that is not in the chord and wants to resolve.
- **Phrase**: a short musical idea, like a sentence.
- **Motif**: a tiny idea (two to four notes) that you repeat and change.
- **Call and response**: a phrase that asks, followed by one that answers.
- **Voice leading**: connecting chords by moving each note a short distance, often a half step.
- **Blue note**: the b5 added to the minor pentatonic to make the blues scale.
- **Vamp**: a short chord loop that repeats, often one or two chords.
- **Bend**: pushing a string to raise its pitch to a target note.
- **Slide**: gliding a finger along a string from one fret to another.

On the fretboard: exercises 46 to 50

EXERCISE 46 — 10 MIN

Three questions on a loop

1. Record yourself strumming Am, F, C, G, four beats each, for one minute, or use a backing track.
2. Before playing, say aloud: key (A minor), chord (first chord), feeling (calm or tense).
3. Play only the A minor pentatonic box 1 over the loop. Use long notes.
4. Repeat three times, each time with a different feeling.



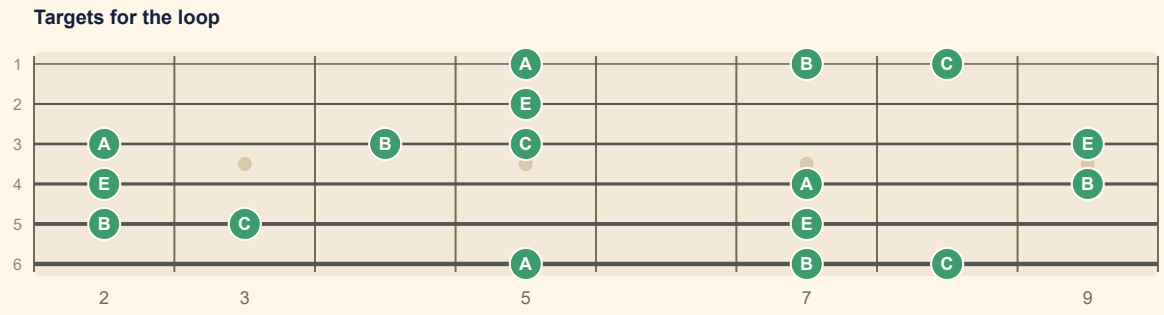
Box 1 of A minor pentatonic for this exercise.

☐ Self-check: can you answer key, chord and feeling at the start of every pass, and do your phrases sound different when the feeling changes?

EXERCISE 47 — 10 MIN

Land on the 3rd

1. Loop Am, F, C, G.
2. Over Am aim for C, over F for A, over C for E, over G for B.
3. Play any notes you like in between, but finish each chord on its target.
4. Record yourself and listen: does each landing sound like it belongs?



The four landing notes: C (Am), A (F), E (C), B (G).

☐ Self-check: can you name the 3rd of every chord before it arrives, and do you land on it on beat one at least three times out of four?

EXERCISE 48 8 MIN

Motif and response

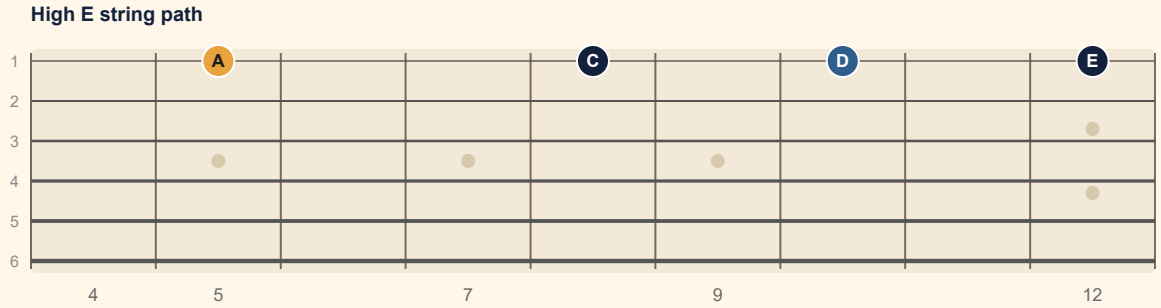
1. Invent a motif of three notes from A minor pentatonic with a rhythm you like.
2. Play it, rest for two beats, then play it again starting on a different note.
3. Change one thing: rhythm, direction or the last note.
4. Make the second phrase end on A (home).

☐ Self-check: can another person hear that the second phrase is related to the first? Does the answer end on home?

EXERCISE 49 10 MIN

Slide between boxes

1. On the high E string, find A (fret 5), C (fret 8), D (fret 10) and E (fret 12).
2. Play A and slide to C, then slide to D, then to E.
3. Add the notes from the nearest box around each stop and keep the phrase going.
4. Do it again in reverse, coming down.



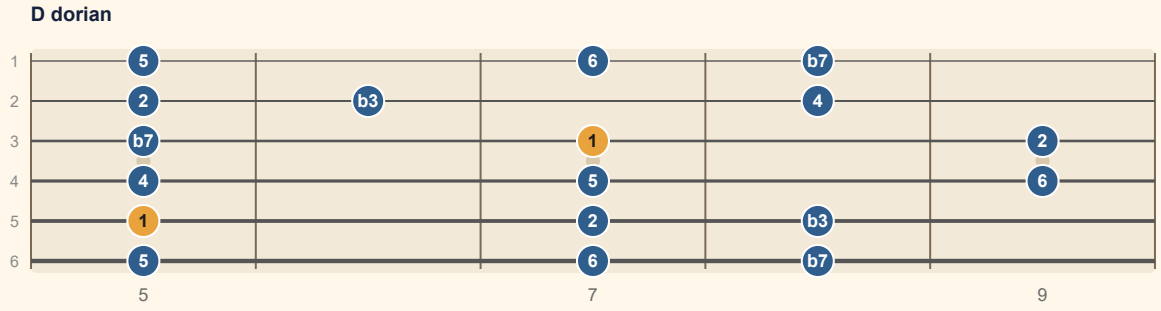
A, C, D and E on the high E string.

☐ Self-check: can you travel from fret 5 to fret 12 without stopping, and name every note you stop on?

EXERCISE 50 12 MIN

Dorian vamp

1. Record a vamp: Dm7 for two bars, then G7 for two bars, looping.
2. Play D Dorian (D E F G A B C) over it.
3. Land on F over Dm7 and on B over G7.
4. Resolve the last phrase on D.



D Dorian: 1 2 b3 4 5 6 b7.

☐ Self-check: can you hear the difference when B arrives over G7, and can you finish a phrase on D?

Next step

You can now improvise with a plan. Choose your favorite loop, record it, and spend twenty minutes a day on the routine above. In Module 11, you will turn what you hear into what you play: working songs out by ear and writing your own.

MODULE 11

Working songs out by ear and writing your own

- Hear distance and hear function: the two skills behind every song you will ever figure out
- Find the key and the root of each chord without tabs
- A 5-step method that works on almost any song
- Turn your own ideas into a finished song: key, progression, melody, structure

By the end: By the end of this module you will be able to take a song you love, find its key, its roots and its chord qualities by ear, and then use the same tools to write something of your own.

> *You have spent ten modules learning to see the fretboard. This last module is about hearing it. Everything you know about intervals, scales, chords and progressions is only useful if your ears can connect to it. The good news: ear skills are not a gift some people are born with. Most players find they improve steadily with a few minutes of the right kind of practice, and the theory you already know gives your ears a map to follow.*

What ear training really is

Forget the idea that ear training means naming notes out of thin air. That rare skill is called perfect pitch, and you do not need it. What working musicians use is **relative pitch**: the ability to hear how one note relates to another. It comes down to two simple skills.

- **Hearing distance.** How far apart are two notes? You met this in Module 2: every interval has its own flavor. A major third sounds bright and settled. A minor third sounds darker. A perfect fifth sounds open and strong.
- **Hearing function.** What job does a note or chord do inside the key? One note feels like home. Another feels like it is pulling back toward home. Another feels like a gentle step away. Once you hear these jobs, you can recognize them in any key.

Distance tells you what happened between two notes. Function tells you why it feels the way it does. A song in G and the same song in D use completely different notes, yet your ear says, "it is the same song." That is because the functions are identical. This is exactly why the number system from Module 8 is so useful: numbers describe function, and function is what your ear actually remembers.

Anchor intervals to tunes you already know

The fastest way to recognize an interval is to link it to the first jump of a melody you know by heart. Sing the tune, stop after the first jump, and ask yourself what that jump felt like. Here are a few that almost everyone knows.

INTERVAL	DIRECTION	TUNE THAT STARTS WITH IT
m3 (minor third)	up	The first jump of the old folk tune Greensleeves
M3 (major third)	up	The first jump of When the Saints Go Marching In
P4 (perfect fourth)	up	The first jump of the Bridal Chorus ("Here comes the bride")
P5 (perfect fifth)	up	The first jump of Twinkle, Twinkle, Little Star
M6 (major sixth)	up	The first jump of My Bonnie Lies Over the Ocean

Use whatever tunes work for you. A reference you chose yourself is stronger than one from a list. Write your own reference tunes on the first page of your notebook (we set one up later in this module).

> **Play it, then sing it**
>
> Always check an interval on the guitar. Play the first note, sing the second one in your head, then play it and see if you were right. Hearing alone is slow; hearing plus playing is how your ears and fingers learn to agree.

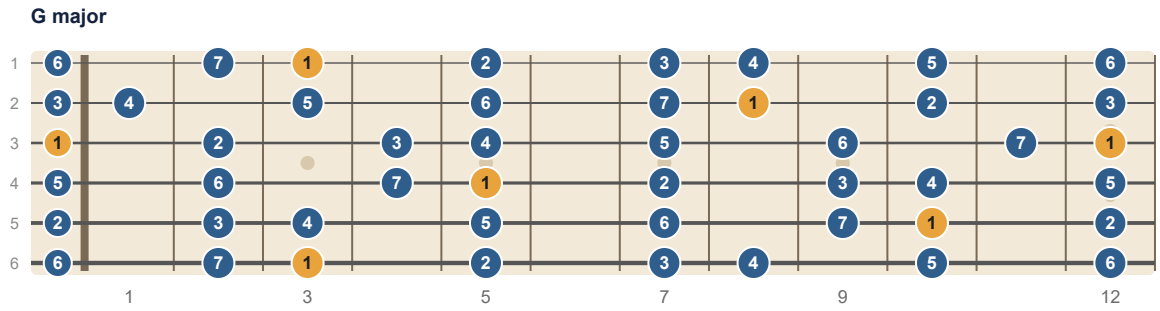
Finding the key of a song by ear

Before you can work out any chord, you need to know where home is. Every song in a major or minor key has a **home note**, also called the tonic, or 1. It is the note that feels like rest. When the song ends, the melody usually lands on it, and your body knows it is over.

Step by step: find the home note

1. Play the song (or sing it from memory) and hum along on any comfortable note.
2. Stop the music suddenly and keep humming. Your voice will drift toward the note that feels like rest. That is your candidate for the home note.
3. On the low E string (string 6), walk up fret by fret: 0, 1, 2, 3 and so on. Play each note while the song is playing or while you hum the tune. Most wrong notes feel tense or sour; the home note feels like it settles into the music.
4. Once you have a candidate, check it. Does the melody end on it? Does the last chord of the song have this note as its root? If both answers are yes, you have it.

Suppose your candidate is G. You will find it on string 6 at fret 3, on string 5 at fret 10, and on string 4 at fret 5. Any of those is a fine home base, and having the three locations is the reason you learned the "three anchors" in Module 1.



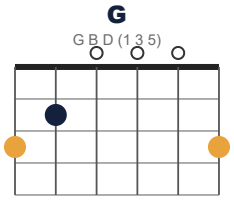
The G major scale with degrees. Home is 1 (the root dots). When you are lost, play up and down this pattern: if your candidate home note is G, every note of the melody should belong to it.

Major or minor? Test with the triad

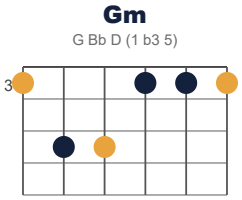
Knowing the home note is half of the answer. The second half is whether the song lives in a major or a minor key. The test is simple and nearly always works.

Play a major triad built on your home note, then a minor triad on the same note. Hold each for a few seconds with the song humming in your head. One of them sounds like it *belongs* and the other sounds like a stranger. For the home note G, the two triads are:

- **G major** = G B D (1, 3, 5). Bright, settled.
- **G minor** = G Bb D (1, b3, 5). Darker, more serious.



G major (G B D): the major-key test chord.



G minor (G Bb D): the minor-key test chord, with the root on string 6, fret 3.

> **If both seem to fit**
>
> Some songs move between major and minor, and some sit in a mode. If neither triad feels perfect, listen to the third in the melody: if you hear a note that sounds like 3 (bright), call it major; if it sounds like b3 (darker), call it minor. Do not worry about being exact on the first try. The next steps will tell you if you were wrong.

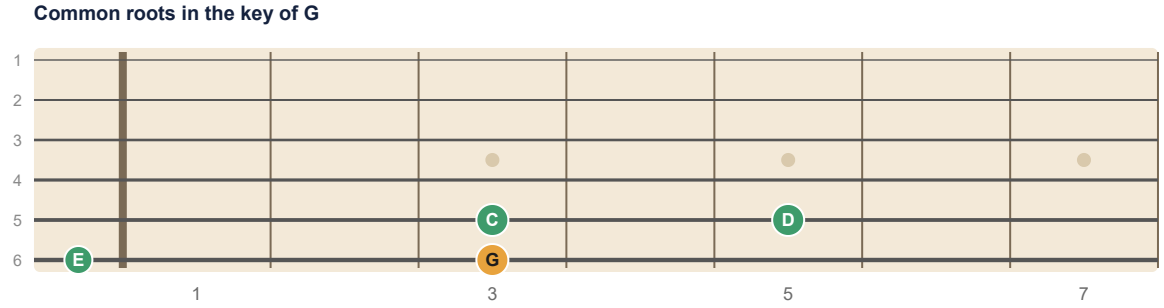
Working out the bass line: finding the roots

Once you know the key, the next job is to find the **root** of each chord. The roots form the bass line of the song, and in most pop, rock, folk and country songs, the bass note is the root. When you can follow the roots, you already know most of what the chords are doing.

How to hear the bass

- Listen to the lowest sound in the recording. If the music sounds thin on your speakers, turn up the bass or use headphones.
- Hum the lowest note in your chest voice, then slide your hum until you match it.
- Find that note on string 6 or string 5 and check it by playing it together with the recording.

Most songs use only a handful of roots. In the key of G, the most common ones sit right under your hand.



G (string 6, fret 3), the E of Em (string 6, open), C (string 5, fret 3) and D (string 5, fret 5). These four roots cover a huge number of songs.

Try this: while the song plays, tap one root per bar and watch for the moment the bass changes. Each change is a new chord. You do not need to know the quality yet. Just write down the roots as note names: G, E, C, D.

The number system, by ear

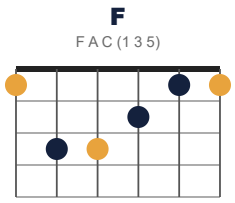
In Module 8 you learned to describe chords with numbers: I, ii, iii, IV, V, vi, vii dim. Now you will learn what they sound like. The goal is to hear "that was a IV" without a guitar in your hands.

NUMBER	WHAT IT IS IN G	WHAT IT FEELS LIKE
I	G major	Home. Rest. Everything settles here.
IV	C major	A lift. Opens the sound up, feels like stepping away from home on purpose.
V	D major	Tension. It wants to go home, and the pull is strong.
vi	E minor	The sad relative. Built from the same family of notes as I, but the mood is darker.
bVII	F major	The rock move. A chord borrowed from outside the key, with a bold, open sound.

Hear a I-IV-V. Play G, C, D, G. The first chord feels like home. The C feels like a step up and away. The D feels like a question that needs an answer, and the return to G answers it. Most players find that after a few days of listening, the pull of V back to I becomes obvious.

Hear a vi. Play G and then E minor. The feeling is that the sun has gone behind a cloud. The notes are from the same family, but the center of gravity moved. Many pop songs go I - vi - IV - V. Play G, Em, C, D and notice how natural it sounds.

Hear a bVII. Play G then F, then back to G. The F is not in the G major scale (G has an F sharp), so it sounds like a surprise, bold and rock-flavored. It is the signature chord of the Mixolydian sound you met in Module 4. If a chord in a song sounds "too cool for the key," it is often a bVII.



F major as a bVII in the key of G (F A C). Root on string 6, fret 1.

> **A shortcut for hearing**
>
> Learn the sound of each number by singing the root before you play the chord. Sing 1 (home), then 4, then 5, then 1 again. Your voice already knows the feeling of the V returning to I even if you have never named it.

A 5-step method to work out any song without tabs

Here is the method this module is built around. It works for an entire song, but it works best one small piece at a time.

1. **Pick one section.** Choose four or eight bars, such as the first verse or the chorus. Loop it. Never start with the whole song.
2. **Find the key.** Hum the home note and test it with a major and a minor triad, as above.
3. **Find the roots.** Follow the bass, one chord per bar at a time. Write them as note names.
4. **Find the quality.** For each root, try the major chord, then the minor chord. Keep the one that fits. Later you can test for sevenths and sus chords.
5. **Test with the melody.** Sing or hum the vocal line over your chords. If a note clashes, one of your chords is probably wrong. Go back to step 3 or 4.

When the section sounds right, convert it to numbers. Four bars of G, Em, C, D become I, vi, IV, V. Now you can play the same section in any key, and you have a pattern you can recognize in other songs.

> **Do not chase every detail**
>
> The first goal is the skeleton: the key, the roots and the main qualities. Passing chords, extra notes and strumming details are a bonus. A rough version that sounds right is far more useful than a perfect version you never finish.

Using a slow-down tool and loops

You do not need to hear a song at full speed to understand it. Most players use software or an app that can **slow a recording down without changing its pitch**, and that can **loop** any short section. Any tool with those two features is fine. The brand does not matter.

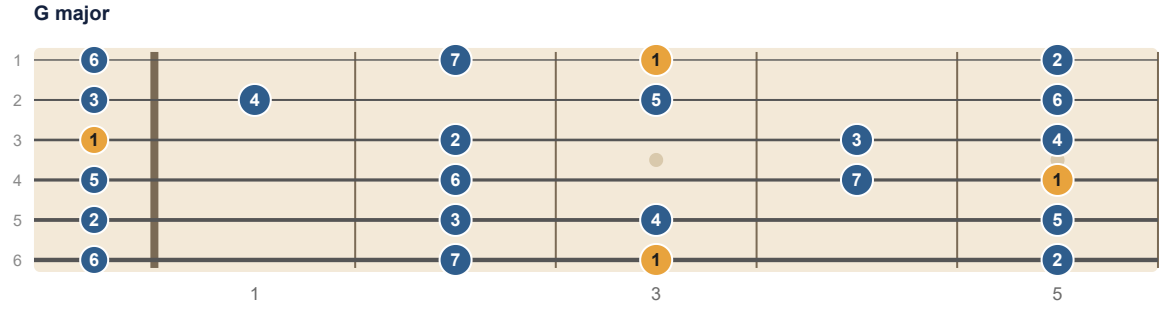
- **Set the loop to the section you are working on.** Two to four bars is plenty.
- **Start slow.** Try 60 to 75 percent speed. If a fast passage still blurs, drop to 50 percent.
- **Bring it back up gradually.** When your version is right at slow speed, raise the speed in small steps until you can play along at full speed.
- **Boost the bass for roots.** Many tools have a simple equalizer. Turn the low end up to hear the bass line, and the high end up to hear the melody.
- **Use headphones.** You hear more detail, and you can play along without muddying the recording.

A loop also helps your memory. After the sixth or seventh pass, your ear starts predicting what comes next. That prediction is hearing function in action.

Transcribing a melody as scale degrees

A melody is easier to learn, remember and move to other keys when you write it as **scale degrees** instead of note names. The number tells you the job of each note: 1 is home, 5 is the strong supporting note, 3 is the color note, 7 is the one that wants to resolve up to 1.

The procedure is the same as for the roots. Find the home note first, then listen to each melody note and ask: is it higher or lower than the previous one, and by how much? Use the major scale in front of you as a ruler.



G major scale with degrees in the first five frets. Use it as a ruler: 1 = G, 2 = A, 3 = B, 4 = C, 5 = D, 6 = E, 7 = F sharp.

A simple four-bar melody might be written 3 5 3 | 6 5 3 | 4 6 4 | 2 7 5. Because it is in degrees, you can play it in any key by changing only the starting point. Here is the same line in three keys. Every note

name was checked against the scale of that key.

DEGREES	IN G	IN C	IN D
3 5 3	B D B	E G E	F# A F#
6 5 3	E D B	A G E	B A F#
4 6 4	C E C	F A F	G B G
2 7 5	A F# D	D B G	E C# A

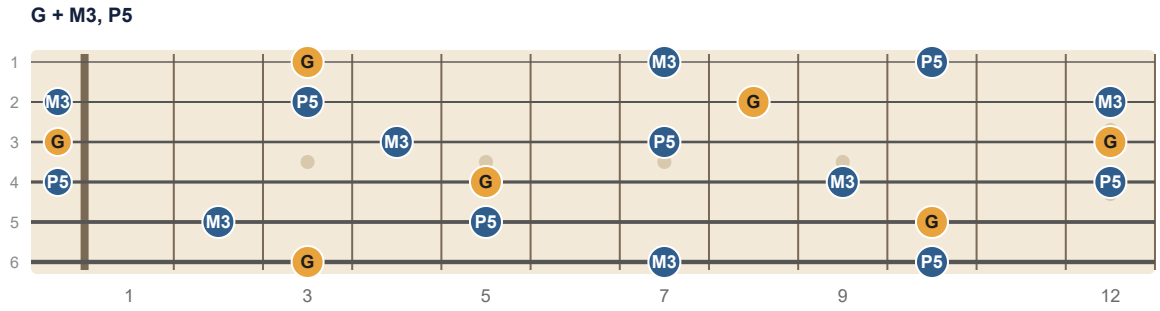
This is the real payoff of ear training combined with theory. Once you hear a melody as numbers, you are no longer tied to one key or one place on the neck.

Training exercises you can do in five minutes

Short, regular practice beats a long session once a week. Choose one of these and do it every day for a few weeks.

Interval of the day

Pick one interval for the day, such as the perfect fifth. Play a note on the guitar, sing the note you think is a fifth above it, then play it to check. Next, play two notes without looking and try to name the interval. Over the week, move through the intervals one by one.



From G: the M3 is B and the P5 is D. Pick one of these, sing it, then check on the neck.

Chord quality quiz

Close your eyes. Play a random major or minor chord (or have a friend do it) and say which it was before you look. Start with only two choices, major and minor. When you reach nine out of ten, add the dominant seventh, then the major seventh and minor seventh. Add one new quality at a time and only when the current set feels easy.

Humming the root

Play any chord, then hum the note you think is its root. Play string 6 or 5 to check. When you can do this in two seconds, you can follow bass lines in real songs without effort.

Writing your own song

Writing a song uses every tool in this book. You will pick a key, choose a progression, build a melody from the scale and shape it into a structure. None of it needs advanced theory. It needs a few decisions and a willingness to try.

1. Pick a key

Choose a key where the chords are comfortable and the melody fits your voice. G, C, D, A and E major, and E and A minor, are all good because they use open chords. Play with a few and see which one feels right for the mood you want.

2. Pick a progression

Use the progressions from Module 8 as a starting point. Four chords are enough. A few reliable choices, written as numbers so they work in any key:

PROGRESSION	MOOD	IN G
I - V - vi - IV	Warm, familiar	G D Em C
I - vi - IV - V	Classic, nostalgic	G Em C D
vi - IV - I - V	Emotional, a little sad	Em C G D
I - bVII - IV - I	Open, rock	G F C G

3. Write a melody from the scale

Use the notes of the key and let the chord tones guide you. Land on a chord tone (1, 3 or 5 of the chord underneath) at the start of a bar or on a long note. Use the other notes of the scale to move between those landing points.

- **Start with a motif.** A motif is a tiny idea, often only three or four notes with a rhythm. Think of it as the seed of the melody.
- **Repeat it.** Repetition is what makes a melody memorable. Play the motif again over the next chord.
- **Change it a little.** Move the motif up or down one step, or change the last note. This is called a sequence, and it keeps repetition from getting boring.
- **Add contrast.** After two or three repeats, do something different: a longer note, a leap, a pause. Contrast gives the ear something to notice.

A melody that only repeats feels flat. A melody that never repeats feels like noise. The sweet spot is repeat, repeat, change.

4. Lyrics and structure in plain English

Most songs are built from a few blocks.

- **Verse.** The story. Each verse has new words but the same chords and melody.
- **Chorus.** The main message. The words and music are the same every time, and it is usually the most memorable part. Many songs lift the energy here.
- **Bridge.** A short contrasting section, usually once, with new chords or a different feel. It gives the listener a break before the final chorus.

A simple, proven layout is: verse, chorus, verse, chorus, bridge, chorus. If you are stuck on lyrics, write the chorus first. Say in one plain sentence what the song is about, then build lines around it. Short lines that you can sing in one breath work best. Do not try to be clever; be honest and specific.

5. Arranging and recording basics

- **Record on your phone first.** A voice memo is enough to capture a song and to hear it from outside.
- **Use a steady beat.** Play along with a metronome or click so your timing is even. If you want to layer parts, you need a steady tempo.
- **Build in layers.** Record the chords, then a simple bass line (the roots), then the melody, then a second guitar part if you like.
- **Change something between sections.** Strum harder on the chorus, play lighter on the verse, add a different rhythm on the bridge. Contrast makes a song feel finished.
- **Leave room.** If the guitar plays everything all the time, there is no space for a voice. Fewer notes often sound better.

A complete worked example in G

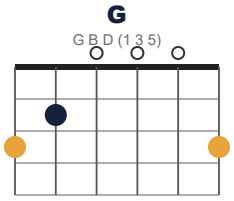
Let us write a short song idea from scratch using the steps above. Every note here has been checked against the key of G major: G A B C D E F#.

Step 1: pick the key

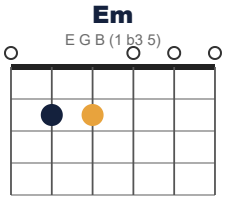
We choose G major. It is comfortable for most voices and has easy open chords.

Step 2: pick the progression

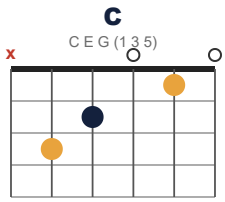
We choose I - vi - IV - V, which in G is G, Em, C, D. This is the verse. Here are the four chords.



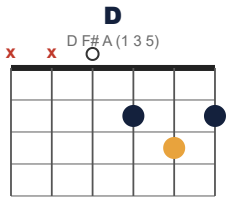
Bar 1: G (I). G B D.



Bar 2: Em (vi). E G B.



Bar 3: C (IV). C E G.



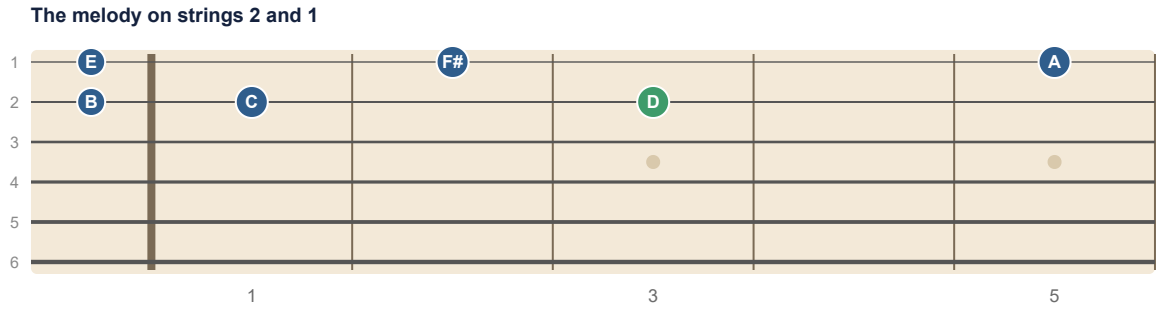
Bar 4: D (V). D F# A.

Step 3: write a motif and build the melody

Our motif is the shape low - high - low using degrees 3, 5, 3: a gentle up-and-back that settles. Short, short, long works well rhythmically: two quick notes, then a held one. We then follow the repeat, repeat, change plan.

BAR	CHORD	MELODY DEGREES	MELODY NOTES	WHY IT WORKS
1	G	3 5 3	B D B	All chord tones of G. This is the motif.
2	Em	6 5 3	E D B	Starts on the root of Em, then steps down. Echoes the motif.
3	C	4 6 4	C E C	The motif moved up one step. All chord tones of C.

BAR	CHORD	MELODY DEGREES	MELODY NOTES	WHY IT WORKS
4	D	2 7 5	A F# D	Steps down through the chord tones of D, setting up the return to G.



All six notes of the melody within the first five frets: B, D, E, C, A and F#. Bar 1 uses B (string 2, open) and D (string 2, fret 3). Bar 4 comes down from A (string 1, fret 5) to F# (string 1, fret 2) and then D.

Play it slowly, bar by bar: B D B, then E D B, then C E C, then A F# D. Strum the chord underneath, or have a friend play it. Sing the degrees as you go: three, five, three; six, five, three; four, six, four; two, seven, five.

Step 4: add a chorus and a bridge

For contrast, the chorus can use C - G - D - Em (IV, I, V, vi), starting on the lift of the IV instead of home. The bridge can use Em - D - C - D (vi, V, IV, V), ending on the V so the return of the verse feels like coming home. Reuse the same motif in each section with small changes: this is what makes the song sound like one piece.

Step 5: test it

Hum the melody over the chords. Check that no note feels wrong. If one does, move it up or down to the nearest chord tone. Then record a voice memo. You have just written a song using nothing but what is in this book.

> **Try it in another key**
>
> Using the table in the transcribing section, play the same four bars in C or D. If your fingers can do it, you truly understand the pattern, not just the notes.

Keep an idea notebook

Ideas are fragile. A short riff you hum in the car or find while warming up can vanish in minutes. A notebook, paper or digital, keeps them alive.

- **Capture first, judge later.** Record ten seconds on your phone the moment an idea shows up.
- **Label it.** Write the date, the key, the progression in numbers, and one line describing the feel.

- **Keep separate lists.** Riffs, progressions, melodies (as degrees), lyric lines and song titles each get their own page.
- **Review weekly.** Every Sunday, listen to the week's ideas and mark one to develop. Most players find that ideas combine in surprising ways when they sit next to each other.
- **Finish small things.** A finished eight-bar idea teaches more than ten unfinished songs.

DATE	KEY	PROGRESSION (NUMBERS)	FEEL	MELODY (DEGREES)
Example	G	I vi IV V	Warm, hopeful	3 5 3 | 6 5 3 | 4 6 4 | 2 7 5

Common mistakes

- **Starting with the whole song.** Work one section at a time. Four bars is a win.
- **Skipping the key.** Guessing chords before you know the home note leads to endless trial and error.
- **Trusting the fingers over the ear.** If your hands say yes but your ear says no, believe your ear.
- **Using full speed from the start.** Slow it down and loop it.
- **Testing only one quality per root.** Always try major and minor on each root.
- **Too many ideas in a single song.** One motif, repeated and varied, beats five unrelated riffs.
- **Waiting for inspiration.** Set a timer for ten minutes and write something every day. Most players find that quantity leads to quality.
- **Throwing away unfinished ideas.** Put them in the notebook. They often become something later.

Module 11 cheat sheet

TOPIC	RULE OF THUMB
Relative pitch	Hear distance (intervals) and function (the job a note or chord does in the key).
Home note	The note that feels like rest. Find it by humming and stopping, then check the last chord.
Major or minor	Play the major triad and the minor triad on the home note. The one that fits is the answer.
Roots	The lowest note of each chord. Walk string 6 or 5 to match the bass.
Numbers	I home, IV lift, V tension, vi sad relative, bVII rock move.
5-step method	Pick a section, find the key, find the roots, find the quality, test with the melody.
Slow-down tool	Loop 2 to 4 bars, start at 60 to 75 percent speed, bring it up gradually.
Melody	Write it as degrees (3 5 3 | 6 5 3). Move to any key by changing the home note.
Writing	Key, progression, motif, repeat, change, contrast. Chorus first if you are stuck.
Structure	Verse, chorus, verse, chorus, bridge, chorus.

NUMBER	IN G	IN C	IN D	IN A	IN E
I	G	C	D	A	E
IV	C	F	G	D	A
V	D	G	A	E	B
vi	Em	Am	Bm	F#m	C#m
bVII	F	Bb	C	G	D

Module 11 glossary

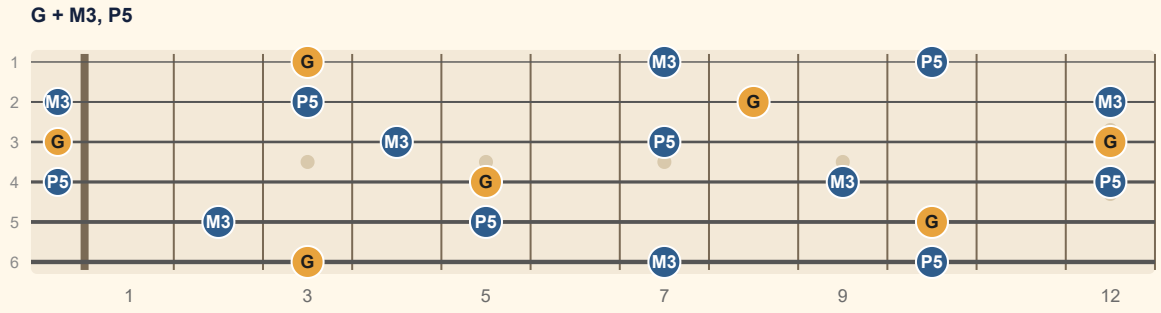
TERM	PLAIN-ENGLISH MEANING
Ear training	Practice that teaches your ears to recognize intervals, chords and melodies.
Relative pitch	Hearing how notes relate to each other, not their absolute names.
Home note (tonic)	The note of the key that feels like rest. Degree 1.
Root	The note a chord is named after and built on. Usually the bass note.
Quality	Whether a chord is major, minor, dominant seventh, and so on.
Scale degree	The number of a note in the scale: 1 to 7.
Function	The job a note or chord does inside a key, such as home or tension.
Transcribe	To work out a piece of music and write it down.
Loop	A short section played over and over.
Motif	A very short musical idea that a melody is built from.
Sequence	A motif repeated at a higher or lower step.
Verse / chorus / bridge	The story section, the main message section and the contrasting section.
Hook	The catchiest part of a song, the part you remember.

On the fretboard: Module 11 exercises

EXERCISE 51 — 5 MIN

Interval of the day

1. Choose one interval from this list: m3, M3, P4, P5, M6.
2. Play G on string 6, fret 3, then sing the note you think is that interval above it.
3. Play the answer on the guitar and see whether your voice matched.
4. Repeat with three different starting notes.



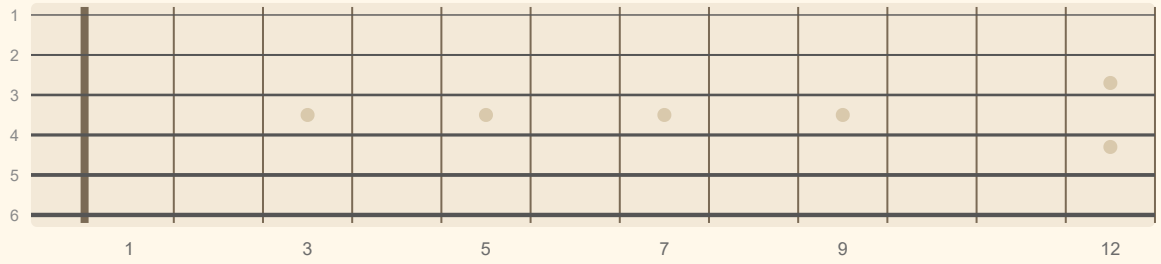
M3 and P5 from G.

☐ Self-check: you can sing the interval before you play it at least 7 times out of 10, and you can name a tune that starts with it. For G, the M3 is B and the P5 is D.

EXERCISE 52 — 10 MIN

Find the home note

1. Choose a song you know well, and play it or sing it.
2. Hum along, stop the music, and keep humming the note that feels like rest.
3. Walk up string 6 from fret 0 until you match it.
4. Play the major triad and then the minor triad on that note. Decide which one fits.



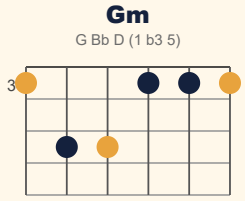
Mark your home note in the places where you can find it on strings 6, 5 and 4.

☐ Self-check: the home note matches the end of the melody, the final chord has the same root, and the triad you chose sounds like it belongs. If you picked G, the choices are G B D or G Bb D.

EXERCISE 53 10 MIN

Chord quality quiz

1. Play a major or minor chord at random without looking at which one you chose (or ask someone to play it).
2. Say major or minor out loud.
3. Check by looking at the third: major has a 3, minor has a b3.
4. Do 20 rounds and count the score.



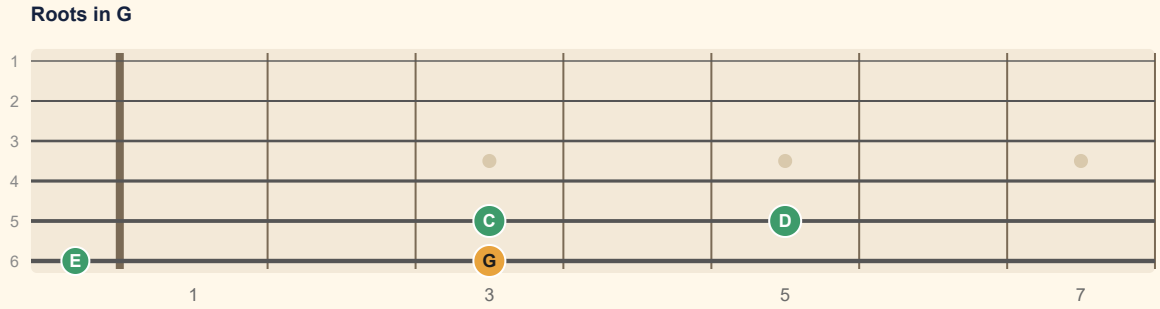
G minor: compare it with G major to hear the b3.

☐ Self-check: you get at least 16 out of 20 right. When you reach that score, add dominant seventh as a third choice.

EXERCISE 54 20 MIN

Work out a four-bar section

1. Pick four bars of a song and loop them at about 70 percent speed.
2. Find the key, then the roots, one per bar, written as note names.
3. Find the quality of each chord with the major and minor test.
4. Write the progression in numbers and sing the melody over your chords.



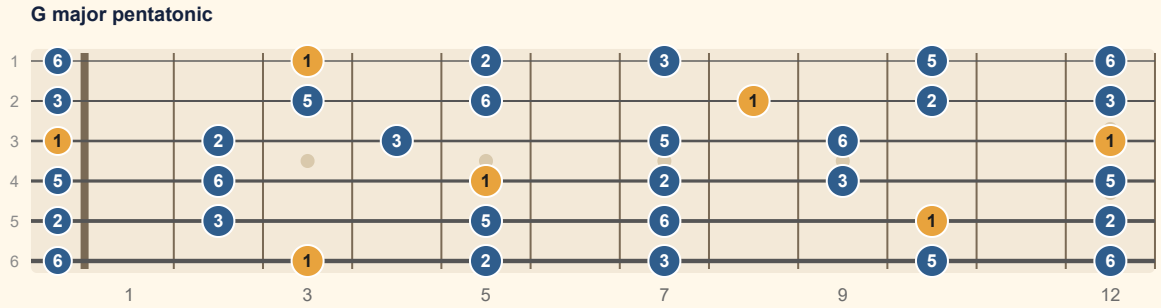
Use these as a check if your song is in G.

☐ Self-check: your four chords sound right when played with the recording, and you have the numbers written down (for example, I vi IV V). If the roots were G, E, C, D, the numbers are I, vi, IV, V.

EXERCISE 55 20 MIN

Write an eight-bar song idea

1. Choose a key and a progression of four chords from this module.
2. Write a three-note motif as scale degrees and play it over the first chord.
3. Repeat it over the second chord, then move it up a step on the third, and end with something different on the fourth.
4. Record the result on your phone and add it to your idea notebook with the date, key and numbers.



G major pentatonic: a very safe set of notes for a first melody.

☐ Self-check: the melody starts and ends on chord tones, you can play it twice in a row from memory, and you have it written as degrees. A safe choice in G is the major pentatonic: G A B D E.

Next step

You have now gone from seeing the fretboard to building scales, chords and progressions, and to hearing and writing music of your own. The only thing left is to keep it going. Open the 90-day plan that comes with this book and follow it a little each day: it brings together the exercises from all eleven modules into a steady routine, with short sessions that fit a busy week.

Today, before you put the guitar down, do just one thing: pick a song you love, loop four bars, and find the home note. That is the first step of everything in this module.

The 26-week practice plan

> *This plan walks you through the whole book in 26 weeks, about six months, at ten minutes a day, five days a week. It is split into two stages of 13 weeks each, about 90 days apiece. You will spend one to three weeks on each module.*

The book contains 55 exercises, numbered 1 to 55, and every module has five. Module m uses the exercises from 5(m-1)+1 to 5m. So Module 1 is exercises 1 to 5, Module 2 is 6 to 10, Module 3 is 11 to 15, and so on until Module 11, which is 51 to 55.

Every day, the same rhythm

- 2 minutes: warm up with the cheat sheet of the current module.
- 6 minutes: work the day's exercise, following the steps.

- 2 minutes: do the self-check, and mark the exercise as done when you pass it.
- Take two days off per week. Rest helps you remember.

Stage 1: weeks 1 to 13

WEEK	MODULE	DAILY FOCUS (DAYS 1 TO 5)	EXERCISES
1	1	Name the open strings, then every natural note on strings 6 and 5. Close each day by naming ten random frets.	1, 2
2	1	Octave shapes between strings, and the anchors that make any note findable in a few seconds.	3, 4
3	1	Days 1 to 3: finish the module. Day 4: retake questions 1 to 3 of the test. Day 5: read the cheat sheet.	5
4	2	Count half steps and whole steps. Find the minor third, major third and perfect fifth from a root.	6, 7, 8
5	2	Hear and play the intervals. Finish the module and review its cheat sheet.	9, 10
6	3	Build the major scale from the pattern W W H W W W H. Start with C, then G.	11, 12
7	3	Spell the major scale from other roots and see how sharps and flats come out of the pattern.	13, 14
8	3	Relative minor. Finish the module and review the cheat sheet.	15
9	4	The seven modes as colors: play Dorian and Mixolydian against a single root note.	16, 17, 18
10	4	Lydian, Phrygian and the others. Review the cheat sheet and compare each mode's degrees with the major scale.	19, 20
11	5	Build triads from the major scale: 1, 3, 5. Name every note of the chords you already play.	21, 22, 23
12	5	Major, minor, sus2, sus4 and diminished triads. Finish the module.	24, 25
13	6	Seventh chords: maj7, 7 and m7. Take the dominant seventh from the 5th degree.	26, 27, 28

Stage 2: weeks 14 to 26

WEEK	MODULE	DAILY FOCUS (DAYS 1 TO 5)	EXERCISES
14	6	Tensions and color tones (the 9, 11 and 13). Finish the module and retake the whole test.	29, 30
15	7	CAGED: the first two shapes of the system and how they connect.	31, 32

WEEK	MODULE	DAILY FOCUS (DAYS 1 TO 5)	EXERCISES
16	7	The other shapes. Chain all five along the neck in one key.	33, 34
17	7	Move the system to a second key. Review the cheat sheet.	35
18	8	Number the chords of a key with Roman numerals. Play I, IV and V in several keys.	36, 37, 38
19	8	The common progressions and why they work. Finish the module.	39, 40
20	9	Read the circle of fifths: neighbors, sharps and flats. Count the sharps or flats in each key.	41, 42, 43
21	9	Modulation: move from one key to a close neighbor.	44, 45
22	10	Improvise with intention using the minor pentatonic scale in one position.	46, 47
23	10	Target notes: land on the chord tones. Connect two positions.	48, 49
24	10	Phrases and space. Finish the module and review the cheat sheet.	50
25	11	Work out a short song by ear. Find its key, its root note and its chords.	51, 52, 53
26	11	Write your own short progression and melody. Finish the module and retake the full test.	54, 55

> **If you fall behind**
>
> Add a week to the module you are on. The plan is a guide, not a deadline. Most players find that finishing the book matters much more than finishing it fast.

The express route

If your time is short, follow the express route: Modules 1, 3, 7 and 10. On the same ten-minutes-a-day schedule, that takes about 12 weeks. Use the same weekly rhythm as above, and go back to the other modules when you want more detail.

WEEKS	MODULE	EXERCISES
1 to 3	1. The fretboard stops being a maze	1 to 5
4 to 6	3. Scales and keys: the master rule	11 to 15
7 to 9	7. CAGED: the whole fretboard opens up	31 to 35
10 to 12	10. Improvising with intention	46 to 50

Master glossary

Every term in the book, in plain words. Entries are in alphabetical order.

TERM	IN PLAIN WORDS
Accidental	A sharp or flat that changes a note by a half step.
Arpeggio	The notes of a chord played one at a time instead of together.
Barre	Using one finger to press several strings across the same fret.
Blues scale	The minor pentatonic scale plus one extra note, the b5: 1 b3 4 b5 5 b7.
Cadence	A short chord move that ends a phrase, like V to I.
CAGED	Five chord shapes (C, A, G, E, D) that link up across the neck and cover every key.
Chord	Three or more notes played together.
Chord tone	A note that belongs to the chord you are playing.
Chromatic	Using all twelve notes, each one fret apart.
Circle of fifths	A circle that shows the twelve keys, each a perfect fifth from its neighbor.
Degree	The number of a note in a scale: 1 is the root, 2 is the next note, and so on.
Diatonic	Belonging to a key. Diatonic chords and notes are the ones that come from the key's scale.
Diminished chord	A chord of 1, b3 and b5. It sounds tense.
Dominant seventh	A chord of 1, 3, 5 and b7, built on the 5th degree of a major key.
Dorian	A minor-sounding mode: 1 2 b3 4 5 6 b7.
Enharmonic	Two names for the same pitch, such as F# and Gb.
Fret	The metal strip on the neck, and by extension the space behind it. One fret is one half step.
Half step	The smallest distance on the guitar: one fret.
Harmonic minor	The natural minor scale with the 7th raised a half step: 1 2 b3 4 5 b6 7.
Interval	The distance between two notes.
Inversion	A chord with a note other than the root as the lowest note.
Key	The home of a song: the scale and the chords that belong to it.
Lick	A short, memorable phrase you can play in a solo.
Locrian	The mode with 1 b2 b3 4 b5 b6 b7. It sounds unstable.

TERM	IN PLAIN WORDS
Lydian	A bright major-sounding mode: 1 2 3 #4 5 6 7.
Major chord	A chord of 1, 3 and 5.
Major scale	The scale built with W W H W W W H, the reference for all the others.
Major seventh	A chord of 1, 3, 5 and 7. It sounds smooth and dreamy.
Melodic minor	The minor scale with a raised 6th and 7th: 1 2 b3 4 5 6 7 (as it is usually played going up).
Minor chord	A chord of 1, b3 and 5.
Minor seventh	A chord of 1, b3, 5 and b7.
Minor scale	The natural minor scale: 1 2 b3 4 5 b6 b7.
Mixolydian	A major-sounding mode with a flat seventh: 1 2 3 4 5 6 b7.
Mode	A scale made by starting a parent scale on a different note.
Modulation	Changing from one key to another within a song.
Muted	A string that is deadened or not played.
Natural	A note that is neither sharp nor flat.
Octave	The same note name 12 frets higher or lower. A distance of eight scale steps.
Open chord	A chord that uses open strings.
Passing note	A note that connects two chord tones on the way through.
Pentatonic	A five-note scale. The minor pentatonic is 1 b3 4 5 b7. The major pentatonic is 1 2 3 5 6.
Perfect fifth	An interval of 7 half steps, such as C to G.
Phrygian	A dark minor-sounding mode: 1 b2 b3 4 5 b6 b7.
Position	A spot on the neck where a scale or chord shape sits under your hand.
Power chord	Two notes: the root and the fifth, and often the octave.
Progression	A series of chords played in order.
Relative minor	The minor key that shares its seven notes with a major key. It starts on the 6th degree.
Resolution	Moving from a tense sound to a stable one.
Riff	A repeating short musical pattern, often the hook of a song.
Root	The note a chord or scale is named after.

TERM	IN PLAIN WORDS
Scale	A set of notes in a fixed order of whole steps and half steps.
Seventh chord	A chord that adds a seventh above the root to a triad.
Sharp and flat	A sharp raises a note one half step. A flat lowers it one half step.
Slash chord	A chord with a note other than the root in the bass, such as C/E.
Sus chord	A chord where the 3rd is replaced. Sus2 is 1 2 5 and sus4 is 1 4 5.
Target note	A note you aim for, usually a chord tone, when improvising.
Tension	A note or chord that makes you want to move to another one.
Tonic	The home chord or note of the key, the I.
Transposing	Moving a song or a shape to a different key.
Triad	A three-note chord built from the 1, 3 and 5.
Tritone	An interval of six half steps, such as B to F.
Whole step	Two half steps: two frets.

Frequently asked questions

Does this work on acoustic and electric guitar?

Yes. The notes, intervals, scales and chords are exactly the same on both. The only differences are feel and sound: an electric usually has lighter strings and a lower string height, which makes barre chords and bends easier. Nothing in this book depends on the type of guitar.

What if I use another tuning?

The diagrams and the note names assume standard tuning: E A D G B E. If you tune down a half step, the shapes stay the same and your pitches all sound a half step lower. The note names in the diagrams will then be one half step off from what you hear. In drop D, only the 6th string is lowered, from E to D, so every diagram is still right except for notes on that string. For the cleanest results, learn this book in standard tuning first.

I am left-handed. Can I use this book?

Yes. All diagrams use the standard right-handed view. If you play left-handed, mirror each diagram left to right, so the nut is on the right and the frets count toward the left. String 6 is still the thickest string and string 1 the thinnest.

Do I need to read music?

No. Nothing here uses standard notation. Notes are named with letters, scale steps with numbers, and everything is drawn as dots on the fretboard. If you can read a chord chart, you can read this book.

Can I use a capo?

Yes. A capo shifts everything up by the number of the fret it is on. A G shape with a capo on the 2nd fret sounds as A, and a capo on the 5th fret turns a C shape into an F. The shapes and the degrees stay the same.

Can I skip modules?

You can, with care. Use the diagnostic test to find your starting point and follow the express route if time is short. Part 1 is the foundation for everything that follows, so if a later module feels hard, go back and finish its cheat sheet.

What if I get stuck on an exercise?

Read the module's glossary, do the exercise again more slowly, and look at the diagram before the text. If it still does not click, move on to the next exercise and return to it in a week. Most players find that concepts that seemed hard become clear when they have more context.

Is ten minutes really enough?

Ten focused minutes done five days a week adds up to over 40 hours across the plan. With practice, many players find that regular short sessions work better than occasional long ones. If you have more time, spend it playing songs and applying what you learned.

What next: keeping your progress

Finishing the book is the beginning, not the end. The ideas become yours only when you use them. Here are three habits that help.

Keep a short journal

A notebook with one line per day is enough. Write the date, the exercise number, and one sentence on what you noticed. Over time it shows you what is getting easier, and it reminds you of what you want to revisit.

- Date and exercise number.
- One thing that clicked.
- One thing to repeat tomorrow.

Record yourself

Use your phone to record yourself every couple of weeks. Play the same thing each time: a progression from Module 8, an improvisation from Module 10, or a song you worked out in Module 11. Listening back is the fastest way to hear your own progress, and the mistakes you hear are the ones you can fix.

Play real songs with your new eyes

- Pick a song you know and name its key and the Roman numerals of its chords.
- Find the chord tones on the neck and play them as a melody.
- Transpose the song to another key using what you learned about shapes and the circle of fifths.
- Write a short progression and a melody of your own.

> **Come back to the test**
>
> Retake the diagnostic test every few months. Most players find that questions that once felt like guesses become quick and easy.

Checkpoints along the way

A plan works better with signposts. Use these checkpoints to make sure each stage has really sunk in before you move on. If you can do the task without looking at the book, you are ready.

AFTER MODULE	CHECKPOINT	YOU ARE READY IF YOU CAN...
1	Name any note	Name the note at any fret of strings 6 and 5 in a few seconds, and find a given note in at least three places.
2	Measure distances	Name the interval between two notes by counting half steps, for example 7 half steps is a perfect fifth.
3	Build a key	Spell the major scale from C, G, D and A without looking, and name the relative minor of each (A minor, E minor, B minor and F# minor).
4	Hear the colors	Name the degrees that separate Dorian, Mixolydian and Lydian from the major scale.
5	Spell chords	Name the notes of C, Am, G and D chords and say which are the 1, 3 and 5.
6	Add color	Explain how Cmaj7, C7 and Cm7 differ from each other by one note.
7	See the whole neck	Play one chord in five places on the neck and say which CAGED shape each place uses.
8	Read a progression	Name the chords of I IV V vi in C and G, and say what the numbers mean.

AFTER MODULE	CHECKPOINT	YOU ARE READY IF YOU CAN...
9	Change key	Use the circle of fifths to find the key one step clockwise (a perfect fifth up) and the key one step counterclockwise.
10	Improvise	Play a short phrase over a chord that lands on a chord tone.
11	Create	Work out the key and the chords of a simple song by ear.

The key checks above rely on some simple facts you can verify yourself. C major has no sharps or flats. G major has one sharp (F#). D major has two (F# and C#). A major has three (F#, C# and G#). Each step clockwise on the circle of fifths adds one sharp, and each step counterclockwise adds one flat: F major has one flat (Bb), and Bb major has two (Bb and Eb).

Common mistakes, and what to do instead

MISTAKE	WHAT TO DO INSTEAD
Rushing through the diagrams to get to the text.	Look at the diagram for a full minute first. Say what you see, then read.
Memorizing shapes without the degrees.	Always say the degrees (1, b3, 5) as you play. They are what let you move a shape to another key.
Skipping the self-check because it feels obvious.	Do it anyway. The moments you feel sure are exactly when it is worth proving it.
Practicing too long and burning out.	Stop at ten minutes. Leaving a little energy for tomorrow keeps the habit alive.
Moving on with a shaky module.	Add a week. A solid Module 3 or Module 7 helps every module that follows.
Playing with a guitar out of tune.	Tune before every session. Intervals only sound like themselves when the strings are in tune.

Mini-projects for each stage

Projects are optional, but they turn exercises into music. Pick one when you finish a part of the book.

AFTER PART	MINI-PROJECT	WHAT YOU PRACTICE
1. SEE	Name every note of a song you know as you play it, on one string at a time.	Finding notes quickly and hearing intervals.
2. BUILD	Write out the chords of two keys (C and G) with the notes of every chord.	Scales, keys and triads.

AFTER PART	MINI-PROJECT	WHAT YOU PRACTICE
3. CONNECT	Play a favorite progression in three different positions of the neck.	CAGED and the Roman numerals.
4. CREATE	Record a one-minute improvisation over a looped progression.	Targets, phrases and space.

A weekly review you can do in five minutes

1. Open the journal and read last week's lines.
2. Pick the exercise that felt hardest and do its self-check again.
3. Play one scale or chord shape from memory and say its degrees out loud.
4. Choose one thing to carry into next week.

Where to go from here

With the book finished, you have the vocabulary to learn from anything: a song, a lesson, a record, a friend. Whenever you hear a new chord or a new scale, you can now ask the right questions. What is the root? What are the degrees? Which shape is it? Which key does it belong to? Those four questions will take you further than any list of memorized patterns.

> **A word on pace**
>
> Every player moves at a different speed. Most players find that the ideas that felt slow at first become the ones they use without thinking. With practice, the neck becomes a map you can read.

Next step

Open the diagnostic test, find your starting module, and give it ten minutes today. Then give it ten minutes tomorrow. Tick off one exercise at a time, and let the plan carry you from there.

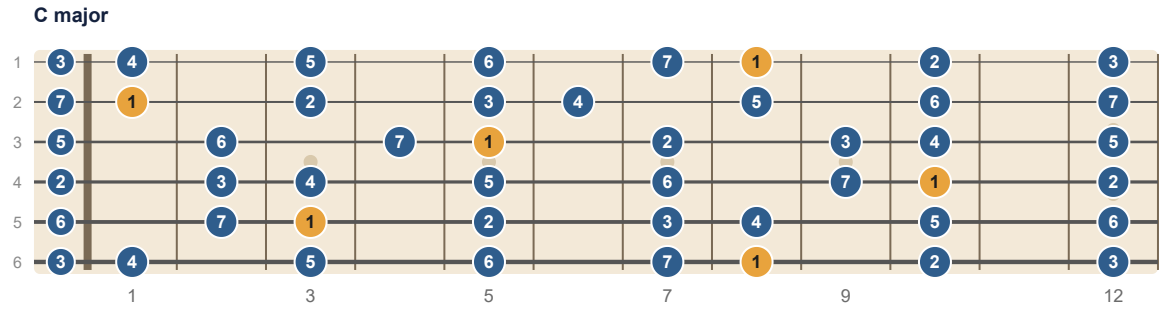
Key Atlas: the 12 major keys on the neck

A one-page map for each major key: the seven notes, the chords that belong to the key, its relative minor, and the scale on the whole neck with the degree of every note. Use it to check your work from any module.

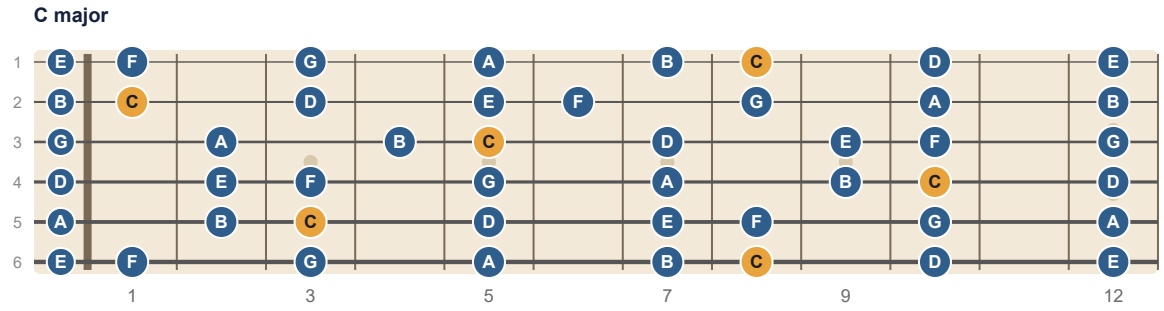
Key of C major

Notes: C D E F G A B | **Relative minor:** Am

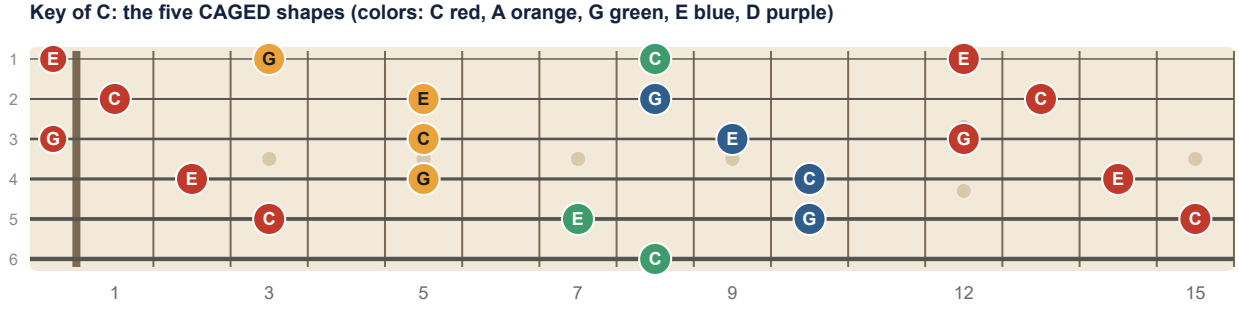
DEGREE	TRIAD	SEVENTH CHORD
I	C	Cmaj7
ii	Dm	Dm7
iii	Em	Em7
IV	F	Fmaj7
V	G	G7
vi	Am	Am7
vii dim	Bdim	Bm7b5



C major with scale degrees (orange = root)



C major with note names

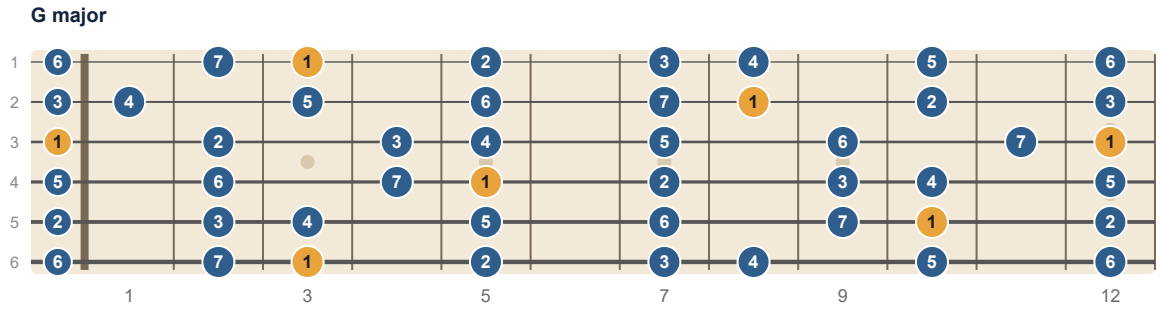


The five CAGED shapes for C

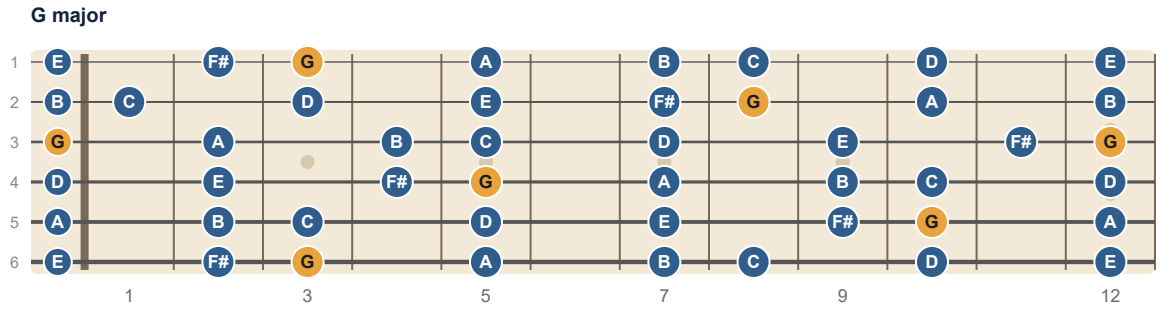
Key of G major

Notes: G A B C D E F# | **Relative minor:** Em

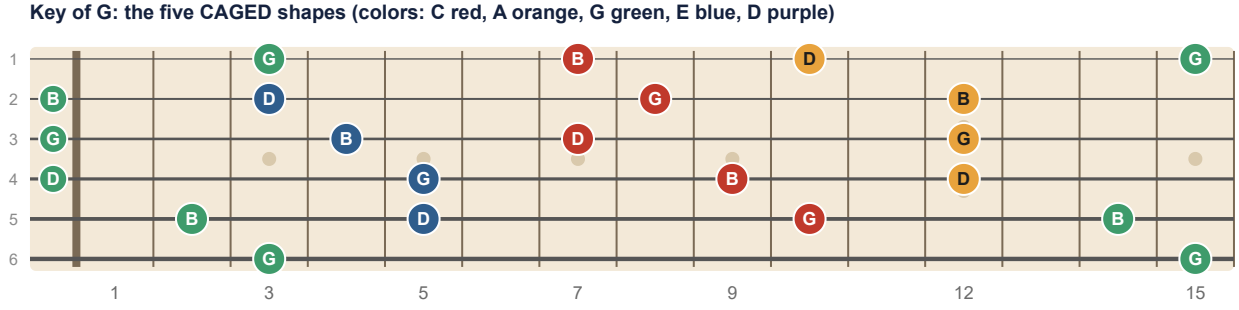
DEGREE	TRIAD	SEVENTH CHORD
I	G	Gmaj7
ii	Am	Am7
iii	Bm	Bm7
IV	C	Cmaj7
V	D	D7
vi	Em	Em7
vii dim	F#dim	F#m7b5



G major with scale degrees (orange = root)



G major with note names

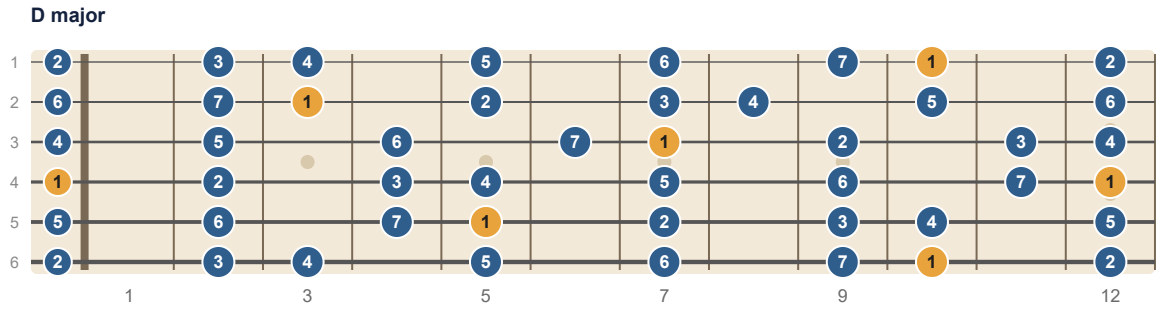


The five CAGED shapes for G

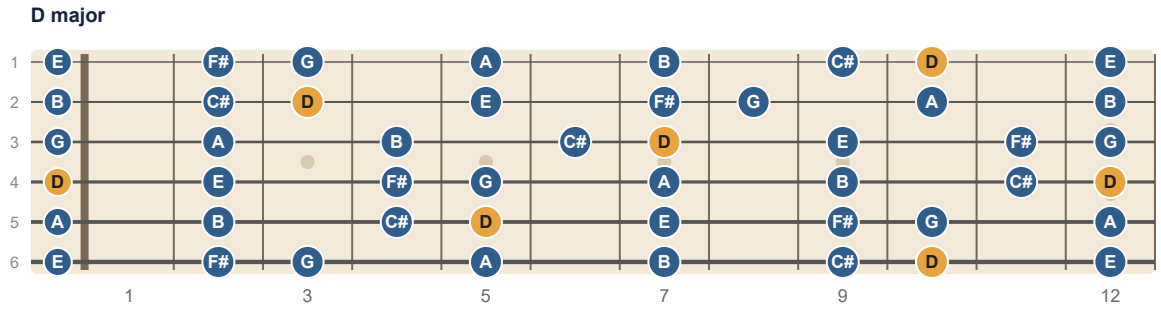
Key of D major

Notes: D E F# G A B C# | **Relative minor:** Bm

DEGREE	TRIAD	SEVENTH CHORD
I	D	Dmaj7
ii	Em	Em7
iii	F#m	F#m7
IV	G	Gmaj7
V	A	A7
vi	Bm	Bm7
vii dim	C#dim	C#m7b5

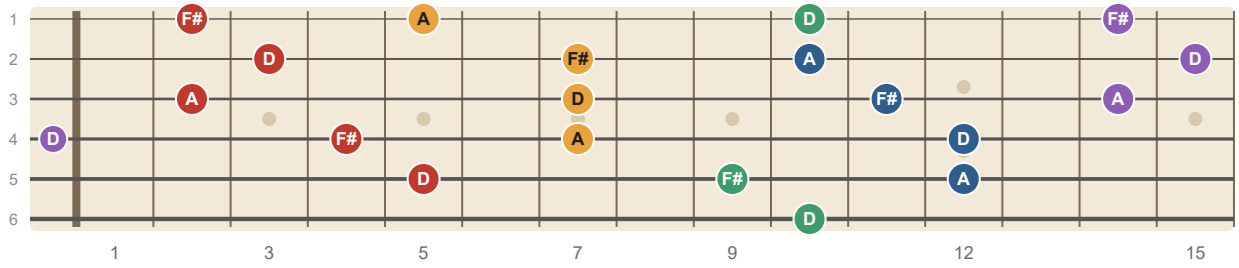


D major with scale degrees (orange = root)



D major with note names

Key of D: the five CAGED shapes (colors: C red, A orange, G green, E blue, D purple)

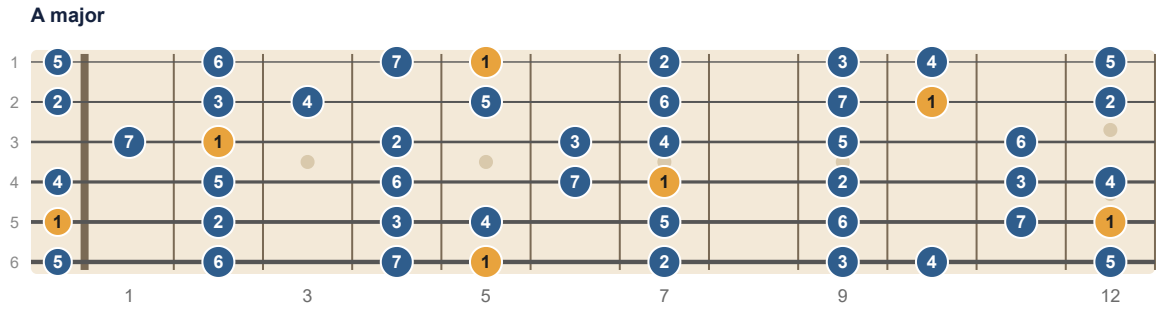


The five CAGED shapes for D

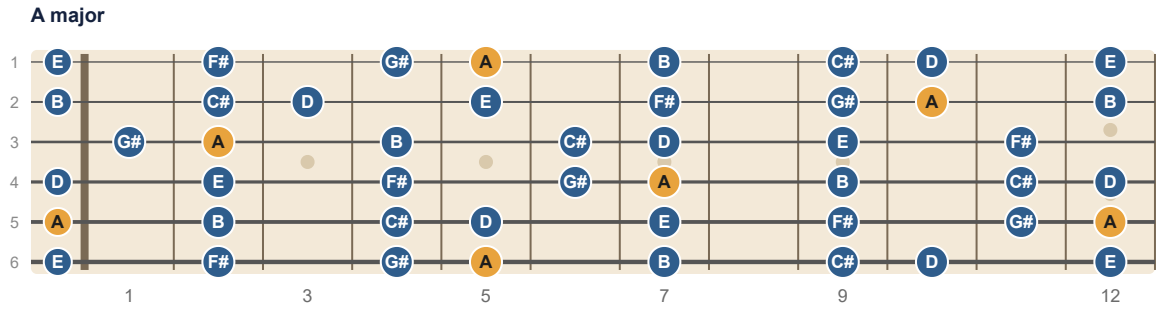
Key of A major

Notes: A B C# D E F# G# | **Relative minor:** F#m

DEGREE	TRIAD	SEVENTH CHORD
I	A	Amaj7
ii	Bm	Bm7
iii	C#m	C#m7
IV	D	Dmaj7
V	E	E7
vi	F#m	F#m7
vii dim	G#dim	G#m7b5

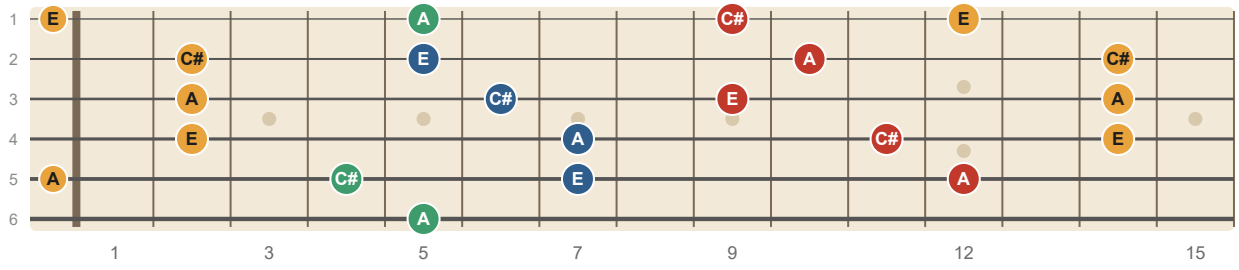


A major with scale degrees (orange = root)



A major with note names

Key of A: the five CAGED shapes (colors: C red, A orange, G green, E blue, D purple)

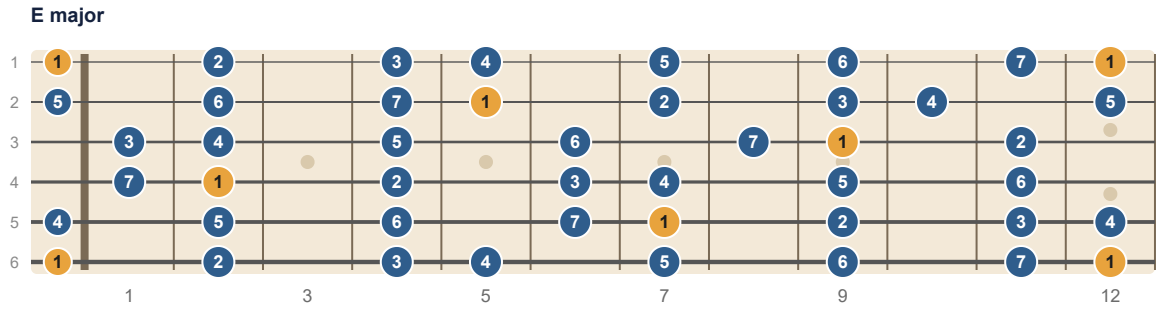


The five CAGED shapes for A

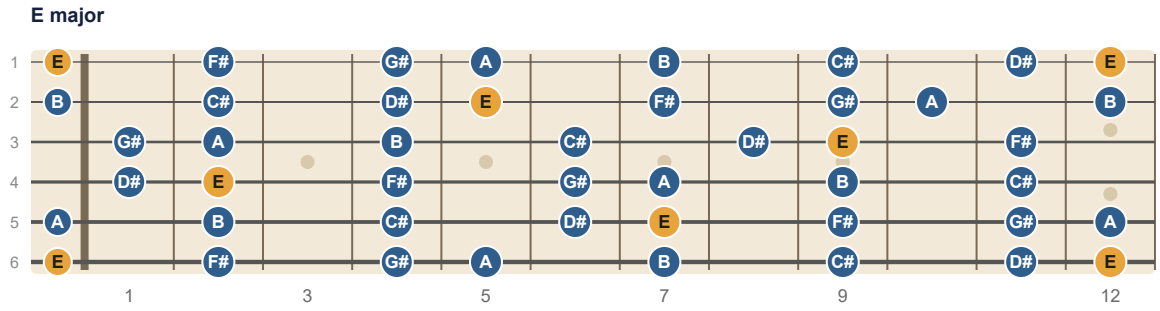
Key of E major

Notes: E F# G# A B C# D# | **Relative minor:** C#m

DEGREE	TRIAD	SEVENTH CHORD
I	E	Emaj7
ii	F#m	F#m7
iii	G#m	G#m7
IV	A	Amaj7
V	B	B7
vi	C#m	C#m7
vii dim	D#dim	D#m7b5

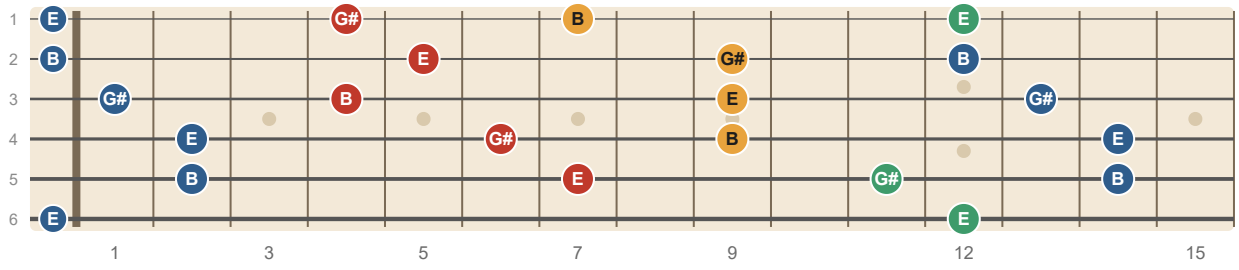


E major with scale degrees (orange = root)



E major with note names

Key of E: the five CAGED shapes (colors: C red, A orange, G green, E blue, D purple)

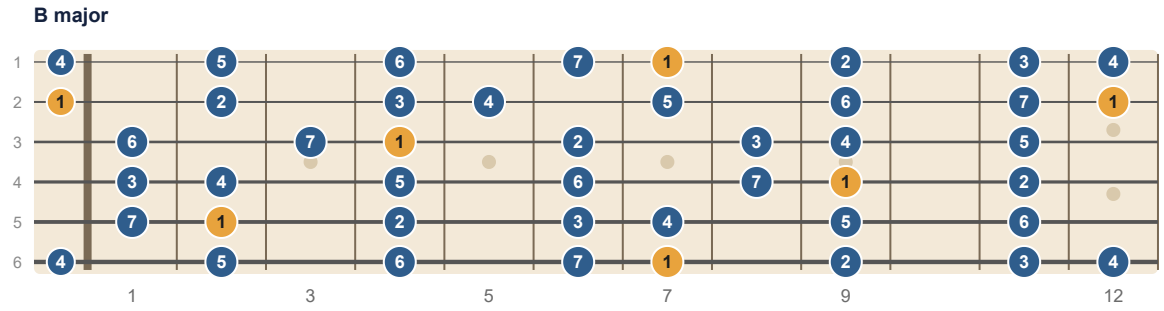


The five CAGED shapes for E

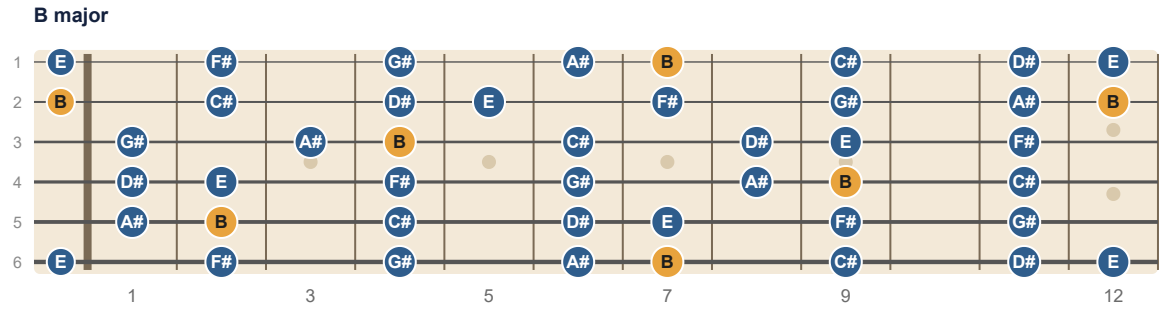
Key of B major

Notes: B C# D# E F# G# A# | **Relative minor:** G#m

DEGREE	TRIAD	SEVENTH CHORD
I	B	Bmaj7
ii	C#m	C#m7
iii	D#m	D#m7
IV	E	Emaj7
V	F#	F#7
vi	G#m	G#m7
vii dim	A#dim	A#m7b5

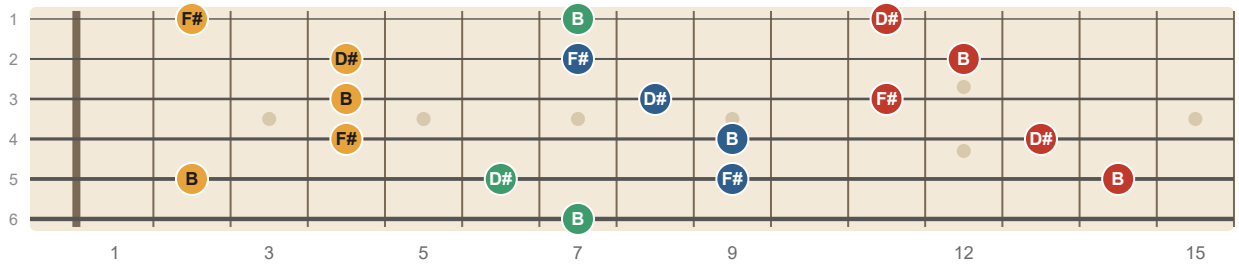


B major with scale degrees (orange = root)



B major with note names

Key of B: the five CAGED shapes (colors: C red, A orange, G green, E blue, D purple)



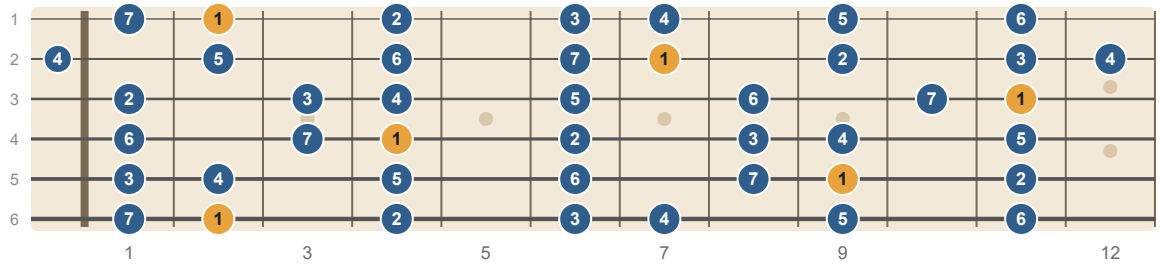
The five CAGED shapes for B

Key of F# major

Notes: F# G# A# B C# D# E# | **Relative minor:** D#m

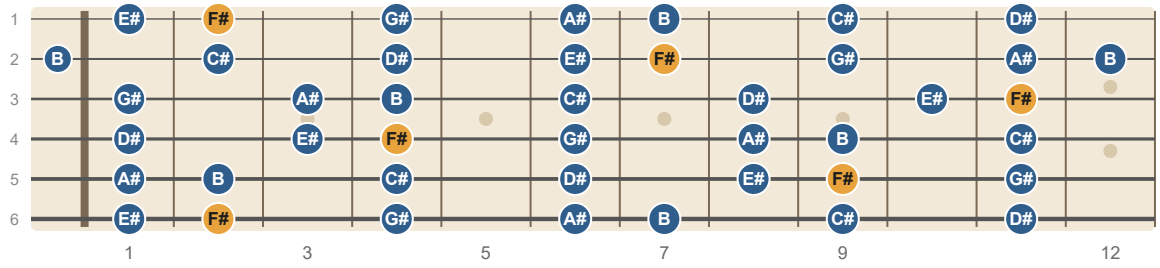
DEGREE	TRIAD	SEVENTH CHORD
I	F#	F#maj7
ii	G#m	G#m7
iii	A#m	A#m7
IV	B	Bmaj7
V	C#	C#7
vi	D#m	D#m7
vii dim	E#dim	E#m7b5

F# major



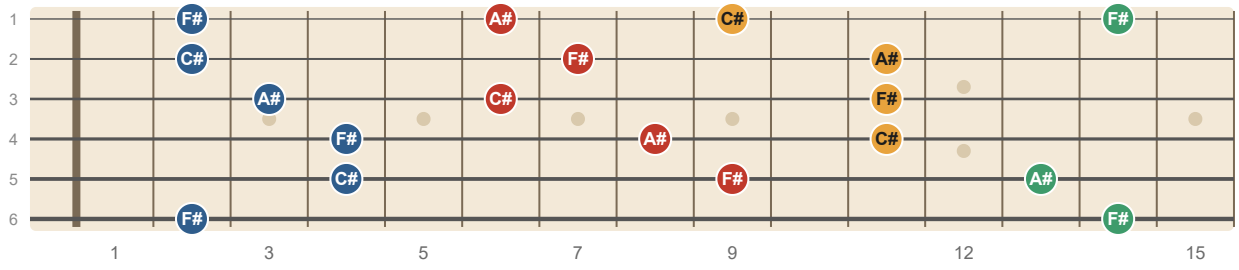
F# major with scale degrees (orange = root)

F# major



F# major with note names

Key of F#: the five CAGED shapes (colors: C red, A orange, G green, E blue, D purple)



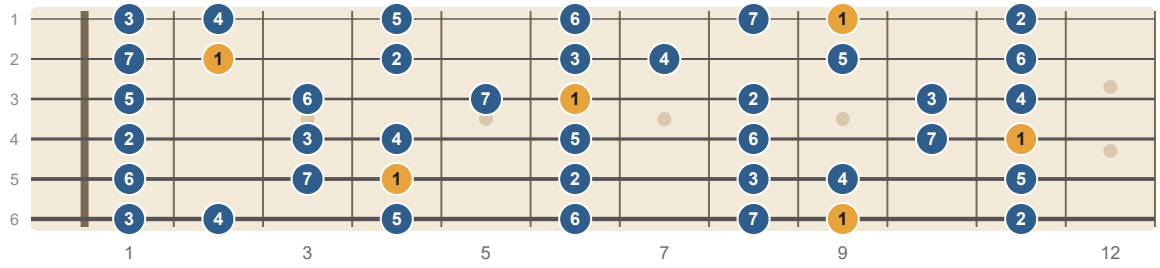
The five CAGED shapes for F#

Key of Db major

Notes: Db Eb F Gb Ab Bb C | **Relative minor:** Bbm

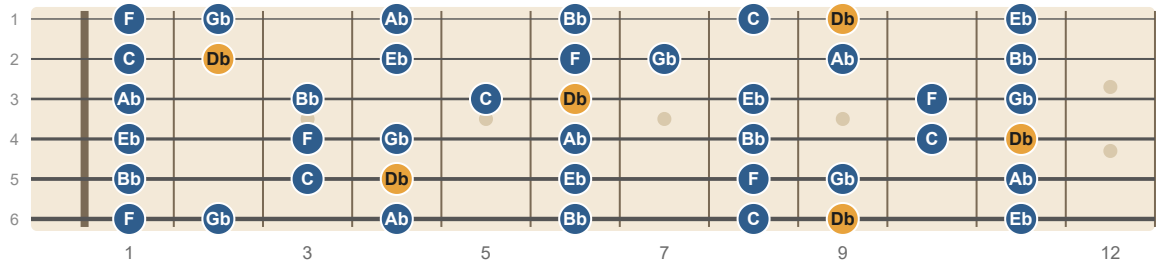
DEGREE	TRIAD	SEVENTH CHORD
I	Db	Dbmaj7
ii	Ebm	Ebm7
iii	Fm	Fm7
IV	Gb	Gbmaj7
V	Ab	Ab7
vi	Bbm	Bbm7
vii dim	Cdim	Cm7b5

Db major



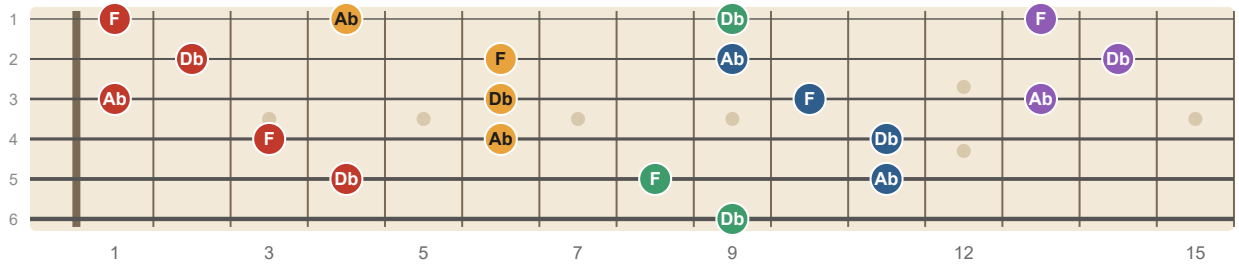
Db major with scale degrees (orange = root)

Db major



Db major with note names

Key of Db: the five CAGED shapes (colors: C red, A orange, G green, E blue, D purple)



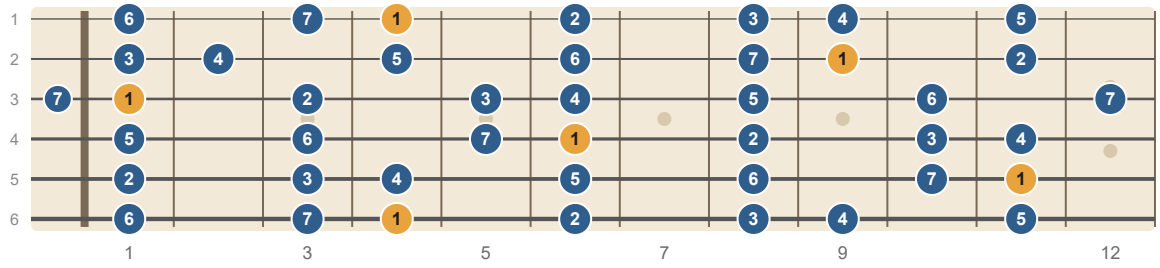
The five CAGED shapes for Db

Key of Ab major

Notes: Ab Bb C Db Eb F G | **Relative minor:** Fm

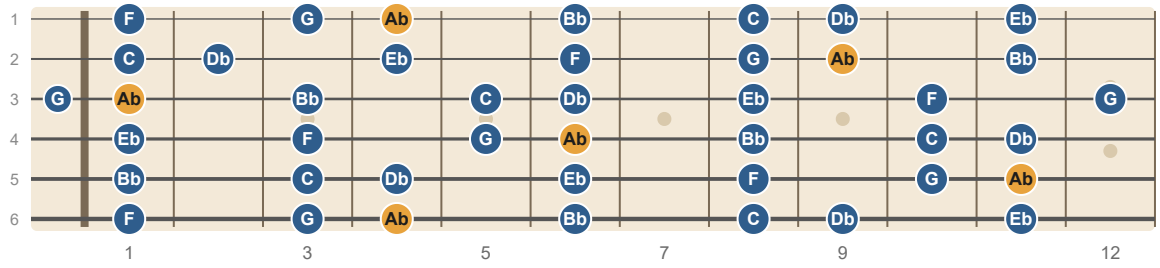
DEGREE	TRIAD	SEVENTH CHORD
I	Ab	Abmaj7
ii	Bbm	Bbm7
iii	Cm	Cm7
IV	Db	Dbmaj7
V	Eb	Eb7
vi	Fm	Fm7
vii dim	Gdim	Gm7b5

Ab major



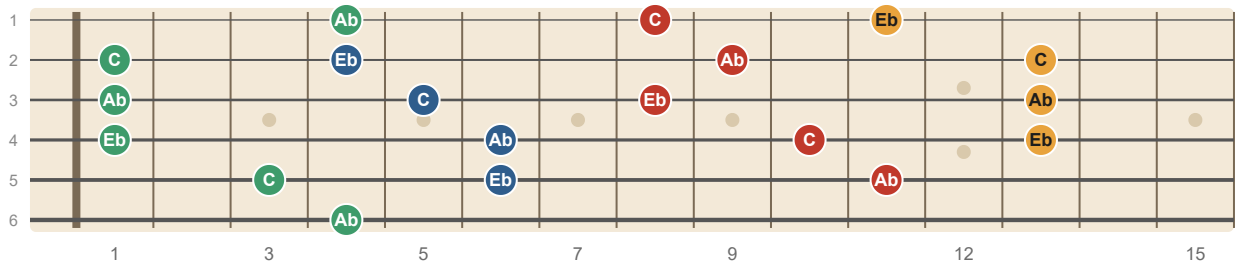
Ab major with scale degrees (orange = root)

Ab major



Ab major with note names

Key of Ab: the five CAGED shapes (colors: C red, A orange, G green, E blue, D purple)



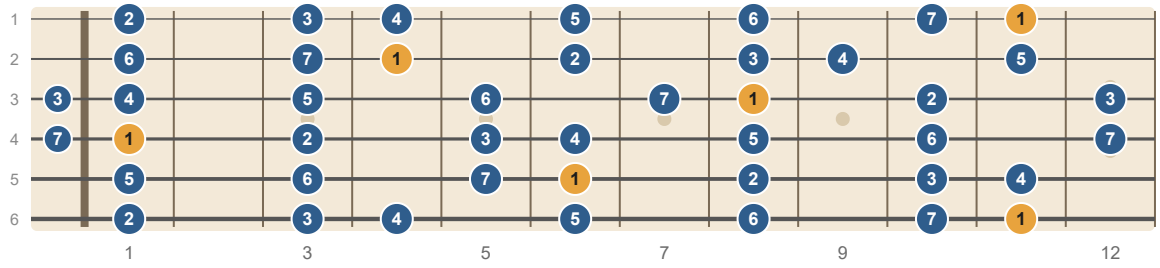
The five CAGED shapes for Ab

Key of Eb major

Notes: Eb F G Ab Bb C D | **Relative minor:** Cm

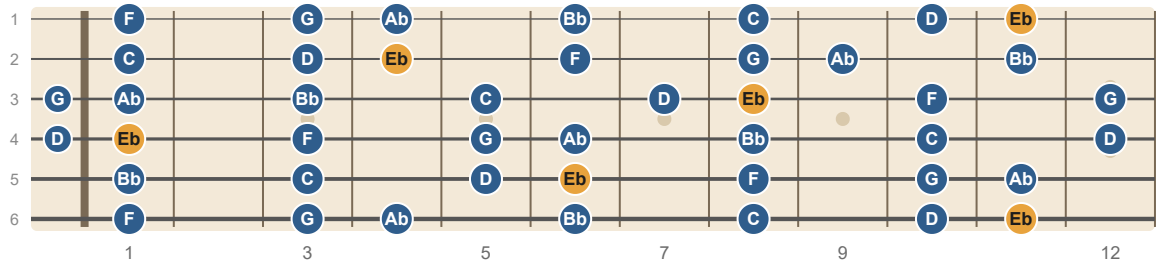
DEGREE	TRIAD	SEVENTH CHORD
I	Eb	Ebmaj7
ii	Fm	Fm7
iii	Gm	Gm7
IV	Ab	Abmaj7
V	Bb	Bb7
vi	Cm	Cm7
vii dim	Ddim	Dm7b5

Eb major



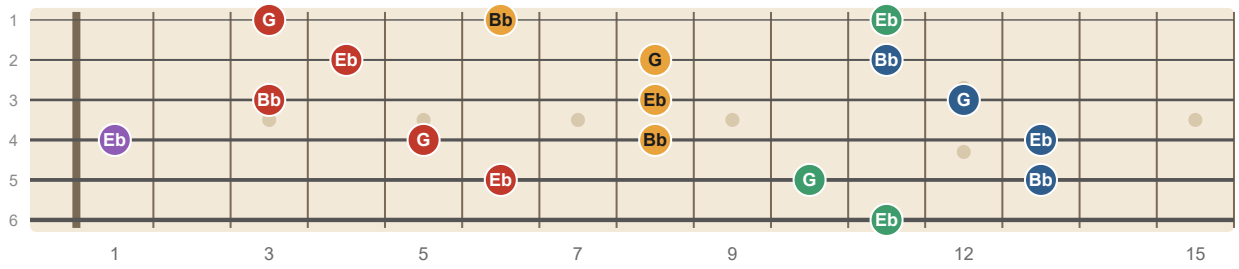
Eb major with scale degrees (orange = root)

Eb major



Eb major with note names

Key of Eb: the five CAGED shapes (colors: C red, A orange, G green, E blue, D purple)

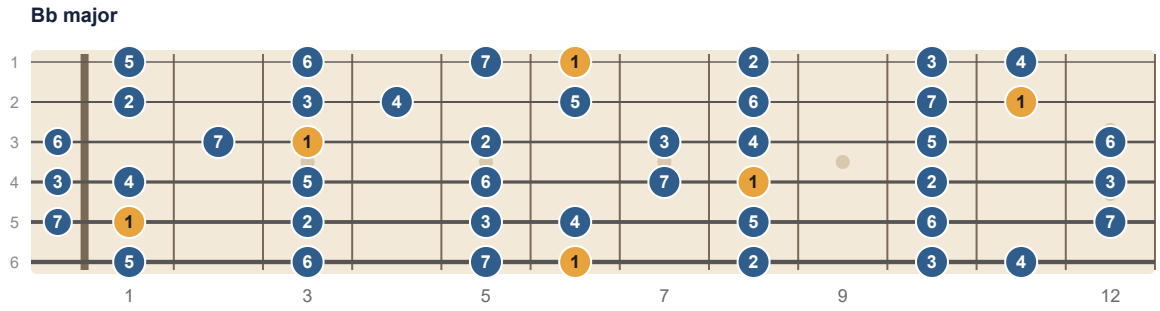


The five CAGED shapes for Eb

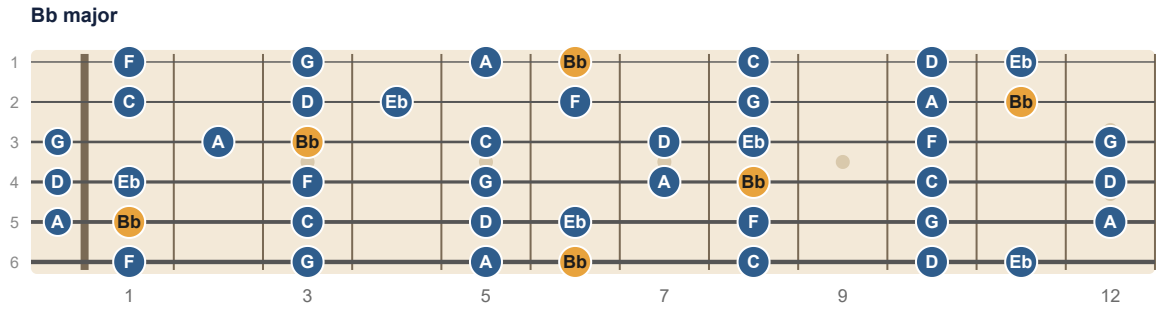
Key of Bb major

Notes: Bb C D Eb F G A | **Relative minor:** Gm

DEGREE	TRIAD	SEVENTH CHORD
I	Bb	Bbmaj7
ii	Cm	Cm7
iii	Dm	Dm7
IV	Eb	Ebmaj7
V	F	F7
vi	Gm	Gm7
vii dim	Adim	Am7b5

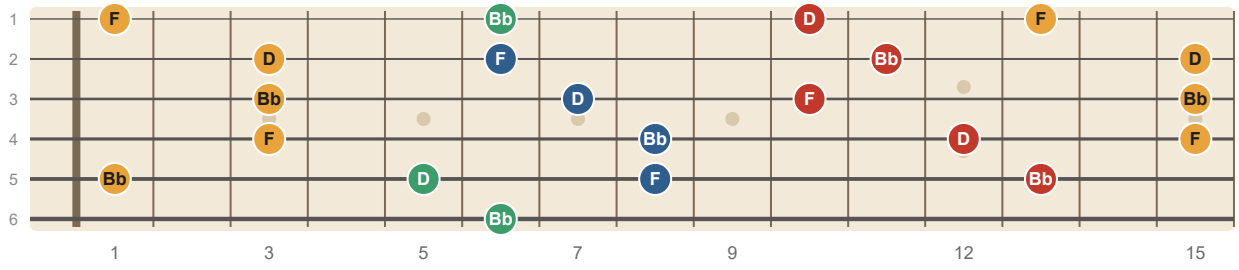


Bb major with scale degrees (orange = root)



Bb major with note names

Key of Bb: the five CAGED shapes (colors: C red, A orange, G green, E blue, D purple)

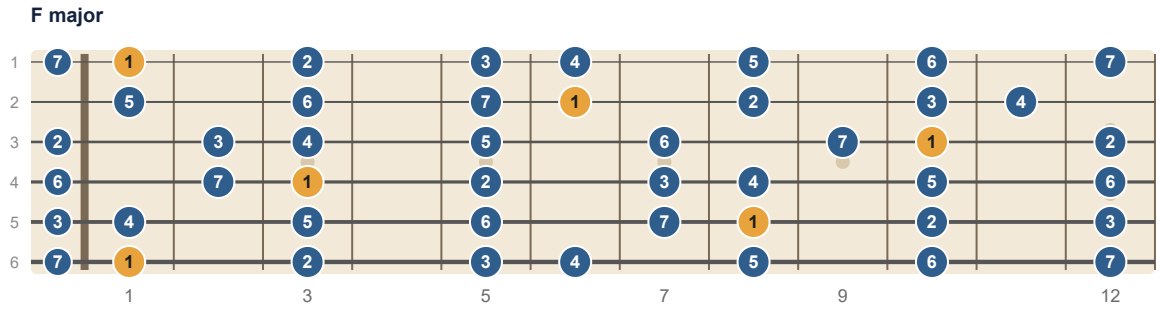


The five CAGED shapes for Bb

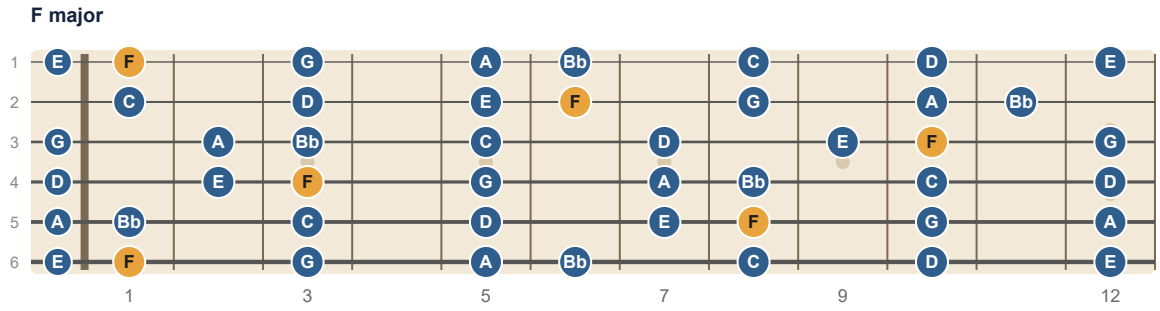
Key of F major

Notes: F G A Bb C D E | **Relative minor:** Dm

DEGREE	TRIAD	SEVENTH CHORD
I	F	Fmaj7
ii	Gm	Gm7
iii	Am	Am7
IV	Bb	Bbmaj7
V	C	C7
vi	Dm	Dm7
vii dim	Edim	Em7b5

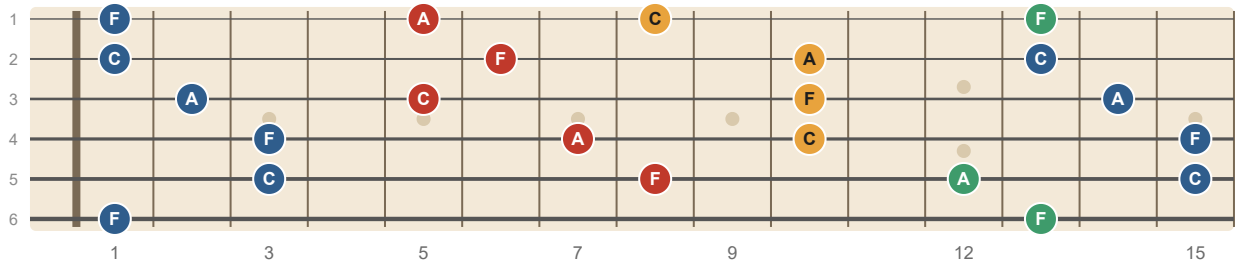


F major with scale degrees (orange = root)



F major with note names

Key of F: the five CAGED shapes (colors: C red, A orange, G green, E blue, D purple)



The five CAGED shapes for F